ENVIRONMENTAL
FINAL GOVERNING STANDARDS

SPAIN

MAY 2014

CNREURAFSWA Environmental
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DEPARTMENT OF THE NAVY
COMMANDER
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Ser N00/400
1 Oct 13

From: Commander Navy Region Europe, Africa, Southwest Asia
To: Commander, U.S. European Command (Attn: J4)

Subj: FINAL ENVIRONMENTAL GOVERNING STANDARDS FOR SPAIN (FGS-SPAIN)

Ref: (a) DoD Instruction 4715.5 of 22 Apr 96
(b) CNREURAFSWA/DLA e-mail of 12 Sept 13
(c) CNREURAFSA/IMCOM e-mail of 13 Sept 13
(d) CNREURAFSWA/HQ USAFE e-mail 23 May 13

Encl: (1) Environmental Final Governing Standards for Spain

1. Request U.S. European Command approval of enclosure (1) for publication. This new version incorporates applicable Spanish environmental laws.

2. Commander Navy Region Europe, Africa, Southwest Asia (CNREURAFSWA) as Environmental Executive Agent (EEA) for Spain, is required to update the FGS-SPAIN in accordance with reference (a). As outlined in references (b) through (d), Headquarters, United States Air Force Europe, Installation Management Command and Defense Logistics Agency have coordinated the updated FGS-SPAIN.

3. My point of contact is Ms. Kathryn Ostapuk at DSN 314-626-2921, commercial 011-39-081-568-2921, or email Kathryn.Ostapuk@eu.navy.mil.

A. E. GAIANI

Copy to:
DLA Installation Support Europe & Africa
IMCOM Environmental Europe
HQ USAFE/A7A

Revalidations and Updates *IAW DODI 4715.5, paragraph 6.3.6.*		
Action	**Date**	**Posted By**
Full Revision	May 2014	CNREURAFSWA

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C1. CHAPTER 1

OVERVIEW

C1.1. PURPOSE

The primary purpose of these Final Governing Standards (FGS) is to provide environmental compliance criteria at United States (U.S.) Department of Defense (DoD) installations in Spain. This document implements DoD Instruction 4715.5, “Management of Environmental Compliance at Overseas Installations,” dated 22 April 1996, and is based upon DoD 4715.05-G, “Overseas Environmental Baseline Guidance Document” (OEBGD), dated 1 May 2007.

C1.2. APPLICABILITY

C1.2.1. These FGS provide environmental compliance criteria applicable to actions for DoD Components at installations located in Spain.

C1.2.2. These FGS represent minimum criteria; DoD Components may impose additional criteria provided those policies and directives do not directly conflict with these FGS.

C1.2.3. Activities and installations shall notify the Environmental Executive Agent (EEA) of any directly conflicting DoD Component policies or directives they discover prior to imposing criteria more protective than provided in these FGS.

C1.2.4. Other than required authorizations for certain activities, DoD Components shall not enter into agreements with Spanish authorities at any level that establishes a criterion for compliance with an environmental criterion different than provided in these FGS without the prior written approval of the EEA.

C1.3. EXEMPTIONS

C1.3.1. These FGS do not apply to:

C1.3.1.1. DoD installations that do not have more than *de minimis* potential to affect the natural environment (e.g., offices whose operations are primarily administrative, including defense attaché offices, security assistance offices, foreign buying offices, and other similar organizations), or for which the DoD Components exercise control only on a temporary or intermittent basis.

C1.3.1.2. Leased, joint use, and similar facilities to the extent that the DoD does not control the instrumentality or operation that a criterion seeks to regulate.

C1.3.1.3. Operations of U.S. military vessels or the operations of U.S. military aircraft, or off-installation operational and training deployments. Off-installation operational deployments include cases of hostilities, contingency operations in hazardous areas, and when U.S. forces are operating as part of a multi-national force not under full control of the United States. Such excepted operations and deployments shall be conducted in accordance with applicable international agreements, other DoD Directives (DoDDs) and DoD Instructions (DoDIs), and environmental annexes incorporated into operation plans or operation orders. However, these FGS apply to support functions for U.S. military vessels and U.S. military aircraft provided by the DoD Components, including management or disposal of off-loaded waste or material.

C1.3.1.4. Facilities and activities associated with the Naval Nuclear Propulsion Program, which are covered under Executive Order (E.O.) 12344 "Naval Nuclear Propulsion Program," and conducted pursuant to 42 United States Code (U.S.C.) 7158.

C1.3.1.5. The determination or conduct of remediation to correct environmental problems caused by the Department of Defense's past activities, conducted in accordance with DoD Instruction (DoDI) 4715.8, "Environmental Remediation Overseas."

C1.3.1.6. Environmental analyses conducted under E.O. 12114, "Environmental Effects Abroad of Major Federal Actions."

C1.4. DEFINITIONS

For purposes of these FGS, unless otherwise indicated, the following definitions apply:

C1.4.1. Criteria and Management Practices. Particular substantive provisions of the OEBGD that are used by the EEA to develop these FGS.

C1.4.2. Existing Facility. Any facility and/or building, source, or project in use or under construction before 1 October 1994, unless it is subsequently substantially modified.

C1.4.3. Final Governing Standards. A comprehensive set of country-specific substantive provisions, typically technical limitations on effluent, discharges, etc., or a specific management practice.

C1.4.4. Host Nation. Throughout this document the term "host nation" refers to Spain.

C1.4.5. Local National. A DoD employee hired under Spanish employment conditions, including local subcontractors' employees.

C1.4.6. New Facility. Any facility and/or building, source, or project with a construction start date on or after 1 October 1994, or a pre-existing facility that has been substantially modified since 1 October 1994.

C1.4.7. Substantial Modification. Any modification to a facility and/or building the cost of which exceeds $1 million, regardless of funding source.

C1.5. ADDITIONAL INFORMATION

C1.5.1. The DoD Components shall establish and implement an environmental audit program to ensure that overseas installations assess compliance with these FGS at least once every 3 years at all major installations.

C1.5.2. DoDI 4715.4, "Pollution Prevention," implements policy, assigns responsibility, and prescribes procedures for implementation of pollution prevention programs throughout the DoD. As a matter of DoD policy, DoDI 4715.4 shall be consulted for particular requirements that apply to activities outside the United States. Pollution prevention shall be considered in developing the criteria and management practices for these FGS. Where economically advantageous and consistent with mission requirements, pollution prevention shall be the preferred means for attaining compliance with these FGS.

C1.5.3. Laboratory analyses necessary to implement these FGS would normally be conducted in a laboratory that has been certified by a U.S. or host nation regulatory authority for the applicable test method. In the absence of a certified laboratory, analyses may also be conducted at a laboratory that has an established reliable record of quality assurance compliance with standards for the applicable test method that are generally recognized by appropriate industry or scientific organizations.

C1.5.4. Unless otherwise specified, all record keeping requirements, including assessments, inspection records, logs, manifests, notices, forms, and formats, are described in DoD 8910.1-M, "DoD Procedures for Management of Information Requirements."

C1.5.5. These FGS do not create any rights or obligations enforceable against the United States, the DoD, or any of its components, nor does it create any standard of care or practice for individuals. Although these FGS refer to DoDDs and DoDIs, it is intended only to coordinate the requirements of those directives/instructions as required to implement the policies found in DoDI 4715.5. These FGS do not change other DoD or service directives/instructions, or alter DoD or service policies.

C1.5.6. References to websites that are not controlled by the DoD are intended for informational purposes, solely to facilitate compliance with the existing FGS in this document. Any other requirement, standard or criteria listed in the website shall not be incorporated by references as a Final Governing Standard. Reference to the website does not establish any obligation to comply with other requirements, standards, or criteria listed in the website. Referenced websites that are not under the control of the DoD may be changed at any time without notice to the DoD.

C1.6. WAIVERS

C1.6.1. If compliance with an FGS criteria at a particular installations or facilities would seriously impair operations, adversely affect relations with Spanish authorities, or require substantial expenditure of funds at an installation that has been identified for closure or at an installation that has been identified for a realignment that would remove the requirement, a DoD Component may ask the EEA to waive the particular criteria. See DoD Instruction 4715.5, Management of Environmental Compliance at Overseas Installations, and EUCOM Compliance Instruction (ECI) 4804.01, Environmental Security, for complete waiver procedures.

C1.7. AUTHORIZATIONS AND REPORTING

C1.7.1. The Spanish Base Commander is the formal representative of the installation and serves as the liaison with host nation authorities. Any formal interaction or interaction of common (U.S. and Spanish) interest with Spanish authorities shall be conducted with the cognizance of the Spanish Base Commander.

C1.7.2. The DoD Components shall not directly obtain authorizations (including, but not limited to licenses and permits) from Spanish authorities. If an authorization is required, DoD Components will assist the Spanish Base Commander by providing reports, records or otherwise helping him/her with their liaison duties.

C1.7.3. If the Spanish Base Commander obtains an authorization on behalf of the DoD Component and the authorization requires a more protective criteria than prescribed in the FGS, the criteria in the authorization shall be the compliance criteria. However, if an authorization allows a less protective criteria, then the FGS will be the compliance criteria unless a waiver is obtained.

C1.7.4. If the installation at issue does not fall under the responsibility of a Spanish Base Commander, the DoD installation may request clarification from the EEA via the chain of command.

C1.7.5. Contractors performing work for DoD must comply with all Spanish laws and regulations including obtaining all necessary licenses, permits, and/or authorizations. Notwithstanding the use of contracts, DoD Components and agencies are required to comply with these FGS unless exempted under Section C1.3, Exemptions.

C1.7.6. Certificates obtained from appropriate Spanish authorities (e.g., tank tightness testing) do not fall within the definition of an authorization for which the SBC has responsibility. Request for such services shall be forwarded directly to the appropriate organizations without involving the SBC.

C1.7.7. Access to Installations & Information by Spanish Authorities. Inspections and non-routine requests for information by Spanish authorities shall be coordinated with the SBC and reported to the EEA via the Component chain-of-command. To the maximum extent possible, U.S. military personnel shall lead the review of DoD Component activities by Spanish authorities during the inspection.

C1.8. LEGAL PROTECTIONS FOR U.S. FORCES EMPLOYEES

C1.8.1. Legal protections afforded DoD personnel investigated or charged by a host nation with alleged offenses arising out of any act or omission done in the performance of official duty vary depending upon the status of the individual under the NATO Status of Forces Agreement (SOFA) and other applicable international agreements. In general, there is a greater ability for the U.S. to assert a primary right to exercise criminal jurisdiction over a military service member of the force as opposed to a member of the civilian component. There is no such right with respect to local national employees. Accordingly, military service members shall assume responsibility under these FGS in lieu of DoD civilians or local national employees where non-compliance may result in a host nation enforcement action.

C1.8.2. Installations, activities, and personnel shall immediately contact their servicing legal office when faced with possible or actual host nation enforcement actions in order to maximize protections afforded to DoD personnel under the NATO SOFA, bilateral agreements, U.S. and DoD regulations, and to secure legal representation for the subjects of the enforcement action (including local national employees). Installations will report possible or actual Spanish enforcement actions to the EEA via their chain-of-command.

C1.8.3. If the host nation authorities seek to question a DoD employee regarding his or her official duties, a representative from installation legal shall accompany the employee. The installation legal representative shall clarify with the host nation authorities whether the employee is questioned as a suspect or a witness, and the employee advised accordingly. It may not be readily apparent whether the host nation legal proceeding is criminal or administrative in nature. This point shall be clarified as soon as possible since different procedures may apply depending on the answer. Installations will report interview by Spanish authorities to the EEA via their chain-of-command.

C1.9. ENVIRONMENTAL EXECUTIVE AGENT

C1.9.1. The EEA for these FGS is the Commander, Navy Region Europe Africa Southwest Asia. Any questions or comments pertaining to these FGS shall be sent to:

Commander, Navy Region Europe Africa Southwest Asia
PSC 817 Box 108
FPO AE 09622
DSN Voice (314) 626-2886
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C2. CHAPTER 2

AIR EMISSIONS

C2.1. SCOPE

This Chapter contains standards for air emissions sources. Criteria addressing open burning of solid waste are contained in Chapter 7, “Solid Waste.” Criteria addressing asbestos are contained in Chapter 15, “Asbestos.”

C2.2. DEFINITIONS

C2.2.1. Coal Refuse. Waste products from coal mining, cleaning, and coal preparation operations (e.g., culm and gob) containing coal, matrix material, clay, and other organic and inorganic material.

C2.2.2. Cold Cleaning Machine. Any device or piece of equipment that contains and/or uses liquid solvent, into which parts are placed to remove soil and other contaminants from the surfaces of the parts or to dry the parts. Cleaning machines that contain and use heated, nonboiling solvent to clean the parts are classified as cold cleaning machines.

C2.2.3. Fossil Fuel. Natural gas, petroleum, coal, and any form of solid, liquid, or gaseous fuel derived from such material for the purpose of creating useful heat.

C2.2.4. Fluorinated Gases (F-gases). Family of man-made gases used in a range of industrial applications as substitute for ozone-depleting substances. F-gases are considered greenhouse gases. They include the following groups: Chlorofluorocarbons (CFC), Halons, Hydrobromofluorocarbons (HBFC), Hydrochlorofluorocarbons (HCFC), Bromochloromethane (BCM), hydrofluorocarbons (HFCs), Perfluorocarbons (PFCs) and Sulfur Hexafluoride (SF_6).

C2.2.5. Freeboard Ratio. The ratio of the solvent cleaning machine freeboard height to the smaller interior dimension (length, width, or diameter) of the solvent cleaning machine.

C2.2.6. Hydrofluorocarbon (HFC). A compound consisting of hydrogen, fluorine, and carbon often used as a replacement for Ozone-Depleting Substances (ODS).

C2.2.7. Incinerator. A stationary or mobile unit dedicated to the thermal treatment of waste (including municipal waste combustion units) with or without recovery of the combustion heat generated. This includes the incineration by oxidation of waste as well as other thermal treatment processes such as pyrolysis, gasification or plasma processes in so far as the substances resulting from the treatment are subsequently incinerated.

C2.2.7.1. Co-incinerator. A stationary or mobile unit which uses waste as a fuel source and where energy generation or production is the main purpose.

C2.2.7.2. Existing Incinerator or Co-incinerator Plant. An incinerator or co-incinerator authorized before 14 June 2003 and in operation before 29 December 2003.

C2.2.8. Motor Vehicle. Any commercially available vehicle that is not adapted to military use which is self-propelled and designed for transporting persons or property on a street or highway, including but not limited to, passenger cars, light duty vehicles, and heavy duty vehicles.

C2.2.9. Municipal Solid Waste (MSW). Any waste produced in housing or commercial units, offices, and services, and any other waste classified as non-hazardous that may be comparable to waste produced by those activities. Similar wastes generated at industrial facilities are also considered municipal wastes. Municipal waste also includes waste from cleaning operations of public roads, green areas, recreational areas and beaches, as well as furniture, household electrical and electronic equipment, abandoned vehicles and wastes proceeding from minor household remodeling.

C2.2.10. Municipal Waste Combustion (MWC) Units. Any equipment that combusts solid, liquid, or gasified municipal solid waste (MSW) including, but not limited to, field-erected MWC units (with or without heat recovery), modular MWC units (starved-air or excess-air), boilers (for example, steam generating units), furnaces (whether suspension-fired, grate-fired, mass-fired, air curtain incinerators, or fluidized bed-fired), and pyrolysis/combustion units. Municipal waste combustion units do NOT include pyrolysis or MWC units located at a plastics or rubber recycling unit, cement kilns that combust MSW, internal combustion engines, gas turbines, or other combustion devices that combust landfill gases collected by landfill gas collection systems.

C2.2.11. Ozone-Depleting Substances (ODS). Those substances listed in Table 2.8.

C2.2.12. Perfluorocarbon (PFC). A compound consisting solely of carbon and fluorine often used as a replacement for ODS.

C2.2.13. Process Heater. A device that is primarily used to heat a material to initiate or promote a chemical reaction in which the material participates as a reactant or catalyst.

C2.2.14. Pyrolysis. The endothermic gasification of hospital waste and/or medical/infectious waste using external energy.

C2.2.15. Stack. Any point in a source covered by criteria contained in C2.3.1., C2.3.2., C2.3.3., C2.3.4., or C2.3.5. designed to emit pollutants.

C2.2.16. Steam/Hot Water Generating Unit. A device that combusts any fuel and produces steam or heats water or any other heat transfer medium. This definition does not include nuclear steam generators or process heaters.

C2.2.17. Substantially-Modified. Any modification to a facility/building, the cost of which exceeds $1 million, regardless of funding source.

C2.2.18. Volatile Organic Compound (VOC). Any organic compound having a vapor pressure ≥ 0.01 kPa at a temperature of 293.15 K , or having a corresponding volatility under the particular conditions of use.

C2.2.19. Vapor Cleaning Machine. A batch or in-line solvent cleaning machine that boils liquid solvent which generates solvent vapor that is used as a part of the cleaning or drying cycle.

C2.2.20. Vehicle Fluff. Vehicle scraps and parts from a scrap vehicle after it has been stripped for recycling.

C2.3. CRITERIA

C2.3.1. Steam/Hot Water Generating Units. Installations that operate steam/hot water generating units shall provide sufficient information to the Spanish Base Commander for authorization prior to their use (see Chapter 1 for the process).

C2.3.1.1. Installations that operate steam/hot water-generating units shall maintain an inventory that includes information on the units, emitted pollutants, any exceedances of emission limits, types of emission controls, and inspection records. This inventory shall be maintained for ≥ 10 years.

C2.3.1.2. Additionally, steam/hot water generating units with a rated thermal input > 20 MW (except hazardous or municipal waste incinerators) shall provide sufficient information to the Spanish Base Commander for a greenhouse gas emissions authorization (see Chapter 1 for the process).

C2.3.1.3. The following standards apply to units that commenced construction and/or operation on or after 1 October 1994 or that were substantially modified since 1 October 1994.

C2.3.1.4. Air Emission Standards. The following criteria apply to steam/hot water generating units as indicated below:

C2.3.1.4.1. Steam/hot water generating units and associated emissions controls, if applicable, must be designed to meet the emission standards for specific-sized units shown in Tables 2.1, 2.2, and 2.3 at all times, except during periods of start up, shut down, malfunction, or when emergency conditions exist.

C2.3.1.4.2. For units combusting liquid or solid fossil fuels, fuel sulfur content (weight %) and higher heating value will be measured and recorded for each new shipment of fuel. Use these data to calculate sulfur dioxide (SO_2) emissions and document compliance with the SO_2 limits in Tables 2.1, 2.2, and 2.3. Alternatively, install a properly calibrated and maintained continuous emissions monitoring system to measure the flue gas for SO_2, NO_X, PM and either oxygen (O_2) or carbon dioxide (CO_2).

C2.3.1.5. Air Emissions Monitoring. Steam/hot water generating units subject to NO_X, SO_2 and PM standards in Tables 2.1, 2.2, and 2.3 shall have a properly calibrated and maintained continuous emissions monitoring system (CEMS) to measure the flue gas as follows:

C2.3.1.5.1. For units with a maximum design heat input capacity > 30 million Btu/hr: Opacity, except that CEMS is not required where gaseous or distillate fuels are the only fuels combusted.

C2.3.1.5.2. For fossil fuel fired units with a maximum design heat input capacity > 100 million Btu/hr: NO_X and either O_2 or CO_2.

C2.3.1.6. Testing of Emission Controls. Periodic testing shall be performed in order to verify the effectiveness of the emission prevention and control measures and compliance with the emission limit values.

C2.3.2. Incinerators. Installations that operate an incinerator for waste treatment (municipal waste, sludge, hazardous waste, or medical waste) shall provide sufficient information to the Spanish Base Commander for authorization (see Chapter 1 for the process). The following criteria apply to the combustion of all types of waste. Refer to Chapter 6, “Hazardous Waste,” for additional information regarding hazardous waste disposal and incineration.

C2.3.2.1. Waste Incinerator. Each waste incinerator shall meet the emission limits specified in Tables 2.4 and 2.5.

C2.3.2.2. Waste Co-incinerator. Each waste co-incinerator shall meet the emission limits, specified in Tables 2.6 and 2.7.

C2.3.2.3. Monitoring Requirements. All waste incinerators and co-incinerators shall be subject to continuous measurements of the following parameters: NO_X, CO, particulate matter, TOC, HCl, HF, SO_2.

C2.3.2.4. Sewage Sludge Incinerators. All sewage sludge incinerators that commenced construction on or after 1 October 1994 or that were substantially modified since 1 October 1994 and that burn > 1 ton per day (tpd) of sewage sludge or > 10% sewage sludge, must also be designed to meet a particulate emission limit of 0.65 g/kg dry sludge (1.30 lb/ton dry sludge) and an opacity limit of 20% at all times, except during periods of start up, shut down, malfunction, or when emergency conditions exist.

C2.3.2.5. Medical Waste Incinerators (MWI). The following standards apply to all units. These requirements do not apply to any portable units (field deployable), pyrolysis units, or units that burn only pathological, low-level radioactive waste, or chemotherapeutic waste. Refer to Chapter 8, “Medical Waste,” for other requirements pertaining to medical waste management.

C2.3.2.5.1. All MWI must be designed and operated according to the following standards:

C2.3.2.5.1.1. Unit design: dual chamber.

C2.3.2.5.1.2. Minimum temperature in primary chamber: 1562-1600°F.

C2.3.2.5.1.3. Minimum temperature in secondary chamber: 1800-2200°F.

C2.3.2.5.1.4. Flue gas temperature and internal combustion chamber for waste containing >1% of halogenated substances (expressed as chlorine): 1,100°C (2,012 °F).

C2.3.2.5.1.5. Minimum residence time in the secondary chamber: 2 seconds.

C2.3.2.5.1.6. Incinerator operators must be trained in accordance with applicable Service requirements.

C2.3.2.5.2. Each incinerator shall be equipped with auxiliary burners that are automatically activated when the flue gas temperature drops below 850 °C (1562 °F) or 1,100 °C (2,012 °F) depending on the waste stream.

C2.3.3. Perchloroethylene (PCE) Dry Cleaning Machines. The following requirements apply to all dry cleaning machines. These criteria do not apply to coin-operated machines.

C2.3.3.1. Emissions from PCE dry cleaning machines installed before 1 October 1994 that use > 2000 gallons per year of PCE (installation wide) in dry cleaning operations, must be controlled with a refrigerated condenser, unless a carbon absorber was already installed. The temperature of the refrigerated condenser must be maintained at ≤ 45 ºF.

C2.3.3.2. All PCE dry cleaning systems installed on or after 1 October 1994 must be of the dry-to-dry design with emissions controlled by a refrigerated condenser. The temperature of the refrigerated condenser must be maintained at ≤ 45 ºF. Dry cleaning machines and control devices must be operated according to manufacturer recommendations.

C2.3.3.3. The total VOCs emission limit for dry cleaning machines shall be ≤ 20 g/kg of cleaned and dried product.

C2.3.3.4. Dry cleaning machines releasing > 10 kg/h (22.05 lb/h) of total organic carbon (TOC) at the final point of discharge shall be equipped with continuously monitoring equipment.

C2.3.4. Chromium Electroplating and Chromium Anodizing Tanks. Electroplating and anodizing tanks must comply with one of the three methods below for controlling chromium emissions. Implement one of the following methods that is most appropriate to suit local conditions:

C2.3.4.1. Option 1: Limit chromium emissions in the ventilation exhaust to 0.015 milligrams per dry standard cubic meter (mg/dscm). Control devices/methods must be operated according to manufacturer recommendations.

C2.3.4.2. Option 2: Use chemical tank additives to prevent surface tension of the electroplating or anodizing bath from exceeding 45 dynes per centimeter (cm) as measured by a stalagmometer or 35 dynes/cm as measured by a tensiometer. Measure the surface tension prior to the first initiation of electric current on a given day and every 4 hours thereafter.

C2.3.4.3. Option 3: Limit chromium emissions to the maximum allowable mass emission rate (MAMER) calculated using the following equation: MAMER = ETSA x K x 0.015 mg/dscm, where: MAMER = the alternative emission rate for enclosed hard chromium electroplating tanks in mg/hr; ETSA = the hard chromium electroplating tank surface area in square feet (ft^2); K = a conversion factor, 425 dscm/(ft^2-hr). Option 3 is ONLY applicable to hard chrome electroplating tanks equipped with an enclosing hood and ventilated at half the rate or less than that of an open surface tank of the same surface area.

C2.3.5. Halogenated Solvent Cleaning Machines. These requirements apply to all solvent cleaning machines that use solvent which contains > 5% by weight: methylene chloride (CAS No. 75-09-2), perchloroethylene (CAS No. 127-18-4), trichloroethylene (CAS No. 79-01-6), 1,1,1-trichloroethane (CAS No. 71-55-6), carbon tetrachloride (CAS No. 56-23-5), chloroform (CAS No. 67-66-3), or any combination of these halogenated solvents.

C2.3.5.1. All cold cleaning machines (remote reservoir and immersion tanks) must be covered when not in use. Additionally, immersion type cold cleaning machines must have either a 1-inch water layer or a freeboard ratio of ≥ 0.75.

C2.3.5.2. All vapor cleaning machines (vapor degreasers) must incorporate design and work practices which minimize the direct release of halogenated solvent to the atmosphere. When the mass flow of the sum of the halogenated organic solvents (as specified in C2.3.5) is ≥ 100 g/h (0.22 lb/h), the emission shall not exceed 20 mg/Nm^3.

C2.3.6. Units Containing ODS Listed In Table 2.8. The following criteria apply to direct atmospheric emissions of ODS listed in Table 2.8.

C2.3.6.1. Except as allowed in C2.3.6.2 and C2.3.6.3, use of the following ODSs, as listed in Table 2.8, is prohibited. Running an existing system without maintenance (e.g., using a refrigerator) would not be classified as use.

C2.3.6.2. Halons shall only be used for critical uses as specified in Table 2.9.

C2.3.6.3. The use of ODSs shall be allowed:

C2.3.6.3.1. In essential laboratory and analytical uses.

C2.3.6.3.2. As halon substitutes in existing fire protection systems specified in Table 2.9 under the following conditions:

C2.3.6.3.2.1. Original halons shall be replaced with the same type of halon.

C2.3.6.3.2.2. Original halons contained in such fire protection systems shall be replaced completely;

C2.3.6.4. Refrigerant Recovery/Recycling. All repairs, including leak repairs or services to appliances, industrial process refrigeration units, air conditioning units, or motor vehicle air conditioners, must be performed using commercially available refrigerant recovery/recycling equipment operated by trained personnel. Refrigerant technicians shall be trained in proper recovery/recycling procedures, leak detection, safety, shipping, and disposal in accordance with recognized industry standards or Spanish equivalent.

C2.3.6.4.1. Recovery Requirements for ODSs. ODSs identified in Table 2.8 shall be recovered as follows using equipment operated by trained personnel:

C2.3.6.4.1.1. Repaired equipment containing ≥ 3 kg (6.6 lbs) of refrigerating fluid charge shall be checked annually for leaks.

C2.3.6.4.1.2. Where methyl bromide is used in soil fumigation, the use of virtually impermeable films for a sufficient time, or other techniques ensuring the same level of environmental protection must be used. All practicable precautionary measures must be taken to prevent and minimize leakage of methyl bromide from fumigation installations and operations.

C2.3.6.4.1.3. Halons and other ODSs recovered shall be returned to the DLA Defense Reserve (located in Germersheim Germany; Zevenaar Netherlands; and Richmond, Virginia) for reprocessing in accordance with the established DoD ODS turn-in procedures (dated September 2012).

C2.3.6.4.2. Use of HCFCs as Refrigerants. Reclaimed or recycled HCFCs may be used until 31 December 2014 for the maintenance and servicing of existing refrigeration, air conditioning, and heat pump equipment provided they have been recovered from such equipment. These uses are prohibited after 31 December 2014.

C2.3.6.5. Refrigerant Venting Prohibition. Any class I or II ODS, HFC, and PFC refrigerant (identified in Table 2.8) shall not be intentionally released in the course of maintaining, servicing, repairing, or disposing of appliances, industrial process refrigeration units, air conditioning units, or motor vehicle air conditioners. *De minimis* releases associated with good faith attempts to recycle or recover ODS, HFC, and PFC refrigerants are not subject to this prohibition.

C2.3.6.6. Refrigerant Leak Monitoring and Repair. Monitor and repair refrigeration equipment for ODS leakage in accordance with the following criteria and repair, if found to be leaking.

C2.3.6.7. Installations containing ODS refrigerant. The following precautionary measures shall be taken to prevent leakage of ODSs included in Table 2.8:

C2.3.6.7.1. All precautionary measures practicable shall be taken to prevent and minimize any leakage and emissions of ODSs.

C2.3.6.7.2. Units containing fluorinated refrigerants with fluid charge of ≥ 300 kg shall be equipped with a leak detection system.

C2.3.6.7.3. When operating refrigeration, air conditioning or heat pump equipment, or fire protection systems that contain ODSs, the stationary equipment or systems shall be checked for leakage at the following frequency:

Fluid charge of controlled substance	Frequency
≥ 3 kg and < 30 kg	12 months
≥ 30 kg and < 300 kg	6 months
≥ 300 kg	3 months

C2.3.6.7.4. Any detected leakage from the above specified equipment or from any other equipment relying on ODSs shall be repaired within 14 days. The equipment or system shall be checked for leakage within 1 month after a leak has been repaired to ensure its effectiveness.

C2.3.6.8. Unit Maintenance. Installation, maintenance and dismantling of refrigeration units shall only be performed by qualified personnel certified in accordance with a recognized U.S. certification body and/or the Spanish equivalent.

C2.3.6.9. Unit Labeling. All refrigeration units shall be placarded with the following information:

C2.3.6.9.1. Name and address of the installation company;

C2.3.6.9.2. Model and serial number of the equipment;

C2.3.6.9.3. Year of manufacture;

C2.3.6.9.4. Date of the next periodic inspection;

C2.3.6.9.5. Type of refrigerant;

C2.3.6.9.6. Fluid charge, expressed in kg;

C2.3.6.9.7. Allowable maximum pressure, expressed in bar; and

C2.3.6.9.8. CE or UL seal of approval.

C2.3.6.9.9. Additionally, units containing fluorinated greenhouse gas refrigerants shall be labeled with the chemical name of the refrigerant and its quantity.

C2.3.6.10. Refrigerant Inventory. DoD installations shall maintain an inventory of all refrigeration units that includes information on the manufacturer and type of unit installed, name of the installation company, date of the first and subsequent periodic inspections, and monitoring and repair records.

C2.3.7. Fluorinated Gases.

C2.3.7.1. Installation, handling and maintenance of the following equipment containing fluorinated gases shall be performed by personnel certified in accordance with a recognized U.S. certification body and/or Spanish equivalent:

C2.3.7.1.1. Refrigeration or air conditioning equipment;

C2.3.7.1.2. Air conditioning systems in motor vehicles; and

C2.3.7.1.3. Fire protection systems that use fluorinated gas as an extinguishing agent.

C2.3.7.2. ODS Fire Suppression Agent (Halon) Venting Prohibition. Halons shall not be intentionally released into the environment while testing, maintaining, servicing, repairing, or disposing of halon-containing equipment or using such equipment for technician training. Halon uses authorized in C2.3.6.2 are exempt from the venting prohibition in the following situations:

C2.3.7.2.1. *De minimis* releases associated with good faith attempts to recycle or recover halons (i.e., release of residual halon contained in fully discharged total flooding fire extinguishing systems).

C2.3.7.2.2. Emergency releases for the legitimate purpose of fire extinguishing, explosion inertion, or other emergency applications for which the equipment or systems were designed.

C2.3.7.2.3. Releases during the testing of fire extinguishing systems if each of the following is true: systems or equipment employing suitable alternative fire extinguishing agents are not available; release of extinguishing agent is essential to demonstrate equipment functionality; failure of system or equipment would pose great risk to human safety or the environment; and, a simulant agent (i.e., substitute product that can perform the same function) cannot be used.

C2.3.8. Motor Vehicles. This criteria applies to DoD-owned motor vehicles as defined in paragraph C2.2.8.

C2.3.8.1. All vehicles shall be inspected as follows to ensure that no tampering with the factory-installed emission control equipment has occurred:

Vehicle Age	Inspection Frequency
0-4 years	every 2 years
after 4 years	annually

C2.3.8.2. CO emission limits for idling gasoline-fueled vehicles that were manufactured in Europe shall not exceed 5% by volume (at 15-20º C (59-68 º F) and 750-760 mm Hg). Motor vehicles manufactured in Europe and equipped with diesel engines shall not exceed the following opacity limits:

Engine Horsepower	Opacity Limit[1] (Absolute Units)
> 200	2.1
>100 and <200	2.4
< 100	2.8

1. The opacity limits are based on measures made at a minimum motor temperature of 60°C (140 ºF).

C2.3.8.3. Use only unleaded gasoline in vehicles that are designed for this fuel.

C2.3.9. Stack Heights. H_g is the good engineering practice stack height necessary to minimize downwash of stack emissions due to aerodynamic influences from nearby structures.

C2.3.9.1. Stacks shall be designed and constructed to heights at least equal to the largest H_g calculated from either of the following two criteria:

C2.3.9.1.1. $H_g = H + 1.5L$, where H is the height of the nearby structure measured from the ground level elevation at the base of the stack, and L is the lesser of height or projected width of the nearby structure(s). A structure is determined to be nearby when the stack is located within 5L of the structure envelope but not > 0.8 km (0.5 mile). This calculation shall be performed for each structure nearby the stack being studied to determine the greatest H_g.

C2.3.9.1.2. H_g is the height demonstrated by a fluid model or a field study, which ensures that the emissions from a stack do not result in ground-level concentrations of any air pollutant related to the combustion activities above the permitted limit values as a result of atmospheric downwash, wakes, or eddy effects created by the source itself, nearby structures, or nearby terrain features ≥ 40% in excess of the maximum ground-level concentrations of any air pollutant experienced in the absence of such atmospheric downwash, wakes, or eddy effects. For purposes of this paragraph, "nearby" means ≤ 0.8 km (0.5 mile), except that the portion of a terrain feature may be considered to be nearby which falls within a distance of up to 10 times the maximum height (H_t) of the feature, not to exceed 2 miles if such feature achieves a height (H_t) 0.8 km from the stack that is ≥ 40% of the good engineering practice stack height determined by the formula provided in C2.3.9.1.1. of this part or 26 meters, whichever is greater, as measured from the ground-level elevation at the base of the stack. The height of the structure or terrain feature is measured from the ground-level elevation at the base of the stack.

C2.3.10. Fuel Composition. The composition of fuels allowed for use in Spain is specified in the following table:

Parameter	Units	Diesel Fuel		Fuel Oil	
		Minimum	Maximum	Minimum	Maximum
Density at 15°C	kg/m^3	820	845	720	775
Viscosity at 40°C	mm^2/s	2.00	4.50	---	---
Total sulfur content	mg/kg	---	10	---	10
Total lead content	g/l	---	---	---	0.005
Water	mg/kg	---	200	---	---

C2.3.10.1. The presence of the metallic additive in fuel is limited to 6 mg of manganese per liter as of 1 January 2011, and 2 mg of manganese per liter as of 1 January 2014.

Table 2.1. Emission Limits for Power Plants with Energy-Generating Potential < 50 MW

Fuel Source	**Emission Limits**		
	Opacity [1] (%)	**SO_2 (mg/Nm^3)**	**Particulate Matter (mg/Nm^3)**
Liquid fuels	20	5,500	200
Solid fuel (anthracite coal)	20	2,400	400
Solid fuel (lignite coal)	20	9,000	400

Notes:
1. The limits shall not exceed 40% opacity or more than one 2-minute period per hour.
2. The units of mg/Nm^3 are at normal physical conditions of 0ºC and 0.1013 Mpa.

These emission limits apply to the activities subjected to notification (group C).

Table 2.2. Emission Limits for Industrial Combustion Facilities
(Steam/Heat Generating Units, Excluding Thermal Plants)

Fuel Source	**Emission Limits**		
	Opacity [1] (%)	**SO_2 (mg/Nm^3)**	**CO (ppm)**
Gas oil or domestic fuel oil	20	1,700	1,445
Heavy #1 fuel oil	40	4,200	1,445
Heavy #2 fuel oil	50	6,800	1,445

Notes:
1. The limits shall not be exceeded more than three times per day, with each period lasting < 10 minutes.
2. The units of mg/Nm^3 are at normal physical conditions of 0ºC (32 ºF) and 0.1013 Mpa.

These emission limits apply to the activities subjected to notification (group C).

Table 2.3. Emission Limit Values
(for new installations with a power capacity of >50 MW authorized after 27 November 2002)

Pollutant	**Thermal input**		
	50 to100 MW	**100 to 300 MW**	**> 300 MW**
Solid fuels (mg/Nm^3) (O_2 content 6 %)			
SO_2: General case Biomass	 850 200	 200 200	 200 200
NO_X : General case Biomass	 400 400	 200 300	 200 200
Particulate matter	50	30	30
Opacity (%)	20		
Liquid fuels (mg/Nm^3) (O_2 content 3 %)			
SO_2	850	400 to 200 (linear decrease)	200
NO_X	400	200	200
Particulate matter	50	30	30
Opacity (%)	20		
Gaseous fuels (mg/Nm^3) (O_2 content 3 %)			
SO_2	35		
NO_X : Natural gas Other gases	 150 200		 100 200
Particulate matter	5		

Table 2.4. Emission Limit Values for Waste Incinerators
(daily average values)

Pollutant	Emission Standards
Particulate matter	10 mg/m^3
Gaseous and vaporous organic substances, expressed as total organic carbon	10 mg/m^3
Hydrogen chloride (HCl)	10 mg/m^3
Hydrogen fluoride (HF)	1 mg/m^3
Sulfur dioxide (SO_2)	50 mg/m^3
Nitrogen monoxide (NO) and nitrogen dioxide (NO_2) expressed as nitrogen dioxide for existing incineration plants with a nominal capacity exceeding 6 tonnes per hour or new incineration plants	200 mg/m^3
Nitrogen monoxide (NO) and nitrogen dioxide (NO_2), expressed as nitrogen dioxide for existing incineration plants with a nominal capacity of ≤6 tonnes per hour	400 mg/m^3

Table 2.5. Heavy Metals Emission Limit Values for Waste Incinerators
(All average values over the sample period of a minimum of 30 minutes and a maximum of 8 hours)

<table>
<tr><th>Pollutant</th><th colspan="2">Total emission limit value</th></tr>
<tr><td>Cadmium and its compounds, expressed as cadmium (Cd)</td><td rowspan="2">Total 0.05 mg/m^3</td><td rowspan="2">Total 0.1 mg/m^3</td></tr>
<tr><td>Thallium and its compounds, expressed as thallium (Tl)</td></tr>
<tr><td>Mercury and its compounds, expressed as mercury (Hg)</td><td>0.05 mg/m^3</td><td>0.1 mg/m^3</td></tr>
<tr><td>Antimony and its compounds, expressed as antimony (Sb)</td><td rowspan="9">Total 0.5 mg/m^3</td><td rowspan="9">Total 1 mg/m^3</td></tr>
<tr><td>Arsenic and its compounds, expressed as arsenic (As)</td></tr>
<tr><td>Lead and its compounds, expressed as lead (Pb)</td></tr>
<tr><td>Chromium and its compounds, expressed as chromium (Cr)</td></tr>
<tr><td>Cobalt and its compounds, expressed as cobalt (Co)</td></tr>
<tr><td>Copper and its compounds, expressed as copper (Cu)</td></tr>
<tr><td>Manganese and its compounds, expressed as manganese (Mn)</td></tr>
<tr><td>Nickel and its compounds, expressed as nickel (Ni)</td></tr>
<tr><td>Vanadium and its compounds, expressed as vanadium (V)</td></tr>
</table>

Table 2.6. Emission Limit Values for Waste Co-incinerators
(daily average values)

Pollutant	< 50 MW	50 to100 MW	100 to 300 MW	> 300 MW
Solid fuels (mg/m^3) (O_2 content 6 %)				
SO_2: General case Indigenous fuels		850 or rate of desulphurization ≥ 90 %	850 to 200 (linear decrease from 100 to 300 MW) or rate of desulphurization ≥ 92 %	200 or rate of desulphurization ≥ 95 %
NO_X		400	300	200
Particulate matter	50	50	30	30
Biomass (mg/m^3) (O_2 content 6 %)				
SO_2		200	200	200
NO_X		350	300	300
Particulate matter	50	50	30	30
Liquid fuels (mg/m^3) (O_2 content 3 %)				
SO_2		850	850 to 200 (linear decrease from 100 to 300 MW)	200
NO_X		400	300	200
Particulate matter	50	50	300	30

Table 2.7. Heavy Metals Emission Limit Values for Waste Co-incinerators

Pollutant	Total emission limit value
Cd + Tl	0.05 mg/m^3
Hg	0.05 mg/m^3
Sb + As + Pb + Cr + Co + Cu + Mn + Ni + V	0.5 mg/m^3
Dioxins and furans	0.1 mg/m^3

Table 2.8. Ozone Depleting Substances

Molecular Formula	Common Name	CAS Number[1]	Chemical Name
Chlorofluorocarbons (CFCs)			
$CFCl_3$	CFC – 11	75-69-4	Trichlorofluoromethane
CF_2Cl_2	CFC – 12	75-71-8	Dichlorodifluoromethane
$C_2F_3Cl_3$	CFC – 113	76-13-1	Trichlorotrifluoroethane
$C_2F_4Cl_2$	CFC – 114	76-14-2	Dichlorotetrafluoroethane
C_2F_5Cl	CFC – 115	76-15-3	Chloropentafluoroethane
Other Fully Halogenated Chlorofluorocarbons			
CF_3Cl	CFC – 13	75-72-9	Chlorotrifluoromethane
C_2FCl_5	CFC – 111	354-56-3	Pentachlorofluoroethane
$C_2F_2Cl_4$	CFC – 112	76-12-0	Tetrachlorodifluoroethane
C_3FCl_7	CFC – 211	422-78-6	Heptachlorofluoropropane
$C_3F_2Cl_6$	CFC – 212	3182-26-1	Hexachlorodifluoropropane
$C_3F_3Cl_5$	CFC – 213	2354-06-5	Pentachlorotrifluoropropane
$C_3F_4Cl_4$	CFC – 214	29255-31-0	Tetrachlorotetrafluoropropane
$C_3F_5Cl_3$	CFC – 215	4259-43-2	Trichloropentafluoropropane
$C_3F_6Cl_2$	CFC – 216	661-97-2	Dichlorohexafluoropropane
C_3F_7Cl	CFC – 217	422-86-6	Chloroheptafluoropropane
Halons			
CF_2BrCl	Halon – 1211	353-59-3	Bromochlorodifluoromethane
CF_3Br	Halon – 1301	75-63-8	Bromotrifluoromethane
$C_2F_4Br_2$	Halon – 2402	124-73-2	Dibromotetrafluoroethane
Carbon Tetrachloride			
CCl_4	Carbon Tetrachloride	56-23-5	Carbon Tetrachloride
1,1,1-trichloroethane			
$C_2H_3Cl_3$	Methyl Chloroform	71-55-6	1,1,1-trichloroethane
Methyl Bromide			
CH_3Br	Methyl Bromide	74-83-9	Methyl Bromide
Hydrobromofluorocarbons			
$CHFBr_2$	N/A		Dibromofluoromethane
CHF_2Br	HBFC-22B1		Bromodifluoromethane
CH_2FBr	N/A		Bromofluoromethane
C_2HFBr_4	N/A		Tetrabromofluoroethane
$C_2HF_2Br_3$	N/A		Tribromodifluoroethane

Molecular Formula	Common Name	CAS Number[1]	Chemical Name
$C_2HF_3Br_2$	N/A		Dibromotrifluoroethane
C_2HF_4Br	N/A		Bromotetrafluoroethane
$C_2H_2FBr_3$	N/A		Tribromofluoroethane
$C_2H_2F_2Br_2$	N/A		Dibromodifluoroethane
$C_2H_2F_3Br$	N/A		Bromotrifluoroethane
$C_2H_3FBr_2$	N/A		Dibromofluoroethane
$C_2H_3F_2Br$	N/A		Bromodifluoroethane
C_2H_4FBr	N/A		Bromofluoroethane
C_3HFBr_6	N/A		Hexabromofluoropropane
$C_3HF_2Br_5$	N/A		Pentabromodifluoropropane
$C_3HF_3Br_4$	N/A		Tetrabromotrifluoropropane
$C_3HF_4Br_3$	N/A		Tribromotetrafluoropropane
$C_3HF_5Br_2$	N/A		Dibromopentafluoropropane
C_3HF_6Br	N/A		Bromohexafluoropropane
$C_3H_2FBr_5$	N/A		Pentabromofluoropropane
$C_3H_2F_2Br_4$	N/A		Tetrabromodifluoropropane
$C_3H_2F_3Br_3$	N/A		Tribromotrifluoropropane
$C_3H_2F_4Br_2$	N/A		Dibromotetrafluoropropane
$C_3H_2F_5Br$	N/A		Bromopentafluoropropane
$C_3H_3FBr_4$	N/A		Tetrabromofluoropropane
$C_3H_3F_2Br_3$	N/A		Tribromodifluoropropane
$C_3H_3F_3Br_2$	N/A		Dibromotrifluoropropane
$C_3H_3F_4Br$	N/A		Bromotetrafluoropropane
$C_3H_4FBr_3$	N/A		Tribromofluoropropane
$C_3H_4F_2Br_2$	N/A		Dibromodifluoropropane
$C_3H_4F_3Br$	N/A		Bromotrifluoropropane
$C_3H_5FBr_2$	N/A		Dibromofluoropropane
$C_3H_5F_2Br$	N/A		Bromodifluoropropane
C_3H_6FBr	N/A		Bromofluoropropane
Hydrochlorofluorocarbons (HCFCs)			
$CHFCl_2$	HCFC – 21		Dichlorofluoromethane
CHF_2Cl	HCFC – 22		Chlorodifluoromethane
CH_2FCl	HCFC – 31		Chlorofluoromethane
C_2HFCl_4	HCFC – 121		Tetrachlorofluoroethane
$C_2HF_2Cl_3$	HCFC – 122		Trichlorodifluoroethane
$C_2HF_3Cl_2$	HCFC – 123		Dichlorotrifluoroethane
C_2HF_4Cl	HCFC – 124		Chlorotetrafluoroethane
$C_2H_2FCl_3$	HCFC – 131		Trichlorofluoroethane
$C_2H_2F_2Cl_2$	HCFC – 132		Dichlorodifluoroethane
$C_2H_2F_3Cl$	HCFC – 133		Chlorotrifluoroethane
$C_2H_3FCl_2$	HCFC – 141		Dichlorofluoroethane
CH_3CFCl_2	HCFC – 141b		1,1-dichloro-1-fluoroethane

Molecular Formula	Common Name	CAS Number[1]	Chemical Name
$C_2H_3F_2Cl$	HCFC – 142		Chlorodifluoroethane
CH_3CF_2Cl	HCFC – 142b		1-chloro-1,1-difluoroethane
C_2H_4FCl	HCFC – 151		Chlorofluoroethane
C_3HFCl_6	HCFC – 221		Hexachlorofluoropropane
$C_3HF_2Cl_5$	HCFC – 222		Pentachlorodifluoropropane
$C_3HF_3Cl_4$	HCFC – 223		Tetrachlorotrifluoropropane
$C_3HF_4Cl_3$	HCFC – 224		Trichlorotetrafluoropropane
$C_3HF_5Cl_2$	HCFC – 225		Dichloropentafluoropropane
$CF_3CF_2CHCl_2$	HCFC – 225ca		3,3-dichloro-1,1,1,2,2-pentafluoropropane
CF_2ClCF_2CHClF	HCFC – 225cb		1,3-dichloro-1,1,2,2,3-pentafluoropropane
C_3HF_6Cl	HCFC – 226		Chlorohexafluoropropane
$C_3H_2FCl_5$	HCFC – 231		Pentachlorofluoropropane
$C_3H_2F_2Cl_4$	HCFC – 232		Tetrachlorodifluoropropane
$C_3H_2F_3Cl_3$	HCFC – 233		Trichlorotrifluoropropane
$C_3H_2F_4Cl_2$	HCFC – 234		Dichlorotetrafluoropropane
$C_3H_2F_5Cl$	HCFC – 235		Chloropentafluoropropane
$C_3H_3FCl_4$	HCFC – 241		Tetrachlorofluoropropane
$C_3H_3F_2Cl_3$	HCFC – 242		Trichlorodifluoropropane
$C_3H_3F_3Cl_2$	HCFC – 243		Dichlorotrifluoropropane
$C_3H_3F_4Cl$	HCFC – 244		Chlorotetrafluoropropane
$C_3H_4FCl_3$	HCFC – 251		Trichlorofluoropropane
$C_3H_4F_2Cl_2$	HCFC – 252		Dichlorodifluoropropane
$C_3H_4F_3Cl$	HCFC – 253		Chlorotrifluoropropane
$C_3H_5FCl_2$	HCFC – 261		Dichlorofluoropropane
$C_3H_5F_2Cl$	HCFC – 262		Chlorodifluoropropane
C_3H_6FCl	HCFC – 271		Chlorofluoropropane
Bromochloromethane			
CH_2BrCl	BCM		Bromochloromethane
Dibromodifluoromethane			
CBr_2F_2	Halon-1202		Dibromodifluoromethane

Note:

1. The American Chemical Society's Chemical Abstracts Service number.

Table 2.9. Critical Uses of Halon

Application				Cut-off date[1] (31 December of the stated year)	End date[2] (31 December of the stated year)
Category of equipment or facility	**Purpose**	**Type of extinguisher**	**Type of Halon**		
On military ground vehicles	For the protection of engine compartments	Fixed system	1301 1211 2402	**2010**	**2035**
	For the protection of crew compartments	Fixed system	1301 2402	**2011**	**2040**
		Portable extinguisher	1301 1211	**2011**	**2020**
On ship military surface ships	For the protection of normally occupied machinery spaces	Fixed system	1301 2402	**2010**	**2040**
	For the protection of normally unoccupied engine spaces	Fixed system	1301 1211 2402	**2010**	**2035**
	For the protection of normally unoccupied electrical compartments	Fixed system	1301 1211	**2010**	**2030**
	For the protection of command centers	Fixed system	1301	**2010**	**2030**
	For the protection of fuel pump rooms	Fixed system	1301	**2010**	**2030**
	For the protection of flammable liquid storage compartments	Fixed system	1301 1211 2402	**2010**	**2030**
	For the protection of aircraft in hangars and maintenance areas	Portable extinguisher	1301 1211	**2010**	**2016**
On military submarines	For the protection of machinery spaces	Fixed system	1301	**2010**	**2040**
	For the protection	Fixed system	1301	**2010**	**2040**

Application				Cut-off date[1] (31 December of the stated year)	End date[2] (31 December of the stated year)
Category of equipment or facility	**Purpose**	**Type of extinguisher**	**Type of Halon**		
	of command centers				
	For the protection of diesel generator spaces	Fixed system	1301	**2010**	**2040**
	For the protection of electrical compartments	Fixed system	1301	**2010**	**2040**
On aircraft	For the protection of normally unoccupied cargo compartments	Fixed system	1301 1211 2402	**2018**	**2040**
	For the protection of cabins and crew compartments	Portable extinguisher	1211 2402	**2014**	**2025**
	For the protection of engine nacelles and auxiliary power units	Fixed system	1301 1211 2402	**2014**	**2040**
	For the inerting of fuel tanks	Fixed system	1301 2402	**2011**	**2040**
	For the protection of lavatory waste receptacles	Fixed system	1301 1211 2402	**2011**	**2020**
	For the protection of dry bays	Fixed system	1301 1211 2402	**2011**	**2040**
In oil, gas and petrochemicals facilities	For the protection of spaces where flammable liquid or gas could be released	Fixed system	1301 2402	**2010**	**2020**
On commercial cargo ships	For the inerting of normally occupied spaces where flammable liquid or gas could be released	Fixed system	1301 2402	**1994**	**2016**
In land-based command and	For the protection of normally	Fixed system	1301 2402	**2010**	**2025**

Application				Cut-off date[1] (31 December of the stated year)	End date[2] (31 December of the stated year)
Category of equipment or facility	**Purpose**	**Type of extinguisher**	**Type of Halon**		
communications facilities Essential to national security	occupied spaces	Portable extinguisher	1211	**2010**	**2013**
	For the protection of normally unoccupied spaces	Fixed system	1301 2402	**2010**	**2020**
At airfields and airports	For crash rescue vehicles	Portable extinguisher	1211	**2010**	**2016**
	For the protection of aircraft in hangars and maintenance areas	Portable extinguisher	1211	**2010**	**2016**
In nuclear power and nuclear research facilities	For the protection of spaces where necessary to minimize risk of dispersion of radioactive matter	Fixed system	1301	**2010**	**2020**
In the Channel Tunnel	For the protection of technical facilities	Fixed system	1301	**2010**	**2016**
	For the protection of power cars and shuttle wagons of Channel Tunnel trains	Fixed system	1301	**2010**	**2020**
Other	For the initial extinguishing by fire brigades where essential to personal safety	Portable extinguisher	1211	**2010**	**2013**
	For the protection of persons by military and police personnel	Portable extinguisher	1211	**2010**	**2013**

C3. CHAPTER 3

DRINKING WATER

C3.1. SCOPE

This Chapter contains criteria for providing potable water.

C3.2. DEFINITIONS

C3.2.1. Action Level. The concentration of a substance in water that establishes appropriate treatment for a water system.

C3.2.2. Appropriate DoD Medical Authority. The medical professional designated by the in-theater DoD Component commander to be responsible for resolving medical issues necessary to provide safe drinking water at the DoD Component's installations.

C3.2.3. Concentration/Time (CT). The product of residual disinfectant concentration, C, in milligrams per liter (mg/L) determined before or at the first customer, and the corresponding disinfectant contact time, T, in minutes. CT values appear in Tables 3.11. through 3.24.

C3.2.4. Conventional Treatment. Water treatment, including chemical coagulation, flocculation, sedimentation, and filtration.

C3.2.5. Diatomaceous Earth Filtration. A water treatment process of passing water through a precoat of diatomaceous earth deposited onto a support membrane while additional diatomaceous earth is continuously added to the feed water to maintain the permeability of the precoat, resulting in substantial particulate removal from the water.

C3.2.6. Direct Filtration. Water treatment, including chemical coagulation, possibly flocculation, and filtration, but not sedimentation.

C3.2.7. Disinfectant. Any oxidant, including but not limited to, chlorine, chlorine dioxide, chloramines, and ozone, intended to kill or inactivate pathogenic microorganisms in water.

C3.2.8. DoD Water System. A public or non-public water system.

C3.2.9. Emergency Assessment. Evaluation of the susceptibility of the water source, treatment, storage and distribution system(s) to disruption of service caused by natural disasters, accidents, and sabotage.

C3.2.10. First Draw Sample. A one-liter sample of tap water that has been standing in plumbing at least six hours and is collected without flushing the tap.

C3.2.11. Haloacetic Acids. The sum of the concentrations in milligrams per liter of the haloacetic acid compounds (monochloroacetic acid, dichloroacetic acid, trichloroacetic acid, monobromoacetic acid, and dibromoacetic acid), rounded to two significant figures after addition.

C3.2.12. Groundwater Under the Direct Influence of Surface Water (GWUDISW). Any water below the surface of the ground with significant occurrence of insects or other microorganisms, algae, or large diameter pathogens such as *Giardia lamblia*; or significant and relatively rapid shifts in water characteristics, such as turbidity, temperature, conductivity, or pH, which closely correlate to climatological or surface water conditions.

C3.2.13. Lead-free. A maximum lead content of 0.2% for solder and flux, and 8.0% for pipes and fittings.

C3.2.14. Lead Service Line. A service line made of lead that connects the water main to the building inlet, and any lead pigtail, gooseneck, or other fitting that is connected to such line.

C3.2.15. Maximum Contaminant Level (MCL). The maximum permissible level of a contaminant in water that is delivered to the free-flowing outlet of the ultimate user of a public water system except for turbidity for which the maximum permissible level is measured after filtration. Contaminants added to the water under circumstances controlled by the user, except those resulting from the corrosion of piping and plumbing caused by water quality, are excluded.

C3.2.16. Maximum Residual Disinfectant Level (MRDL). The level of a disinfectant added for water treatment measured at the consumer's tap, which may not be exceeded without the unacceptable possibility of adverse health effects.

C3.2.17. Non-Public Water System (NPWS). A system that does not meet the definition of a public water system (for example, a well serving a building with fewer than 25 people).

C3.2.18. Point-of-Entry (POE) Treatment Device. A treatment device applied to the drinking water entering a facility to reduce contaminants in drinking water throughout the facility.

C3.2.19. Point-of-Use (POU) Treatment Device. A treatment device applied to a tap to reduce contaminants in drinking water at that tap.

C3.2.20. Potable Water. Produced, purchased, or blended water that has been examined and treated to meet the standards in this Chapter, and has been approved as potable by the appropriate DoD medical authority. It includes water used for drinking, cooking, food preparation, and other domestic uses, regardless of its origin or the manner in which it is distributed to the consumer.

C3.2.21. Public Water System (PWS). A system for providing piped water to the public for human consumption, if such system has at least 15 service connections or regularly serves a daily average of at least 25 individuals at least 60 days of the year. This also includes any collection, treatment, storage, and distribution facilities under control of the operator of such systems, and any collection or pretreatment storage facilities not under such control that are used primarily in connection with such systems. A PWS is either a "community water system" or a "non-community system":

C3.2.21.1. Community Water System (CWS). A PWS that has at least 15 service connections used by year-round residents, or which regularly serves at least 25 year-round residents.

C3.2.21.2. Non-Community Water System (NCWS). A PWS that serves the public, but does not serve the same people year-round.

C3.2.22. Non-transient, Non-community Water System (NTNCWS). A PWS that supplies water to at least 25 of the same people at least six months per year, but not year-round. Examples include schools, factories, office buildings, and hospitals that have their own water systems.

C3.2.23. Transient, Non-Community Water System (TNCWS). A PWS that provides water to at least 25 persons (but not the same 25 persons) at least six months per year. Examples include but are not limited to gas stations, motels, and campgrounds that have their own water sources.

C3.2.24. Sanitary Survey. An on-site review of the water source, facilities, equipment, operation, and maintenance of a public water system to evaluate the adequacy of such elements for producing and distributing potable water.

C3.2.25. Slow Sand Filtration. Water treatment process where raw water passes through a bed of sand at a low velocity (1.2 ft/hr), resulting in particulate removal by physical and biological mechanisms.

C3.2.26. Total Trihalomethanes. The sum of the concentration in milligrams per liter of chloroform, bromoform, dibromochloromethane, and bromodichloromethane.

C3.2.27. Underground Injection. A subsurface emplacement through a bored, drilled, driven or dug well where the depth is greater than the largest surface dimension, whenever the principal function of the well is emplacement of any fluid.

C3.2.28. Vulnerability Assessment. The process the commander uses to determine the susceptibility to attack from the full range of threats to the security of personnel, family members, and facilities, which provide a basis for determining antiterrorism measures that can protect personnel and assets from terrorist attacks.

C3.2.29. Wellhead Protection Area. A protected surface and subsurface zone surrounding a well or well field for the purpose of preventing contaminants from reaching the well water.

C3.3. CRITERIA

C3.3.1. General Requirements. DoD installations that intend to withdraw and distribute water shall provide sufficient information to the Spanish Base Commander for authorization (see Chapter 1 for the process).

C3.3.2. DoD water systems, regardless of whether they produce or purchase water, shall:

C3.3.2.1. Maintain a map/drawing of the complete potable water system.

C3.3.2.2. Update the potable water system master plan at least every 5 years.

C3.3.2.3. Protect all water supply aquifers (groundwater) and surface water sources from contamination by suitable placement and construction of wells, by suitable placing of the new intake (heading) to all water treatment facilities, by siting and maintaining septic systems and onsite treatment units, and by appropriate land use management on DoD installations.

C3.3.2.4. Establish a wellhead protection area around the well or well field as part of the authorization process in consultation with the Spanish Base Commander to include a list of prohibited activities (see Chapter 1 for the process). A placard shall also be affixed to the withdrawal point indicating its use for drinking water extraction.

C3.3.2.5. Conduct sanitary surveys of the water system at least every 3 years for systems using surface water, and every 5 years for systems using groundwater, or as warranted, including review of required water quality analyses. Off-installation surveys will be coordinated with Spanish Base Commander.

C3.3.2.6. Provide proper treatment for all water sources as follows:

C3.3.2.6.1. Surface water supplies, including GWUDISW, shall conform to the surface water treatment requirements set forth in Table 3.1-a.and 3.1-b and characterized as Class A1, A2, or A3 according to the requirements in Table 3.1-b. After the initial characterization, the source water shall be periodically analyzed to assess compliance with the source water quality requirements for that class.

C3.3.2.6.2. Groundwater supplies, at a minimum, must be disinfected.

C3.3.2.6.3. Treatment additives in DoD-produced water (from groundwater or surface water) shall not exceed the maximum doses listed in Table 3.1-c.

C3.3.2.7. Maintain a continuous positive pressure of at least 20 pounds per square inch (psi) in the water distribution system.

C3.3.2.8. Perform water distribution system operation and maintenance practices consisting of:

C3.3.2.8.1. Cleaning and disinfection activities prior to start-up of a distribution system;

C3.3.2.8.2. Maintenance of a disinfectant residual throughout the water distribution system (except where determined unnecessary by the appropriate DoD medical authority);

C3.3.2.8.3. Proper procedures for repair and replacement of mains (including disinfection and bacteriological testing);

C3.3.2.8.4. An effective annual water main flushing program;

C3.3.2.8.5. Proper operation and maintenance of storage tanks and reservoirs; and

C3.3.2.8.6. Maintenance of distribution system appurtenances (including hydrants and valves).

C3.3.2.9. Establish an effective cross connection control and backflow prevention program.

C3.3.2.10. Manage underground injection on DoD installations to protect underground water supply sources. At a minimum, conduct monitoring to determine the effects of any underground injection wells on nearby groundwater supplies. DoD installations that intend to conduct underground injection shall provide sufficient information to the Spanish Base Commander for authorization (see Chapter 1 for the process).

C3.3.2.11. Develop and update as necessary an emergency contingency plan to ensure the provision of potable water despite interruptions from natural disasters and service interruptions. At a minimum, the plan will include:

C3.3.2.11.1. Plans, procedures, and identification of equipment that can be implemented or utilized in the event of an intentional or un-intentional disruption;

C3.3.2.11.2. Identification of key personnel;

C3.3.2.11.3. Procedures to restore service;

C3.3.2.11.4. Procedures to isolate damaged lines;

C3.3.2.11.5. Identification of alternative water supplies; and

C3.3.2.11.6. Installation public notification procedures.

C3.3.2.12. Use only lead-free and aluminum-free pipe, solder, flux, and fittings in the installation or repair of water systems and plumbing systems for drinking water. Provide installation public notification concerning the lead content of materials used in distribution or plumbing systems, or the corrosivity of water that has caused leaching, which indicates a potential health threat if exposed to leaded water, and remedial actions which shall be taken.

C3.3.2.13. Regularly analyze potable water for organoleptic (odor, taste, color, turbidity), microbiological (coliforms) and chemical parameters in accordance with criterion C3.3.2.

C3.3.2.14. Maintain records showing monthly operating reports for at least 3 years, and records of bacteriological results for at least 5 years, and chemical results for at least 10 years. Installations shall also maintain the following records:

C3.3.2.14.1. Analytical methods used for each parameter analyzed and units of measure;

C3.3.2.14.2. Analytical laboratory used; location, date, and time of sampling; classification of the water (potable or non-potable); altered parameters (if any) and concentration; and

C3.3.2.14.3. Date, type, and magnitude of non-compliance events; duration; corrective measures; and observations.

C3.3.2.15. All chemical and biological analytical results shall be provided to the Spanish Base Commander upon request.

C3.3.2.16. Document corrective actions taken to correct breaches of criteria and maintain such records for at least 3 years. Cross connection and backflow prevention testing and repair records should be kept for at least 10 years.

C3.3.2.17. Conduct vulnerability assessments, which include, but are not limited to, a review of:

C3.3.2.17.1. Pipes and constructed conveyances, physical barriers, water collection, pretreatment, treatment, storage, and distribution facilities, electronic, computer, or other automated systems utilized by the PWS;

C3.3.2.17.2. Use, storage, or handling of various chemicals; and

C3.3.2.17.3. Operation and maintenance of the water storage, treatment, and distribution systems.

C3.3.3. Testing Requirements

C3.3.3.1. DoD water systems will, by independent testing or by validated supplier testing, ensure conformance with the following criteria. Selection of public or private laboratories shall be limited to those conducting drinking water analysis in accordance with C1.5.3 (Chapter 1, “Overview”).

C3.3.3.2. The testing methods utilized shall be in accordance with DoD approved test methods (Tables 3.25 to 3.27). Alternatively, the appropriate DoD medical authority may approve the use of Spanish test methods in Table 3.28-a so long as the detection limits are, at a minimum, as good as their DoD counterparts. The method detection limits for certain parameters are provided in Table 3.28-b.

C3.3.3.3. Total Bacteria Requirements

C3.3.3.3.1. An installation responsible for a PWS or NPWS will conduct a bacteriological monitoring program to ensure the safety of water provided for human consumption and allow evaluation with the total coliform-related MCL. The MCL is based only on the presence or absence of total coliforms. The presence of total coliform bacteria may indicate, but not specify the presence of fecal coliforms, Enterococci, or Clostridium perfringens. The MCL is $\leq$ 5% positive samples per month for a system examining $\geq$ 40 samples a month, and $\leq$ 1 positive sample per month when a system analyzes $<$ 40 samples per month. Further, the

MCL is exceeded whenever a routine sample is positive for fecal coliforms or *E. coli*, *Enterococci*, and *Clostridium* perfringens, or any repeat sample is positive for total coliforms.

C3.3.3.3.2. Each system must develop a written, site-specific monitoring plan and collect routine samples according to Table 3.2., "Total Coliform Monitoring Frequency."

C3.3.3.3.3. Systems with initial samples testing positive for total coliforms, *Enterococci* or *Clostridium* perfingens, will collect repeat samples as soon as possible, preferably the same day. Repeat sample locations are required at the same tap as the original sample plus an upstream and downstream sample, each within five service connections of the original tap. Any additional repeat sampling which shall be required will be performed according to the appropriate DoD medical authority. Monitoring will continue until total coliforms, *Enterococci* or *Clostridium* perfingens, are no longer detected.

C3.3.3.3.4. When any routine or repeat sample tests positive for total coliforms, it will be tested for fecal coliform or *E. coli*. Fecal-type testing can be foregone on a total coliform positive sample if fecal or *E. coli* is assumed to be present.

C3.3.3.3.5. If a system has exceeded the MCL for total coliforms, the installation will complete the notification in subsection C3.3.4. to:

C3.3.3.3.5.1. The appropriate DoD medical authority, as soon as possible, but in no case later than the end of the same day the command responsible for operating the PWS is notified of the result.

C3.3.3.3.5.2. The installation public as soon as possible, but not later than 72 hours after the system is notified of the test result that an acute risk to public health may exist.

C3.3.3.4. Inorganic Chemical Requirements

C3.3.3.4.1. An Installation responsible for a PWS shall ensure that the water distributed for human consumption does not exceed applicable limitations set out in Table 3.3, "Inorganic Chemical MCLs." Except for nitrate, nitrite, and total nitrate/nitrite, for systems monitored quarterly or more frequently, a system is out of compliance if the annual running average concentration of an inorganic chemical exceeds the MCL. For systems monitored annually or less frequently, a system is out of compliance if a single sample exceeds the MCL. For nitrate, nitrite, and total nitrate/nitrite, system compliance is determined by averaging the single sample that exceeds the MCL with its confirmation sample; if this average exceeds the MCL, the system is out of compliance.

C3.3.3.4.2. Systems will be monitored for inorganic chemicals at the frequency set in Table 3.4 a, "Inorganics Monitoring Requirements."

C3.3.3.4.3. If a system is out of compliance, the installation will complete the notification in paragraph C3.3.4. as soon as possible. If the nitrate, nitrite, or total nitrate and nitrite MCLs are exceeded, then this is considered an acute health risk and the installation will complete the notification to:

C3.3.3.4.3.1. The appropriate DoD medical authority as soon as possible, but in no case later than the end of the same day the command responsible for operating the PWS is notified of the result. Additionally, the Spanish Base Commander shall be notified within 24 hours of the detected non-compliance.

C3.3.3.4.3.2. The installation public as soon as possible, but not later than 72 hours after the system is notified of the test result. If the installation is only monitoring annually on the basis of direction from the appropriate DoD medical authority, it will immediately increase monitoring in accordance with Table 3.4-a, "Inorganics Monitoring Requirements," until remedial actions are completed and authorities determine the system is reliable and consistent.

C3.3.3.4.4. The MCL for arsenic applies to CWS and NTNCWS.

C3.3.3.5. Fluoride Requirements

C3.3.3.5.1. An installation commander responsible for a PWS shall ensure that the fluoride content of drinking water does not exceed the MCL of 4 mg/L as stated in Table 3.3., "Inorganic Chemical MCLs."

C3.3.3.5.2. Systems will be monitored for fluoride by collecting one treated water sample annually at the entry point to the distribution system for surface water systems, and once every three years for groundwater systems. Daily monitoring is recommended for systems practicing fluoridation using the criteria in Table 3.5., “Recommended Fluoride Concentrations at Different Temperatures.”

C3.3.3.5.3. If any sample exceeds the MCL, the installation will complete the notification in paragraph C3.3.4. as soon as possible, but in no case later than 24 hours after the violation.

C3.3.3.6. Lead and Copper Requirements

C3.3.3.6.1. DoD CWS and NTNCWS will comply with action levels (distinguished from the MCL) of 0.015 mg/L and 0.010 mg/L for lead (applicable until 31 December 2013, and after 1 January 2014, respectively) and 1.3 mg/L for copper to determine if corrosion control treatment, public education, and removal of lead service lines, if appropriate, are required. Actions are triggered if the respective lead or copper levels are exceeded in > 10% of all sampled taps.

C3.3.3.6.2. Affected DoD systems will conduct monitoring in accordance with Table 3.6., “Monitoring Requirements for Lead and Copper Water Quality Parameters.” High risk sampling sites will be targeted by conducting a materials evaluation of the distribution system. Sampling sites will be selected as stated in Table 3.6.

C3.3.3.6.3. If an action level is exceeded, the installation will collect additional water quality samples specified in Table 3.6., “Monitoring Requirements for Lead and Copper Water Quality Parameters,” within 24 hours to confirm the exceedance. Optimal corrosion control treatment will be pursued. If action levels are exceeded after implementation of applicable corrosion control and source water treatment, lead service lines will be replaced if the lead service lines cause the lead action level to be exceeded. The installation commander will implement an education program for persons present on the installation within 60 days and will complete the notification in paragraph C3.3.4. as soon as possible, but in no case later than 24 hours after the confirmation sampling.

C3.3.3.7. Synthetic Organics Requirements

C3.3.3.7.1. An installation responsible for a PWS system shall ensure that synthetic organic chemicals in water distributed to people do not exceed the limitations delineated in Table 3.7., “Synthetic Organic Chemical MCLs.” For systems monitored quarterly or more frequently, a system is out of compliance if the annual running average concentration of an organic chemical exceeds the MCL. For systems monitored annually or less frequently, a system is out of compliance if a single sample exceeds the MCL.

C3.3.3.7.2. Systems will be monitored for synthetic organic chemicals according to the schedule stated in Table 3.8, “Synthetic Organic Chemical Monitoring Requirements”.

C3.3.3.7.3. If a system is out of compliance, the notification set out in paragraph C3.3.4. shall be completed as soon as possible, but in no case later than 24 hours after the confirmation sampling. The installation will immediately begin quarterly monitoring and will increase quarterly monitoring if the level of any contaminant is at its detection limit but less than its MCL, as noted in Table 3.8., “Synthetic Organic Chemical Monitoring Requirements,” and will continue until the installation commander determines the system is back in compliance, and all necessary remedial measures have been implemented.

C3.3.3.8. Disinfectant/Disinfection Byproducts (DDBP) Requirements

C3.3.3.8.1. An installation responsible for a CWS and NTNCWS that adds a disinfectant (oxidant, such as chlorine, chlorine dioxide, chloramines, or ozone) to any part of its treatment process (to include the addition of disinfectant by a local water supplier) will:

C3.3.3.8.1.1. Ensure that the MCL of 0.08 mg/L for total trihalomethanes (TTHM), the MCL of 0.06 mg/L for haloacetic acids (HAA5), the MCL of 1.0 mg/L for chlorite, and the MCL of 0.01 mg/L for bromate are met in drinking water.

C3.3.3.8.1.2. Ensure that the maximum residual disinfectant level (MRDL) of 4.0-mg/L for free residual chlorine, the MRDL of 4.0 mg/L (measured as combined total chlorine) , for chloramines when ammonia is added during chlorination, and the MRDL of 0.8 mg/L for chlorine dioxide are met in drinking water. Operators may increase residual disinfectant levels of chlorine or chloramines (but not chlorine dioxide) in the distribution system to a level and for a time necessary to protect public health to address specific microbiological contamination water contamination, or cross-connections.

C3.3.3.8.2. Such systems that add a disinfectant will monitor TTHM and HAA5 in accordance with Table 3.9., “Disinfectant/Disinfection Byproducts Monitoring Requirements.” Additional disinfectant and disinfection byproduct monitoring for systems that utilize chlorine dioxide, chloramines, or ozone are also included in Table 3.9.

C3.3.3.8.3. For TTHM and HAA5 a system is noncompliant when the running annual average of quarterly averages of all samples taken in the distribution system, computed quarterly, exceed the MCL for TTHM, 0.080 mg/L, or the MCL for HAA5, 0.060 mg/L. Refer to Table 3.9. for chlorine, chloramine, and chlorine dioxide compliance requirements. If a system is out of compliance as described in Table 3.9., the installation will accomplish the notification requirements outlined in paragraph C3.3.4. as soon as possible, but in no case later than 24 hours after the violation, and undertake remedial measures.

C3.3.3.9. Radionuclide Requirements

C3.3.3.9.1. An installation responsible for a CWS will test the system for conformance with the applicable radionuclide limits contained in Table 3.10., “Radionuclide MCLs and Monitoring Requirements.”

C3.3.3.9.2. Systems will perform radionuclide monitoring as stated in 3.10.

C3.3.3.9.3. If the average annual MCL for gross alpha activity for radium is exceeded, the installation will complete the notification according to the procedures in paragraph C3.3.4. Notification to the Spanish Base Commander will be completed in accordance with paragraph C3.3.4. as soon as possible, but in no case later than 24 hours after the confirmation sampling. Monitoring will continue until remedial actions are completed and the average annual concentration no longer exceeds the respective MCL. Continued monitoring for gross alpha-related contamination will occur quarterly, while gross beta-related monitoring will be monthly. If any gross beta MCL is exceeded, the major radioactive components will be identified.

C3.3.3.10. Surface Water Treatment Requirements. DoD water systems that use surface water sources or GWUDISW will meet the surface water treatment requirements delineated in Table 3.1-a. If the turbidity readings in Table 3.1-a are exceeded, the installation will complete the notification in paragraph C3.3.4. as soon as possible, but in no case later than 7 days after the confirmation sampling and undertake remedial action. Surface water and GWUDISW systems that make changes to their disinfection practices (e.g., a change in disinfectant or application point) in order to meet DDBP requirements (C3.3.3.8.), will ensure that protection from microbial pathogens is not compromised.

C3.3.3.11. Non-Public Water Systems. DoD NPWSs will be monitored for total coliforms, at a minimum, and disinfectant residuals periodically.

C3.3.3.12. Alternative Water Supplies. DoD installations will, if necessary, only utilize alternative water sources, including POE/POU treatment devices and bottled water supplies, which are approved by the installation commander.

C3.3.3.13. Filter Backwash Requirements. To prevent microbes and other contaminants from passing through and into finished drinking water, DoD PWSs will ensure that recycled streams (i.e., recycled filter backwash water, sludge thickener supernatant, and liquids from dewatering processes) are treated by direct and conventional filtration processes. This requirement only applies to DoD PWSs that:

C3.3.3.13.1. Use surface water or GWUDISW;

C3.3.3.13.2. Use direct or conventional filtration processes; and

C3.3.3.13.3. Recycle spent filter backwash water, sludge thickener supernatant, or liquids from dewatering processes.

C3.3.4. Notification Requirements. When a DoD water system is out of compliance as set forth in the preceding criteria, the appropriate DoD medical authority and installation personnel (U.S. Base Commander and Spanish Base Commander) shall be notified as soon as possible but in no case later than the end of the same day of the detection of the violation. Persons present on the installation shall be notified as soon as possible but in no case later than 14 days after the violation. The notice will provide a clear and readily understandable explanation of the violation, any potential adverse health effects, the population at risk, the steps being taken to correct the violation, the necessity for seeking an alternative water supply, if any, and any preventive measures the consumer shall take until the violation is corrected. The appropriate DoD medical authority will coordinate notification of Spanish authorities in cases where off-installation populations are at risk.

C3.3.5. System Operator Requirements. DoD installations shall ensure that personnel are appropriately certified to operate DoD water systems. At a minimum, certification shall be for the class of water treatment plant operated. Certification shall meet the requirements of a recognized U.S. certification body such as the Association of Boards of Certification and/or Spanish equivalent.

C3.3.6. Labeling of Water Taps. DoD installations shall label all non-potable water taps in public buildings (e.g., administrative offices, work areas, and schools). The sign/label shall be in English; if Spanish-speaking workers are present in the workplace, the sign/label shall also be in Spanish.

Table 3.1-a. Surface Water Treatment Requirements

1. Unfiltered Systems

a. Systems which use unfiltered surface water or GUDISW will analyze the raw water for total coliforms or fecal coliforms at least weekly and for turbidity at least daily, and must continue as long as the unfiltered system is in operation. If the total coliforms and/or fecal coliforms exceed 100/100 milliliters (mL) and 20/100 mL, respectively, in excess of 10% of the samples collected in the previous 6 months, appropriate filtration must be applied. Appropriate filtration must also be applied if turbidity of the source water immediately prior to the first or only point of disinfectant application exceeds 1 Nephelometric Turbidity Units (NTU).
b. Disinfection must achieve at least 99.9% (3-log) inactivation of *Giardia lamblia* cysts and 99.99% (4-log) inactivation of viruses by meeting applicable CT values, as shown in Tables 3.11 through 3.24.
c. Disinfection systems must have redundant components to ensure uninterrupted disinfection during operational periods.
d. Disinfectant residual monitoring immediately after disinfection is required once every four hours that the system is in operation. Disinfectant residual measurements in the distribution system will be made at least daily.
e. Disinfectant residual of water entering the distribution system cannot be <0.2 mg/L for >4 hours.
f. Water in a distribution system with a heterotrophic bacteria concentration ≤500/mL measured as heterotrophic plate count is considered to have a detectable disinfectant residual for the purpose of determining compliance with the Surface Water Treatment Requirements.
g. If disinfectant residuals in the distribution system are undetected in more than 5% of monthly samples for 2 consecutive months, appropriate filtration must be implemented.

2. Filtered Systems

a. Filtered water systems will provide a combination of disinfection and filtration that achieves a total of 99.9% (3-log) removal of *Giardia lamblia* cysts and 99.99% (4-log) removal of viruses.
b. The turbidity of filtered water will be monitored at least once every four hours. The turbidity of filtered water for direct and conventional filtration systems will not exceed 0.5 NTU (1 NTU for slow sand and diatomaceous earth filters) in 95% of the analyses in a month, with a maximum of 5 NTU.
c. Disinfection must provide the remaining log-removal of *Giardia lamblia* cysts and viruses not obtained by the filtration technology applied.*
d. Disinfection residual maintenance and monitoring requirements are the same as those for unfiltered systems.

*Proper conventional treatment typically removes 2.5-log *Giardia*/ 2.0-log viruses. Proper direct filtration and diatomaceous earth filtration remove 2.0-log *Giardia*/ 1.0-log viruses. Slow sand filtration removes typically removes 2.0-log *Giardia*/ 2.0-log viruses. Less log-removal may be assumed if treatment is not properly applied.

3. SW or GWUDISW systems will provide at least 99% (2-log) removal of Cryptosporidium. A system is considered to be compliant with the Cryptosporidium removal requirements if:

a. For conventional and direct filtration systems, the turbidity level of the system's combined filter effluent water does not exceed 0.3 NTU in at least 95% of the measurements taken each month and at no time exceeds 1 NTU.
b. For slow sand and diatomaceous earth filtration plants, the turbidity level of the system's combined filter effluent water does not exceed 1 NTU in at least 95% of measurements taken each month and at no time exceeds 5 NTUs.
c. For alternative systems, the system demonstrates to the appropriate medical authority that the alternative filtration technology, in combination with disinfection treatment, consistently achieves 3-log removal and/or inactivation of *Giardia lamblia* cysts, 4-log removal and/or inactivation of viruses, and 2-log removal of Cryptosporidium oocysts.
d. For unfiltered systems, the system continues to meet the source water monitoring requirements noted in 1a above to remain unfiltered.

4. **Individual Filter Effluent Monitoring**. Conventional or direct filtration systems must continuously monitor (every 15 minutes) the individual filter turbidity for each filter used at the system. Systems with two or fewer filters may monitor combined filter effluent turbidity continuously, in lieu of individual filter turbidity monitoring. If a system exceeds 1.0 NTU in two consecutive measurements for three months in a row (for the same filter), the installation must conduct a self assessment of the filter within 14 days. The self-assessment must include at least the following components: assessment of filter performance; development of a filter profile; identification and prioritization of factors limiting filter performance; assessment of the applicability of corrections; and preparation of a self-assessment report. If a system exceeds 2.0 NTU (in two consecutive measurements 15 minutes apart) for two months in a row, a Comprehensive Performance Evaluation (CPE) must be conducted within 90 days by a third party.

5. **Covers for Finished Water Storage Facilities.** Installations must physically cover all finished water reservoirs, holding tanks, or storage water facilities.

Table 3.1-b. Quality Standards for Surface Water used for Drinking Water

Parameter	Units	Threshold Limits by Category		
		A1	A2	A3
pH	---	(6.5-8.5)	(5.5-9)	(5.5-9)
Color	scale Pt	20	100	200
Conductivity	μs/cm	(1,000)	(1,000)	(1,000)
Suspended solids	mg/L	(25)	---	---
Temperature	°C	25	25	25
BOD_5	mg/L	(3)	(5)	(7)
COD	mg/L	---	---	(30)
Total coliforms (37°C)	100 mL	(50)	(5,000)	(50,000)
Fecal coliform	100 mL	(20)	(2,000)	(20,000)
Fecal streptococci	100 mL	(20)	(1,000)	(10,000)
Salmonella	---	---	absent in 5,000 mL	absent in 1,000 mL
Ammonia	mg/L	(0.05)	1.5	4
Arsenic	mg/L	0.05	0.05	0.1
Barium	mg/L	0.1	1	1
Boron	mg/L	(1)	(1)	(1)
Cadmium	mg/L	0.005	0.005	0.005
Chlorides	mg/L	(200)	(200)	(200)
Chloroform Extractable substances	mg/L SEC	(0.1)	(0.2)	(0.5)
Chromium (total)	mg/L	0.05	0.05	0.05
Copper	mg/L	0.05	(0.05)	(1)
Cyanide	mg/L	0.05	0.05	0.05
Detergents (lauril-sulfate)	mg/L	(0.2)	(0.2)	(0.5)
Dissolved iron	mg/L	0.3	2	(1)
Dissolved or emulsified Hydrocarbons	mg/L	0.05	0.2	1
Dissolved oxygen	% saturation	(70)	(50)	(30)
Fluoride (2)	mg/L	1.5	0.7/1.7	0.7/1.7
Lead	mg/L	0.05	0.05	0.05
Manganese	mg/L	(0.05)	(0.1)	1
Mercury	mg/L	0.001	0.001	0.001
Nitrates	mg/L	50	50	50
Nitrogen Kjedhal	mg/L	(1)	(2)	(3)
PAH	mg/L	0.0002	0.0002	0.001
Pesticides	mg/L	0.001	0.0025	0.005
Phenols	mg/L as C_6H_5OH	0.001	0.005	0.1
Phosphates	mg/L	(0.4)	(0.7)	(0.7)
Selenium	mg/L	0.01	0.01	0.01
Sulfates	mg/L	250	250	250

Parameter	Units	Threshold Limits by Category		
		A1	A2	A3
Zinc	mg/L	3	5	5

Notes:

1. Values within parenthesis shall be considered as guidance values.
2. Limit based on temperature [0.7: elevated temperature; 1.7: low temperature]

Class	Required Treatment
A1	Simple physical treatment and disinfection (e.g., fast filtration and disinfection)
A2	Normal physical and chemical treatment plus disinfection (e.g., pre-chlorination, coagulation, flocculation, decantation, filtration, and final chlorination)
A3	Intensive physical and chemical treatment plus disinfection (e.g., chlorination to break point, coagulation, flocculation, decantation, filtration, adsorption (activated carbon), and disinfection (ozone or final chlorination))

Instructions:

If a raw water sample does not meet the value of a given class, then the source water falls into the next class for treatment purposes. Surface water with concentrations or values above threshold limit A3 cannot be used for production of drinking water. Exceptions must be authorized by the appropriate DoD medical authority.

Acceptance quality criteria are as follows:

- If 95% of the samples conform to established quality values
- If 90% of the samples conform to established quality values and 5% of the non-conforming samples comply with the following conditions:

    1. Results do not differ by more than 50% for all parameters except for temperature, pH, dissolved oxygen, and microbiologic parameters;
    2. Health is not threatened; and
    3. Consecutive samples taken with a statistical frequency do not differ for the corresponding parameters.

Table 3.1-c. Allowable Drinking Water Additives

Name	UNE-EN Standard	CAS	EINECS	Main functions
Acetic acid	13194	64-19-7	203-56-48	Denitrification
Hydrochloric acid	939	7647-01-0	231-595-7	pH corrective, regenerating resins, Chlorine dioxide precursor
Phosphoric acid (Orthophosphoric acid)	974	7664-38-2	231-633-2	Corrosion inhibitor
Hexafluorsilicic acid	12175	16961-83-4	241-034-8	Fluoridation
Sulfuric acid	899	7664-93-9	231-639-5	pH corrective
Phosphonic acid and its salts	15040	32545-75-8 6419-19-8 2809-21-4 15827-60-8 1429-50-1 5995-42-6 37971-36-1 23605-74-5	251-094-7 229-146-5 220-552-8 239-931-4 215-851-5 227-833-4 253-733-5 245-781-0	Anti-fouling
Sodium alginate	1405	9005-38-3	232-68-01	Coagulant / flocculant
Modified starches	1406	9005-25-8 56780-58-6 9063-38-1	232-679-6	Coagulant / flocculant
Granulated activated aluminum (Aluminium oxide)	13753	1344-28-1	---	Coagulant / flocculant
Sodium aluminate (Aluminum oxide and sodium)	882	11138-49-1	234-391-6	Coagulant / flocculant
Expanded Aluminum silicate	12905	---	---	Filtration
Liquid ammonia	12126	7664-41-7	231-635-3	Chloramination pre-treatment
Ammonia	12122	1336-21-6	215-647-6	Chloramination pre-treatment
Anthracite	12909	---	---	Filtration
Manganese Greensand	12911	90387-66-9 1313-13-9	291-341-6 215-202-6	Filtration
Quartz gravel and sand	12904	---	---	Filtration
Baryta	12912	13462-86-7	236-664-5	Filtration
Bentonite	13754	1302-78-9	201-108-5	Flocculation adjuvant, adsorbent
Calcium bis(dihydrogen phosphate)	1204	7758-11-4	231-837-1	Corrosion inhibitor
Calcium hydroxide/calcium oxide	12518	1305-62-0 1305-78-8	215-137-3 215-138-9	pH corrective, remineralization, co-precipitation and water softening, reduction of aggressivity
Activated carbon in dust	12903	7440-44-0	231-153-3	Adsorbent

Name	UNE-EN Standard	CAS	EINECS	Main functions
Granular activated carbon - reactivated	12915-2	7440-44-0	231-153-3	Adsorbent
Granular activated carbon - virgin	12915-1	7440-44-0	231-153-3	Adsorbent
Pyrolised carbon	12907	---	---	Filtration
Calcium carbonate coated with manganese dioxide	14368	1313-13-9 471-34-1	215-202-6 207-439-9	Filtration
Calcium carbonate	1018	1317-65-3	215-279-6	pH corrective, remineralization, reduction of aggressivity
Sodium carbonate	897	497-19-8	207-838-8	pH and alkalinity corrective
Sodium chlorate	15028	7775-09-9	231-887-4	Chlorine dioxide precursor
Sodium chlorite	938	7758-19-2	231-836-6	Chlorine dioxide precursor
Iron (III) chloride sulfate	891	12410-14-9	235-649-0	Coagulant
Aluminum chloride, Hydroxy-aluminum chloride, Aluminum hydroxy chloride sulfate	881	7446-70-0 1327-41-9 14215-15-7 39290-78-3	231-208-1 215-477-2 238-071-7 254-400-7	Coagulant / flocculant
Iron (III) and Aluminum chloride, Hydroxy-aluminum and iron (III) chloride	935	7446-70-0 7705-08-0 1327-41-9 14215-15-7	231-208-1 231-729-4 215-477-2 238-071-7	Coagulant / flocculant
Ammonium chloride	1421	12125-02-9	235-186-4	Chloramination pre-treatment
Iron (III) chloride	888	7705-08-0 10025-77-1	231-729-4	Coagulant / flocculant
Sodium chloride	973	7647-14-5	231-598-3	Regenerating resins
Sodium chloride	14805	7647-14-5	231-598-3	Electrochemical generation of chlorine using technology without membrane
Potassium dihydrogen phosphate	1201	7778-77-0	231-913-4	Corrosion inhibitor
Zinc dihydrogen phosphate in solution	1197	13598-37-3	237-067-2	Corrosion inhibitor
Sodium dihydrogen phosphate	1198	7758-80-7	231-449-2	Corrosion inhibitor
Sodium dihydrogen pyrophosphate	1205	7758-16-9	231-835-0	Anti-scaling
Carbon dioxide	936	124-38-9	204-696-9	pH corrective, remineralization, reduction of aggressivity
Manganese dioxide	13752	1313-13-9	215-202-6	Filtration
Sodium disulphite	12121	7681-57-4	231-673-0	Reducing agent
Dolomite half calcined	1017	471-34-1	207-439-9	pH corrective
Sodium fluoride	12173	7681-49-4	231-667-8	Fluorination
Tripotassium phosphate	1203	7778-53-2	231-907-1	Corrosion inhibitor
Trisodium phosphate	1200	7601-54-9	231-509-8	Corrosion inhibitor

Name	UNE-EN Standard	CAS	EINECS	Main functions
Garnet	12910	---	---	Filtration
Sodium hexafluorsilicate	12174	16893-85-9	240-934-8	Fluorination
Sodium hydrogen carbonate	898	144-55-8	205-633-8	pH and alkalinity corrective
Potassium hydrogen phosphate	1202	7758-11-4	231-834-5	Corrosion inhibitor
Sodium hydrogen phosphate	1199	7558-79-4	231-448-7	Corrosion inhibitor
Sodium bisulfite	12120	7631-90-5	231-548-0	Reducing agent
Sodium hydroxide	896	1310-73-2	215-185-5	pH and alkalinity corrective
Iron covered with active granular alumina	14369	1344-28-1 10028-22-5	215-691-6 233-072-9	Filtration, absorbent
Iron (III) oxide-hydroxide	15029	20344-49-4	243-746-4	Absorbent
Oxygen	12876	7782-44-7	231-956-9	Oxygenation, ozonization pre-treatment
Ozone	1278	10028-15-6	---	Ozonization
Pearlite in dust	12914	---	---	Filtration
Potassium permanganate	12672	7722-64-7	231-76-03	Oxidant
Sodium permanganate	15482	10101-50-5	233-251-1	Oxidant
Pumice stone	12906	---	---	Filtration
Tetrapotassium Pyrophosphate	1207	7320-34-5	230-785-7	Anti-scaling
Tetrasodium Pyrophosphate	1206	7722-88-5	231-767-1	Corrosion inhibitor
Poly(Dimethyl ammonium chloride)	1408	26062-79-3	---	Coagulant / flocculant
Calcium and sodium polyphosphate	1208	23209-59-8	245-490-9	Corrosion inhibitor, anti-scaling
Sodium polyphosphate	1212	68915-31-1	272-808-3	Corrosion inhibitor
Polyphosphates	15041	---	---	Anti-fouling
Aluminum polyhydroxy chlorosilicate	885	94894-80-1	215-477-2 215-475-1 231-598-3	Coagulant / flocculant
Aluminum polyhydroxy chloride and aluminum polyhydroxy chloro sulfate	883	1327-41-9 12042-91-0 10284-64-7 39290-78-3	215-477-2 234-933-1 233-632-2 254-400-7	Coagulant / flocculant
Aluminum polyhydroxy sulfate silicate	886	131148-05-5	259-881-7 215-475-1 231-820-9	Coagulant / flocculant
Sodium silicate	1209	1344-09-8	215-687-4	Coagulant / flocculant, Corrosion inhibitor
Iron (III) and Aluminum sulfate	887	10043-01-3 10028-22-5	233-135-0 233-072-9	Coagulant
Aluminum sulfate	878	10043-01-3 16828-11-8 7784-31-8	233-135-0	Coagulant / flocculant
Ammonium sulfate	12123	7783-20-2	213-984-1	Chloramination pre-

Name	UNE-EN Standard	CAS	EINECS	Main functions
				treatment
Copper sulfate	12386	7758-98-7 7758-99-7	231-847-6	Algicide
Iron (II) sulfate	889	7782-63-0 7720-78-7	231-753-5	Coagulant
Iron (III) sulfate - liquid	890	10028-22-5	233-072-9	Coagulant
Iron (III) sulfate - solid	14664	10028-22-5	233-072-9	Coagulant
Sodium sulfite	12124	7757-83-7	231-821-4	Reducing agent
Diatom soil in dust	12913	61790-53-2 90053-39-3 68855-54-9	293-303-4	Filtration
Sodium thiosulfate	12125	7772-98-7 10102-17-7	231-867-5	Reducing agent
Potassium Tripolyphosphate	1211	13845-36-8	237-574-9	Anti-scaling
Sodium Tripolyphosphate	1210	7758-29-4	231-838-7	Anti-scaling
BIOCIDES				
Chlorine	937	7782-50-5	231-959-5	---
Sulfur dioxide	1019	7446-09-5	231-195-2	---
Chlorine dioxide	12671	10049-04-4	233-162-8	---
Calcium hypochlorite	900	7778-54-3	231-908-7	---
Sodium hypochlorite	901	7681-52-9	231-668-3	---
Hydrogen peroxide	902	7722-84-1	231-765-0	---
Potassium peroxymonosulfate	12678	70693-62-8	274-778-7	---
Trichloroisocyanuric acid	12933	87-90-1	201-782-8	---
Sodium dichloroisocyanurate anhydrous	12931	2893-78-9	2-207-67-7	---
Sodium dichloroisocyanurate dihydrate	12932	51580-86-0	---	---

Table 3.2. Total Coliform Monitoring Frequency

Population Served	Number of Samples [1]	Population Served	Number of Samples [1]
< 500	1	59,001 to 70,000	70
501 to 2,500	2	70,001 to 83,000	80
2,501 to 3,300	3	83,001 to 96,000	90
3,301 to 4,100	4	96,001 to 130,000	100
4,101 to 4,900	5	130,001 to 220,000	120
4,901 to 5,800	6	220,001 to 320,000	150
5,801 to 6,700	7	320,001 to 450,000	180
6,701 to 7,600	8	450,001 to 600,000	210
7,601 to 8,500	9	600,001 to 780,000	240
8,501 to 12,900	10	780,001 to 970,000	270
12,901 to 17,200	15	970,001 to 1,230,000	300
17,201 to 21,500	20	1,230,001 to 1,520,000	330
21,501 to 25,000	25	1,520,001 to 1,850,000	360
25,001 to 33,000	30	1,850,001 to 2,270,000	390
33,001 to 41,000	40	2,270,001 to 3,020,000	420
41,001 to 50,000	50	3,020,001 to 3,960,000	450
50,001 to 59,000	60	3,960,001 or more	480

Notes:
1. Minimum Number of Routine Samples Per Month.

Note: Systems that use groundwater, serve < 4,900 people, and collect samples from different sites, may collect all samples on a single day. All other systems must collect samples at regular intervals throughout the month.

Table 3.3. Inorganic Chemical MCLs

Parameter	Units	MCL
Organoleptic parameters		
Color	mg/L Pt/Co	15
Odor	--	3 at 25°C
Taste	--	3 at 25°C
Physical-chemical parameters		
Aluminum (Al)	mg/L	0.2
Ammonium (NH_4)	mg/L	0.5
Antimony (Sb)[1]	mg/L	0.005
Arsenic (As)[1]	mg/L	0.010
Asbestos[1]	fibers/L (>10 µm)	7 million
Barium (Ba)	mg/L	2.0
Beryllium (Be)[1]	mg/L	0.004
Boron (B)	mg/L	1
Cadmium (Cd)[1]	mg/L	0.005
Chlorides (Cl^-)	mg/L	250
Chromium (Cr)[1]	mg/L	0.05
Conductivity	uS/cm^{-1} at 20ºC	2,500
Cyanides (CN)[1]	mg/L	0.05
Fluoride	mg/L	4.0
Hydrogen ion concentration pH	--	6.5 – 9.5
Iron (Fe)	mg/L	0.2
Manganese (Mn)	mg/L	0.05
Mercury (Hg)[1]	mg/L	0.001
Nickel (Ni)[1]	mg/L	0.02
Nitrate (as N)	mg/L	10
Nitrite (as N)	mg/L	0.5
Oxidability (O_2)	mg/L	5
Selenium (Se)[1]	mg/L	0.01
Sodium (Na)	mg/L	200
Sulfates (SO_4)	mg/L	250
Thallium	mg/L	0.002
Total Nitrite and Nitrate (as N)	mg/L	10

Notes:

1. MCLs apply to CWS and NTNCWS systems.

Table 3.4-a. Inorganics Monitoring Requirements (refer to Table 3.4-b for Check/Audit Monitoring Requirements)

Parameter	Groundwater Baseline Requirement [1]	Surface Water Baseline Requirement	Trigger That Increases Monitoring [2]	Reduced Monitoring
Color	Check/Audit	Check/Audit	---	---
Odor	Check/Audit	Check/Audit	---	---
Taste	Check/Audit	Check/Audit	---	---
Aluminum	Check/Audit	Check/Audit	---	---
Ammonium	Check/Audit	Check/Audit	---	---
Antimony	Audit Monitoring	Audit Monitoring	>MCL	---
Arsenic	Based on population served: ≤5,000:1 sample / 3 yr >5,000: Audit Monitoring	Based on population served: ≤5,000:Annual sample >5,000: Audit Monitoring	>MCL	---
Asbestos	1 sample / 9 yr	1 sample / 9 yrs	>MCL	Yes [3]
Barium	1 sample / 3 yr	Annual sample	>MCL	---
Beryllium	1 sample / 3 yr	Annual sample	>MCL	---
Boron	Audit Monitoring	Audit Monitoring	---	---
Cadmium	Based on population served: ≤5,000:1 sample / 3 yr >5,000: Audit Monitoring	Based on population served: ≤5,000:Annual sample >5,000: Audit Monitoring	>MCL	---
Chlorides	Audit Monitoring	Audit Monitoring	---	---
Chromium	Based on population served: ≤5,000:1 sample / 3 yr >5,000: Audit Monitoring	Based on population served: ≤5,000:Annual sample >5,000: Audit Monitoring	>MCL	---
Conductivity	Check Monitoring	Check Monitoring	---	--
Corrosivity [8]	Once	Once	---	---
Cyanide	Based on population served: ≤5,000:1 sample / 3 yr >5,000: Audit Monitoring	Based on population served: ≤5,000:1 sample / 3 yr >5,000: Audit Monitoring	>MCL	---
Fluoride	Audit Monitoring	Audit Monitoring	>MCL	
Free carbon dioxide	Audit Monitoring	Audit Monitoring	---	---
Hydrogen ion concentration (pH)	Check monitoring[8]	Check monitoring[8]	---	---
Iron	Check/Audit	Check/Audit	---	---

Parameter	Groundwater Baseline Requirement [1]	Surface Water Baseline Requirement	Trigger That Increases Monitoring [2]	Reduced Monitoring
Manganese	Audit Monitoring	Audit Monitoring	---	---
Mercury	Based on population served: ≤5,000:1 sample / 3 yr >5,000: Audit Monitoring	Based on population served: ≤5,000:1 sample / 3 yr >5,000: Audit Monitoring	>MCL	---
Nickel	Based on population served: ≤5,000:1 sample / 3 yr >5,000: Audit Monitoring	Based on population served: ≤5,000:1 sample / 3 yr >5,000: Audit Monitoring	>MCL	---
Nitrate	Audit Monitoring [4]	Quarterly [4] or Audit Monitoring (if more frequent)	>50% MCL [5]	Yes [6]
Nitrite	Check/Audit [4]	Quarterly [4] or Check/Audit (if more frequent)	>50% MCL [5]	Yes [7]
Total Nitrate/Nitrite	Annual sample	Quarterly	>50% Nitrite MCL	---
Oxidizability	Audit Monitoring	Audit Monitoring	---	---
Residual chlorine	Check Monitoring	Check Monitoring	---	---
Selenium	Based on population served: ≤5,000:1 sample / 3 yr >5,000: Audit Monitoring	Based on population served: ≤5,000:1 sample / 3 yr >5,000: Audit Monitoring	>MCL	---
Sodium [11]	Based on population served: ≤5,000:1 sample / 3 yr >5,000: Audit Monitoring	Based on population served: ≤5,000:1 sample / 3 yr >5,000: Audit Monitoring	---	---
Sulfates	Audit Monitoring	Audit Monitoring	---	---
Thallium	1 sample / 3 yr	Annual sample	>MCL	---

Notes:

1. Samples shall be taken as follows: groundwater systems shall take a minimum of one sample at every entry point to the distribution system, which is representative of each well after treatment; surface water systems shall take at least one sample at every entry point to the distribution system after any application of treatment or in the distribution system at a point, which is representative of each source after the treatment.
2. Increased quarterly monitoring requires a minimum of 2 samples per quarter for groundwater systems and at least 4 samples per quarter for surface water systems.
3. Necessity for analysis is predicated upon a sanitary survey conducted by the PWS.

4. Any sampling point with an analytical value $\geq$ 0.5 mg/L as N, (50% of the Nitrite MCL) must begin sampling for nitrate and nitrite separately. Since nitrite readily converts to nitrate, a system can conclude that if the total nitrate/nitrite value of a sample is less than half of the nitrite MCL, then the value of nitrite in the sample would also be below half of its MCL.
5. Increased quarterly monitoring shall be undertaken for nitrate and nitrate if a sample is > 50% of the MCL.
6. The appropriate DoD medical authority may reduce repeat sampling frequency for surface water systems to annually if after 1 year results are < 50% of MCL.
7. The appropriate DoD medical authority may reduce repeat sampling frequency to 1 annual sample if results are 50% of MCL.
8. PWSs shall be analyzed within 1 year of the effective date of country-specific FGS to determine the corrosivity entering the distribution system. Two samples (one mid-winter and one mid-summer) will be collected at the entry point of the distribution system for systems using surface water and GWUDISW. One sample will be collected for systems using only groundwater. Corrosivity characteristics of the water shall include measurements of pH, calcium, hardness, alkalinity, temperature, total dissolved solids, and calculation of the Langelier Saturation Index.

Table 3.4-b. Monitoring Frequencies based on Volume (refer to Table 3.4-a)

Volume of water treated / distributed (m^3/day)	Check monitoring		Audit monitoring
	Exit of treatment plant (samples/year)	Distribution network (samples/year)	Exit of treatment plant and distribution network (samples/year)
< 100	1	1	Determined by DoD medical authority
> 100 - 1,000	2	2	1
> 1,000 to 10,000 for Audit Monitoring	2 for each 1,000 m^3/day and part thereof	1+1 for each 1,000 m^3/day and part thereof	1 for each 5,000 m^3/day and part thereof
> 10,000 - 100,000	---	---	2+1 for each 20,000 m^3/day and part thereof
> 100,000	---	---	5+1 for 50,000 m^3/day and part thereof

Table 3.5. Recommended Fluoride Concentrations at Different Temperatures

Annual Average of Maximum Daily Air Temperatures (°F)	Control Limits (mg/L)		
	Lower	Optimum	Upper
50.0 - 53.7	0.9	1.2	1.~~7~~2
53.8 - 58.3	0.8	1.1	1.~~5~~2
58.4 - 63.8	0.8	1.0	1.~~3~~2
63.9 - 70.6	0.7	0.9	1.2
70.7 - 79.2	0.7	0.8	1.0
79.3 - 90.5	0.6	0.7	0.8

Table 3.6. Monitoring Requirements for Lead and Copper Water Quality Parameters

Population Served	# of Sites for Standard Monitoring [1, 2]	# of Sites for Reduced Monitoring [3]	# of Sites for Water Quality Parameters [4]
>100,000	100	50	25
10,001 - 100,000	60	30	10
3,301 - 10,000	40	20	3
501 - 3,300	20	10	2
101 - 500	10	5	1
<100	5	5	1

Notes:

1. Every 6 months for lead and copper.
2. Sampling sites shall be based on a hierarchical approach. For CWS, priority will be given to single family residences which contain copper pipe with lead solder installed after 1982, contain lead pipes, or are served by lead service lines; then, structures, including multi-family residences with the foregoing characteristics; and finally, residences and structures with copper pipe with lead solder installed before 1983. For NTNCWS, sampling sites will consist of structures that contain copper pipe with lead solder installed after 1982, contain lead pipes, and/or are served by lead service lines. First draw samples will be collected from a cold water kitchen or bathroom tap; non-residential samples will be taken at an interior tap from which water is typically drawn for consumption.
3. Annually for lead and copper if action levels are met during each of 2 consecutive 6-month monitoring periods. Any small or medium-sized system (<50,000) that meets the lead and copper action levels during three consecutive years may reduce the monitoring for lead and copper from annually to once every three years. Annual or triennial sampling will be conducted during the four warmest months of the year.
4. This monitoring must be conducted by all large systems (>50,000). Small and medium sized systems must monitor water quality parameters when action levels are exceeded. Samples will be representative of water quality throughout the distribution system and include a sample from the entry to the distribution system. Samples will be taken in duplicate for pH, alkalinity, calcium, conductivity or total dissolved solids, and water temperatures to allow a corrosivity determination (via a Langelier saturation index or other appropriate saturation index); additional parameters are orthophosphate when a phosphate inhibitor is used and silica when a silicate inhibitor is used.

Table 3.7. Synthetic Organic Chemical MCLs

Parameter[2]	MCL (mg/L)	Detection Limit (mg/L)
Pesticides/PCBs		
Alachlor	0.0001	0.0002
Aldrin	0.00003	---
Aldicarb	0.0001	0.0005
Aldicarb sulfone	0.0001	0.0008
Aldicarb sulfoxide	0.0001	0.0005
Atrazine	0.0001	0.0001
Benzo[a]pyrene (as total PAH)	0.000010	0.00002
Carbofuran	0.0001	0.0009
Chlordane	0.0001	0.0002
Dalapon	0.2	---
2,4-D	0.0001	0.0001
1,2-Dibromo-3-chloropropane (DBCP)	0.0001	0.00002
Di (2-ethylhexyl) adipate	0.4	0.0006
Di (2-ethylhexyl) phthalate	0.006	0.0006
Dieldrin	0.00003	---
Dinoseb	0.0001	0.0002
Diquat	0.0001	0.0004
Endrin	0.0001	0.00002
Endothall	0.0001	0.009
Ethylene dibromide (EDB) (1,2-Dibromoethane)	0.00005	0.00001
Glyphosphate	0.0001	0.006
Heptachlor	0.00003	0.00004
Heptachlorepoxide	0.00003	0.00002
Hexachlorobenzene	0.001	0.0001
Hexachlorocyclopentadiene	0.05	0.0001
Lindane	0.0001	0.00002
Methoxychlor	0.0001	0.0001
Oxamyl (Vydate)	0.0001	0.002
PCBs (as decachlorobiphenyls)	0.0001	0.0001
Pentachlorophenol	0.0001	0.00004
Picloram	0.0001	0.0001
Polycyclic Aromatic Hydrocarbons	0.0001	---
Simazine	0.0001	0.00007
2,3,7,8-TCDD (Dioxin)	0.00000003	0.000000005
Toxaphene	0.0001	0.0001
2,4,5-TP (Silvex)	0.0001	0.0002
Total Pesticides[1]	0.0005	---

Parameter[2]	MCL (mg/L)	Detection Limit (mg/L)
Volatile Organic Chemicals		
Benzene	0.001	0.0005
Carbon tetrachloride (Tetrachloromethane)	0.005	0.0005
o-Dichlorobenzene (1,2-Dichlorobenzene)	0.6	0.0005
cis-1,2-Dichloroethylene (cis-1,2-Dichloroethene)	0.07	0.0005
trans-1,2-Dichloroethylene (trans-1,2-Dichloroethene)	0.1	0.0005
1,1-Dichloroethylene (1,1-Dichloroethene)	0.007	0.0005
1,1,1-Trichloroethane	0.2	0.0005
1,2-Dichloroethane	0.003	0.0005
Dichloromethane	0.005	---
1,1,2-Trichloroethane	0.005	---
1,2,4-Trichloro-benzene	0.07	---
1,2-Dichloropropane	0.005	0.0005
Ethylbenzene	0.7	0.0005
Hydrocarbons (dissolved or emulsified; after extraction); mineral oil	0.01	---
Monochlorobenzene (Chlorobenzene)	0.1	0.0005
para-Dichlorobenzene (1,4-Dichlorobenzene)	0.075	0.0005
Styrene	0.1	0.0005
Tetrachloroethylene (Tetrachloroethene)	0.005	0.0005
Trichloroethylene (Trichloroethene)	0.005	0.0005
Trihalomethanes (total)	0.08	---
Toluene	1.0	0.0005
Vinyl chloride	0.0005	0.0005
Xylene (total)	10	0.0005
Other Organics		
Acrylamide	0.05% dosed at 1 ppm[2]	
Epihydrochlorin	Treatment technique 0.01% dosed at 20 ppm[2]	
Phenols (as C_6H_5OH)	0.0005	---

Note:

1. The MCL for Total Pesticides excludes dalapon.
2. Only applies when adding these polymer flocculants to the treatment process. No sampling is required; the system certifies that dosing is within specified limits.

Table 3.8. Synthetic Organic Chemical Minimum Monitoring Requirements

Parameter	Base Requirement [1]		Trigger for more monitoring [2]	Reduced Monitoring
	Groundwater	Surface Water		
VOCs	Quarterly	Quarterly	> 0.0005 mg/L	Yes [3, 4]
Pesticides/PCBs	4 quarterly samples/3 years during most likely period for their presence		> Detection limit [6]	Yes [4, 5]

Notes:

1. Groundwater systems shall take a minimum of one sample at every entry point which is representative of each well after treatment; surface water systems will take a minimum of one sample at every entry point to the distribution system at a point which is representative of each source after treatment. For CWS, monitoring compliance is to be, met within 1 year of the publishing of the FGS; for NTNCW, compliance is to be met within 2 years of the publishing of the FGS.
2. Increased monitoring requires a minimum of 2 quarterly samples for groundwater systems, and at least 4 quarterly samples for surface water systems.
3. Repeat sampling frequency may be reduced to annually after 1 year of no detection, and every 3 years after three rounds of no detection.
4. Monitoring frequency may be reduced if warranted based on a sanitary survey of the PWS.
5. Detection limits noted in Table 3.7, or as determined by the best available testing methods.
6. Repeat sampling frequency may be reduced to the following if after one round of no detection: systems >3,300 reduce to a minimum of 2 quarterly samples in one year during each repeat compliance period, or systems <3,300 reduce to a minimum of 1 sample every 3 years.
7. Compliance is based on an annual running average for each sample point for systems monitoring quarterly or more frequently; for systems monitoring annually or less frequently, compliance is based on a single sample, unless the appropriate DoD medical authority requests a confirmation sample. A system is out of compliance if any contaminant exceeds the MCL.

Table 3.9. Disinfectant/Disinfection Byproducts Monitoring Requirements

Source Water Type	Population Served by System	Analyte & Frequency of Samples	Number of Samples
Surface Water (SW) or Groundwater Under the Direct Influence of Surface Water (GWUDISW)	10,000 or more	TTHM & HAA5 – Quarterly [1,2]	4 [1,2,3]
SW or GWUDISW	Serving 500 to 9,999	TTHM & HAA5 - Quarterly [4]	1 [5,6]
SW or GWUDISW	499 or less	TTHM & HAA5 - Yearly	1 [7,8]
Ground Water (GW)	10,000 or more	TTHM & HAA5 – Quarterly [9]	1 [10,11]
GW	9,999 or less	TTHM & HAA5 - Yearly [12]	1 [13,14]
		Chlorite - Daily & Monthly [15,16,17,18]	
		Bromate – Monthly [19,20]	
		Chlorine [21,22]	
		Chloramines [23,24]	
		Chlorine Dioxide [25,26,27]	
		TOC [28]	

Notes:

1. For TTHM and HAA5, a DoD system using surface water or GWUDISW that treats its water with a chemical disinfectant must collect the number of samples listed above. One of the samples must be taken at a location in the distribution system reflecting the maximum residence time of water in the system. The remaining samples shall be taken at representative points in the distribution system.
2. To be eligible for reduced monitoring, a system must meet all of the following conditions: a) the annual average for TTHM is no more than 0.040 mg/L; b) the annual average for HAA5 is no more than 0.030 mg/L; c) at least one year of routine monitoring has been completed; and d) the annual average source water total organic carbon level is no more than 4.0 mg/L prior to treatment. Systems may then reduce monitoring of TTHM and HAA5 to one sample per treatment plant per quarter. Systems remain on the reduced schedule as long as the average of all samples taken in the year is no more than 0.060 mg/L for TTHM and 0.045 mg/L for HAA5. Systems that do not meet these levels must revert to routine monitoring the following quarter.
3. A system is noncompliant if the running annual average for any quarter exceeds the TTHM MCL, 0.080 mg/L or the HAA5 MCL, 0.060 mg/L.
4. One sample must be collected per treatment plant in the system at the point of maximum residence time in the distribution system.
5. Systems meeting the eligibility requirements in Note 2 may reduce monitoring frequency to one sample per treatment plant per year. Sample must be taken at the point of maximum residence time in the distribution system and during the month of warmest water temperature. Systems remain on the reduced schedule as long as the average of all samples taken in the year is no more than 0.060 mg/L for TTHM and 0.045 mg/L for HAA5. Systems that do not meet these levels must revert to routine (quarterly) monitoring the following quarter.

6. A system is noncompliant if the annual average of all samples taken that year exceeds the TTHM MCL, 0.080 mg/L or the HAA5 MCL, 0.060 mg/L.
7. Sample must be taken at the point of maximum residence time in the distribution system and during the month of warmest water temperature. If annual sample exceeds MCL (TTHM or HAA5) the system must increase monitoring to one sample per treatment plant per quarter at the point of maximum residence time. The system may return to routine monitoring if the annual average of quarterly samples is no more than 0.060 mg/L for TTHM and 0.045 mg/L for HAA5.
8. No reduced monitoring schedule is available. Noncompliance exists when the annual sample (or average of annual samples is conducted) exceeds the TTHM MCL, 0.080 mg/L or if the HHA5 concentration exceeds the MCL, 0.060 mg/L.
9. For TTHM and HAA5, a DoD system using only ground water NOT under the influence of surface water that treats its water with a chemical disinfectant must collect the number of samples listed above. Samples must be taken at a location in the distribution system reflecting the maximum residence time of water in the system.
10. System may reduce monitoring to one sample per treatment plant per year if the system meets all of the following conditions: a) the annual average for TTHM is no more than 0.040 mg/L; b) the annual average for HAA5 is no more than 0.030 mg/L; and c) at least one year of routine monitoring has been completed. Sample must be taken at the point of maximum residence time in the distribution system and during the month of warmest water temperature. Systems remain on the reduced schedule as long as the average of all samples taken in the year is no more than 0.060 mg/L for TTHM and 0.045 mg/L for HAA5. Systems that do not meet these levels must revert to routine monitoring the following quarter.
11. Noncompliance exists when the annual average of quarterly averages of all samples, compounded quarterly, exceeds the TTHM MCL, 0.080 mg/L or the HHA5 the MCL, 0.060 mg/L.
12. For TTHM and HAA5, a DoD system using only ground water NOT under the influence of surface water that treats its water with a chemical disinfectant must collect the number of samples listed above. One sample per treatment plant must be taken at a location in the distribution system reflecting the maximum residence time of water in the system and during the month of warmest water temperature. If the sample exceeds the MCL, the system must increase monitoring to quarterly.
13. System may reduce monitoring to one sample per three-year monitoring cycle if the system meets all the following conditions: a) the annual average for TTHM is no more than 0.040 mg/L; b) the annual average for HAA5 is no more than 0.030 mg/L; and c) at least one year of routine monitoring has been completed. Sample must be taken at the point of maximum residence time in the distribution system and during the month of warmest water temperature. Systems remain on the reduced schedule as long as the average of all samples taken in the year is no more than 0.060 mg/L for TTHM, and 0.045 mg/L for HAA5. Systems that do not meet these levels must revert to routine monitoring. Systems on increased monitoring may return to routine monitoring if the annual average of quarterly samples does not exceed 0.060 mg/L for TTHM and 0.045 mg/L for HAA5.
14. Noncompliance exists when the annual sample (or average of annual samples) exceeds the TTHM MCL, 0.080 mg/L or the HHA5 the MCL, 0.060 mg/L.
15. For systems using chlorine dioxide for disinfection or oxidation, daily samples are taken for chlorite at the entrance to the distribution system for chlorite. The monthly chlorite samples are collected within the distribution system, as follows: one as close as possible to the first customer, one in a location representative of average residence time, and one as close as possible to the end of the distribution system (reflects maximum residence time within the distribution system).
16. Additional monitoring is required when a daily sample exceeds the chlorite MCL, 1.0 mg/L. A three-sample set (following the monthly sample set protocol) is required to be collected the following day. Further distribution system monitoring will not be required in that month unless the chlorite concentration at the entrance to the distribution system again exceeds the MCL, 1.0 mg/L.
17. For chlorite, systems may reduce routine distribution system monitoring from monthly to quarterly if the chlorite concentration in all samples taken in the distribution system is below the MCL, 1.0 mg/L, for a period of one year and the system has not been required to conduct any additional monitoring. Daily samples must still be collected. Monthly sample set monitoring resumes when if any one daily sample exceeds the MCL, 1.0 mg/L.
18. Noncompliance for chlorite exists if the average concentration of any three-sample set (i.e., one monthly sample set from within the distribution system) exceeds the MCL, 1.0 mg/L.

19. Systems using ozone for disinfection or oxidation are required to take at least one sample per month from the entrance to the distribution system for each treatment plant in the system using ozone under normal operating conditions. Systems may reduce monitoring from monthly to once per quarter if the system demonstrates that the yearly average raw water bromide concentration is <0.05 mg/L based upon monthly measurements for one year.
20. Noncompliance is based on a running yearly average of samples, computed quarterly, that exceeds the MCL, 0.01 mg/L.
21. Chlorine samples must be measured at the same points in the distribution system and at the same time as total coliforms. Not withstanding the MRDL, operators may increase residual chlorine levels in the distribution system to a level and for a time necessary to protect public health to address specific microbiological contamination problems.
22. Noncompliance is based on a running yearly average of monthly averages of all samples, computed quarterly, exceeds the MRDL, 4.0 mg/L.
23. Chloramine samples (as either total chlorine or combined chlorine) must be measured at the same points in the distribution system and at the same time as total coliforms. Not withstanding the MRDL, operators may increase residual chlorine levels in the distribution system to a level and for a time necessary to protect public health to address specific microbiological contamination problems.
24. Noncompliance is based on a running yearly average of monthly averages of all samples, computed quarterly, exceeds the MRDL, 4.0 mg/L.
25. For systems using chlorine dioxide for disinfection or oxidation, samples must be taken daily at the entrance to the distribution system. If the MRDL, 0.8 mg/L, is exceeded, three additional samples must be taken the following day as follows: one as close as possible to the first customer, one in a location representative of average residence time, and one as close as possible to the end of the distribution system (reflects maximum residence time within the distribution system). Systems not using booster chlorination systems after the first customer must take three samples in the distribution system as close as possible to the first customer at intervals of not <6 hours.
26. If any daily sample from the distribution system exceeds the MRDL and if one or more of the three samples taken the following day from within the distribution system exceeds the MRDL, the system is in violation of the MRDL and must issue public notification in accordance with paragraph C3.3.4. If any two consecutive daily samples exceed the MRDL but none of the distribution samples exceed the MRDL, the system is in violation of the MRDL. Failure to monitor at the entrance to the distribution system on the day following an exceedance of the chlorine dioxide MRDL is also an MRDL violation.
27. The MRDL for chlorine dioxide may NOT be exceeded for short periods to address specific microbiological contamination problems.
28. Systems that use conventional filtration treatment must monitor each treatment plant water source for TOC on a monthly basis. Samples must be taken from the source water prior to treatment and the treated water not later than the point of combined filter effluent turbidity monitoring. Source water alkalinity must also be monitored at the same time. Surface water and GWUDISW systems with average treated water TOC of <2.0 mg/L for two consecutive years, or <1.0 mg/L for one year, may reduce TOC and alkalinity to one paired sample per plant per quarter.

Table 3.10. Radionuclide MCLs and Monitoring Requirements

Parameter	MCL
Total indicative dose	0.10 mSv/yr
Gross Alpha[1]	2.7 pCi/L
Combined Radium-226 and -228	5 pCi/L
Beta Particle and Photon Radioactivity[2]	4 mrem/yr
Remaining Beta	27 pCi/L
Uranium	30 µg/L
Tritium	2702 pCi/L

Notes:

1. Gross alpha activity includes radium-226, but excludes radon and uranium.
2. Beta particle and photon activity is also referred to as gross beta activity from manmade radionuclides.

Monitoring Requirements:

- All CWSs using ground water, surface water, or systems using both ground and surface water must sample at every point (i.e., sampling points) to the distribution system that is representative of all sources being used under normal operating conditions.
- For gross alpha activity and radium-226 and radium-228, systems will be tested once every 4 years. Testing will be conducted using an annual composite of 4 consecutive quarterly samples or the average of four samples obtained at quarterly intervals at a representative point in the distribution system.
- Gross alpha only may be analyzed if activity is <5 picoCuries per liter (pCi/L). Where radium-228 may be present, radium-226 and/or -228 analyses shall be performed when activity is >2 pCi/L. If the average annual concentration is less than half the MCL, analysis of a single sample may be substituted for the quarterly sampling procedure. A system with two or more sources having different concentrations of radioactivity shall monitor source water in addition to water from a free-flowing tap. If the installation introduces a new water source, these contaminants will be monitored within the first year after introduction.
- Gross alpha, remaining beta and tritium shall be monitored according to Table 3.4-b.

Table 3.11. CT Values for Inactivation of *Giardia* Cysts by Free Chlorine at 0.5°C or Lower*

Chlorine Concentration (mg/L)	pH ≤ 6 Log Inactivations						pH = 6.5 Log Inactivations						pH = 7.0 Log Inactivations						pH = 7.5 Log Inactivations					
	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0
≤0.4	23	46	69	91	114	137	27	54	82	109	136	163	33	65	98	130	163	195	40	79	119	158	198	237
0.6	24	47	71	94	118	141	28	56	84	112	140	168	33	67	100	133	167	200	40	80	120	159	199	239
0.8	24	48	73	97	121	145	29	57	86	115	143	172	34	68	103	137	171	205	41	82	123	164	205	246
1	25	49	74	99	123	148	29	59	88	117	147	176	35	70	105	140	175	210	42	84	127	169	211	253
1.2	25	51	76	101	127	152	30	60	90	120	150	180	36	72	108	143	179	215	43	86	130	173	216	259
1.4	26	52	78	103	129	155	31	61	92	123	153	184	37	74	111	147	184	221	44	89	133	177	222	266
1.6	26	52	79	105	131	157	32	63	95	126	158	189	38	75	113	151	188	226	46	91	137	182	228	273
1.8	27	54	81	108	135	162	32	64	97	129	161	193	39	77	116	154	193	231	47	93	140	186	233	279
2	28	55	83	110	138	165	33	66	99	131	164	197	39	79	118	157	197	236	48	95	143	191	238	286
2.2	28	56	85	113	141	169	34	67	101	134	168	201	40	81	121	161	202	242	50	99	149	198	248	297
2.4	29	57	86	115	143	172	34	68	103	137	171	205	41	82	124	165	206	247	50	99	149	199	248	298
2.6	29	58	88	117	146	175	35	70	105	139	174	209	42	84	126	168	210	252	51	101	152	203	253	304
2.8	30	59	89	119	148	178	36	71	107	142	178	213	43	86	129	171	214	257	52	103	155	207	258	310
3	30	60	91	121	151	181	36	72	109	145	181	217	44	87	131	174	218	261	53	105	158	211	263	316

Chlorine Concentration (mg/L)	pH = 8 Log Inactivations						pH = 8.5 Log Inactivations						pH = 9.0 Log Inactivations					
	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0
≤0.4	46	92	139	185	231	277	55	110	165	219	274	329	65	130	195	260	325	390
0.6	48	95	143	191	238	286	57	114	171	228	285	342	68	136	204	271	339	407
0.8	49	98	148	197	246	295	59	118	177	236	295	354	70	141	211	281	352	422
1	51	101	152	203	253	304	61	122	183	243	304	365	73	146	219	291	364	437
1.2	52	104	157	209	261	313	63	125	188	251	313	376	75	150	226	301	376	451
1.4	54	107	161	214	268	321	65	129	194	258	323	387	77	155	232	309	387	464
1.6	55	110	165	219	274	329	66	132	199	265	331	397	80	159	239	318	398	477
1.8	56	113	169	225	282	338	68	136	204	271	339	407	82	163	245	326	408	489
2	58	115	173	231	288	346	70	139	209	278	348	417	83	167	250	333	417	500
2.2	59	118	177	235	294	353	71	142	213	284	355	426	85	170	256	341	426	511
2.4	60	120	181	241	301	361	73	145	218	290	363	435	87	174	261	348	435	522
2.6	61	123	184	245	307	368	74	148	222	296	370	444	89	178	267	355	444	533
2.8	63	125	188	250	313	375	75	151	226	301	377	452	91	181	272	362	453	543
3	64	127	191	255	318	382	77	153	230	307	383	460	92	184	276	368	460	552

*$CT_{99.9}$ =CT for 3 log inactivation.

Table 3.12. CT Values for Inactivation of *Giardia* Cysts by Free Chlorine at 5.0°C*

Chlorine Concentration (mg/L)	pH ≤ 6 Log Inactivations						pH = 6.5 Log Inactivations						pH = 7.0 Log Inactivations						pH = 7.5 Log Inactivations					
	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0
≤0.4	16	32	49	65	81	97	20	39	59	78	98	117	23	46	70	93	116	139	28	55	83	111	138	166
0.6	17	33	50	67	83	100	20	40	60	80	100	120	24	48	72	95	119	143	29	57	86	114	143	171
0.8	17	34	52	69	86	103	20	41	61	81	102	122	24	49	73	97	122	146	29	58	88	117	146	175
1	18	35	53	70	88	105	21	42	63	83	104	125	25	50	75	99	124	149	30	60	90	119	149	179
1.2	18	36	54	71	89	107	21	42	64	85	106	127	25	51	76	101	127	152	31	61	92	122	153	183
1.4	18	36	55	73	91	109	22	43	65	87	108	130	26	52	78	103	129	155	31	62	94	125	156	187
1.6	19	37	56	74	93	111	22	44	66	88	110	132	26	53	79	105	132	158	32	64	96	128	160	192
1.8	19	38	57	76	95	114	23	45	68	90	113	135	27	54	81	108	135	162	33	65	98	131	163	196
2	19	39	58	77	97	116	23	46	69	92	115	138	28	55	83	110	138	165	33	67	100	133	167	200
2.2	20	39	59	79	98	118	23	47	70	93	117	140	28	56	85	113	141	169	34	68	102	136	170	204
2.4	20	40	60	80	100	120	24	48	72	95	119	143	29	57	86	115	143	172	35	70	105	139	174	209
2.6	20	41	61	81	102	122	24	49	73	97	122	146	29	58	88	117	146	175	36	71	107	142	178	213
2.8	21	41	62	83	103	124	25	49	74	99	123	148	30	59	89	119	148	178	36	72	109	145	181	217
3	21	42	63	84	105	126	25	50	76	101	126	151	30	61	91	121	152	182	37	74	111	147	184	221

Chlorine Concentration (mg/L)	pH = 8 Log Inactivations						pH = 8.5 Log Inactivations						pH = 9.0 Log Inactivations					
	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0
≤0.4	33	66	99	132	165	198	39	79	118	157	197	236	47	93	140	186	233	279
0.6	34	68	102	136	170	204	41	81	122	163	203	244	49	97	146	194	243	291
0.8	35	70	105	140	175	210	42	84	126	168	210	252	50	100	151	201	251	301
1	36	72	108	144	180	216	43	87	130	173	217	260	52	104	156	208	260	312
1.2	37	74	111	147	184	221	45	89	134	178	223	267	53	107	160	213	267	320
1.4	38	76	114	151	189	227	46	91	137	183	228	274	55	110	165	219	274	329
1.6	39	77	116	155	193	232	47	94	141	187	234	281	56	112	169	225	281	337
1.8	40	79	119	159	198	238	48	96	144	191	239	287	58	115	173	230	288	345
2	41	81	122	162	203	243	49	98	147	196	245	294	59	118	177	235	294	353
2.2	41	83	124	165	207	248	50	100	150	200	250	300	60	120	181	241	301	361
2.4	42	84	127	169	211	253	51	102	153	204	255	306	61	123	184	245	307	368
2.6	43	86	129	172	215	258	52	104	156	208	260	312	63	125	188	250	313	375
2.8	44	88	132	175	219	263	53	106	159	212	265	318	64	127	191	255	318	382
3	45	89	134	179	223	268	54	108	162	216	270	324	65	130	195	259	324	389

*$CT_{99.9}$ =CT for 3 log inactivation.

Table 3.13. CT Values for Inactivation of *Giardia* Cysts by Free Chlorine at 10°C*

Chlorine Concentration (mg/L)	pH ≤ 6 Log Inactivations						pH = 6.5 Log Inactivations						pH = 7.0 Log Inactivations						pH = 7.5 Log Inactivations					
	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0
≤0.4	12	24	37	49	61	73	15	29	44	59	73	88	17	35	52	69	87	104	21	42	63	83	104	125
0.6	13	25	38	50	63	75	15	30	45	60	75	90	18	36	54	71	89	107	21	43	64	85	107	128
0.8	13	26	39	52	65	78	15	31	46	61	77	92	18	37	55	73	92	110	22	44	66	87	109	131
1	13	26	40	53	66	79	16	31	47	63	78	94	19	37	56	75	93	112	22	45	67	89	112	134
1.2	13	27	40	53	67	80	16	32	48	63	79	95	19	38	57	76	95	114	23	46	69	91	114	137
1.4	14	27	41	55	68	82	16	33	49	65	82	98	19	39	58	77	97	116	23	47	70	93	117	140
1.6	14	28	42	55	69	83	17	33	50	66	83	99	20	40	60	79	99	119	24	48	72	96	120	144
1.8	14	29	43	57	72	86	17	34	51	67	84	101	20	41	61	81	102	122	25	49	74	98	123	147
2	15	29	44	58	73	87	17	35	52	69	87	104	21	41	62	83	103	124	25	50	75	100	125	150
2.2	15	30	45	59	74	89	18	35	53	70	88	105	21	42	64	85	106	127	26	51	77	102	128	153
2.4	15	30	45	60	75	90	18	36	54	71	89	107	22	43	65	86	108	129	26	52	79	105	131	157
2.6	15	31	46	61	77	92	18	37	55	73	92	110	22	44	66	87	109	131	27	53	80	107	133	160
2.8	16	31	47	62	78	93	19	37	56	74	93	111	22	45	67	89	112	134	27	54	82	109	136	163
3	16	32	48	63	79	95	19	38	57	75	94	113	23	46	69	91	114	137	28	55	83	111	138	166

Chlorine Concentration (mg/L)	pH = 8 Log Inactivations						pH = 8.5 Log Inactivations						pH = 9.0 Log Inactivations					
	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0
≤0.4	25	50	75	99	124	149	30	59	89	118	148	177	35	70	105	139	174	209
0.6	26	51	77	102	128	153	31	61	92	122	153	183	36	73	109	145	182	218
0.8	26	53	79	105	132	158	32	63	95	126	158	189	38	75	113	151	188	226
1	27	54	81	108	135	162	33	65	98	130	163	195	39	78	117	156	195	234
1.2	28	55	83	111	138	166	33	67	100	133	167	200	40	80	120	160	200	240
1.4	28	57	85	113	142	170	34	69	103	137	172	206	41	82	124	165	206	247
1.6	29	58	87	116	145	174	35	70	106	141	176	211	42	84	127	169	211	253
1.8	30	60	90	119	149	179	36	72	108	143	179	215	43	86	130	173	216	259
2	30	61	91	121	152	182	37	74	111	147	184	221	44	88	133	177	221	265
2.2	31	62	93	124	155	186	38	75	113	150	188	225	45	90	136	181	226	271
2.4	32	63	95	127	158	190	38	77	115	153	192	230	46	92	138	184	230	276
2.6	32	65	97	129	162	194	39	78	117	156	195	234	47	94	141	187	234	281
2.8	33	66	99	131	164	197	40	80	120	159	199	239	48	96	144	191	239	287
3	34	67	101	134	168	201	41	81	122	162	203	243	49	97	146	195	243	292

*$CT_{99.9}$ =CT for 3 log inactivation.

Table 3.14. CT Values for Inactivation of *Giardia* Cysts by Free Chlorine at 15°C*

Chlorine Concentration (mg/L)	pH ≤ 6 Log Inactivations						pH = 6.5 Log Inactivations						pH = 7.0 Log Inactivations						pH = 7.5 Log Inactivations					
	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0
≤0.4	8	16	25	33	41	49	10	20	30	39	49	59	12	23	35	47	58	70	14	28	42	55	69	83
0.6	8	17	25	33	42	50	10	20	30	40	50	60	12	24	36	48	60	72	14	29	43	57	72	86
0.8	9	17	26	35	43	52	10	20	31	41	51	61	12	24	37	49	61	73	15	29	44	59	73	88
1	9	18	27	35	44	53	11	21	32	42	53	63	13	25	38	50	63	75	15	30	45	60	75	90
1.2	9	18	27	36	45	54	11	21	32	43	53	64	13	25	38	51	63	76	15	31	46	61	77	92
1.4	9	18	28	37	46	55	11	22	33	43	54	65	13	26	39	52	65	78	16	31	47	63	78	94
1.6	9	19	28	37	47	56	11	22	33	44	55	66	13	26	40	53	66	79	16	32	48	64	80	96
1.8	10	19	29	38	48	57	11	23	34	45	57	68	14	27	41	54	68	81	16	33	49	65	82	98
2	10	19	29	39	48	58	12	23	35	46	58	69	14	28	42	55	69	83	17	33	50	67	83	100
2.2	10	20	30	39	49	59	12	23	35	47	58	70	14	28	43	57	71	85	17	34	51	68	85	102
2.4	10	20	30	40	50	60	12	24	36	48	60	72	14	29	43	57	72	86	18	35	53	70	88	105
2.6	10	20	31	41	51	61	12	24	37	49	61	73	15	29	44	59	73	88	18	36	54	71	89	107
2.8	10	21	31	41	52	62	12	25	37	49	62	74	15	30	45	59	74	89	18	36	55	73	91	109
3	11	21	32	42	53	63	13	25	38	51	63	76	15	30	46	61	76	91	19	37	56	74	93	111
Chlorine Concentration (mg/L)	**pH = 8 Log Inactivations**						**pH = 8.5 Log Inactivations**						**pH = 9.0 Log Inactivations**											
	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0						
≤0.4	17	33	50	66	83	99	20	39	59	79	98	118	23	47	70	93	117	140						
0.6	17	34	51	68	85	102	20	41	61	81	102	122	24	49	73	97	122	146						
0.8	18	35	53	70	88	105	21	42	63	84	105	126	25	50	76	101	126	151						
1	18	36	54	72	90	108	22	43	65	87	108	130	26	52	78	104	130	156						
1.2	19	37	56	74	93	111	22	45	67	89	112	134	27	53	80	107	133	160						
1.4	19	38	57	76	95	114	23	46	69	91	114	137	28	55	83	110	138	165						
1.6	19	39	58	77	97	116	24	47	71	94	118	141	28	56	85	113	141	169						
1.8	20	40	60	79	99	119	24	48	72	96	120	144	29	58	87	115	144	173						
2	20	41	61	81	102	122	25	49	74	98	123	147	30	59	89	118	148	177						
2.2	21	41	62	83	103	124	25	50	75	100	125	150	30	60	91	121	151	181						
2.4	21	42	64	85	106	127	26	51	77	102	128	153	31	61	92	123	153	184						
2.6	22	43	65	86	108	129	26	52	78	104	130	156	31	63	94	125	157	188						
2.8	22	44	66	88	110	132	27	53	80	106	133	159	32	64	96	127	159	191						
3	22	45	67	89	112	134	27	54	81	108	135	162	33	65	98	130	163	195						

*$CT_{99.9}$ =CT for 3 log inactivation.

Table 3.15. CT Values for Inactivation of *Giardia* Cysts by Free Chlorine at 20°C*

Chlorine Concentration (mg/L)	pH ≤ 6 Log Inactivations						pH = 6.5 Log Inactivations					
	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0
≤0.4	6	12	18	24	30	36	7	15	22	29	37	44
0.6	6	13	19	25	32	38	8	15	23	30	38	45
0.8	7	13	20	26	33	39	8	15	23	31	38	46
1	7	13	20	26	33	39	8	16	24	31	39	47
1.2	7	13	20	27	33	40	8	16	24	32	40	48
1.4	7	14	21	27	34	41	8	16	25	33	41	49
1.6	7	14	21	28	35	42	8	17	25	33	42	50
1.8	7	14	22	29	36	43	9	17	26	34	43	51
2	7	15	22	29	37	44	9	17	26	35	43	52
2.2	7	15	22	29	37	44	9	18	27	35	44	53
2.4	8	15	23	30	38	45	9	18	27	36	45	54
2.6	8	15	23	31	38	46	9	18	28	37	46	55
2.8	8	16	24	31	39	47	9	19	28	37	47	56
3	8	16	24	31	39	47	10	19	29	38	48	57

Chlorine Concentration (mg/L)	pH = 7.0 Log Inactivations						pH = 7.5 Log Inactivations					
	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0
≤0.4	9	17	26	35	43	52	10	21	31	41	52	62
0.6	9	18	27	36	45	54	11	21	32	43	53	64
0.8	9	18	28	37	46	55	11	22	33	44	55	66
1	9	19	28	37	47	56	11	22	34	45	56	67
1.2	10	19	29	38	48	57	12	23	35	46	58	69
1.4	10	19	29	39	48	58	12	23	35	47	58	70
1.6	10	20	30	39	49	59	12	24	36	48	60	72
1.8	10	20	31	41	51	61	12	25	37	49	62	74
2	10	21	31	41	52	62	13	25	38	50	63	75
2.2	11	21	32	42	53	63	13	26	39	51	64	77
2.4	11	22	33	43	54	65	13	26	39	52	65	78
2.6	11	22	33	44	55	66	13	27	40	53	67	80
2.8	11	22	34	45	56	67	14	27	41	54	68	81
3	11	23	34	45	57	68	14	28	42	55	69	83

Chlorine Concentration (mg/L)	pH = 8 Log Inactivations						pH = 8.5 Log Inactivations						pH = 9.0 Log Inactivations					
	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0
≤0.4	12	25	37	49	62	74	15	30	45	59	74	89	18	35	53	70	88	105
0.6	13	26	39	51	64	77	15	31	46	61	77	92	18	36	55	73	91	109
0.8	13	26	40	53	66	79	16	32	48	63	79	95	19	38	57	75	94	113
1	14	27	41	54	68	81	16	33	49	65	82	98	20	39	59	78	98	117
1.2	14	28	42	55	69	83	17	33	50	67	83	100	20	40	60	80	100	120
1.4	14	28	43	57	71	85	17	34	52	69	86	103	21	41	62	82	103	123
1.6	15	29	44	58	73	87	18	35	53	70	88	105	21	42	63	84	105	126
1.8	15	30	45	59	74	89	18	36	54	72	90	108	22	43	65	86	108	129
2	15	30	46	61	76	91	18	37	55	73	92	110	22	44	66	88	110	132
2.2	16	31	47	62	78	93	19	38	57	75	94	113	23	45	68	90	113	135
2.4	16	32	48	63	79	95	19	38	58	77	96	115	23	46	69	92	115	138
2.6	16	32	49	65	81	97	20	39	59	78	98	117	24	47	71	94	118	141
2.8	17	33	50	66	83	99	20	40	60	79	99	119	24	48	72	95	119	143
3	17	34	51	67	84	101	20	41	61	81	102	122	24	49	73	97	122	146

*$CT_{99.9}$ =CT for 3 log inactivation.

Table 3.16. CT Values for Inactivation of *Giardia* Cysts by Free Chlorine at 25°C*

Chlorine Concentration (mg/L)	pH ≤ 6 Log Inactivations						pH = 6.5 Log Inactivations						pH = 7.0 Log Inactivations						pH = 7.5 Log Inactivations					
	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0
≤0.4	4	8	12	16	20	24	5	10	15	19	24	29	6	12	18	23	29	35	7	14	21	28	35	42
0.6	4	8	13	17	21	25	5	10	15	20	25	30	6	12	18	24	30	36	7	14	22	29	36	43
0.8	4	9	13	17	22	26	5	10	16	21	26	31	6	12	19	25	31	37	7	15	22	29	37	44
1	4	9	13	17	22	26	5	10	16	21	26	31	6	12	19	25	31	37	8	15	23	30	38	45
1.2	5	9	14	18	23	27	5	11	16	21	27	32	6	13	19	25	32	38	8	15	23	31	38	46
1.4	5	9	14	18	23	27	6	11	17	22	28	33	7	13	20	26	33	39	8	16	24	31	39	47
1.6	5	9	14	19	23	28	6	11	17	22	28	33	7	13	20	27	33	40	8	16	24	32	40	48
1.8	5	10	15	19	24	29	6	11	17	23	28	34	7	14	21	27	34	41	8	16	25	33	41	49
2	5	10	15	19	24	29	6	12	18	23	29	35	7	14	21	27	34	41	8	17	25	33	42	50
2.2	5	10	15	20	25	30	6	12	18	23	29	35	7	14	21	28	35	42	9	17	26	34	43	51
2.4	5	10	15	20	25	30	6	12	18	24	30	36	7	14	22	29	36	43	9	17	26	35	43	52
2.6	5	10	16	21	26	31	6	12	19	25	31	37	7	15	22	29	37	44	9	18	27	35	44	53
2.8	5	10	16	21	26	31	6	12	19	25	31	37	8	15	23	30	38	45	9	18	27	36	45	54
3	5	11	16	21	27	32	6	13	19	25	32	38	8	15	23	31	38	46	9	18	28	37	46	55

Chlorine Concentration (mg/L)	pH = 8 Log Inactivations						pH = 8.5 Log Inactivations						pH = 9.0 Log Inactivations					
	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0	0.5	1.0	1.5	2.0	2.5	3.0
≤0.4	8	17	25	33	42	50	10	20	30	39	49	59	12	23	35	47	58	70
0.6	9	17	26	34	43	51	10	20	31	41	51	61	12	24	37	49	61	73
0.8	9	18	27	35	44	53	11	21	32	42	53	63	13	25	38	50	63	75
1	9	18	27	36	45	54	11	22	33	43	54	65	13	26	39	52	65	78
1.2	9	18	28	37	46	55	11	22	34	45	56	67	13	27	40	53	67	80
1.4	10	19	29	38	48	57	12	23	35	46	58	69	14	27	41	55	68	82
1.6	10	19	29	39	48	58	12	23	35	47	58	70	14	28	42	56	70	84
1.8	10	20	30	40	50	60	12	24	36	48	60	72	14	29	43	57	72	86
2	10	20	31	41	51	61	12	25	37	49	62	74	15	29	44	59	73	88
2.2	10	21	31	41	52	62	13	25	38	50	63	75	15	30	45	60	75	90
2.4	11	21	32	42	53	63	13	26	39	51	64	77	15	31	46	61	77	92
2.6	11	22	33	43	54	65	13	26	39	52	65	78	16	31	47	63	78	94
2.8	11	22	33	44	55	66	13	27	40	53	67	80	16	32	48	64	80	96
3	11	22	34	45	56	67	14	27	41	54	68	81	16	32	49	65	81	97

*$CT_{99.9}$ =CT for 3 log inactivation.

Table 3.17. CT Values for Inactivation of Viruses by Free Chlorine

Temperature (°C)	Log Inactivation 2.0 pH		Log Inactivation 3.0 pH		Log Inactivation 3.0 pH	
	6-9	10	6-9	10	6-9	10
0.5	6	45	9	66	12	90
5	4	30	6	44	8	60
10	3	22	4	33	6	45
15	2	15	3	22	4	30
20	1	11	2	16	3	22
25		1	1	11	2	15

Table 3.18. CT Values for Inactivation of *Giardia* Cysts by Chlorine Dioxide

Inactivation	Temperature (°C)					
	≤1	5	10	15	20	25
0.5-log	10	4.3	4	3.2	2.5	2
1-log	21	8.7	7.7	6.3	5	3.7
1.5-log	32	13	12	10	7.5	5.5
2-log	42	17	15	13	10	7.3
2.5-log	52	22	19	16	13	9
3-log	63	26	23	19	15	11

Table 3.19. CT Values for Inactivation of Viruses by Free Chlorine Dioxide pH 6-9

Removal	Temperature (°C)					
	≤1	5	10	15	20	25
2-log	8.4	5.6	4.2	2.8	2.1	1.4
3-log	25.6	17.1	12.8	8.6	6.4	4.3
4-log	50.1	33.4	25.1	16.7	12.5	8.4

Table 3.20. CT Values for Inactivation of *Giardia* Cysts by Ozone

Inactivation	Temperature (°C)					
	≤1	5	10	15	20	25
0.5-log	0.48	0.32	0.23	0.16	0.12	0.08
1-log	0.97	0.63	0.48	0.32	0.24	0.16
1.5-log	1.5	0.95	0.72	0.48	0.36	0.24
2-log	1.9	1.3	0.95	0.63	0.48	0.32
2.5-log	2.4	1.6	1.2	0.79	0.60	0.40
3-log	2.9	1.9	1.43	0.95	0.72	0.48

Table 3.21. CT Values for Inactivation of Viruses by Free Ozone

Inactivation	Temperature (°C)					
	≤1	5	10	15	20	25
2-log	0.9	0.6	0.5	0.3	0.25	0.15
3-log	1.4	0.9	0.8	0.5	0.4	0.25
4-log	1.8	1.2	1.0	0.6	0.5	0.3

Table 3.22. CT Values for Inactivation of *Giardia* Cysts by Chloramine pH 6-9

Inactivation	Temperature (°C)					
	≤1	5	10	15	20	25
0.5-log	635	365	310	250	185	125
1-log	1,270	735	615	500	370	250
1.5-log	1,900	1,100	930	750	550	375
2-log	2,535	1,470	1,230	1,000	735	500
2.5-log	3,170	1,830	1,540	1,250	915	625
3-log	3,800	2,200	1,850	1,500	1,100	750

Table 3.23. CT Values for Inactivation of Viruses by Chloramine

Inactivation	Temperature (°C)					
	≤1	5	10	15	20	25
2-log	1,243	857	643	428	321	214
3-log	2,063	1,423	1,067	712	534	356
4-log	2,883	1,988	1,491	994	746	497

Table 3.24. CT Values for Inactivation of Viruses by UV

Log Inactivation	
2.0	3.0
21	36

Table 3.25. DoD-Approved Inorganic Chemical Analytical Test Methods

Parameter	Methodology [13]	EPA	ASTM [3]	SM [4]	Other
Alkalinity	Titrimetric Electrometric titration		D1067-92B	2320 B	I-1030-85 [5]
Aluminum		200.7 [2] 200.8 [2] 200.9 [2]	3120 B 3113 B 3111 D		
Antimony	ICP-Mass Spectrometry Hydride-Atomic Absorption Atomic Absorption (Platform) Atomic Absorption (Furnace)	200.8 [2] 200.9 [2] 	 D-3697-92 	 3113 B	
Arsenic [14]	Inductively Coupled Plasma ICP-Mass Spectrometry Atomic Absorption (Platform) Atomic Absorption (Furnace) Hydride Atomic Absorption	200.7 [2] 200.8 [2] 200.9[2] 	 D-2972-93C D-2972-93B	3120 B 3113 B 3114 B	
Asbestos	Transmission Electron Microscopy Transmission Electron Microscopy	100.1 [9] 100.2[10]			
Barium	Inductively Coupled Plasma ICP-Mass Spectrometry Atomic Absorption (Direct) Atomic Absorption (Furnace)	200.7 [2] 200.8 [2] 		3120 B 3111 D 3113 B	
Beryllium	Inductively Coupled Plasma ICP-Mass Spectrometry Atomic Absorption (Platform) Atomic Absorption (Furnace)	200.7 [2] 200.8 [2] 200.9 [2] 	 D3645-93B	3120 B 3113 B	
Cadmium	Inductively Coupled Plasma ICP-Mass Spectrometry Atomic Absorption (Platform) Atomic Absorption (Furnace)	200.7 [2] 200.8 [2] 200.9 [2] 		 3113 B	
Calcium	EDTA Titrimetric Atomic Absorption (Direct Aspiration) Inductively-Coupled Plasma	 200.7 [2]	D511-93A D511-93B 	3500-Ca D 3111 B 3120 B	
Chloride		300.0 [6]	D4327-91 4500-Cl-D D512-89B	4110 B 4500-Cl-B	
Chromium	Inductively Coupled Plasma ICP-Mass Spectrometry Atomic Absorption (Platform) Atomic Absorption (Furnace)	200.7 [2] 200.8 [2] 200.9 [2] 		3120 B 3113 B	
Color				2120 B	

Parameter	Methodology [13]	EPA	ASTM [3]	SM [4]	Other
Copper	Atomic Absorption (Furnace) Atomic Absorption (Direct Aspiration) ICP ICP - Mass Spectrometry Atomic Absorption (Platform)	 200.7 [2] 200.8 [2] 200.9 [2]	D1688-95C D1688-95A	3113 B 3111 B 3120 B	
Conductivity Conductance			D1125-95A	2510 B	
Cyanide	Manual Distillation followed by: Spectrophotometric (Amenable) Spectrophotometric Manual Semi-automated Selective Electrode	 335.4 [6]	D2036-91A D2036-91B D2036-91A	4500-CN-C 4500-CN-G 4500-CN-E 4500-CN-F	 I-3300-85 [5]
Foaming Agents				5540C	
Fluoride	Ion Chromatography Manual Distill (Color SPADNS) Manual Electrode Automated Electrode Automated Alizarin	300.0 [6]	D4327-91 D1179-93B	4110 B 4500-F-B, D 4500-F-C 4500-F-E	 380-75WE [11] 129-71W [11]
Iron		200.7 [2] 200.9 [2]		3120 B 3111 B 3113 B	
Lead	Atomic Absorption (Furnace) ICP-Mass Spectrometry Atomic Absorption (Platform) Differential Pulse Anodic Stripping Voltammetry	 200.8 [2] 200.9 [2]	D3559-95D	3113 B	 1001 [15]
Magnesium	Atomic Absorption ICP Complexation Titrimetric Methods	 200.7 [2]	D 511-93 B D 511-93 A	3111 B 3120 B 3500-Mg E	
Manganese		200.7 [2] 200.8 [2] 200.9 [2]		3120 B 3111 B 3113 B	
Mercury	Manual (Cold Vapor) Automated (Cold Vapor) ICP-Mass Spectrometry	245.1 [2] 245.2 [1] 200.8 [2]	D3223-91	3112 B	

Parameter	Methodology [13]	EPA	ASTM [3]	SM [4]	Other
Nickel	Inductively Coupled Plasma	200.7 [2]		3120 B	
	ICP-Mass Spectrometry	200.8 [2]			
	Atomic Absorption (Platform)	200.9 [2]			
	Atomic Absorption (Direct)			3111 B	
	Atomic Absorption (Furnace)			3113 B	
Nitrate	Ion Chromatography	300.0 [6]	D4327-91	4110 B	B-1011 [8]
	Automated Cadmium Reduction	353.2 [6]	D3867-90A	4500-NO3- F	
	Ion Selective Electrode			4500-NO3-D	601 [7]
	Manual Cadmium Reduction		D3867-90B	4500-NO3- E	
Nitrite	Ion Chromatography	300.0 [6]	D4327-91	4110 B	B-1011 [8]
	Automated Cadmium Reduction	353.2 [6]	D3867-90A	4500-NO3-F	
	Manual Cadmium Reduction		D3867-90B	4500-NO3- E	
	Spectrophotometric			4500-NO2-B	
Odor				2150 B	
Ortho-phosphate[12]	Colorimetric (Automated, Ascorbic Acid)	365.1 [6]		4500-P F	
	Colorimetric (Ascorbic Acid, Single Reagent)		D515-88A	4500-P E	
	Colorimetric (Phosphomolybdate)				I-1602-85 [5]
	Automated-Segmented Flow				I-2601-90 [5]
	Automated Discrete				I-2598-85 [5]
	Ion Chromatography	300.0 [6]	D4327-91	4110 B	
pH	Electrometric	150.1 [1] 150.2 [1]	D1293-95	4500-H+ B	
Selenium	Hydride-Atomic Absorption		D3859-93A	3114 B	
	ICP-Mass Spectrometry	200.8 [2]			
	Atomic Absorption (Platform)	200.9 [2]			
	Atomic Absorption (Furnace)		D3859-93B	3113 B	
Silica	Colorimetric, Molybdate Blue				I-1700-85 [5]
	Automated-Segmented Flow				I-2700-85 [5]
	Colorimetric Molybdosilicate		D859-95	4500-Si D	
Silver		200.7 [2]		3120 B	I-3720-85 [16]
		200.8 [2]		3111 B	
		200.9 [2]		3113 B	
Sodium	Inductively-Coupled Plasma	200.7 [2]			

Parameter	Methodology [13]	EPA	ASTM [3]	SM [4]	Other
	Atomic Absorption (Direct Aspiration)			3111 B	
Sulfate		300.0 [16] 375.2 [16]	D4327-91 D516-90	4110 B 4500-SO42-F 4500-SO42- C, D 4500-SO42-E	
Temperature	Thermometric			2550	
Thallium	ICP-Mass Spectrometry Atomic Absorption (Platform)	200.8 [2] 200.9 [2]			
Zinc		200.7 [2] 200.8 [2]		3120 B 3111 B	

Notes:

The procedures shall be done in accordance with the documents listed below. Copies of the documents may be obtained from the sources listed below. Information regarding obtaining these documents can be obtained from the Safe Drinking Water Hotline at 800-426-4791. Documents may be inspected at EPA's Drinking Water Docket, 401 M Street, SW., Washington, DC 20460 (Telephone: 202-260-3027); or at the Office of Federal Register, 800 North Capitol Street, NW., Suite 700, Washington, DC.

1. "Methods for Chemical Analysis of Water and Wastes'', EPA/600/4-79/020, March 1983. Available at NTIS, PB84- 128677.
2. "Methods for the Determination of Metals in Environmental Samples--Supplement I'', EPA/600/R-94/111, May 1994. Available at NTIS, PB95-125472.
3. Annual Book of ASTM Standards, 1994 and 1996, Vols. 11.01 and 11.02, American Society for Testing and Materials. The previous versions of D1688-95A, D1688-95C (copper), D3559-95D (lead), D1293-95 (pH), D1125-91A (conductivity) and D859-94 (silica) are also approved. These previous versions D1688-90A, C; D3559-90D, D1293- 84, D1125-91A and D859-88, respectively are located in the Annual Book of ASTM Standards, 1994, Vols. 11.01. Copies may be obtained from the American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428.
4. 18th and 19th editions of Standard Methods for the Examination of Water and Wastewater, 1992 and 1995, respectively, American Public Health Association; either edition may be used. Copies may be obtained from the American Public Health Association, 1015 Fifteenth Street NW, Washington, DC 20005.
5. Method I-2601-90, Methods for Analysis by the U.S. Geological Survey National Water Quality Laboratory-- Determination of Inorganic and Organic Constituents in Water and Fluvial Sediments, Open File Report 93-125, 1993; For Methods I-1030-85; I-1601-85; I-1700-85; I-2598-85; I-2700-85; and I-3300-85 See Techniques of Water Resources Investigation of the U.S. Geological Survey, Book 5, Chapter A-1, 3rd ed., 1989; Available from Information Services, U.S. Geological Survey, Federal Center, Box 25286, Denver, CO 80225-0425.
6. "Methods for the Determination of Inorganic Substances in Environmental Samples'', EPA/600/R-93/100, August 1993. Available at NTIS, PB94-120821.
7. The procedure shall be done in accordance with the Technical Bulletin 601 "Standard Method of Test for Nitrate in Drinking Water'', July 1994, PN 221890-001, Analytical Technology, Inc. Copies may be obtained from ATI Orion, 529 Main Street, Boston, MA 02129.
8. Method B-1011, "Waters Test Method for Determination of Nitrite/Nitrate in Water Using Single Column Ion Chromatography,'' August 1987. Copies may be obtained from Waters Corporation, Technical Services Division, 34 Maple Street, Milford, MA 01757.

9. Method 100.1, "Analytical Method For Determination of Asbestos Fibers in Water'', EPA/600/4-83/043, EPA, September 1983. Available at NTIS, PB83-260471.
10. Method 100.2, "Determination of Asbestos Structure Over 10-µm In Length In Drinking Water'', EPA/600/R-94/134, June 1994. Available at NTIS, PB94-201902.
11. Industrial Method No. 129-71W, "Fluoride in Water and Wastewater'', December 1972, and Method No. 380- 75WE, "Fluoride in Water and Wastewater'', February 1976, Technicon Industrial Systems. Copies may be obtained from Bran & Luebbe, 1025 Busch Parkway, Buffalo Grove, IL 60089.
12. Unfiltered, no digestion or hydrolysis.
13. Because MDLs reported in EPA Methods 200.7 and 200.9 were determined using a 2X preconcentration step during sample digestion, MDLs determined when samples are analyzed by direct analysis (i.e., no sample digestion) will be higher. For direct analysis of cadmium and arsenic by Method 200.7, and arsenic by Method 3120 B sample preconcentration using pneumatic nebulization may be required to achieve lower detection limits. Preconcentration may also be required for direct analysis of antimony, lead, and thallium by Method 200.9; antimony and lead by Method 3113 B; and lead by Method D3559-90D unless multiple in-furnace depositions are made.
14. If ultrasonic nebulization is used in the determination of arsenic by Methods 200.7, 200.8, or SM 3120 B, the arsenic must be in the pentavalent state to provide uniform signal response. For methods 200.7 and 3120 B, both samples and standards must be diluted in the same mixed acid matrix concentration of nitric and hydrochloric acid with the addition of 100 µL of 30% hydrogen peroxide per 100ml of solution. For direct analysis of arsenic with method 200.8 using ultrasonic nebulization, samples and standards must contain one mg/L of sodium hypochlorite.
15. The description for Method Number 1001 for lead is available from Palintest, LTD, 21 Kenton Lands Road, P.O.
16. Method I-3720-85, Techniques of Water Resources Investigation of the U.S. Geological Survey, Book 5, Chapter A-1, 3rd ed., 1989; Available from Information Services, U.S. Geological Survey, Federal Center, Box 25286, Denver, CO 80225-0425.

Table 3.26. DoD- Approved Organic Chemicals Analytical Test Methods

Parameter	Method
Alachlor [1]	507, 525.2, 508.1, 505, 551.1
Atrazine [1]	507, 525.2, 508.1, 505, 551.1
Benzene	502.2, 524.2
Benzo(a)pyrene	525.2, 550, 550.1
Carbofuran	531.1, 6610
Carbon tetrachloride (Tetrachloromethane)	502.2, 524.2, 551.1
Chlordane	508, 525.2, 508.1, 505
Monochlorobenzene (Chlorobenzene)	502.2, 524.2
2,4-D4 (as acid, salts and esters)	515.2, 555, 515.1, 515.3, D5317-93
Dalapon	552.1, 515.1, 552.2, 515.3
Dibromochloropropane (DBCP)	504.1, 551.1
o-Dichlorobenzene (1,2-Dichlorobenzene)	502.2, 524.2
p-Dichlorobenzene (1,4-Dichlorobenzene)	502.2, 524.2
1,2-Dichloroethane	502.2, 524.2
1,1-Dichloroethylene (1,1-Dichloroethene)	502.2, 524.2
cis-Dichloroethylene (cis-1,2-Dichloroethene)	502.2, 524.2
trans-Dichloroethylene (trans-1,2-Dichloroethene)	502.2, 524.2
Dichloromethane	502.2, 524.2
1,2-Dichloropropane	502.2, 524.2
Di(2-ethylhexyl)adipate	506, 525.2
Di(2-ethylhexyl)phthalate	506, 525.2
Dinoseb [3]	515.2, 555, 515.1, 515.3
Dioxin (2,3,7,8-TCDD)	1613
Diquat	549.2
Endothall	548.1
Endrin	508, 525.2, 508.1, 505, 551.1
Ethylbenzene	502.2, 524.2
Ethylene dibromide (EDB) (1,2-Dibromoethane)	504.1, 551.1
Glyphosate	547, 6651
Heptachlor	508, 525.2, 508.1, 505, 551.1
Heptachlor Epoxide	508, 525.2, 508.1, 505, 551.1
Hexachlorobenzene	508, 525.2, 508.1, 505, 551.1
Hexachlorocyclopentadiene	508, 525.2, 508.1, 505, 551.1
Lindane	508, 525.2, 508.1, 505, 551.1
Methoxychlor	508, 525.2, 508.1, 505, 551.1
Oxamyl	531.1, 6610
PCBs [2] (as decachlorobiphenyl) (as Aroclors)	508A 508.1, 508, 525.2, 505
Pentachlorophenol	515.2, 525.2, 555, 515.1, 515.3, D5317-93
Picloram [3]	515.2, 555, 515.1, 515.3, D5317-93
Silvex (2,4,5-TP [3])	515.2, 555, 515.1, 515.3, D5317-93
Simazine [1]	507, 525.2, 508.1, 505, 551.1

Parameter	Method
Styrene	502.2, 524.2
Tetrachloroethylene (Tetrachloroethene)	502.2, 524.2, 551.1
Toluene	502.2, 524.2
Total Trihalomethanes	502.2, 524.2, 551.1
Toxaphene	508, 508.1, 525.2, 505
1,2,4-Trichlorobenzene	502.2, 524.2
1,1,1-Trichloroethane	502.2, 524.2, 551.1
1,1,2-Trichloroethane	502.2, 524.2, 551.1
Trichloroethylene (Trichloroethene)	502.2, 524.2, 551.1
Vinyl chloride	502.2, 524.2
Xylenes (total)	502.2, 524.2

Notes:

1. Substitution of the detector specified in Method 505, 507, 508 or 508.1 for the purpose of achieving lower detection limits is allowed as follows. Either an electron capture or nitrogen phosphorous detector may be used provided all regulatory requirements and quality control criteria are met.
2. PCBs are qualitatively identified as Aroclors and measured for compliance purposes as decachlorobiphenyl. Users of Method 505 may have more difficulty in achieving the required detection limits than users of Methods 508.1, 525.2 or 508.
3. Accurate determination of the chlorinated esters requires hydrolysis of the sample as described in EPA Methods 515.1, 515.2, 515.3 and 555, and ASTM Method D 5317-93.

Method 508A and 515.1 are in Methods for the Determination of Organic Compounds in Drinking Water, EPA/600/4-88-039, December 1988, Revised, July 1991.

Methods 547, 550 and 550.1 are in Methods for the Determination of Organic Compounds in Drinking Water--Supplement I, EPA/600-4-90-020, July 1990. Methods 548.1, 549.1, 552.1 and 555 are in Methods for the Determination of Organic Compounds in Drinking Water--Supplement II, EPA/600/R-92-129, August 1992.

Methods 502.2, 504.1, 505, 506, 507, 508, 508.1, 515.2, 524.2 525.2, 531.1, 551.1 and 552.2 are in Methods for the Determination of Organic Compounds in Drinking Water--Supplement III, EPA/600/R-95-131, August 1995. Method 1613 is titled “Tetra-through Octa-Chlorinated Dioxins and Furans by Isotope-Dilution HRGC/HRMS’’, EPA/821-B-94-005, October 1994.

These documents are available from the National Technical Information Service, NTIS PB91- 231480, PB91-146027, PB92-207703, PB95-261616 and PB95-104774, U.S. Department of Commerce, 5285 Port Royal Road, Springfield, Virginia 22161. The toll-free number is 800-553-6847.

Method 6651 shall be followed in accordance with Standard Methods for the Examination of Water and Wastewater, 18th edition, 1992 and 19th edition, 1995, American Public Health Association (APHA); either edition may be used.

Method 6610 shall be followed in accordance with the Supplement to the 18th edition of Standard Methods for the Examination of Water and Wastewater, 1994 or with the 19th edition of Standard Methods for the Examination of Water and Wastewater, 1995, APHA; either publication may be used. The APHA documents are available from APHA, 1015 Fifteenth Street NW., Washington, D.C. 20005. Other required analytical test procedures germane to the conduct of these analyses are contained in Technical Notes on Drinking Water Methods, EPA/600/R-94-173, October 1994, NTIS PB95-104766. EPA Methods 515.3 and 549.2 are available from U.S. Environmental Protection Agency, National Exposure Research Laboratory (NERL)-Cincinnati, 26 West Martin Luther King Drive, Cincinnati, OH 45268. ASTM Method D 5317-93 is available in the Annual Book of ASTM Standards, 1996, Vol. 11.02, American Society for Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428, or in any edition published after 1993.

Table 3.27. DoD-Approved Coliform and Turbidity Analytical Test Methods

Parameter	Methodology	Citation [1]
Total Coliform [2]	Total Coliform Fermentation Technique [3, 4, 5] Total Coliform Membrane Filter Technique [6] ONPG-MUG Test [7]	9221 A, B, C 9222 A, B, C 9223
Fecal Coliforms [2]	Fecal Coliform Procedure [8] Fecal Coliform Filter Procedure	9221 E 9222 D
Heterotrophic bacteria [2]	Pour Plate Method	9215 B
Turbidity	Nephelometric Method Nephelometric Method Great Lakes Instruments	2130 B 180.1 [9] Method 2 [10]

Notes:

The procedures shall be done in accordance with the documents listed below. Copies of the documents may be obtained from the sources listed below. Information regarding obtaining these documents can be obtained from the Safe Drinking Water Hotline at 800-426-4791. Documents may be inspected at EPA's Drinking Water Docket, 401 M Street, SW, Washington, D.C. 20460 (Telephone: 202-260-3027); or at the Office of the Federal Register, 800 North Capitol Street, NW, Suite 700, Washington, D.C. 20408.

1. Except where noted, all methods refer to Standard Methods for the Examination of Water and Wastewater, 18th edition, 1992 and 19th edition, 1995, American Public Health Association, 1015 Fifteenth Street NW, Washington, D.C. 20005; either edition may be used.
2. The time from sample collection to initiation of analysis may not exceed 8 hours. Systems must hold samples at <10 °C during transit.
3. Lactose broth, as commercially available, may be used in lieu of lauryl tryptose broth, if the system conducts at least 25 parallel tests between this medium and lauryl tryptose broth using the water normally tested, and this comparison demonstrates that the false-positive rate and false-negative rate for total coliform, using lactose broth, is <10%.
4. Media shall cover inverted tubes at least one-half to two-thirds after the sample is added.
5. No requirement exists to run the completed phase on 10% of all total coliform-positive confirmed tubes.
6. MI agar also may be used. Preparation and use of MI agar is set forth in the article, "New medium for the simultaneous detection of total coliform and Escherichia coli in water'' by Brenner, K.P., et al., 1993, Appl. Environ. Microbiol. 59:3534-3544. Also available from the Office of Water Resource Center (RC-4100), 401 M Street SW, Washington, D.C. 20460, EPA 600/J-99/225.
7. The ONPG-MUG Test is also known as the Autoanalysis Colilert System.
8. A-1 Broth may be held up to three months in a tightly closed screw cap tube at 4 °C.
9. "Methods for the Determination of Inorganic Substances in Environmental Samples'', EPA/600/R-93/100, August 1993. Available at NTIS, PB94-121811.
10. GLI Method 2, "Turbidity'', November 2, 1992, Great Lakes Instruments, Inc., 8855 North 55th Street, Milwaukee, Wisconsin 53223.

Tables 3.28-a. Spanish Analytical Methods

Organoleptic Parameters	
Color	Photometric method calibrated according to Pt/Co scale
Taste	Successive dissolution, measurements performed between 12-25ºC
Odor	Successive dissolution, measurements performed between 12-25ºC
Physical-Chemical Parameters	
Aluminum	Atomic absorption Absorption spectrophotometry
Calcium	Atomic absorption Compleximetry
Chloride	Mohr method Titration
Conductivity	Electrometry
Dry Residues	Desiccation at 180ºC and weighing
Free Carbon Dioxide	Acidimetry
Magnesium	Atomic absorption
Oxygen	Winkler method Method with specific electrodes
pH	Electrometry
Potassium	Atomic absorption
Radioactivity	Measurement of the Total-α and Total-β activity of the sample
Sodium	Atomic absorption
Silicon	Absorption spectrophotometry
Sulfate	Gravimetry Compleximetry Spectrophotometry
Temperature	Thermometry
Total Carbonate Hardness	Compleximetry
Turbidity	Silicon method, Formacine method, or Secchi method
Microbiologic Parameters	
Fecal Coliforms	Multiple tube fermentation. Confirmation media tube removal. Counting according to most probable quantity or membranes filtration and growing in appropriate media like gelose. Lactosed with tergitol gelose, endo gelose, or 0.4% teepol broth, removal and identification of suspected colonies. Incubation at 44ºC.
Total Coliforms	• Multiple tube fermentation. Confirmation media tube removal. Counting according to most probable quantity or membranes filtration and growing in appropriate media like gelose. Lactosed with tergitol gelose, endo gelose, or 0.4% teepol broth, removal and edentification of suspected colonies. Incubation at 37ºC. • Detection and counting of coliform bacteria and Escherichia coli in drinking water by membrane filtration using Chromogenic Coliform Agar method. • Detection and counting of coliform bacteria and Escherichia coli in

	drinking water by Most Probable Number in liquid medium using Defined Substrate Technology.
Fecal Streptococcus	Method with sodium acid (Litsky). Counting according to most probable number.
Sulfite-Reducing Clostridia	• Counting of the spores after sample heating until 80ºC according to the following procedures: • Growing in glucose medium with iron sulfite and counting of the black halo colonies • Filtration with membrane, deposition of the inverted filter over a medium with glucose, sulfite and iron, covered with gelose, and counting of black colonies. • Distribution in tubs with DRCM (Differential Reinforced Clostridia Medium), removal of the black tubes in a chatoyant milk media, and counting according to most probable number.
Total Germs Counting	Incorporation inoculation in nutritive gelose

Table 3.28-b. Performance Parameters of Analytical Methods

Parameter	Trueness % of parametric value	Precision % of parametric value	Detection Limit % of parametric value
Aluminium	10	10	10
Ammonium	10	10	10
Antimony	25	25	25
Arsenic	10	10	10
Benzo(a)pyrene	25	25	25
Benzene	52	25	25
Boron	10	10	10
Bromate	25	25	25
Cadmium	10	10	10
Chloride	10	10	10
Chromium	10	10	10
Conductivity	10	10	10
Copper	10	10	10
Cyanide	10	10	10
1,2-dichloroethane	25	25	10
Fluoride	11	10	10
Iron	10	10	10
Lead	10	10	10
Manganese	10	10	10
Mercury	20	20	20
Nickel	10	10	10
Nitrate	10	10	10
Nitrite	10	10	10
Oxidisability	25	10	10
Pesticides	25	25	25
Polycyclic aromatic hydrocarbons	25	25	25
Selenium	10	10	10
Sodium	10	10	10
Sulfate	10	10	10
Tetrachloroethene	25	25	10
Trichloroethene	25	25	10
Trihalomethanes (total)	25	25	10

C4. CHAPTER 4

WASTEWATER

C4.1. SCOPE

This Chapter contains criteria to control and regulate discharges of wastewater into surface waters. This includes, but is not limited to, storm water runoff associated with industrial activities, domestic and industrial wastewater discharges, and pollutants from industrial dischargers.

C4.2. DEFINITIONS

C4.2.1. Best Management Practices (BMP). Practical practices and procedures that will minimize or eliminate the possibility of pollution being introduced into waters of Spain.

C4.2.2. Biochemical Oxygen Demand (BOD_5). The five-day measure of the dissolved oxygen used by microorganisms in the biochemical oxidation of organic matter. The pollutant parameter is biochemical oxygen demand (i.e., biodegradable organics in terms of oxygen demand).

C4.2.3. Carbonaceous BOD_5 ($CBOD_5$). The five-day measure of the pollutant parameter, $CBOD_5$. This test can substitute for the BOD_5 testing which suppresses the nitrification reaction/component in the BOD_5 test.

C4.2.4. Daily Discharge. The "discharge of a pollutant" measured during a calendar day or any 24-hour period that reasonably represents the calendar day for purposes of sampling. For pollutants with limitations expressed in units of mass, the "daily discharge" is calculated as the total mass of the pollutant discharged over the day. For pollutants with limitations expressed in other units of measurement (e.g., concentration) "daily discharge" is calculated as the average measurement of the pollutant over the day.

C4.2.5. Direct Discharge. Discharge of treated or untreated wastewater to continental waters of Spain.

C4.2.6. Discharge of a Pollutant. Any addition of any pollutant or combination of pollutants to waters of Spain from any "point source."

C4.2.7. Domestic Wastewater Treatment System (DWTS). Any DoD or Spanish facility designed to treat wastewater before its discharge to waters of Spain and in which the majority of such wastewater is made up of domestic sewage.

C4.2.8. Effluent Limitation. Any restriction imposed on quantities, discharge rates, and concentrations of pollutants that are ultimately discharged from point sources into waters of Spain.

C4.2.9. Indirect Discharge. Discharge of treated or untreated wastewater to a Wastewater Treatment Plant (WWTP).

C4.2.10. Industrial Activities Associated with Storm Water. Activities that may contribute pollutants to storm water runoff or drainage during wet weather events. (See Table 4.5., “Best Management Practices.”)

C4.2.11. Industrial Wastewater. Any wastewater generated from commercial or manufacturing industrial activities or from development or recovery activities.

C4.2.12. Interference. Any addition of any pollutant or combination of pollutant discharges that inhibits or disrupts the DWTS, its treatment processes or operations, or its sludge handling processes, use or disposal.

C4.2.13. Maximum Daily Discharge Limitation. The highest allowable daily discharge based on volume as well as concentration.

C4.2.14. Municipal Wastewater Treatment Plant (WWTP). HN-owned and operated facility designed to remove biological or chemical waste products from wastewater prior to discharge to Waters of Spain.

C4.2.15. Point Source. Any discernible, confined, and discrete conveyance, including, but not limited to, any pipe, ditch, channel, tunnel, conduit, well, discrete fissure, container, or rolling stock; but not including vessels, aircraft, or any conveyance that merely collects natural surface flows of precipitation.

C4.2.16. Pollutant. Includes, but is not limited to, the following: dredged spoil; solid waste; incinerator residue; filter backwash; sewage; garbage; sewage sludge; munitions; chemical waste; biological material; radioactive material; heat; wrecked or discarded equipment; rock; sand; cellar dirt; and industrial, municipal, and agricultural waste discharged into water.

C4.2.17. Regulated Facilities. Those facilities for which criteria are established under this Chapter.

C4.2.18. Storm Water. Run-off and drainage from wet weather events such as rain, snow, ice, sleet, or hail.

C4.2.19. Total Suspended Solids (TSS). The pollutant parameter total filterable suspended solids.

C4.2.20. Waters of Spain. Waters included in the following categories:

C4.2.20.1. Continental Waters . Any water body within the Spanish territory other than coastal waters; this includes groundwater and surface waters (e.g., underground aquifers, rivers, streams, lakes, reservoirs, springs, irrigation channels).

C4.2.20.2. Coastal Waters. The zone between the coastline of low waters and a line parallel to the coastline that is approximately 12 miles (20 km) from the shore.

C4.2.20.3. Exclusions to Waters of Spain. Domestic or industrial waste treatment systems, including treatment ponds or lagoons designed to meet the requirements of this Chapter, are not waters of Spain. This exclusion applies only to manmade bodies of water that were neither originally waters of Spain nor resulted from impoundment of waters of Spain.

C4.3. CRITERIA

C4.3.1. Effluent Limitations for-Wastewater Dischargers

C4.3.1.1. DoD installations discharging wastewater into Waters of Spain or a Spanish domestic WWTP shall provide sufficient information to the Spanish Base Commander for authorization prior to the start of the discharge for new facilities or as soon as possible for existing facilities (see Chapter 1 for the process).

C4.3.1.2. Discharges to Spanish Continental Waters. Domestic wastewater discharges to Spanish continental waters shall comply with the limit values provided in Table 4.1.

C4.3.1.3. Discharges to Coastal Waters. Discharges to coastal waters shall comply with the limit values provided in Table 4.2. Prior to 1 March of each year, an annual discharge declaration shall be submitted to the Spanish Base Commander that includes the following information:

C4.3.1.3.1. Site location;

C4.3.1.3.2. Discharge characteristics;

C4.3.1.3.3. Annual discharge volume;

C4.3.1.3.4. Average monthly discharge flow rate;

C4.3.1.3.5. WWTP capacity;

C4.3.1.3.6. Monitoring results report;

C4.3.1.3.7. Evaluation of the discharge effects on the receptor medium;and

C4.3.1.3.8. Incidents occurred during the past year.

C4.3.1.4. Discharges to Domestic WWTPs. Discharges to Spanish domestic WWTPs shall comply with the limit values provided in Table 4.3.

C4.3.1.5. Monitoring. Monitoring requirements apply to all regulated facilities. Sampling and analysis shall be performed in accordance with the frequency established in Table 4.4, “Monitoring Requirements.” Samples shall be collected at the point of discharge to the Waters of Spain.

C4.3.1.6. Recordkeeping Requirements. The following monitoring and recordkeeping requirements are BMPs and apply to all facilities. Retain records for three years.

C4.3.1.6.1. The effluent, concentration, or other measurement specified for each regulated parameter.

C4.3.1.6.2. The daily volume of effluent discharge from each point source.

C4.3.1.6.3. Test procedures for the analysis of pollutants.

C4.3.1.6.4. The date, exact place, and time of sampling and/or measurements.

C4.3.1.6.5. The name of the person who performed the sampling and/or measurements.

C4.3.1.6.6. The date of analysis.

C4.3.1.7. Complaint System. A system for investigating water pollution complaints from individuals or Spanish water pollution control authorities will be established, involving the EEA, as appropriate.

C4.3.2. Effluent Limitations For Non-Categorical Industrial Indirect Dischargers

C4.3.2.1. Effluent Limits. Installations discharging pollutants to Spanish domestic WWTPs shall comply with the effluent limits provided in Tables 4.1, 4.2, and 4.3.

C4.3.2.2. The following discharges are prohibited:

C4.3.2.2.1. The discharge to a DoD-operated WWTP or a Spanish domestic WWTP of any type of waste, either hazardous and non-hazardous (solid, viscous, liquid, or gaseous), classified as such based on the criteria established in Chapter 6 , “Hazardous Waste,” Chapter 7 , “Solid Waste,” and Chapter 8, “Medical Waste,” that would result in an obstruction or disruption to the WWTP flow.

C4.3.2.2.2. Discharge to a DoD-operated WWTP or a Spanish domestic WWTP of any toxic and poisonous solid, liquid or gas, that may present a health and safety risk to domestic WWTP personnel and impede the correct functioning and maintenance of the domestic WWTP.

C4.3.2.2.3. Discharge to a DoD-operated WWTP or a Spanish domestic WWTP of any radioactive or isotope wastes in concentrations that may damage the domestic WWTP or present a health and safety risk to domestic WWTP personnel.

C4.3.2.2.4. Discharges that cause difficulties for the proper functioning of the domestic WWTP (e.g., containing organic solvents, dyes, paints, varnishes, pigments, non-biodegradable detergents, odoriferous compounds, etc.).

C4.3.2.2.5. Corrosivity. The discharge of pollutants with the potential to be structurally corrosive to the municpal WWTP is prohibited. In addition, no discharge of wastewater at a pH <6.0 is allowed.

C4.3.2.2.6. Oil and Grease. The discharge of the following oils that can pass through or cause interference to the domestic WWTP is prohibited: petroleum oil, non-biodegradable cutting oil, and products of mineral oil origin.

C4.3.2.2.7. Trucked and Hauled Waste. The discharge of trucked and hauled waste into any WWTP is prohibited. These wastes shall be handled, treated, and disposed of in accordance with Chapter 6 , “Hazardous Waste” or Chapter 7 , “Solid Waste,” as applicable.

C4.3.2.2.8. Heat. Heat in amounts that inhibit biological activity in the domestic WWTP resulting in interference, but in no case in such quantities that the temperature of the process water at the domestic WWTP exceeds 40°C (104°F).

C4.3.2.3. Spills and Batch Discharges (slugs). Activities or installations that have a significant potential for spills or batch discharges will develop a slug prevention plan. Each plan must contain the following minimum requirements:

C4.3.2.3.1. Description of discharge practices, including non-routine batch discharges;

C4.3.2.3.2. Description of stored chemicals;

C4.3.2.3.3. Plan for immediately notifying the DWTS of slug discharges and discharges that would violate prohibitions under this Chapter, including procedures for subsequent written notification within five days;

C4.3.2.3.4. Necessary practices to prevent accidental spills. This would include proper inspection and maintenance of storage areas, handling and transfer of materials, loading and unloading operations, control of plant site runoff, and worker training;

C4.3.2.3.5. Proper procedures for building containment structures or equipment;

C4.3.2.3.6. Necessary measures to control toxic organic pollutants and solvents; and

C4.3.2.3.7. Proper procedures and equipment for emergency response, and any subsequent plans necessary to limit damage suffered by the treatment plant or the

C4.3.2.4. Complaint System. A system for investigating water pollution complaints from Spanish water pollution control authorities will be established, involving the EEA as appropriate.

C4.3.2.5. Wastewater Sampling Location. A wastewater sampling location shall be identified and communicated to the Spanish Base Commander prior to discharging to the Spanish public sewer system.

C4.3.3. Effluent Limitations for Categorical Industrial Dischargers (Direct or Indirect)

C4.3.3.1. Installations with categorical industrial discharges shall comply with the discharge limits described in C4.3.3.3.8 through C4.3.3.3.11, and the monitoring requirements described in C4.3.3.4.

C4.3.3.2. All installations which have activities that fall into any of the industrial categories listed below must comply with the following effluent limitations.

C4.3.3.3. Electroplating. The following discharge standards apply to electroplating operations in which metal is electroplated on any basis material and to related metal finishing operations as set forth in the various subparts. These standards apply whether such operations are conducted in conjunction with electroplating, independently, or as part of some other operation. Electroplating subparts are identified as follows:

C4.3.3.3.1. Electroplating of Common Metals. Discharges of pollutants in process waters resulting from the process in which a material is electroplated with copper, nickel, chromium, zinc, tin, lead, cadmium, iron, aluminum, or any combination thereof.

C4.3.3.3.2. Electroplating of Precious Metals. Discharges of pollutants in process waters resulting from the process in which a material is plated with gold, silver, iridium, palladium, platinum, rhodium, ruthenium, or any combination thereof.

C4.3.3.3.3. Anodizing. Discharges of pollutants in process waters resulting from the anodizing of ferrous and nonferrous materials.

C4.3.3.3.4. Metal Coatings. Discharges of pollutants in process waters resulting from the chromating, phosphating, or immersion plating on ferrous and nonferrous materials.

C4.3.3.3.5. Chemical Etching and Milling. Discharges of pollutants in process waters resulting from the chemical milling or etching of ferrous and nonferrous materials.

C4.3.3.3.6. Electroless Plating. Discharges of pollutants in process waters resulting from the electroless plating of a metallic layer on a metallic or nonmetallic substrate.

C4.3.3.3.7. Printed Circuit Board Manufacturing. Discharges of pollutants in process waters resulting from the manufacture of printed circuit boards, including all manufacturing operations required or used to convert an insulating substrate to a finished printed circuit board.

C4.3.3.3.8. The following discharge standards apply to facilities in the above electroplating subparts which discharge <38,000 liters per day (10,000 gallons per day):

Pollutant	Discharge Limit (mg/L)
Cyanide, amenable	5.0
Lead	0.6

C4.3.3.3.9. The following discharge standards apply to facilities in the above electroplating subparts that discharge ≥ 38,000 liters per day (10,000 gallons per day):

Pollutant	Discharge Limit (mg/L)
Cyanide, total	1.0
Copper	3.0
Nickel	3.0
Chrome	3.0
Zinc	4.2
Lead	0.6
Total Metals	10.5

C4.3.3.3.10. In addition to the above standards, facilities that electroplate precious metals and that discharge ≥ 38,000 liters per day (10,000 gallons per day) must comply with the following standard:

Pollutant	Discharge Limit (mg/L)
Silver	0.5

C4.3.3.3.11. The following discharge limits shall be met if the wastewater contains cadmium, regardless of the volume of wastewater discharged:

Pollutant	Maximum Discharge Load	Discharge Limit (mg/L)
Cadmium	0.3 g of Cd discharged / kg of Cd used	0.2

C4.3.3.4. Monitoring. Monitoring of categorical industrial dischargers (including both sampling and analysis) will be accomplished quarterly and will include all parameters that are specified in the paragraph of this Chapter dealing with industrial dischargers. Samples shall be collected at the point of discharge to Waters of Spain or a Spanish domestic WWTP prior to any mixing with the receiving water.

C4.3.4. Storm Water Management

C4.3.4.1. Develop and implement storm water pollution prevention (P2) plans (SWPPP) for activities listed in Table 4.5, "Best Management Practices." Update the SWPPP annually using in-house resources.

C4.3.4.2. Employee Training. Personnel who handle hazardous substances or perform activities that could contribute pollution in wet weather events, shall be trained in appropriate BMPs. Such training shall stress P2 principles and awareness of possible pollution sources, including, non-traditional sources such as sediment, nitrates, pesticides, and fertilizers.

C4.3.4.3. The direct discharge of storm water into groundwater is prohibited. Storm water which does not have the potential of being contaminated or is conveyed to a separate storm water sewer system can be discharged to surface water or onto soil without prior authorization.

Potentially contaminated storm water can be discharged to surface water or soil if monitoring indicates that the discharge meets the established limits. Otherwise, the discharge shall be treated and shall comply with the criteria established in C4.3.1.

C4.3.4.4. Storm water that is discharged to a combined sewer system shall follow the criteria established in C4.3.1.

C4.3.5. Septic System. Discharge to a septic system of wastewater containing industrial pollutants in levels that will inhibit biological activity is prohibited. Known discharges of industrial pollutants to existing septic systems shall be eliminated, and appropriate actions shall be taken to eliminate contamination. Siting of such systems is addressed in Chapter 3, “Drinking Water.”

C4.3.6. Sludge, Grease, and Oil Disposal. All sludge, grease, and oil produced during the treatment of wastewater will be disposed in accordance with the guidance under Chapter 6, “Hazardous Waste” or Chapter 7, “Solid Waste,” as appropriate.

C4.3.7. Wastewater Treatment System Operator Certification. DoD installations shall ensure that personnel operating DoD WWTPs are appropriately certified for the class of WWTP operated. Certification shall meet the requirements of a recognized U.S. certification body such as the Association of Boards of Certification or Spanish equivalent.

Table 4.1. Effluent Limits for Discharges of Domestic Wastewater to Continental Waters

Parameters	Limit	Minimum percentage of reduction [1]	Methodology
BOD_5 at 20 ºC	25 mg/L O_2	70 – 90	Homogenized, unfiltered, undecanted sample. Determination of dissolved oxygen before and after five-day incubation at 20 °C ± 1 °C, in complete darkness. Addition of a nitrification inhibitor.
COD	125 mg/L O_2	75	Homogenized, unfiltered, undecanted sample Potassium dichromate.
TSS	35 mg/L	90	Filtering of a representative sample through a 0.45 μm filter membrane. Drying at 105 °C and weighing.
Total Phosphorus [2]	2 mg/L	80	Molecular absorption spectrophotometry.
Total Nitrogen [2]	15 mg/L	70 – 80	Molecular absorption spectrophotometry.

1 Reduction in relation to the load of the influent.
2 Requirements for discharges from domestic WWTP to sensitive areas which are subject to eutrophication.

Table 4.2. Effluent Limits for Domestic and Industrial Wastewater to Coastal Waters in the Andalusian Region

Parameter	Units	Monthly Average	Daily Average	Single Value	Application to domestic/industrial wastewater
General Parameters					
pH	ND	5.5 – 9.5			Domestic/Industrial
Color [1]	ND	1:40			Domestic/Industrial
Dissolved Solids	mg/L	300	400	500	Domestic/Industrial
Sedimentable Materials	ml/L	2	3	4	Domestic/Industrial
Coarse Material	ND	Absent			Domestic/Industrial
Floating solids	ND	Absent			Domestic/Industrial
BOD_5	mg/L	300	400	500	Domestic/Industrial
COD	mg/L	450	600	750	Domestic/Industrial
Total Organic Carbon	mg/L	150	200	250	Domestic/Industrial
Turbidity	NTU	150	250	400	Domestic/Industrial
Temperature [2]	ºC	Increase of ±3º			Domestic/Industrial
Total Residual Chlorine	mg/L	0.2	0.5	1	Domestic/Industrial
Aluminum	mg/L	3	6	10	Industrial
Aldehydes	mg/L	1	2	3	Industrial
Nitrates	mg of O_3/L	75	100	150	Domestic/Industrial
Non-polar Hydrocarbons	mg/L	15	20	40	Industrial
Oils and Greases	mg/L	25	40	75	Industrial
Organochlorated	kg AOX/TAD[3]	1	1	1	Industrial
Pesticides	mg/L	0.4	1.2	2.5	Industrial
Phenols	mg/L	3	15	15	Industrial
Polycyclic Aromatic Hydrocarbons	mg/L	0.01	0.02	0.05	Industrial
Sulfides	mg/L	1	2	4	Industrial
Sulfites	mg/L	1	2	4	Domestic/Industrial
Surfactants	mg/L	5	20	50	Domestic/Industrial
Toxicity (equitox)	ND	50	50	50	Industrial
Hazardous Parameters					
Aldrin and derivatives	mg/L	0.002	0.01	0.02	Industrial
Ammonia (as NH_4^+)	mg/L	60	80	100	Industrial
Arsenic	mg/L	1	3	5	Industrial
Cadmium	mg/L	0.2	0.4	1	Industrial
Carbon Tetrachloride	mg/L	1.5	3	6	Industrial
Chloroform	mg/L	1	2	4	Industrial
Chrome IV	mg/L	0.2	0.4	0.5	Industrial
Chrome Total	mg/L	0.5	2	4	Industrial
Copper	mg/L	0.5	2.5	4	Industrial
Cyanides	mg/L	0.5	1	2	Industrial
DDT	mg/L	0.2	0.4	0.8	Industrial
1,2-Dichloroethane (EDC) [5]	mg/L	2.5	5	10	Industrial
Fluorides	mg/L	10	15	20	Industrial
Hexachlorocyclohexane	mg/L	2	4	8	Industrial

Parameter	Units	Monthly Average	Daily Average	Single Value	Application to domestic/industrial wastewater
General Parameters					
(HCH)					
Hexachlorbenzene (HCB)	mg/L	1 [6] and 1.9 [7]	2 [6] and 3.8 [7]	4 [6] and 7.6 [7]	Industrial
Hexachlorobutadiene (HCBD)	mg/L	1.5	3	6	Industrial
Lead	mg/L	0.5	1	2	Industrial
Mercury	mg/L	0.05	0.2 [4] and 0.1 [2]	0.2 [4] and 0.1 [5]	Industrial
Nickel	mg/L	3	6	10	Industrial
Pentachlorophenol	mg/L	1	2	3	Industrial
Perchlorethylene	mg/L	1.25	2.5	5	Industrial
Selenium	mg/L	0.05	0.1	0.2	Industrial
Tin	mg/L	10	15	20	Industrial
Titanium	mg/L	1	3	5	Industrial
Total Phosphorus	mg/L	40	50	60	Industrial
Trichloroethylene	mg/L	0.5	1	2	Industrial
Trichlorobenzene	mg/L	1	2	4	Industrial
Zinc	mg/L	3	6	10	Industrial

Notes:

1 Not perceptible over a thickness of 10 cm, with the indicated dilution in >10% of the reference value in Co-Pt units.
2 At a distance of 100 m from the discharge point and at a depth of 1 m.
3 TAD: Ton of dry mash.
4 Sector of the electrolysis of the chlorides that uses mercury cathodes cells.
5 Other sectors excluding the electrolysis of the alkaline chlorides.
6 Production and transformation of HCB.
7 Production of perchlorethylene and carbon tetrachloride by means of percolation.

Table 4.3. Effluent Limits for Discharges to Spanish Domestic WWTPs (Municipalities of Morón de la Frontera and Rota)

Parameter	Units	Limit value		Application to domestic/industrial wastewater
		Rota	Morón de la Frontera	
Physical parameters				
pH	NA	≥ 6 and ≤ 9	≥ 6 and ≤ 9	Domestic/Industrial
Conductivity	µS/cm	5,000	2,000	Domestic/Industrial
Solids sedimentable in one hour	mg/L	10	10	Domestic/Industrial
Suspended matter	mg/L	1,000	300	Domestic/Industrial
Temperature	ºC	40	40	Domestic/Industrial
Color		Biodegradable in the treatment plant	ND	Domestic/Industrial
Chemical parameters				
Oils and Greases	mg/L	200	150	Industrial
Aluminum	mg/L	10	10	Industrial
Arsenic	mg/L	0.70	0.70	Industrial
Barium	mg/L	12	12	Industrial
Boron	mg/L	2	2	Industrial
Cadmium	mg/L	0.70	0.70	Industrial
Total cyanides	mg/L	1.50	1	Industrial
Zinc	mg/L	10	10	Industrial
Chlorides	mg/L	ND	1,600	Industrial
Dissolved copper	mg/L	0.50	0.50	Industrial
Total copper	mg/L	3	3	Industrial
Hexavalent chromium	mg/L	0.60	0.60	Industrial
Total chrome	mg/L	3	3	Industrial
BOD_5	mg/L	1,000	300	Domestic/Industrial
Biodegradable detergents	mg/L	10	10	Domestic/Industrial
COD	mg/L	1,750	550	Domestic/Industrial
Ecotoxicity	Equitox/m^3	15	15	Industrial
Tin	mg/L	2	2	Industrial
Phenols	mg/L	3	3	Industrial
Fluorides	mg/L	9	9	Industrial
Total phosphorus	mg/L	ND	75	Industrial
Phosphates	mg/L	100	100	Industrial
Iron	mg/L	25	10	Industrial
Manganese	mg/L	3	2	Industrial
Mercury	mg/L	0.20	0.10	Industrial
Molybdenum	mg/L	1	1	Industrial
Nickel	mg/L	3	3	Industrial

Parameter	Units	Limit value		Application to domestic/industrial wastewater
		Rota	Morón de la Frontera	
Nitrates	mg/L	80	80	Domestic/Industrial
Kjeldahl nitrogen	mg/L	ND	75	Industrial
Ammoniacal nitrogen	mg/L	25	25	Industrial
Silver	mg/L	ND	0.50	Industrial
Lead	mg/L	1.20	1.20	Industrial
Selenium	mg/L	1	1	Industrial
Sulfates	mg/L	500	500	Industrial
Total sulfides	mg/L	5	5	Industrial
Total metals	mg/L	130	ND	Industrial
Total metals without Fe and Zn	mg/L	30	ND	Industrial
TOC	mg/L	300	300	Industrial
Gaseous parameters				
Ammonia (NH_3)	cm^3 of gas/m^3 of air	25	25	Industrial
Hydrocyanic acid (CNH)	cm^3 of gas/m^3 of air	2	2	Industrial
Chloride (Cl_2)	cm^3 of gas/m^3 of air	0.25	0.25	Industrial
Sulfur dioxide (SO_2)	cm^3 of gas/m^3 of air	2	2	Industrial
Carbon Monoxide (CO)	cm^3 of gas/m^3 of air	15	15	Industrial
Hydrogen Sulfide (SH_2)	cm^3 of gas/m^3 of air	10	10	Industrial

ND: Not defined.
NA: Not applicable.

Table 4.4. Monitoring Requirements

Plant Capacity (MGD)	Monitoring Frequency
0.001 - 0.99	Monthly
1.0 - 4.99	Weekly
> 5.0	Daily

Table 4.5. Best Management Practices

Activity	Best Management Practice
Aircraft Ground Support Equipment Maintenance	Perform maintenance/repair activities inside Use drip pans to capture drained fluids Cap hoses to prevent drips and spills
Aircraft/runway deicing	Perform anti-icing before the storm Put critical aircraft in hangars/shelters
Aircraft/vehicle fueling operations	Protect fueling areas from the rain Provide spill response equipment at fueling station
Aircraft/vehicle maintenance & repair	Perform maintenance/repair activities inside Use drip pans to capture drained fluids
Aircraft/vehicle washing	Capture wash water and send to wastewater treatment plant Treat wash water with oil water separator before discharge
Bulk fuel storage areas	Use dry camlock connectors to reduce fuel loss Capture spills with drip pans when breaking connections Curb fuel transfer areas, treat with oil water separator
Construction activities	Construct sediment dams/silt fences around construction sites
Corrosion control activities	Capture solvent/soaps used to prepare aircraft for painting Perform corrosion control activities inside
Hazardous material storage	Store hazardous materials inside or under cover Reduce use of hazardous materials
Outdoor material storage areas	Cover and curb salt, coal, urea piles Store product drums inside or under cover Reduce quantity of material stored outside
Outdoor painting/depainting operations	Capture sandblasting media for proper disposal Capture paint clean up materials (thinners, rinsates)
Pesticide operations	Capture rinse water when mixing chemicals Store spray equipment inside
Power production	Capture leaks and spills from power production equipment using drip pans, etc.
Vehicle storage yards	Check vehicles in storage for leaks and spills Use drip pans to capture leaking fluids

C5. CHAPTER 5

HAZARDOUS MATERIAL

C5.1. SCOPE

This Chapter contains criteria for the storage, handling, and disposition of hazardous materials. It does not cover solid or hazardous waste, underground storage tanks, petroleum storage, and related spill contingency and emergency response requirements, which are covered under other Chapters. This FGS does not cover munitions, radioactive material, or the transportation of hazardous material.

Information and requirements for the transportation of hazardous material can be found in the Joint Transportation and Traffic Management, 30 August 2002. The International Maritime Dangerous Goods (IMDG) Code and appropriate DoD and Component instructions provide requirements for international maritime transport of hazardous materials originating from DoD installations. (OEBGD) International air shipments of hazardous materials originating from DoD installations are subject to International Civil Aviation Organization Technical Instructions or DoD Component guidance, including Air Force Interservice Manual 24-204(I), Army Technical Order (TO) 38-250, NAVSUP PUB 505, MCO P4030.19I, and DLAI 4145.3, DCMAD1, Ch3.4 (HM24) "Defense Materiel Disposition Manual". For further guidance, contact the EUCOM Dangerous Goods Advisor.

C5.2. DEFINITIONS

C5.2.1. EU List of Hazardous Materials. Official list of hazardous materials registered for use in Europe. A copy of the list is provided as Addendum 1.

C5.2.2. Hazardous Chemical Warning Label. A label, tag, or marking on a container that provides the following information:

C5.2.2.1. Identification/name of hazardous chemicals;

C5.2.2.2. Appropriate hazard warnings; and

C5.2.2.3. The name and address of the manufacturer, importer, or other responsible party; and which is prepared in accordance with DoDI 6050.05.

C5.2.3. Hazardous Material. Any substance, compound, mixture, or material that is capable of posing an unreasonable risk to health, safety, or the environment if improperly handled, stored, issued, transported, labeled, or disposed because it displays a characteristic listed in Table 5.1., "Typical Hazardous Materials Characteristics," or the material is listed in Addendum 1, "EU List of Hazardous Materials" or Appendix A, "List of Hazardous Substances & Materials." Munitions are excluded.

C5.2.4. Hazardous Material Information Resource System (HMIRS). The computer-based information system developed to accumulate, maintain and disseminate important information on hazardous material used by the Department of Defense in accordance with DoDI 6050.05.

C5.2.5. Hazardous Material Shipment. Any movement of hazardous material in a DoD land vehicle, either from an installation to a final destination off the installation, or from a point of origin off the installation to a final destination on the installation, in which certification of the shipment is involved.

C5.2.6. Safety Data Sheet (SDS). A form prepared by manufacturers or importers of chemical products to communicate to users the chemical and physical properties and the hazardous effects of a particular product.

C5.3. CRITERIA

C5.3.1. Storage and handling of hazardous materials will adhere to the DoD Component policies, including Joint Service Publication on Storage and Handling of Hazardous Materials. Defense Logistics Agency Instruction (DLAI) 4145.11, Army Technical Manual (TM) 38-410, Naval Supply Publication (NAVSUP PUB) 573, Air Force Joint Manual (AFJMAN) 23-209, and Marine Corps Order (MCO) 4450.12A “Storage and Handling of Hazardous Materials”provide additional guidance on the storage and handling of hazardous materials.

C5.3.1.1. Installations that store hazardous materials in quantities ≥ than the volumes listed in Table 5.2 shall also provide sufficient information to the Spanish Base Commander on the types and quantities of stored materials for authorization (see Chapter 1 for the process).

C5.3.2. Hazardous material dispensing areas will be properly maintained. Drums/containers must not be leaking. Drip pans/absorbent materials will be placed under containers as necessary to collect drips or spills. Container contents will be clearly marked. Dispensing areas will be located away from catch basins and floor/storm drains to prevent potential discharges to the sewer system. The dispensing area shall also be designed and operated to ensure appropriate segregation of incompatible hazardous materials.

C5.3.2.1. Appropriate chemical warning signs shall be posted for each hazardous material stored in the area. The responsibility for ensuring that appropriate signs are posted resides with the operator of the hazardous material storage area. For further guidance, operators should consult with Fire & Emergency Services (F&ES).

C5.3.3. Each installation will maintain a master listing of all storage locations for hazardous material as well as an inventory of all hazardous materials contained therein. (See paragraph C18.3.2.)

C5.3.4. Each SDS shall be in English or the predominant language in the work place. If local national workers are present in the workplace, the SDS shall also be in Spanish. Until a Spanish SDS is available, bilingual training will be provided on the contents of the English SDS. Each SDS shall contain at least the following information.

C5.3.4.1. The identity used on the label.

C5.3.4.1.1. If the hazardous chemical is a single substance, its chemical and common name;

C5.3.4.1.2. If the hazardous chemical is a mixture that has been tested as a whole to determine its hazards, the chemical and common name(s) of the ingredients that contribute to these known hazards, and the common name(s) of the mixture itself; or

C5.3.4.1.3. If the hazardous chemical is a mixture that has not been tested as a whole:

C5.3.4.1.3.1. The chemical and common name(s) of all ingredients that have been determined to be health hazards, and that comprise ≥ 1% of the composition, except that chemicals identified as carcinogens shall be listed if the concentrations are ≥ 0.1%;

C5.3.4.1.3.2. The chemical and common name(s) of all ingredients that have been determined to be health hazards, and that comprise < 1% (0.1% for carcinogens) of the mixture, if there is evidence that the ingredient(s) could be released from the mixture in concentrations that would exceed an established Occupational Safety and Health Administration (OSHA)-permissible exposure limit, or could present a health hazard to employees; and

C5.3.4.1.3.3. The chemical and common name(s) of all ingredients that have been determined to present a physical hazard when present in the mixture.

C5.3.4.2. Physical and chemical characteristics of the hazardous chemical (such as vapour pressure, flash point);

C5.3.4.3. The physical hazards of the hazardous chemical, including the potential for fire, explosion, and reactivity;

C5.3.4.4. The health hazards of the hazardous chemical, including signs and symptoms of exposure, and any medical conditions that are generally recognized as being aggravated by exposure to the chemical;

C5.3.4.5. The primary route(s) of entry (inhalation, skin absorption, ingestion, etc.);

C5.3.4.6. The appropriate occupational exposure limit recommended by the chemical manufacturer, importer, or employer preparing the SDS, where available;

C5.3.4.7. Whether the hazardous chemical has been found to be a potential carcinogen;

C5.3.4.8. Any generally applicable precautions for safe handling and use that are known to the chemical manufacturer, importer, or employer preparing the SDS, including appropriate hygienic practices, protective measures during repair and maintenance of contaminated equipment, and procedures for clean-up of spills and leaks;

C5.3.4.9. Any generally applicable control measures that are known to the chemical manufacturer, importer, or employer preparing the SDS, such as appropriate engineering controls, work practices, or personal protective equipment;

C5.3.4.10. Emergency and first aid procedures;

C5.3.4.11. The date of preparation of the SDS or the last change to it; and

C5.3.4.12. The name, address and telephone number of the chemical manufacturer, importer, employer, or other responsible party preparing or distributing the SDS who can provide additional information on the hazardous chemical and appropriate emergency procedures, if necessary.

C5.3.5. SDSs shall be reviewed to determine whether additional handling and storage requirements are required for a specific hazardous material.

C5.3.6. Each work center will maintain a file of SDSs for each hazardous material procured, stored, or used at the work center. SDSs that are not contained in the HMIRS and those SDSs prepared for locally purchased items shall be incorporated into the HMIRS. A file of SDS information not contained in the HMIRS shall be maintained on site.

C5.3.7. All containers of hazardous materials on DoD installations shall have manufacturer-generated warning labels or a Hazardous Chemical Warning Label in accordance with DoD Instruction 6050.05 and have SDS information either available on site or in HMIRS in accordance with DoD Instruction 6050.05 and other DoD Component instructions. These requirements apply throughout the life-cycle of these materials.

C5.3.7.1. All hazardous materials labels shall be provided in English and Spanish, if Spanish workers are present in the workplace.

C5.3.8. DoD installations will reduce the use of hazardous materials where practical through resource recovery, recycling, source reduction, acquisition, or other minimization strategies in accordance with Service guidance on improved hazardous material management processes and techniques.

C5.3.9. All excess hazardous material will be processed through the Defense Logistics Agency (DLA) Disposition Services in accordance with the procedures in DoD 4160.21-M "Defense Materiel Disposition Manual." DLA Disposition Services will only donate, transfer, or sell hazardous material to environmentally responsible parties. This paragraph is not intended to prohibit the transfer of usable hazardous material between DoD activities participating in a regional or local pharmacy or exchange program.

C5.3.10. All personnel who use, handle, or store hazardous materials will be trained in accordance with DoDI 6050.05, "DoD Hazard Communication (HAZCOM) Program" and other DoD Component instructions.

C5.3.10.1. Spanish personnel who use, handle, or store hazardous materials shall also receive training in accordance with Spanish requirements. Consult with the Safety and/or Occupational Health Department.

C5.3.11. The installation must prevent the unauthorized entry of persons or livestock into the hazardous materials storage area.

C5.3.12. Prohibited or Restricted-use Products. The use of materials listed in Table 5.3 is either prohibited or restricted in Spain, under specified conditions, with the exception of mission critical materials.

C5.3.12.1. Use of perfluorooctane sulfonates (PFOS)-based Fire-fighting foams (e.g., some aqueous film forming foams (AFFF)) that were commercially available in the European Union before 27 December 2006 is prohibited.

Table 5.1. Typical Hazardous Materials Characteristics

1. The item is a physical hazard, a hazard to human health and to the environment, including hazards to the ozone layer. Physical hazards include explosives, flammable gas, flammable aerosols, oxidizing gases, gases under pressure, flammable liquids and solids, self-reactive substances and mixtures, pyrophoric liquids and solids, self-heating substances and mixtures, substances and mixtures which in contact with water emit flammable gases, oxidizing liquids and solids, organic peroxides, and corrosives to metal. Health hazards include acute dermal, oral and inhalation toxicity, skin corrosion/irritation, serious eye damage/irritation, respiratory or skin sensitization, germ cell mutagenicity, carcinogenicity, reproductive toxicity, specific target organ toxicity-single exposure or repeated exposure, aspiration hazard. Environmental hazards include hazards to the aquatic environment and hazards to the ozone layer.

   The above-mentioned hazard characteristics are defined as follows:

   I. Physical Hazards:

   a) Explosive. An explosive substance (or mixture) is a solid or liquid which is in itself capable by chemical reaction of producing gas at such a temperature and pressure and at such a speed as to cause damage to the surroundings.
   b) Flammable Gases. Flammable gas means a gas having a flammable range in air at 20°C and a standard pressure of 101.3 kPa.
   c) Flammable Aerosols. Aerosols should be considered for classification as a Flammable Aerosol if they contain any component classified as flammable according to the GHS criteria for flammable liquids, flammable gases, or flammable solids.
   d) Oxidizing Gases. Oxidizing gas means any gas which may, generally by providing oxygen, cause or contribute to the combustion of other material more than air does.
   e) Gases Under Pressure. Gases under pressure are gases that are contained in a receptacle at a pressure not less than 280 Pa at 20°C or as a refrigerated liquid. This endpoint covers four types of gases or gaseous mixtures to address the effects of sudden release of pressure or freezing which may lead to serious damage to people, property, or the environment independent of other hazards the gases may pose.
   f) Flammable Liquids. Flammable liquid means a liquid having a flash point of ≤ 60°C.
   g) Flammable Solids. Flammable solids are solids that are readily combustible, or may cause or contribute to fire through friction. Readily combustible solids are powdered, granular, or pasty substances which are dangerous if they can be easily ignited by brief contact with an ignition source, such as a burning match, and if the flame spreads rapidly.
   h) Self-Reactive Substances. Self-reactive substances are thermally unstable liquids or solids liable to undergo a strongly exothermic thermal decomposition even without participation of oxygen (air). This definition excludes materials classified as explosive, organic peroxides or as oxidizing.
   i) Phosphoric Liquids. A pyrophoric liquid is a liquid which, even in small quantities, is liable to ignite within five minutes after coming into contact with air.

j) Phosphoric Solids. A pyrophoric solid is a solid which, even in small quantities, is liable to ignite within five minutes after coming into contact with air.
k) Self-Heating Substances. A self-heating substance is a solid or liquid, other than a pyrophoric substance, which, by reaction with air and without energy supply, is liable to self-heat. This endpoint differs from a pyrophoric substance in that it will ignite only when in large amounts (kilograms) and after long periods of time (hours or days).
l) Substances which on Contact with Water Emit Flammable Gases. Substances that, in contact with water, emit flammable gases are solids or liquids which, by interaction with water, are liable to become spontaneously flammable or to give off flammable gases in dangerous quantities.
m) Oxidizing Liquids. An oxidizing liquid is a liquid which, while in itself not necessarily combustible, may, generally by yielding oxygen, cause or contribute to the combustion of other material.
n) Oxidizing Solids. An oxidizing solid is a solid which, while in itself not necessarily combustible, may, generally by yielding oxygen, cause or contribute to the combustion of other material.
o) Organic Peroxides. An organic peroxide is an organic liquid or solid which contains the bivalent -O-O- structure and may be considered a derivative of hydrogen peroxide, where one or both of the hydrogen atoms have been replaced by organic radicals. The term also includes organic peroxide formulations (mixtures).
p) Substances Corrosive to Metal. A substance or a mixture that by chemical action will materially damage, or even destroy, metals is termed 'corrosive to metal'.

II. Health Hazards:

a) Acute Toxicity. Five categories have been included in the Acute Toxicity scheme from which the appropriate elements relevant to transport, consumer, worker and environment protection can be selected. Substances are assigned to one of the five toxicity categories on the basis of LD_{50} (oral, dermal) or LC_{50} (inhalation).
b) Skin Corrosion. Skin corrosion means the production of irreversible damage to the skin following the application of a test substance for up to 4 hours.
c) Skin Irritation. Skin irritation means the production of reversible damage to the skin following the application of a test substance for up to 4 hours.
d) Eye Effects:
    i. Serious Eye Damage. Serious eye damage means the production of tissue damage in the eye, or serious physical decay of vision, following application of a test substance to the front surface of the eye, which is not fully reversible within 21 days of application.
    ii. Eye Irritation. Eye irritation means changes in the eye following the application of a test substance to the front surface of the eye, which are fully reversible within 21 days of application.
e) Sensitization:
    iii. Respiratory Sensitizer. Respiratory sensitizer means a substance that induces hypersensitivity of the airways following inhalation of the substance.
    iv. Skin Sensitizer. Skin sensitizer means a substance that will induce an

allergic response following skin contact.

f) Germ Cell Mutagenicity. Mutagen means an agent giving rise to an increased occurrence of mutations in populations of cells and/or organisms. A mutation means permanent change in the amount or structure of the genetic material in a cell.

g) Carcinogenicity. Carcinogen means a chemical substance or a mixture of chemical substances that induce cancer or increase its incidence.

h) Reproductive Toxicity. Reproductive toxicity includes adverse effects on sexual function and fertility in adult males and females, as well as developmental toxicity in offspring. Materials, which cause concern for the health of breastfed children, have a separate category, Effects on or Via Lactation.

i) Target Organ System Toxicity (TOST). Single Exposure & Repeated Exposure. TOST includes any significant health effects that can impair function, reversible and irreversible, immediate and/or delayed are included in the non-lethal TOST. Narcotic effects and respiratory tract irritation are considered to be TOSTs.

j) Aspiration Hazard. Aspiration toxicity includes severe acute effects such as chemical pneumonia, varying degrees of pulmonary injury or death following aspiration. Aspiration is the entry of a liquid or solid directly through the oral or nasal cavity, or indirectly from vomiting, into the trachea and lower respiratory system.

III. Environmental Hazards:

a) Hazardous to the Aquatic Environment:

i. Acute Aquatic Toxicity. Acute aquatic toxicity means the intrinsic property of a substance to cause injury to an aquatic organism in a short-term exposure to that substance.

ii. Chronic Aquatic Toxicity. Chronic aquatic toxicity means the potential or actual properties of a substance to cause adverse effects to aquatic organisms during exposures that are determined in relation to the lifecycle of the organism.

b) Hazardous to the Ozone Layer. Substance hazardous to the ozone layer means a substance which, on the basis of the available evidence concerning its properties and its predicted or observed environmental fate and behavior may present a danger to the structure and/or the functioning of the stratospheric ozone layer.

Note: The classification system described above is immediately applicable to substances and will be effective for mixtures starting 1 June 2015.

2. The item and/or its disposal is regulated by the host nation because of its hazardous nature.

3. The item has a flashpoint below 93° C (200° F) closed cup, or is subject to spontaneous heating or is subject to polymerization with release of large amounts of energy when handled, stored, and shipped without adequate control.

4. The item is a flammable solid or is an oxidizer or is a strong oxidizing or reducing agent with a standard reduction potential of > 1.0 volt or < -1.0 volt.

5. In the course of normal operations, accidents, leaks, or spills, the item may produce dusts, gases, fumes, vapors, mists, or smokes with one or more of the above characteristics.

6. The item has special characteristics that, in the opinion of the manufacturer or the DoD Components, could cause harm to personnel if used or stored improperly.

Table 5.2. Hazardous Material Storage Volumes that Require Authorization

Product Class	Description	Storage Volume that Requires Authorization* Liters	Gallons
Combustible and Flammable Liquids:			
Class A	Liquefied products with an absolute vapor pressure (at 15ºC) > 98 kPa. These products are divided into two subclasses: A1 - products stored in liquid phase at temperatures < 0ºC A2 - products stored in liquid phase in other conditions	–	–
Class B	Products with a flashpoint < 55ºC, that are not included in Class A. Depending on their flashpoint, they are divided into two subclasses: B1 - products whose flashpoint is < 38ºC B2 - products whose flashpoint is ≥ 38ºC	≥ 300	≥ 79.2
Class C	Products with a flashpoint > 55ºC and <100ºC	≥ 3,000	≥ 792.5
Class D	Products with a flashpoint >100ºC	≥ 10,000	≥ 2641.7
Corrosive Materials:			
Class A	Very Corrosive Substances - Substances that cause necrosis when applied over the intact skin of an animal for < 3 minutes	≥ 200	≥ 52.8
Class B	Corrosive Substances - Substances that cause necrosis when applied over the intact skin of an animal for 3 to 60 minutes.	≥ 400	≥ 105.7
Class C	Less Corrosive Substances - Substances that cause necrosis when applied over the intact skin of an animal for 1 to 4 hours. Products that can cause corrosion of steel or aluminum at a rate of 6.25 mm/year at a temperature of 55 ºC are also in this class.	≥ 1,000	≥ 264.2
Toxic Materials:			
Class T+	Very toxic	≥ 100	≥ 26.4
Class T	Toxic	≥ 250	≥ 66
Class Xn	Harmful	≥ 1,000	≥ 264.2

* The volume applies to a single storage area, storage facility, or storage building.

Table 5.3. Hazardous Material Use Restrictions

Material	Conditions of Restriction
- Polychlorinated biphenyls (PCB) except mono- and dichlorinated biphenyls - Polychlorinated terphenyls (PCTs) - Mixtures, including waste oils, with a PCB or PCT content > 0,005 % by weight.	Use is prohibited.
Chloro-1-ethylene (monomer vinyl chloride)	May not be used as aerosol propellant.
Mercury compound	May not be used as substances and constituents of mixtures intended for use: (a) to prevent the fouling by micro-organisms, plants or animals of: - the hulls of boats, - cages, floats, nets and any other appliances or equipment used for fish or shellfish farming, - any totally or partially submerged appliances or equipment; (b) in the preservation of wood; (c) in the impregnation of heavy-duty industrial textiles and yarn intended for their manufacture; (d) in the treatment of industrial waters, irrespective of their use.
Arsenic compounds	May not be used as substances and constituents of mixtures intended for use: (a) To prevent the fouling by micro-organisms, plants or animals of: - the hulls of boats, - cages, floats, nets and any other appliances or equipment used for fish or shellfish farming, - any totally or partially submerged appliances or equipment; (b) in the preservation of wood. Furthermore, wood so treated may not be placed on the market; (c) however: (i) Relating to the substances and mixtures in the preservation of wood: these may only be used in industrial installations using vacuum or pressure to impregnate wood if they are solutions of inorganic compounds of the copper, chromium, arsenic (CCA) type C. Wood so treated may not be placed on the market before fixation of the preservative is completed. (ii) Relating to wood treated with CCA solutions in industrial installations according to point (i): this may be placed on the market for professional and industrial use provided that the structural integrity of the wood is required for human or livestock safety and skin contact by the general public during its service life is unlikely: - as structural timber in public and agricultural

Material	Conditions of Restriction
	buildings, office buildings, and industrial premises, - in bridges and bridgework, - as constructional timber in freshwater areas and brackish waters e.g. jetties and bridges, - as noise barriers, - in avalanche control, - in highway safety fencing and barriers, - as debarked round conifer livestock fence posts, - in earth retaining structures, - as electric power transmission and telecommunications poles, - as underground railway sleepers. (iii) Treated wood referred to under points(i) and (ii) may not be used: - in residential or domestic constructions, whatever the purpose, - in any application where there is a risk of repeated skin contact, - in marine waters, - for agricultural purposes other than for livestock fence posts and structural uses in accordance with point (ii), - in any application where the treated wood may come into contact with intermediate or finished products intended for human and/or animal consumption. May not be used as substances and constituents of mixtures intended for use in the treatment of industrial waters, irrespective of their use.
Organostannic compounds	May not be used as substances and constituents of mixtures intended for use in the treatment of industrial waters.
Cadmium and its compounds	1. May not be used to give color to finished products manufactured from the substances and mixtures listed below: - polyvinyl chloride (PVC) - polyurethane (PUR) - low-density polyethylene (ld PE), with the exception of low-density polyethylene used for the production of colored master batch - cellulose acetate (CA) - cellulose acetate butyrate (CAB) - epoxy resins - melamine – formaldehyde (MF) - urea – formaldehyde (UF) - unsaturated polyesters (UP) - polyethylene terephthalate (PET) - polybutylene terephthalate (PBT)

Material	Conditions of Restriction
	- transparent/general-purpose polystyrene - acrylonitrile methylmethacrylate (AMMA) - cross-linked polyethylene (VPE) - high-impact polystyrene - polypropylene (PP) 2. May not be used for cadmium plating metallic products or components of the products used in the sectors/applications listed below. - equipment and machinery for cooling and freezing
Monomethyl – tetrachlorodiphenyl methane Trade name: Ugilec 141	Use is prohibited.
Monomethyl-dichloro-diphenyl methane Trade name: Ugilec 121, Ugilec 21	Use is prohibited.
Monomethyl-dibromo-diphenyl methane Trade name: DBBT	Use is prohibited.
Substances and mixtures containing one or more of the following substances: (a) creosote (b) creosote oil (c) distillates (coal tar), naphthalene oils (d) creosote oil, acenaphthene fraction (e) distillates (coal tar), upper (f) anthracene oil (g) tar acids, coal, crude (h) creosote, wood (i) low temperature tar oil, alkaline	May not be used in the treatment of wood.
Chloroform, Carbon tetrachloride; 1,1,2 Trichloroethane; 1,1,2,2 Tetrachloroethane; 1,1,1,2 Tetrachloroethane; Pentachloroethane; 1,1 Dichloroethylene; 1,1,1 Trichloroethane	May not be used in concentrations ≥ 0.1 % by weight in substances and mixtures placed on the market for sale to the general public and/or in diffusive applications such as in surface cleaning and cleaning of fabrics. The packaging of such substances and mixtures containing them in concentrations ≥ 0.1 % shall be legible and indelibly marked as follows: 'For use in industrial installations only'.
Hexachloroethane	May not be used in the processing of nonferrous metals.
Diphenylether, pentabromo derivative $C_{12}H_5Br_5O$	May not be used as a substance or as a constituent of substances or of mixtures in concentrations > 0.1 % by mass.
Diphenylether, octabromo derivative $C_{12}H_2Br_8O$	May not used as a substance or as a constituent of substances or of mixtures in concentrations >0.1 % by mass.
Nonylphenol $C_6H_4(OH)C_9H_{19}$; Nonylphenol ethoxylate $(C_2H_4O)_nC_{15}H_{24}O$	May not used as a substance or constituent of mixtures in concentrations ≥ 0.1 % by mass for the following purposes: (1) industrial and institutional cleaning except: - controlled closed dry cleaning systems where the washing liquid is recycled or incinerated, - cleaning systems with special treatment where

Material	Conditions of Restriction
	the washing liquid is recycled or incinerated; (2) domestic cleaning; (3) textiles and leather processing except: - processing with no release into waste water, - systems with special treatment where the process water is pre-treated to remove the organic fraction completely prior to biological waste water treatment (degreasing of sheepskin); (5) metal working except: - uses in controlled closed systems where the washing liquid is recycled or incinerated.
Trichlorobenzene	May not be used as a substance or constituent of mixtures in a concentration ≥ 0.1 % by mass for all uses.
Aldrin, Chlordane, Dieldrin, Endrin, Heptachlor, Hexachlorobenzene, Mirex, Toxaphene, Polychlorinated Biphenyls (PCB), DDT (1,1,1-trichloro-2,2-bis(4-chlorophenyl) ethane), Chlordecone, Hexabromobiphenyl, HCH, including lindane	Use is prohibited.

C6. CHAPTER 6

HAZARDOUS WASTE

C6.1. SCOPE

C6.1.1. This Chapter contains criteria for a comprehensive management program to ensure that hazardous waste is identified, stored, transported, treated, disposed, and recycled in an environmentally sound manner.

C6.2. DEFINITIONS

C6.2.1. Acute Hazardous Waste. Those wastes listed in Appendix A, "List of Hazardous Substances & Materials." with a U.S. Environmental Protection Agency (USEPA) waste number with the "P" designator, or those hazardous wastes in Appendix A with Hazard Code "H".

C6.2.2. Disposal. Any activity listed in Appendix B.2 (e.g., the discharge, deposit, injection, dumping, spilling, leaking, or placing of any hazardous waste into or on any land or water that would allow the waste or constituent to enter the environment). Proper disposal effectively mitigates hazards to human health and the environment.

C6.2.3. DoD Hazardous Waste Generator. The Department of Defense considers a generator to be the installation, or activity on an installation, that produces a hazardous waste.

C6.2.4. Elementary Neutralization. A process of neutralizing a HW, that is hazardous only because of the corrosivity characteristic. It must be accomplished in a tank, transport vehicle, or container.

C6.2.5. European List of Wastes. A categorical list of wastes and associated classification codes. The waste classification system assigns a unique six-digit code to each type of waste, which must be used on all documents related to the management of that waste (e.g., shipping, transportation, record-keeping, etc.). The list is provided as Addendum 2.

C6.2.6. Hazardous Constituent. A chemical compound listed by name in Appendix A, "List of Hazardous Substances & Materials," or that possesses the characteristics described in Appendix B.1.

C6.2.7. Hazardous Waste (HW). Any waste which displays hazardous properties and is denoted with an asterisk (*) next to the six-digit code in the European List of Wastes.

C6.2.8. Hazardous Waste Accumulation Point (HWAP). A shop, site, or other work center where hazardous wastes are accumulated until removed to a Hazardous Waste Storage Area (HWSA) or shipped for treatment or disposal. An HWAP may be used to accumulate ≤ 208 liters (55 gallons) of hazardous waste, or 1 liter (1 quart) of acute hazardous waste, from each waste stream. The HWAP must be at or near the point of generation and under the control of the operator.

C6.2.9. Hazardous Waste Generation. Any act or process that produces hazardous waste (HW) as defined in this Guide.

C6.2.10. Hazardous Waste Landfill. An authorized land disposal facility for hazardous waste.

C6.2.11. Hazardous Waste Log. A listing of HW deposited and removed from an HWSA. Information such as the waste type, volume, location, and storage removal dates shall be recorded.

C6.2.12. Hazardous Waste Profile Sheet (HWPS). A document that identifies and characterizes the waste by providing user's knowledge of the waste, and/or lab analysis, and details the physical, chemical, and other descriptive properties or processes that created the hazardous waste. The DLA Disposition Services Form 1930 is typically used for this purpose. Form 1930 may be modified after coordination with DLA Disposition Services or alternate disposal agent, while ensuring that the content remains the same.

C6.2.13. Hazardous Waste Storage Area (HWSA). One or more locations on a DoD installation where HW is collected and stored ≤ 6 months prior to shipment for treatment or disposal. An HWSA may store > 208 liters (55 gallons) of a HW stream, and > 1 liter (1 quart) of an acute HW stream.

C6.2.14. Hazardous Waste Storage Area Manager. A person, or agency, on the installation assigned the operational responsibility for receiving, storing, inspecting, and general management of the installation's HWSA or HWSA program.

C6.2.15. Land Disposal. Placement in or on the land, including, but not limited to, land treatment, facilities, surface impoundments, underground injection wells, salt dome formations, salt bed formations, underground mines or caves.

C6.2.16. Treatment. Any method, technique, or process, excluding elementary neutralization, designed to change the physical, chemical, or biological characteristics or composition of any hazardous waste that would render such waste non-hazardous, or less hazardous; safer to transport, store, or dispose of; or amenable for recovery, amenable for storage, or reduced in volume.

C6.2.17. Unique Identification Number. A number assigned to generators of hazardous waste to identify the generator and used to assist in tracking the waste from point of generation to ultimate disposal. The number shall be the Spanish Hazardous Waste Producer Identification Number, which shall be obtained via the Spanish Base Commander. Additionally, the Unit Identification Code (UIC) or the DoD Activity Address Code (DoDAAC) may be used for DoD internal purposes.

C6.2.18. Used Oil. Any oil or other waste petroleum, oil, or lubricant (POL) product that has been refined from crude oil, or is synthetic oil, that has been used. A used oil mixed with hazardous waste is a hazardous waste and will be managed as such. A used oil is considered a hazardous waste when packaged and tendered for disposal, rather than recycling.

C6.2.19. Used Oil Burned for Energy Recovery. Used oil that is burned for energy recovery is termed "used oil fuel." Used oil fuel includes any fuel produced from used oil by processing, blending, or other treatment.

C6.3. CRITERIA

C6.3.1. DoD Hazardous Waste Generators

C6.3.1.1. Hazardous Waste Determination and Characterization. Generators shall identify and characterize the wastes generated at their site using their knowledge of the materials and processes that generated the waste, or through laboratory analysis of the waste. Generators shall identify inherent hazardous characteristics associated with a waste in terms of physical properties (e.g., solid, liquid, contained gases), chemical properties (e.g., chemical constituents, technical or chemical name) and/or other descriptive properties (e.g., hazardous properties identified in the European List of Wastes). The properties defining the characteristics shall be measurable by standardized and available testing protocols.

C6.3.1.1.1. Generators producing >10,000 kg of HW per year shall submit sufficient information to the Spanish Base Commander for authorization (see Chapter 1 for the process). The generator shall maintain an inventory of the types and quantities of hazardous waste generated and submit an annual declaration of the volumes of HW disposed of to the Spanish Base Commander. The annual declaration shall include: the origin and quantities of wastes generated, the period of their onsite storage, and the means of their disposal. Each HW annual declaration shall include the Environmental Identification Number (NIMA) and shall be mantained for a period of 5 years.

C6.3.1.1.2. Generators producing < 10,000 kg of HW per year shall submit sufficient information to the Spanish Base Commander for authorization (see Chapter 1 for the process). The generator shall maintain an inventory of the types and quantities of hazardous waste generated.

C6.3.1.2. Hazardous Waste Profile Sheet (HWPS). An HWPS will be used to identify each hazardous waste stream. The HWPS must be updated by the generator, as necessary, to reflect any new waste streams or process modifications that change the character of the hazardous waste being handled at the storage area.

C6.3.1.3. Each generator will use a unique identification number for all recordkeeping, reports, and manifests for hazardous waste.

C6.3.1.4. Pre-Transportation Requirements

C6.3.1.4.1. Transportation

C6.3.1.4.1.1. When transporting HW via commercial or military transportation on Spanish public roads and highways, HW generators will prepare off-installation HW shipments in compliance with packaging and labeling requirements included in Chapter 5, “Hazardous Material.” Requirements may include placarding, marking, containerization, and labelling. Hazardous waste designated for transport will be prepared in accordance with the Joint Publication AER 55-355, USAFE Instruction 24-201, 1 May 2003, and U.S. Naval Forces Europe (USNAVEUR) Instruction 4600.7F, “Joint Transportation and Traffic Management Regulation, 30 August 2002.

C6.3.1.4.1.2. When transporting HW via military vehicle on Spanish public roads and highways, generators will ensure compliance with Service regulations for the transport of hazardous materials and, if required by applicable international agreement (Status of Forces Agreement (SOFA), basing, etc.), Spanish transportation regulations.

C6.3.1.4.2. Manifesting. All HW leaving the installation will be accompanied by a Spanish waste manifest (Documento de Control y Seguimiento) to ensure a complete audit trail from point of origin to ultimate disposal. The manifest will include the information listed below. A copy of the manifest shall be kept by the waste generator for a period of 5 years. This manifest shall include:

C6.3.1.4.2.1. Generator's name, address, and telephone number;

C6.3.1.4.2.2. Generator's unique identification number;

C6.3.1.4.2.3. Transporter's name, address, and telephone number;

C6.3.1.4.2.4. Destination name, address, and telephone number;

C6.3.1.4.2.5. Description of waste (physical state and waste code);

C6.3.1.4.2.6. Total quantity of waste;

C6.3.1.4.2.7. Date of shipment;

C6.3.1.4.2.8. Date of receipt; and

C6.3.1.4.2.9. Acceptance Document (*Solicitud de Admisión*) number.

C6.3.1.4.3. Generators will maintain an audit trail of HW from the point of generation to disposal. Generators using DLA Disposition Services disposal services will obtain a signed copy of the waste manifest from the initial DLA Disposition Services recipient of the waste, at which time the DLA Disposition Services will assume responsibility. A generator, as provided in a host-tenant agreement, that uses the HW management and/or disposal program of a DoD Component that has a different unique identification number (see definition C6.2.17.) will obtain a signed copy of the manifest from the receiving component, at which time the receiving component will assume responsibility for subsequent storage, transfer, and disposal of the waste.

C6.3.2. Hazardous Waste Accumulation Point (HWAP)

C6.3.2.1. An HWAP is defined in paragraph C6.2.8. Each HWAP must be designed and operated to provide appropriate segregation for different waste streams, including those that are chemically incompatible. Each HWAP will have warning signs (National Fire Protection Association or appropriate international sign) appropriate for the waste being accumulated at that site.

C6.3.2.2. An HWAP will comply with the storage limits in paragraph C6.2.8. When these limits have been reached or when an individual storage container is full, the generator shall make arrangements within five working days to move the HW to an HWSA or ship it off-site for treatment or disposal. Treatment or disposal arrangements must include submission of all appropriate turn-in documents to initiate the removal (e.g., DD 1348-1A) to appropriate authorities responsible for removing the HW (e.g., DLA Disposition Services). Wastes intended to be recycled or used for energy recovery (for example, used oil or antifreeze) are exempt from the 208-liter (55-gallons)/1-liter (1-Quart) volume accumulation limits, but must be transported off-site to a final destination facility within six months of when the container has reached capacity.

C6.3.2.3. All criteria of paragraph C6.3.4., "Use and Management of Containers," apply to HWAPs with the exception of subparagraph C6.3.4.1.5.

C6.3.2.4. The following provisions of paragraph C6.3.5., "Recordkeeping Requirements," apply to HWAPs: C6.3.5.1. ("Turn-in Documents"), C6.3.5.5. ("Manifests"), and C6.3.5.6. ("Waste Analysis/Characterization Records").

C6.3.2.5. Personnel Training. Personnel assigned HWAP duty must successfully complete appropriate HW training necessary to perform their assigned duties. At a minimum, this must include pertinent waste handling and emergency response procedures. Generic HW training requirements are described in paragraph C6.3.9.

C6.3.3. Hazardous Waste Storage Area (HWSA).

C6.3.3.1. Location Standards. To the maximum extent possible, all HWSAs will be located to minimize the risk of release due to seismic activity, floods, or other natural events. For facilities located where they may face such risks, the installation spill prevention and control plan must address the risk.

C6.3.3.2. Design and Operation of HWSAs. HWSAs must be designed, constructed, maintained, and operated to minimize the possibility of a fire, explosion, or any unplanned release of HW or HW constituents to air, soil, groundwater or surface water that could threaten human health or the environment. Hazardous waste shall not be stored > 6 months in a HWSA. Installations shall submit sufficient information to the Spanish Base Commander for authorization to store HW for a period > 6 months (see Chapter 1 for the process).

C6.3.3.3. Waste Analysis & Verification

C6.3.3.3.1. Waste Analysis Plan. The HWSA manager, in conjunction with the installation(s) served, will develop a plan to determine how and when wastes are to be analyzed. The waste analysis plan will include procedures for characterization and verification testing of both on-site and off-site hazardous waste. The plan shall include: parameters for testing and rationale for choosing them, frequency of analysis, test methods, and sampling methods.

C6.3.3.3.2. Maintenance of Waste Analysis File. The HWSA must have, and keep on file, an HWPS for each waste stream that is stored at each HWSA.

C6.3.3.3.3. Waste Verification. Generating activities will provide identification of incoming waste on the HWPS to the HWSA manager. Prior to accepting the waste, the HWSA manager will:

C6.3.3.3.3.1. Inspect the waste to ensure it matches the description provided.

C6.3.3.3.3.2. Ensure that no waste is accepted for storage unless an HWPS is provided, or is available and properly referenced.

C6.3.3.3.3.3. Request a new HWPS from the generator if there is reason to believe that the process generating the waste has changed;

C6.3.3.3.4. Analyze waste shipments in accordance with the waste analysis plan to determine whether it matches the waste description on the accompanying manifest and documents; and

C6.3.3.3.4.1. Reject shipments that do not match the accompanying waste descriptions unless the generator provides an accurate description.

C6.3.3.4. Security

C6.3.3.4.1. General. The installation must prevent the unknowing entry, and minimize the possibility for unauthorized entry, of persons or livestock onto the HWSA grounds.

C6.3.3.4.2. Security System Design. An acceptable security system for an HWSA consists of either:

C6.3.3.4.2.1. A 24-hour surveillance system (e.g., television monitoring or surveillance by guards or other designated personnel) that continuously monitors and controls entry into the HWSA; or

C6.3.3.4.2.2. An artificial or natural barrier (e.g., a fence in good repair or a fence combined with a cliff) that completely surrounds the HWSA, combined with a means to control entrance at all times (e.g., an attendant, television monitors, locked gate, or controlled roadway access).

C6.3.3.4.3. Required Signs. A sign with the legend "Danger Unauthorized Personnel Keep Out," must be posted at each entrance to the HWSA, and at other locations, in sufficient numbers to be seen from any approach to the HWSA. The legend must be written in English and in any other language predominant in the area surrounding the installation, and must be legible from a distance of ≥ 25 feet. Existing signs with a legend other than "Danger Unauthorized Personnel Keep Out," may be used if the legend on the sign indicates that only authorized personnel are allowed to enter the HWSA, and that entry can be dangerous. Each HWSA shall also have warning signs (National Fire Protection Association or equivalent international signs) appropriate for the waste being stored at the HWSA.

C6.3.3.5. Required Aisle Space. Aisle space must allow for unobstructed movement of personnel, fire protection equipment, spill control equipment, and decontamination equipment to any area of facility operation during an emergency, and shall be ≥ 2 m (6.56 ft) between containers of incompatible classes of hazardous wastes, unless adequate spill containment or other physical barriers are present. Containers must not obstruct an exit.

C6.3.3.6. Access to Communications or Alarm System

C6.3.3.6.1. General. Whenever HW is being poured, mixed, or otherwise handled, all personnel involved in the operation must have immediate access to an internal alarm or emergency communication device, either directly or through visual or voice contact with another person.

C6.3.3.6.2. If there is only one person on duty at the HWSA premises, that person must have immediate access to a device, such as a telephone (immediately available at the scene of operation) or a hand-held two-way radio, capable of summoning external emergency assistance.

C6.3.3.7. Required Equipment. All HWSAs must be equipped with the following:

C6.3.3.7.1. An internal communications or alarm system capable of providing immediate emergency instruction (voice or signal) to HWSA personnel.

C6.3.3.7.2. A device, such as an intrinsically safe telephone (immediately available at the scene of operations) or a hand-held two-way radio, capable of summoning emergency assistance from installation security, fire departments, or emergency response teams.

C6.3.3.7.3. Portable fire extinguishers, fire control equipment appropriate to the material in storage (including special extinguishing equipment as needed, such as that using foam, inert gas, or dry chemicals), spill control equipment, and decontamination equipment.

C6.3.3.7.4. Water at adequate volume and pressure to supply water hose streams, foam-producing equipment, automatic sprinklers, or water spray systems.

C6.3.3.7.5. Readily available personal protective equipment appropriate to the materials stored, and eyewash and shower facilities.

C6.3.3.7.6. Testing and Maintenance of Equipment. All HWSA communications alarm systems, fire protection equipment, spill control equipment, and decontamination equipment, where required, must be maintained to ensure its proper operation in time of emergency.

C6.3.3.8. General Inspection Requirements

C6.3.3.8.1. General. The installation must inspect the HWSA for malfunctions and deterioration, operator errors, and discharges that may be causing, or may lead to, a release of HW constituents to the environment or threat to human health. The inspections must be conducted often enough to identify problems in time to correct them before they harm human health or the environment.

C6.3.3.8.2. Types of Equipment Covered. Inspections must include all equipment and areas involved in storage and handling of HW, including all containers and container storage areas, tank systems and associated piping, and all monitoring equipment, safety and emergency equipment, security devices, and operating and structural equipment (such as dikes and sump pumps) that are important to preventing, detecting, or responding to environmental or human health hazards.

C6.3.3.8.3. Inspection Schedule. Inspections must be conducted according to a written schedule that is kept at the HWSA. The schedule must identify the types of problems (e.g., malfunctions or deterioration) that are to be looked for during the inspection (e.g., inoperative sump pump, leaking fitting, or eroding dike).

C6.3.3.8.4. Frequency of Inspections. Minimum frequencies for inspecting containers and container storage areas are found in subparagraph C6.3.4.1.5. Minimum frequencies for inspecting tank systems are found in subparagraph C6.3.7.5.2. For equipment not covered by those paragraphs, inspection frequency shall be based on the rate of possible deterioration of the equipment and probability of an environmental or human health incident if the deterioration or malfunction or any operator error goes undetected between inspections. Areas subject to spills, such as loading and unloading areas, must be inspected daily when in use.

C6.3.3.8.5. Remedy of Problems Revealed by Inspection. The installation must remedy any deterioration or malfunction of equipment or structures that the inspection reveals on a schedule, which ensures that the problem does not lead to an environmental or human health hazard. Where a hazard is imminent or has already occurred, action must be taken immediately.

C6.3.3.8.6. Maintenance of Inspection Records. The installation must record inspections in an inspection log or summary, and keep the records for at least three years from the date of inspection. At a minimum, these records must include the date and time of inspection, the name of the inspector, a notation of the observations made, and the date and nature of any repairs or other remedial actions.

C6.3.3.9. Personnel Training. Personnel assigned HWSA duty must successfully complete an appropriate HW training program in accordance with the training requirements in paragraph C6.3.9. Additionally, standard operating procedures for HW storage and handling shall be available at the HWSA.

C6.3.3.10. Storage Practices

C6.3.3.10.1. Compatible Storage. The storage of ignitable, reactive, or incompatible wastes must be handled so that it does not threaten human health or the environment. Dangers resulting from improper storage of incompatible wastes include generation of extreme heat, fire, explosion, and generation of toxic gases.

C6.3.3.10.2. General Requirements for Ignitable, Reactive, or Incompatible Wastes. Each HWSA shall be designed and operated to provide appropriate segregation for incompatible wastes. The HWSA manager must take precautions to prevent accidental ignition or reaction of ignitable waste (i.e., substances with a flashpoint <60°C) or reactive waste (i.e., substances that release toxic gases when in contact with water, air, or an acid/base). This waste must be separated and protected from sources of ignition or reaction including but not limited to: open flames, smoking, cutting and welding, hot surfaces, frictional heat, sparks (static, electrical, or mechanical), spontaneous ignition (e.g., from heat-producing chemical reactions), and radiant heat. While ignitable or reactive waste is being handled, the HWSA personnel must confine smoking and open flame to specially designated locations. "No Smoking" signs, or the appropriate icon, must be conspicuously placed wherever there is a hazard from ignitable or reactive waste. In areas where access by non-English speaking persons is expected, the "No Smoking" legend must be written in English and in any other language predominant in the area. Water reactive waste cannot be stored in the same area as flammable and combustible liquid.

C6.3.3.11. Closure and Closure Plans

C6.3.3.11.1. Closure. At closure of an HWSA, HW and HW waste residues must be removed from the containment system, including remaining containers, liners, and bases. Closure shall be done in a manner which eliminates or minimizes the need for future maintenance or the potential for future releases of HW and according to the Closure Plan.

C6.3.3.11.2. Closure Plan. Closure plans will be developed before a new HWSA is opened. Each existing HWSA will also develop a Closure Plan. The Closure Plan will be implemented concurrent with the decision to close the HWSA. The Closure Plan will include: estimates of the storage capacity of the HW, steps to be taken to remove or decontaminate all waste residues, and estimate of the expected date for closure.

C6.3.4. Use and Management of Containers

C6.3.4.1. Container Handling and Storage. To protect human health and the environment, the following guidelines will apply when handling and storing HW containers.

C6.3.4.1.1. Containers holding HW will be in good condition, free from severe rusting, bulging, or structural defects.

C6.3.4.1.2. Containers used to store HW, including overpack containers, must be compatible with the materials stored.

C6.3.4.1.3. Management of Containers

C6.3.4.1.3.1. A container holding HW must always be closed during storage, except when it is necessary to add or remove waste.

C6.3.4.1.3.2. A container holding HW must not be opened, handled, or stored in a manner which may rupture the container or cause it to leak.

C6.3.4.1.3.3. Containers of flammable liquids must be grounded when transferring flammable liquids from one container to the other.

C6.3.4.1.4. Containers holding HW will be marked with a HW label which includes the waste code per the European List of Wastes, the generator’s unique identification number, the packaging date (i.e., the date the container was sealed), and the hazard class of the waste contained (flammable, corrosive, etc.). The label shall be firmly affixed to the container and shall be a minimum of 10 cm x 10 cm (3.94 in x 3.94 in). The label shall be in English and Spanish.

C6.3.4.1.5. Areas where containers are stored must be inspected weekly for leaking and deteriorating containers as well as deterioration of the containment system caused by corrosion or other factors. Secondary containment systems will be inspected for defects and emptied of accumulated releases or retained storm water.

C6.3.4.2. <u>Containment</u>. Container storage areas must have a secondary containment system meeting the following:

C6.3.4.2.1. Must be sufficiently impervious to contain leaks, spills, and accumulated precipitation until the collected material is detected and removed.

C6.3.4.2.2. The secondary containment system must have sufficient capacity to contain 10% of the volume of stored containers or the volume of the largest container, whichever is greater.

C6.3.4.2.3. Storage areas that store containers holding only wastes that do not contain free liquids need not have a containment system as described in subparagraph C6.3.4.2.1., provided the storage area is sloped or is otherwise designed and operated to drain and remove liquid resulting from precipitation, or the containers are elevated or are otherwise protected from contact with accumulated liquid.

C6.3.4.2.4. Rainwater captured in secondary containment areas shall be inspected and/or tested prior to release. The inspection or testing must be reasonably capable of detecting contamination by the HW in the containers. Contaminated water shall be treated as HW until determined otherwise.

C6.3.4.3. <u>Special Requirements for Ignitable or Reactive Waste</u>. Areas that store containers holding ignitable or reactive waste must be located ≥ 15 meters (50 feet) inside the installation's boundary.

C6.3.4.4. <u>Special Requirements for Incompatible Wastes</u>

C6.3.4.4.1. Incompatible wastes and materials must not be placed in the same container.

C6.3.4.4.2. Hazardous waste must not be placed in an unwashed container that previously held an incompatible waste or material.

C6.3.4.4.3. A storage container holding HW that is incompatible with any waste or other materials stored nearby in other containers, piles, open tanks, or surface impoundments, must be separated from the other materials or protected from them by means of a dike, berm, wall, or other device.

C6.3.5. <u>Recordkeeping Requirements</u>

C6.3.5.1. Turn-in Documents. Turn-in documents, e.g., DD 1348-1A or manifests, must be maintained for 5 years.

C6.3.5.2. Hazardous Waste Log. A written HW log will be maintained at the HWSA to record all HW handled and shall consist of the following:

C6.3.5.2.1. Name/address of generator;

C6.3.5.2.2. Description, waste code per the European List of Wastes, and hazard class of the hazardous waste;

C6.3.5.2.3. Number and types of containers;

C6.3.5.2.4. Quantity of hazardous waste;

C6.3.5.2.5. Date stored;

C6.3.5.2.6. Storage location; and

C6.3.5.2.7. Disposition data, to include: dates received, sealed, and transported, and transporter used.

C6.3.5.3. The HW log will be available to emergency personnel in the event of a fire or spill. Logs will be maintained until closure of the installation or for a minimum of 5 years after the date of the last entry, whichever comes later.

C6.3.5.4. Inspection Logs. Records of inspections shall be maintained for a period of 3 years.

C6.3.5.5. Manifests. Manifests of incoming and outgoing hazardous wastes will be retained for a period of 5 years. Completed manifests specifying the treatment and/or disposal method and location shall also be maintained for a period of 5 years.

C6.3.5.6. Waste Analysis/Characterization Records. These records will be retained until 3 years after closure of the HWSA.

C6.3.5.7. The installation will maintain records, identified in subparagraphs C6.3.5.1., C6.3.5.5., and C6.3.5.6. for all HWAPs on the installation.

C6.3.6. Contingency Plan

C6.3.6.1. Each installation will have a contingency plan that describes actions to be taken to contain and clean up spills and releases of HW in accordance with the provisions of Chapter 18, “Spill Prevention and Response Planning.”

C6.3.6.2. A current copy of the installation contingency plan must be:

C6.3.6.2.1. Maintained at each HWSA and HWAP, (HWAPs need maintain only portions of the contingency plan that are pertinent to their facilities and operation); and

C6.3.6.2.2. Submitted to all police departments, fire departments, hospitals, and emergency response teams identified in the plan, and upon which the plan relies to provide emergency services. Contingency Plans shall be available in both English and Spanish.

C6.3.7. Tank Systems. The following criteria apply to all storage tanks containing HW. See Chapter 19, "Underground Storage Tanks," for criteria dealing with underground storage tanks containing POLs and hazardous substances.

C6.3.7.1. Application. The requirements of this subparagraph apply to HWSAs that use tank systems for storing or treating HW. Tank systems that are used to store or treat HW that contain no free liquids and are situated inside a building with an impermeable floor are exempted from the requirements in subparagraph C6.3.7.4., Containment and Detection of Releases. Tank systems, including sumps that serve as part of a secondary containment system to collect or contain releases of HW, are exempted from the requirements in subparagraph C6.3.7.4.

C6.3.7.2. Assessment of the Integrity of an Existing Tank System. For each existing tank system that does not have secondary containment meeting the requirements of subparagraph C6.3.7.4., installations must determine annually whether the tank system is leaking or is fit for use. Installations must obtain, and keep on file at the HWSA, a written assessment of tank system integrity reviewed and certified by a competent authority.

C6.3.7.3. Design and Installation of New Tank Systems or System Components. Managers of HWSAs installing new tank systems or system components must obtain a written assessment, reviewed and certified by a competent authority attesting that the tank system has sufficient structural integrity and is acceptable for storing and treating HW. The assessment must show that the foundation, structural support, seams, connections, and pressure controls (if applicable) are adequately designed and that the tank system has sufficient structural strength, compatibility with the waste(s) to be stored or treated, and corrosion protection to ensure that it will not collapse, rupture, or fail.

C6.3.7.4. Containment and Detection of Releases. To prevent the release of HW or hazardous constituents to the environment, secondary containment that meets the requirements of this subparagraph must be:

C6.3.7.4.1. Provided for all tank systems or components;

C6.3.7.4.2. Designed, installed, and operated to prevent any migration of wastes or accumulated liquid out of the system to the soil, groundwater, or surface water at any time during the use of the tank system; and capable of detecting and collecting releases and accumulated liquid until the collected material is removed; and

C6.3.7.4.3. Constructed to include one or more of the following: a liner external to the tank, a vault, or double-walled tank.

C6.3.7.5. General Operating Requirements

C6.3.7.5.1. Hazardous wastes or treatment reagents must not be placed in a tank system if they could cause the tank, its ancillary equipment, or the containment system to rupture, leak, corrode, or otherwise fail.

C6.3.7.5.2. The installation must inspect and log at least once each operating day:

C6.3.7.5.2.1. The above-ground portions of the tank system, if any, to detect corrosion or releases of waste;

C6.3.7.5.2.2. Data gathered from monitoring and leak detection equipment (e.g., pressure or temperature gauges, monitoring wells) to ensure that the tank system is being operated according to its design; and

C6.3.7.5.2.3. The construction materials and the area immediately surrounding the externally accessible portion of the tank system, including the secondary containment system (e.g., dikes) to detect erosion or signs of releases of HW (e.g., wet spots, dead vegetation).

C6.3.7.5.3. The installation must inspect cathodic protection systems to ensure that they are functioning properly. The proper operation of the cathodic protection system must be confirmed within 6 months after initial installation and annually thereafter. All sources of impressed current must be inspected and/or tested, as appropriate, or at least every other month. The installation manager must document the inspections in the operating record of the HWSA.

C6.3.7.6. Response to Leaks or Spills and Disposition of Leaking or Unfit-For-Use Tank Systems. A tank system or secondary containment system from which there has been a leak or spill, or that is unfit for use, must be removed from service immediately and repaired or closed. Installations must satisfy the following requirements:

C6.3.7.6.1. Cessation of use; prevention of flow or addition of wastes. The installation must immediately stop the flow of HW into the tank system or secondary containment system and inspect the system to determine the cause of the release.

C6.3.7.6.2. Containment of visible releases to the environment. The installation must immediately conduct an inspection of the release and, based on that inspection:

C6.3.7.6.2.1. Prevent further migration of the leak or spill to soil or surface water;

C6.3.7.6.2.2. Remove and properly dispose of any contaminated soil or surface water;

C6.3.7.6.2.3. Remove free product to the maximum extent possible; and

C6.3.7.6.2.4. Continue monitoring and mitigating for any additional fire and safety hazards posed by vapors or free products in subsurface structures.

C6.3.7.6.3. Make required notifications and reports following the procedures established in Chapter 18, “Spill Prevention and Response Planning.”

C6.3.7.7. Closure. At closure of a tank system, the installation must remove or decontaminate HW residues, contaminated containment system components (liners, etc.), contaminated soil to the extent practicable, and structures and equipment.

C6.3.8. Standards for the Management of Used Oil and Lead-Acid Batteries

C6.3.8.1. Installations that generate, treat, or dispose of >500 liters (132.1 gallons) of used oil per year shall maintain records that include: total volume of used oil generated; volume recycled, burned for energy recovery, and/or disposed of each year; the process from which the used oil originated from; and the disposal dates. These records shall be submitted to the Spanish Base Commander, if requested.

C6.3.8.2. Used Oil Burned for Energy Recovery. Installations that intend to burn used oil as fuel shall submit sufficient information to the Spanish Base Commander for authorization (see Chapter 1 for the process). Used oil burned in combustion facilities shal have a PCB/PCT concentration <50 ppm. Used oil fuel may be burned only in the following devices and shall comply with the air emission criteria provided in Chapter 2, “Air Emissions”:

C6.3.8.2.1. Industrial furnaces.

C6.3.8.2.2. Boilers that are identified as follows:

C6.3.8.2.2.1. Industrial boilers located on the site of a facility engaged in a manufacturing process where substances are transformed into new products, including the component parts of products, by mechanical or chemical processes;

C6.3.8.2.2.2. Utility boilers used to produce electric power, steam, heated or cooled air, or other gases or fluids;

C6.3.8.2.2.3. Used oil-fired space heaters provided that:

6.3.8.2.2.3.1 The heater burns only used oil that the installation generates;

6.3.8.2.2.3.2 The heater is designed to have a maximum capacity of ≤ 0.5 million BTU per hour; and

6.3.8.2.2.3.3 The combustion gases from the heater are properly vented to the ambient air.

C6.3.8.3. Prohibitions on Dust Suppression or Road Treatment. Used oil, HW, or used oil contaminated with any HW will not be used for dust suppression or road treatment.

C6.3.8.4. Used Batteries and Accumulators. Lead-acid, NiCd, mercury-containing batteries and accumulators that are to be recycled will be managed as hazardous material. Those batteries that cannot be recycled will be managed as HW.

C6.3.9. Hazardous Waste Training

C6.3.9.1. Application. Personnel and their supervisors who are assigned duties involving actual or potential exposure to HW must successfully complete an appropriate training program prior to assuming those duties. Personnel assigned to such duty after the effective date of this FGS must work under direct supervision until they have completed appropriate training. Additional guidance is contained in DoDI 6050.05, DoD Hazard Communication Program.

C6.3.9.2. Refresher Training. All personnel performing HW duties must successfully complete annual refresher HW training.

C6.3.9.3. Training Contents and Requirements. The training program must:

C6.3.9.3.1. Include sufficient information to enable personnel to perform their assigned duties and fully comply with pertinent HW requirements.

C6.3.9.3.2. Be conducted by qualified trainers who have completed an instructor training program in the subject, have comparable academic credentials, or experience.

C6.3.9.3.3. Be designed to ensure that facility personnel are able to respond effectively to emergencies by familiarizing them with emergency procedures, emergency equipment, and emergency systems.

C6.3.9.3.4. Address the following areas, in particular for personnel whose duties include HW handling and management:

C6.3.9.3.4.1. Emergency procedures (response to fire/explosion/spills; use of communications/alarm systems; body and equipment clean up);

C6.3.9.3.4.2. Drum/container handling/storage; safe use of HW equipment; proper sampling procedures;

C6.3.9.3.4.3. Employee Protection, to include Personal Protective Equipment (PPE), safety and health hazards, hazard communication, worker exposure; and

C6.3.9.3.4.4. Recordkeeping, security, inspections, contingency plans, storage requirements, and transportation requirements.

C6.3.9.4. Documentation of Training. Installations must document all HW training for each individual assigned duties involving actual or potential exposure to HW. Updated training records on personnel assigned duties involving actual or potential exposure to HW must be kept by the HWSA manager or the responsible installation office and retained for at least three years after termination of duty of these personnel.

C6.3.10. Hazardous Waste Disposal

C6.3.10.1. Use of DLA Disposition Services. All DoD HW shall normally be disposed of through the DLA Disposition Services. A decision not to use the DLA Disposition Services for HW disposal may be made in accordance with DoDD 4001.1, Installation Management, to best accomplish the installation mission, but shall be concurred with by the component chain of command to ensure that installation contracts and disposal criteria are at least as protective as criteria used by the DLA Disposition Services.

C6.3.10.2. The DoD Components must ensure that wastes generated by DoD operations and considered hazardous under either U.S. law or Spanish law are not disposed of in Spain unless the disposal is conducted in accordance with FGS and the following:

C6.3.10.2.1. When HW cannot be disposed of in accordance with this FGS within Spain, it will either be retrograded to the United States or, if permissible under international agreements, transferred to another country outside the United States where it can be disposed of in an environmentally sound manner and in compliance with FGS applicable to the country of disposal, if any exist. Transshipment of HW to a country other than the United States for disposal must be approved by, at a minimum, the DUSD(I&E). [Note: delegated to DLA on 25 May 1995.]

C6.3.10.2.2. The determination of whether particular DoD-generated HW may be disposed of in Spain will be made by the EEA, in coordination with the unified combatant commander, the Director of Defense Logistics Agency, other relevant DoD Components, and the Chief of the U.S. Diplomatic Mission.

C6.3.10.3. Disposal Procedures

C6.3.10.3.1. The determination of whether HW may be disposed of in a host nation must include consideration of whether the means of treatment and/or containment technologies employed in the Spanish program, as enacted and enforced, effectively mitigate the hazards of such waste to human health and the environment, and must consider whether the Spanish program includes:

C6.3.10.3.1.1. An effective system for tracking the movement of HW to its ultimate destination.

C6.3.10.3.1.2. An effective system for granting authorization or permission to those engaged in the collection, transportation, storage, treatment, and disposal of HW.

C6.3.10.3.1.3. Appropriate standards and limitations on the methods that may be used to treat and dispose of HW.

C6.3.10.3.1.4. Standards designed to minimize the possibility of fire, explosion, or any unplanned release or migration of HW or its constituents to air, soil, surface, or groundwater.

C6.3.10.3.2. The EEA must also be satisfied, either through reliance on the Spanish regulatory system and/or provisions in the disposal contracts, that:

C6.3.10.3.2.1. Persons and facilities in the waste management process have demonstrated the appropriate level of training and reliability; and

C6.3.10.3.2.2. Effective inspections, monitoring, and recordkeeping will take place.

C6.3.10.4. Spanish facilities that either store, treat, or dispose of DoD-generated waste must be evaluated and approved by qualified personnel as being in compliance with Spanish regulatory requirements. This evaluation and approval may consist of having a valid authorization or Spanish equivalent for the HW that will be handled.

C6.3.10.5. Hazardous waste will be recycled or reused to the maximum extent practical. Safe and environmentally acceptable methods will be used to identify, store, prevent leakage, and dispose of HW, to minimize risks to health and the environment.

C6.3.10.5.1. Installations that intend to conduct on site recycling/recovery operations listed in Appendix B.3 shall submit sufficient information to the Spanish Base Commander for authorization.

C6.3.10.6. Land Disposal Requirements. Hazardous wastes will only be land-disposed when there is a reasonable degree of certainty that there will be no migration of hazardous constituents from the disposal site for as long as the wastes remain hazardous. Hazardous waste may be land-disposed only in facilities meeting the following criteria:

C6.3.10.6.1. The land disposal facility has an impermeable liner and a leachate collection system. The liner will be of natural or man-made materials and restrict the downward or lateral escape of HW, hazardous constituents, or leachate. The landfill impermeable layer (base and walls) shall be $\geq$ 5 m (16.4 ft) thick with permeability $\leq 10^{-7}$ cm/sec. A top soil cover of $\geq$ 1 m (3.28 ft) shall also be present.

C6.3.10.6.2. The land disposal facility has a monitoring program capable of determining the facility's impact on the quality of soil and water in the aquifers underlying the facility.

C6.3.10.7. Incinerator Standards. This subparagraph applies to incinerators that incinerate HW as well as boilers and industrial furnaces that burn HW for any recycling purposes.

C6.3.10.7.1. Incinerators used to dispose of HW must be licensed or permitted by a competent Spanish authority or approved by the EEA. This license, permit, or approval must comply with the criteria listed in subparagraph C6.3.10.7.2. Installations that intend to operate a hazardous waste incinerator shall submit sufficient information to the Spanish Base Commander for authorization (see Chapter 1 for the process).

C6.3.10.7.2. A license, permit, or EEA approval for incineration of HW must require the incinerator to be designed to include appropriate equipment as well as to be operated according to management practices (including proper combustion temperature, waste feed rate, combustion gas velocity, and other relevant criteria) to effectively destroy hazardous constituents and control harmful emissions. A permitting, licensing, or approval scheme that would require an incinerator to achieve the standards set forth in either subparagraphs C6.3.10.7.2.1. or C6.3.10.7.2.2. is acceptable.

C6.3.10.7.2.1. The incinerator achieves a destruction and removal efficiency of 99.99% for the organic hazardous constituents that represent the greatest degree of difficulty of incineration in each waste or mixture of waste. The incinerator must minimize carbon monoxide in stack exhaust gas, minimize emission of particulate matter, and emit ≤ 1.8 Kg (4 pounds) of hydrogen chloride per hour.

C6.3.10.7.2.2. The incinerator has demonstrated, as a condition for obtaining a license, permit, or EEA approval, the ability to effectively destroy the organic hazardous constituents that represent the greatest degree of difficulty of incineration in each waste or mixture of waste to be burned. For example, this standard may be met by requiring the incinerator to conduct a trial burn, submit a waste feed analysis and detailed engineering description of the facility, and provide any other information that may be required to enable the competent Spanish authority or the EEA to conclude that the incinerator will effectively destroy the principal organic hazardous constituents of each waste to be burned.

C6.3.10.7.2.3. The incinerator shall be designed, equipped, and operated to meet the following conditions:

6.3.10.7.2.3.1 Flue gas temperature and internal combustion chamber: 850 °C (1,562 °F);

6.3.10.7.2.3.2 Flue gas temperature and internal combustion chamber for waste containing >1% of halogenated substances (expressed as chlorine): 1,100°C (2,012 °F);

6.3.10.7.2.3.3 O_2 content in wet flue gas: 6% volume; and

6.3.10.7.2.3.4 Contact time: 2 seconds.

C6.3.10.8. Treatment Technologies. DoD waste generators who intend to treat (recycle, recover, or dispose of) their hazardous waste on their installation shall consult with the EEA (via the Component chain of command) and shall provide sufficient information to the Spanish Base Commander for authorization (see Chapter 1 for the process). The treatment technologies listed below and in Appendix B.3 are provided as baseline treatment/disposal technologies for use in determining suitability of Spanish disposal alternatives.

C6.3.10.8.1. Organics

C6.3.10.8.1.1. Incineration in accordance with the requirements of subparagraph C6.3.10.7.1.

C6.3.10.8.1.2. Fuel substitution where the units are operated such that destruction of hazardous constituents are at least as efficient, and hazardous emissions are no greater than those produced by incineration.

C6.3.10.8.1.3. Biodegradation. Wastes are degraded by microbial action. Such units will be operated under aerobic or anaerobic conditions so that the concentrations of a representative compound or indicator parameter (e.g., total organic carbon) has been substantially reduced in concentration. The level to which biodegradation must occur and the process time vary depending on the HW being biodegraded.

C6.3.10.8.1.4. Recovery. Wastes are treated to recover organic compounds. This will be done using, but not limited to, one or more of the following technologies: distillation; thin film evaporation; steam stripping; carbon adsorption; critical fluid extraction; liquid extraction; precipitation/crystallization, or phase separation techniques, such as decantation, filtration, and centrifugation when used in conjunction with one of the above techniques.

C6.3.10.8.1.5. Chemical Degradation. The wastes are chemically degraded in such a manner to destroy hazardous constituents and control harmful emissions.

C6.3.10.8.2. Heavy Metals

C6.3.10.8.2.1. Stabilization or Fixation. Wastes are treated in such a way that soluble heavy metals are fixed by oxidation/reduction, or by some other means that renders the metals immobile in a landfill environment.

C6.3.10.8.2.2. Recovery. Wastes are treated to recover the metal fraction by thermal processing, precipitation, exchange, carbon absorption, or other techniques that yield non-hazardous levels of heavy metals in the residuals.

C6.3.10.8.3. Reactives. Any treatment that changes the chemical or physical composition of a material so it no longer exhibits the characteristic for reactivity defined in Appendix B.1.

C6.3.10.8.4. Corrosives. Corrosive wastes as defined in Appendix B.1will be neutralized to a pH value between 6.0 and 9.0. Other acceptable treatments include recovery, incineration, chemical or electrolytic oxidation, chemical reduction, or stabilization.

C6.3.10.8.5. Batteries. Mercury, nickel-cadmium, lithium, and lead-acid batteries will be processed in accordance with subparagraphs C6.3.10.8.2.1. or C6.3.10.8.2.2. to stabilize, fix or recover heavy metals, as appropriate, and in accordance with subparagraph C6.3.10.8.4. to neutralize any corrosives before disposal.

C6.3.10.9. DoD generators of HW shall not treat HW at the point of generation except for elementary neutralization. This shall not preclude installations from treating HW in accord with subparagraphs C6.3.10.7. and C6.3.10.8.

C6.3.11. Scrap Vehicles. Scrap vehicles shall be turned in to an authorized collection center for decontamination, recycling or destruction. A destruction certificate shall be issued to the vehicle holder upon receipt of the vehicle at the collection center.

C6.3.11.1. Installations that intend to operate a scrap vehicle collection center shall provide sufficient information to the Spanish Base Commander for authorization (see Chapter 1 for the process). Note: In most cases, scrap vehicles are classified as hazardous waste and are therefore subject to the hazardous waste storage restrictions in the FGS.

C6.3.11.2. Scrap vehicle collection centers shall be equipped with:

C6.3.11.2.1. Impermeable surfaces for appropriate areas with spill containment, decanters, and cleanser/degreasers; and

C6.3.11.2.2. Oil-water separators.

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C7. CHAPTER 7

SOLID WASTE

C7.1. SCOPE

This Chapter contains criteria to ensure that solid wastes are identified, classified, collected, transported, stored, treated, and disposed of safely and in a manner protective of human health and the environment. These criteria apply to residential and commercial solid waste generated at the installation level. These criteria are part of integrated waste management. Policies concerning the recycling portion of integrated waste management are found in DoDI 4715.4, “Pollution Prevention,” and service solid waste management manuals. The criteria in this Chapter deal with general solid waste. Criteria for specific types of solid waste that require special precautions are located in Chapter 6, “Hazardous Waste,” Chapter 8, “Medical Waste Management,” Chapter 11, “Pesticides,” and Chapter 14, “Polychlorinated Biphenyls.”

C7.2. DEFINITIONS

C7.2.1. Bulky Waste. Large items of solid waste such as household appliances, furniture, large auto parts, trees, branches, stumps, and other oversize wastes whose large size precludes or complicates their handling by normal solid wastes collection, processing, or disposal methods.

C7.2.2. By-product. All products that are not primary products and have another use or consumption. By-product that can be re-used directly without prior processing or treatment is not considered waste, provided that the characteristics of the by-product are suitable for the specific re-use with no prejudice for human health and/or the environment.

C7.2.3. Collection. The act of consolidating solid wastes (or materials that have been separated for the purpose of recycling) from various locations, including the preliminary sorting and storage before transfer to a waste treatment facility.

C7.2.4. Commercial Solid Waste. All types of solid wastes generated by stores, offices, restaurants, warehouses, and other non-manufacturing activities, excluding residential and industrial wastes.

C7.2.5. Construction and Demolition Waste. The waste building materials, packaging, and rubble resulting from construction, remodeling, repair and demolition operations on pavements, houses, commercial buildings, and other structures. This waste is included in the European List of Wastes (see waste group 17) provided in Addendum 2.

C7.2.6. Daily Cover. Soil that is spread and compacted or synthetic material that is placed on the top and side slopes of compacted solid waste at least at the end of each operating day to control vectors, fire, moisture, and erosion and to assure an aesthetic appearance. Mature compost or other natural material may be substituted for soil if soil is not reasonably available in the vicinity of the landfill and the substituted material will control vectors, fire, moisture, and erosion and will assure an aesthetic appearance.

C7.2.7. Final Cover. A layer of soil, mature compost, other natural material (or synthetic material with an equivalent minimum permeability) that is applied to the landfill after completion of a cell or trench, including a layer of material that will sustain native vegetation, if any.

C7.2.8. Food Waste. The organic residues generated by the handling, storage, sale, preparation, cooking, and serving of foods, commonly called garbage.

C7.2.9. Generation. The act or process of producing solid waste.

C7.2.10. Hazardous Waste. Refer to Chapter 6, "Hazardous Waste."

C7.2.11. Incinerator. Refer to Chapter 2, "Air Emissions."

C7.2.12. Industrial Solid Waste. The solid waste generated by industrial processes and manufacturing.

C7.2.13. Land Application Unit. An area where wastes are applied onto or incorporated into the soil surface (excluding manure spreading operations) for agricultural purposes or for treatment or disposal.

C7.2.14. Landfill. An unpaved and uncovered solid waste disposal area. On-site waste containment areas are considered a landfill if the storage periods exceed one year for waste intended for disposal or two years for waste intended for recycling/recovery.

C7.2.15. Lower Explosive Limit. The lowest % by volume of a mixture of explosive gases in air that will propagate a flame at 25°C and atmospheric pressure.

C7.2.16. Municipal Solid Waste (MSW). Any waste produced in housing or commercial units, offices, and services, and any other waste classified as non-hazardous that may be comparable to waste produced by those activities. Similar wastes generated at industrial facilities are also considered municipal wastes. Municipal waste also includes waste from cleaning operations of public roads, green areas, recreational areas and beaches, as well as furniture, household electrical and electronic equipment, abandoned vehicles and wastes proceeding from minor household remodeling.

C7.2.17. Municipal Solid Waste Landfill (MSWLF) Unit. A discrete area of land or an excavation, on or off an installation, that receives household waste, and that is not a land application unit, surface impoundment, injection well, or waste pile. An MSWLF unit also may receive other types of wastes, such as commercial solid waste and industrial waste.

C7.2.18. Open Burning. Burning of solid wastes in the open, such as in an open dump.

C7.2.19. Open Dump. A land disposal site at which solid wastes are disposed of in a manner that does not protect the environment, is susceptible to open burning, and is exposed to the elements, vectors, and scavengers.

C7.2.20. Packaging. Any product used for holding, protecting, handling, distributing, and presenting goods (from raw materials to finished products) in any phase of the production, distribution, and consumption chain. Includes reusable and disposable packaging, direct sell or primary packaging, collective or secondary packaging, and transportation or tertiary packaging.

C7.2.21. Residential Solid Waste. The wastes generated by normal household activities, including, but not limited to, food wastes, rubbish, ashes, and bulky wastes.

C7.2.22. Rubbish. A general term for solid waste, excluding food wastes and ashes, taken from residences, commercial establishments, and institutions.

C7.2.23. Service Solid Waste Management Manual. Naval Facility Manual of Operation (NAVFAC MO) 213, Air Force Instruction (AFI) 32-7042, Army TM 5-634 “Solid Waste Management”, or their successor documents.

C7.2.24. Sludge. The accumulated semi-liquid suspension of settled solids deposited from wastewaters or other fluids in tanks or basins. It does not include solids or dissolved material in domestic sewage or other significant pollutants in water resources, such as silt, dissolved or suspended solids in industrial wastewater effluent, dissolved materials in irrigation return flows, or other common water pollutants. This includes sludge from wastewater treatment plants (WWTPs) and residual sludge from septic tanks used for the treatment of wastewater.

C7.2.25. Solid Wastes. Garbage, refuse, sludge, and other discarded materials, including solid, semi-solid, liquid, and contained gaseous materials resulting from industrial and commercial operations and from community activities. It does not include solids or dissolved material in domestic sewage or other significant pollutants in water resources, such as silt, dissolved or suspended solids in industrial wastewater effluent, dissolved materials in irrigation return flows, or other common water pollutants.

C7.2.26. Solid Waste Storage Container. A receptacle used for the temporary storage of solid waste while awaiting collection.

C7.2.27. Storage. The interim containment of solid waste after generation and prior to collection for ultimate recovery or disposal.

C7.2.28. Vector. A carrier that is capable of transmitting a pathogen from one organism to another.

C7.2.29. Waste. Any substance, material, or object that the waste generator (producer) discards, intends to discard, or is required to discard, including material intended for recycling. A complete list of waste streams with the corresponding Waste Code is provided in Addendum 2, “European List of Wastes.”

C7.2.30. Yard Waste. Grass and shrubbery clippings, tree limbs, leaves, and similar organic materials commonly generated in residential yard maintenance (also known as green waste).

C7.3. CRITERIA

C7.3.1. DoD solid wastes shall be treated, stored, and disposed of in authorized facilities operated by Spanish registered companies that have been evaluated against paragraphs C7.3.13., C7.3.15., and C7.3.16. These evaluated facilities will be used to the maximum extent practical.

C7.3.1.1. Installations generating > 1,000 metric tons of solid waste per year shall provide sufficient information to the Spanish Base Commander for inclusion in the Spanish registry of waste producers.

C7.3.1.2. Installations shall maintain an inventory of solid waste generated for a period of at least 3 years.

C7.3.2. Installations will cooperate with the Spanish Base Commander and Spanish municipal officials, to the extent possible, in the solid waste management planning process.

C7.3.3. Installations will develop and implement a solid waste management strategy to reduce solid waste disposal. This strategy could include recycling, composting, and waste minimization efforts.

C7.3.4. All solid wastes or materials that have been separated for the purpose of recycling shall be stored in such a manner that they do not constitute a fire, health or safety hazard or provide food or harborage for vectors, and shall be contained or bundled to avoid spillage. The storage of solid waste shall not exceed a period of 1 year if the waste is destined for disposal. The storage of solid waste that has been separated for recycling shall not exceed 2 years, otherwise it shall comply with the landfill requirements (see C7.3.12. and C7.3.13.).

C7.3.5. DoD units collecting bulky waste (e.g., refrigerators, freezers, televisions, washing machines, air conditioners, computers, etc.) shall return the waste appliances to their point of purchase if feasible. If not feasible, they shall be turned over to permitted collection points (via DLA Disposition Services). Scrap vehicles/trailers and large vehicle parts shall be handled in accordance with Chapter 6, “Hazardous Waste”. See C7.3.17. for further requirements on the disposal of used tires. Some bulky wastes contain hazardous wastes and are subject to the same requirements as hazardous waste (e.g., storage, manifesting, disposal requirements, etc.).

C7.3.6. Storage of bulky wastes will include, but will not be limited to, removing all doors from large household appliances and covering the items to reduce both the problems of an attractive nuisance, and the accumulation of solid waste and water in and around the bulky items. Bulky wastes will be screened for the presence of ozone depleting substances as defined in Chapter 2, “Air Emissions,” or hazardous constituents as defined in Chapter 6, “Hazardous Waste.” Readily detachable or removable hazardous waste will be segregated and disposed of in accordance with Chapters 6, 14, and 15 of this document.

C7.3.6.1. Electronic waste storage areas shall be covered and equipped with impermeable surfaces and secondary containment.

C7.3.7. In the design of all buildings or other facilities that are constructed, modified, or leased after the effective date of this FGS, there will be provisions for storage in accordance with these guidelines that will accommodate the volume of solid waste anticipated. Storage areas will be easily cleaned and maintained, and will allow for safe, efficient collection.

C7.3.8. Storage containers shall be leakproof, waterproof, and vermin-proof, including sides, seams and bottoms, and be durable enough to withstand anticipated usage and environmental conditions without rusting, cracking, or deforming in a manner that would impair serviceability. Storage containers shall have functional lids.

C7.3.9. Containers shall be stored on a firm, level, well-drained surface that is large enough to accommodate all of the containers and that is maintained in a clean, spillage-free condition.

C7.3.10. Recycling programs will be instituted on DoD installations in accordance with the policies in DoDI 4715.4, "Pollution Prevention," and the installation's solid waste management strategy (see C7.3.3). The recycling program shall also include the means for separate collection of packaging waste, which shall be disposed of with authorized companies for recycling, re-use, or recovery.

C7.3.11. Installations will not initiate new or expand existing waste landfill units without approval of the Combatant Commander with responsibility for the area where the landfill would be located, and only after justification that unique circumstances mandate a new unit.

C7.3.12. All DoD facilities operating a solid waste landfill shall submit sufficient information to the Spanish Base Commander for authorization (see Chapter 1 for the process). All DoD MSWLF units authorized or operational after 15 July 2001 will be designed and operated in a manner that incorporates the following broad factors, whereas all MSWLF units authorized or operational before 15 July 2001 shall be upgraded to incorporate the following factors to the extent feasible:

C7.3.12.1. Location restrictions with regard to airport safety (i.e., bird hazards), floodplains, wetlands, aquifers, seismic zones, unstable areas, residential and recreational areas, and protected areas;

C7.3.12.2. Procedures for excluding hazardous waste;

C7.3.12.3. Cover material criteria (e.g., daily cover), disease vector control, explosive gas control, air quality criteria (e.g., no open burning), access requirements, liquids restrictions, and record keeping requirements; and

C7.3.12.4. Inspection program.

C7.3.12.5. Liner and leachate collection system designed consistent with location to prevent groundwater contamination that would adversely affect human health and to collect precipitation and surface water runoff.

C7.3.12.6. A groundwater monitoring system unless the installation operating the landfill, after consultation with the EEA, determines that there is no reasonable potential for migration of hazardous constituents from the MSWLF to the uppermost aquifer during the active life of the facility and the post-closure care period.

C7.3.12.7. The landfill base and walls shall consist of a mineral layer with a permeability factor $K \leq 1.0 \times 10^{-9}$ m/s and a thickness ≥ 1 m (3.3 ft). Where the geological barrier does not naturally meet the above conditions, it shall be completed artificially and reinforced by other means giving an equivalent protection. An artificially established geological barrier shall be $\geq$ 0.5 m (1.6 ft) thick.

C7.3.13. Installations operating MSWLF units will:

C7.3.13.1. Use standard sanitary landfill techniques of spreading and compacting solid wastes to ensure stability and minimize slippages and placing daily cover over disposed solid waste at the end of each operating day.

C7.3.13.2. Establish criteria for unacceptable wastes based on site-specific factors such as hydrology, chemical and biological characteristics of the waste, available alternative disposal methods, environmental and health effects, and the safety of personnel.

C7.3.13.3. Implement a program to detect and prevent the disposal of hazardous wastes, infectious wastes, PCBs, and wastes determined unsuitable for the specific MSWLF unit.

C7.3.13.4. Investigate options for composting of MSW as an alternative to landfilling or treatment prior to landfilling.

C7.3.13.5. Prohibit open burning at a MSWLF.

C7.3.13.6. Develop procedures for dealing with yard waste and construction debris that keeps it out of MSWLF units to the maximum extent possible (e.g., composting, recycling).

C7.3.13.7. Operate the MSWLF unit in a manner to protect the health and safety of personnel associated with the operation.

C7.3.13.8. Maintain conditions that are unfavorable for the harboring, feeding, and breeding of disease vectors.

C7.3.13.9. Appropriate measures shall be taken in order to control the accumulation and migration of landfill gases. Landfill gases shall be collected from all landfills receiving biodegradable waste. The accumulated gases shall be treated and utilized to produce energy or if this is not feasible, they shall be flared in a manner that is protective of the environment and human health.

C7.3.13.10. Ensure that methane gas generated by the MSWLF unit does not exceed 25% of the lower explosive limit for methane in structures on or near the MSWLF.

C7.3.13.11. Operate in an aesthetically acceptable manner.

C7.3.13.12. Operate in a manner to protect the surrounding environment (in particular, groundwater and surface water).

C7.3.13.13. Control public access to landfill facilities. The system of control and access to each landfill shall include measures to detect and discourage illegal dumping.

C7.3.13.14. Prohibit the disposal of bulk or non-containerized liquids, infectious waste, potentially explosive, corrosive, flammable, oxidizing waste, as well as whole and shredded used tires, except bicycle tires or tires with an outside diameter > 1,400 mm (55 in).

C7.3.13.15. Reduce disposal of biodegradable waste on landfills.

C7.3.13.16. Implement a control and monitoring program in accordance with Table 7.1. Any significant adverse environmental effects shall be reported to the Spanish Base Commander.

C7.3.13.17. Maintain records on the preceding criteria for a minimum period of 5 years.

C7.3.13.18. During closure and post-closure operations, installations will:

C7.3.13.18.1. Install a final cover system that is designed to minimize infiltration and erosion.

C7.3.13.18.2. Ensure that the infiltration layer is composed of a minimum of 0.5 m (19.70 in) of earthen material, geotextiles, or a combination thereof, that have a permeability ≤ to the permeability of any bottom liner system or natural subsoil present, or a permeability ≤ 1.0×10^{-7}cm/sec, whichever is less.

C7.3.13.18.3. Ensure that the final layer consists of a minimum of 21 cm (8 in) of earthen material that is capable of sustaining native plant growth.

C7.3.13.18.4. Revegetate the final cap with native plants that are compatible with the landfill design, including the liner.

C7.3.13.18.5. Prepare a written Closure Plan, that shall be submitted to the Spanish Base Commander, that includes, at a minimum, a description of the monitoring and maintenance activities required to ensure the integrity of the final cover, a description of the planned uses of the site during the post-closure period, plans for continuing (during the post-closure period) leachate collection, groundwater monitoring, and methane monitoring, and a survey plot showing the exact site location. The plan will be kept as part of the installation's permanent records. The post-closure period of responsibility for the installation operating the MSWLF, including the monitoring program established in C7.3.13.16., will be a minimum of 30 years.

C7.3.14. Open burning will not be the regular method of solid waste disposal. Where burning is the method, authorized Spanish incinerators meeting air quality requirements of Chapter 2, “Air Emissions,” will be used.

C7.3.15. A composting facility that is located on a DoD installation and that processes annually > 454 metric tonnes (5000 tons) of sludge from a domestic wastewater treatment plant (see Chapter 4, “Wastewater”) shall submit sufficient information to the Spanish Base

Commander for authorization (see Chapter 1 for the process) and will comply with the following criteria:

C7.3.15.1. Operators must maintain a record of the characteristics of the waste composted, sewage sludge, and other materials, such as nutrient or bulking agents being composted, including the source and volume or weight of the material.

C7.3.15.1.1. Access to the facility must be controlled. All access points must be secured when the facility is not in operation.

C7.3.15.1.2. By-products, including residuals and materials that can be recycled, must be stored to prevent vector intrusion and aesthetic degradation. Materials that are not composted must be removed periodically.

C7.3.15.1.3. Run-off water that has come in contact with composted waste, materials stored for composting, or residual waste must be diverted to a leachate collection and treatment system.

C7.3.15.1.4. The temperature and retention time for the material being composted must be monitored and recorded.

C7.3.15.1.5. Periodic analysis of the compost must be completed for the following parameters: percentage of total solids, volatile solids as a percentage of total solids, pH, ammonia, nitrate, nitrogen, total phosphorous, cadmium, chromium, copper, lead, nickel, zinc, mercury, and PCBs.

C7.3.15.1.5.1. If compost is used for agricultural purposes (see C7.3.16.1.), the sludge analysis shall also include organic matter and total nitrogen. Analysis shall be conducted at the end of the production process, every 6 months, and any time there is a major change in material composition. If no significant changes occur, or if the capacity of the wastewater treatment plant is <300 kg BOD_5/day, the analysis shall be performed only once per year.

C7.3.15.1.5.2. If the compost is to be incinerated or disposed of in a landfill, the required analysis of the compost characteristics shall be performed, in accordance with this chapter and Chapter 6 "Hazardous Waste."

C7.3.15.1.6. Compost must be produced by a process to further reduce pathogens. Two such acceptable methods are:

C7.3.15.1.6.1. Windrowing, which consists of an unconfined composting process involving periodic aeration and mixing to maintain aerobic conditions during the composting process; and

C7.3.15.1.6.2. The enclosed vessel method, which involves mechanical mixing of compost under controlled environmental conditions. The retention time in the vessel must be at least 72 hours with the temperature maintained at 55°C. A stabilization period of at least 7 days must follow the decomposition period.

C7.3.16. Classification and Use of Compost from DoD Composting Facilities. Compost produced at a composting facility that is located on a DoD installation and that processes sludge from a domestic wastewater treatment plant (see Chapter 4,“Wastewater”) must meet the maximum contaminant concentrations in Table 7.2.

C7.3.16.1. Compost distributed for agricultural purposes shall meet the requirements of Tables 7.2 and 7.3.

C7.3.16.2. Compost that fails to meet the standards in Table 7.2 shall be disposed of in accordance with this chapter (i.e., by landfilling or incineration) or in accordance with Chapter 6 “Hazardous Waste,” if the compost is classified as a hazardous waste.

C7.3.17. Used tires storage areas shall include the following:

C7.3.17.1. Restricted access;

C7.3.17.2. Protection from the elements and vectors;

C7.3.17.3. Surface of the storage area and access roads (if unpaved) shall be appropriately compacted and maintained, and equipped with a storm water drainage system;

C7.3.17.4. Used tires shall be stacked to a maximum height of 3 m (9.8 ft), or up to a maximum height of 6 m (19.7 ft) if they are stored in silos, and shall be stored in a safe manner in order to avoid potential hazard to human health, the facility or equipment due to collapse. The storage shall not exceed the period of 1 year and the quantity of 30 metric tonnes (33 tons).

C7.3.17.5. Divided into independent cells with maximum individual capacities ≤ 1,000 m^3 (1,308 yd^3) in order to minimize the spread of fire. Each cell will be separated by fire lanes and equipped with emergency and fire-fighting protection.

Table 7.1. Landfill Monitoring Data and Collection Frequencies

Meteorological data	
Volume of precipitation	Daily
Temperature	Daily
Direction and force of prevailing wind	Daily
Evaporation	Daily
Atmospheric humidity (14.00 h CET)	Daily
Emission data (water, leachate and gas control)	
Leachate volume	Monthly
Leachate composition	Quarterly
Volume and composition of surface water	Quarterly
Potential gas emissions and atmospheric pressure (CH_4, CO_2, O_2, H_2S, H_2, etc.)	Monthly
Groundwater monitoring	
Level of groundwater	Every six months
Groundwater composition	Site specific frequency (1,2)
Topography of the site	
Structure and composition of landfill body	Yearly
Settling behavior of the level of the landfill body	Yearly

Notes:
1. Frequency shall be based on velocity of groundwater flow.
2. If a trigger level is reached, repeat sampling shall be performed for verification. If the level is confirmed, a contingency plan shall be implemented.
   Trigger levels: When analysis of a groundwater sample shows a significant change in water quality, it will be assumed that a significant adverse environmental effect to the groundwater has occurred. Trigger levels that take into account the specific hydrogeological formations and groundwater quality in the location of the landfill shall be included in the landfill authorization.

Table 7.2. Maximum Concentrations of Heavy Metals in Sludge after Composting

Parameter	Maximum Concentration in Sludge (mg/kg)
PCBs	1
Cadmium	10
Chromium	1,000
Copper	500
Lead	500
Mercury	5
Nickel	100
Zinc	1,000

Table 7.3. Maximum Concentrations of Heavy Metals in Soil for Agricultural Application of Sludge

Parameter	Maximum Concentration of Heavy Metals in Soils to Receive Sludge (mg/kg of dry soil)		Maximum Allowable Amount of Sludge Applied per Year (kg/Ha/year)
	Soils with pH < 7	Soils with pH > 7	
Cadmium	1	3	3
Chromium	100	150	150
Copper	50	210	210
Lead	50	300	300
Mercury	1	1.5	1.5
Nickel	30	112	112
Zinc	150	450	450

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C8. CHAPTER 8

MEDICAL WASTE

C8.1. SCOPE

This Chapter contains criteria for the management of medical waste at medical, dental, research and development, and veterinary facilities generated in the diagnosis, treatment, or immunization of human beings or animals or in the production or testing of biologicals subject to certain exclusions. This waste also includes mixtures of medical waste and hazardous waste. It does not apply to what would otherwise be household waste.

C8.2. DEFINITIONS

C8.2.1. Cytostatic Wastes. Any waste derived from cytostatic medicines that are either toxic, carcinogenic, mutagenic, or teratogenic.

C8.2.2. Infectious Agent. Any organism (such as a virus or bacterium) that is capable of being communicated by invasion and multiplication in body tissues and capable of causing disease or adverse health impacts in humans.

C8.2.3. Infectious Medical Waste. Waste produced by medical and dental treatment facilities that is specially managed because it has the potential for causing disease in humans and may pose a risk to both individuals or community health if not managed properly, and that includes the following classes:

C8.2.3.1. Microbiology waste, including cultures and stocks of etiologic agents which, due to their species, type, virulence, or concentration, are known to cause disease in humans.

C8.2.3.2. Pathology waste, including human tissues and organs, amputated limbs or other body parts, fetuses, placentas, and similar tissues from surgery, delivery, or autopsy procedures. Animal carcasses, body parts, blood, and bedding from contaminated animals are also included.

C8.2.3.3. Human blood and blood products (including serum, plasma, and other blood components), items contaminated with liquid or semi-liquid blood or blood products and items saturated or dripping with blood or blood products, and items caked with blood or blood products, that are capable of releasing these materials during handling.

C8.2.3.4. Potentially infectious materials, including human body fluids such as semen, vaginal secretions, cerebrospinal fluid, pericardial fluid, pleural fluid, peritoneal fluid, amniotic fluid, saliva in dental procedures, any body fluid that is visibly contaminated with blood, and all body fluids in situations where it is difficult or impossible to differentiate between body fluids.

C8.2.3.5. Sharps, including hypodermic needles, syringes, biopsy needles, and other types of needles used to obtain tissue or fluid specimens, needles used to deliver intravenous solutions, scalpel blades, pasteur pipettes, specimen slides, cover slips, glass petri plates, and broken glass potentially contaminated with infectious waste.

C8.2.3.6. Infectious waste from isolation rooms, but only including those items that were contaminated or likely to have been contaminated with infectious agents or pathogens, including excretion exudates and discarded materials contaminated with blood.

C8.2.4. Noninfectious Medical Waste. Waste created that does not require special management because it has been determined to be incapable of causing disease in humans or which has been treated to render it noninfectious.

C8.2.5. Treatment. Any method, technique, or process designed to change the physical, chemical, or biological character or composition of any infectious hazardous or infectious waste so as to render such waste non-hazardous, or less hazardous; safer to transport, store, or dispose of; or amenable for recovery, amenable for storage, or reduced in volume. Treatment methods for infectious waste must eliminate infectious agents so that they no longer pose a hazard to persons who may be exposed.

C8.3. CRITERIA

C8.3.1. Infectious medical waste will be separated if practical from other hazardous and noninfectious waste at the point of origin.

C8.3.2. Mixtures of infectious medical wastes and hazardous wastes will be handled as infectious hazardous waste under DoD 4160.21-M "Defense Materiel Disposition Manual," and are the responsibility of the generating DoD Component. Priority will be given to the hazard that presents the greatest risk. Defense Logistics Agency (DLA) has no responsibility for this type of property until it is rendered noninfectious as determined by the appropriate DoD medical authority.

C8.3.3. Solid waste that is classified as a hazardous waste in accordance with Appendix B.1 and/or Addendum 2 will be managed in accordance with the criteria in Chapter 6, "Hazardous Waste."

C8.3.4. Mixtures of other solid waste and infectious medical waste will be handled as infectious medical waste.

C8.3.5. Radioactive medical waste will be segregated from other waste streams and managed in accordance with Service Directives. Radioactive waste disposed of in Spain shall be turned in to the National Radioactive Waste Company (*Empresa Nacional de Residuos Radiactivos, S.A. "ENRESA"*).

C8.3.6. Infectious medical waste will be segregated, transported, and stored in color-coded bags or receptacles a minimum of 3 mils thick having such durability, puncture resistance, and burst strength as to prevent rupture or leaks during ordinary use.

C8.3.7. All bags or receptacles used to segregate, transport or store infectious medical waste will be clearly marked with the universal biohazard symbol and word “BIOHAZARD” and either “*Residuo Biologico*” or “*Residuo Medico*” in English and Spanish, and will include markings that identifies the generator, date of generation, and the contents.

“BIOHAZARD”
“Residuo Biologico OR *Residuo Medico* / Medical Waste”

C8.3.7.1. Containers for cytostatic waste shall be labeled with the following words and the universal symbol for cytostatic products:

“*Residuos contaminados químicamente – Citostáticos* / Chemically contaminated wastes – Cytostatic”

C8.3.8. Sharps will only be discarded into rigid receptacles. Needles will not be clipped, cut, bent, or recapped before disposal.

C8.3.9. Infectious medical waste will be transported and stored to minimize human exposure, and will not be placed in chutes or dumbwaiters.

C8.3.10. Infectious medical waste will not be compacted unless converted to noninfectious medical waste by treatment as described in paragraph C8.3.17. Sufficient information for all treatment of infectious waste shall be provided to the Spanish Base Commander (see Chapter 1 for the process). Containers holding sharps will not be compacted.

C8.3.11. All anatomical pathology waste (i.e., body parts) must be placed in containers lined with plastic bags that comply with paragraph C8.3.6, and shall only be disposed of by burial in a designated area after being treated for disposal by incineration or cremation.

C8.3.12. Blood, blood products, and other liquid infectious wastes shall be handled as follows:

C8.3.12.1. They shall be placed in receptacles having puncture resistance and burst strength to prevent rupture or leaks, and shall be disposed of as hazardous waste, in accordance with C8.3.17. Blood and liquid non-infectious waste in quantities < 100 ml may be discharged into a sewer system connection (sinks, drains, etc.). Disposal of bulk blood in quantities > 100 ml, other blood products, and other liquid infectious waste into a sewer system connection (sinks, drains, etc.) is prohibited

C8.3.12.2. Suction canister waste from operating rooms will be sealed into leak-proof containers and incinerated.

C8.3.13. All personnel handling infectious medical waste will wear appropriate protective apparel or equipment such as gloves, coveralls, masks, and goggles sufficient to prevent the risk of exposure to infectious agents or pathogens. The appropriate DoD Medical Authority shall determine the required type of protective apparel or equipment based on a risk evaluation of the activities.

C8.3.14. If infectious medical waste cannot be treated on-site, it will be managed during storage as follows:

C8.3.14.1. Infectious medical waste will be maintained in a nonputrescent state, using periodic disinfection and refrigeration as necessary.

C8.3.14.2. Infectious medical waste with multiple hazards (i.e., infectious hazardous waste or infectious radioactive waste) will be segregated from the general infectious waste stream.

C8.3.14.3. Infectious medical waste (other than sharps containers) collected in intermediate storage areas (e.g. laboratory) within the medical facility shall be transported to the designated storage area within 48 to 72 hours of being generated. Sharps containers can be used until full, at which time they shall be transported to the storage area. Infectious medical waste shall follow the storage requirements listed below:

Infectious Medical Waste Quantity	**Temperature**		
	> 4°C (39°F)	**≤ 4°C(39°F)**	**≤ -18°C (-0.4°F)**
>200 kg (440 lbs)	3 days	10 days	30 days
20 – 200 kg (44 – 440 lbs)	3 days	20 days	60 days
2 – 20 kg (4.4 – 44 lbs)	3 days	30 days	90 days
<2 kg (4.4 lbs)	3 days	30 days	120 days

C8.3.15. Storage sites must be:

C8.3.15.1. Specifically designated;

C8.3.15.2. Equipped with engineering controls to prevent any discharge of infectious medical waste to the sewer system;

C8.3.15.3. Constructed to prevent entry of insects, rodents, and other pests;

C8.3.15.4. Prevent access by unauthorized personnel; and

C8.3.15.5. Marked on the outside with the universal biohazard symbol and the word "BIOHAZARD" in both English and Spanish.

C8.3.16. Bags and receptables containing infectious medical waste must be placed into rigid or semi-rigid, leak-proof containers before being transported off-site.

C8.3.17. Infectious medical waste must be treated in accordance with Table 8.1., "Treatment and Disposal Methods for Infectious Medical Waste," and the following before disposal:

C8.3.17.1. Sterilizers must maintain the temperature at 121°C (250°F) for at least 30 minutes at 15 psi. The operating conditions for each sterilization cycle (e.g., temperature, pressure, and time) shall be monitored and recorded in a logbook (see C8.3.20).

C8.3.17.2. The effectiveness of sterilizers must be checked at least weekly using *Bacillus stearo thermophilus* spore strips or an equivalent biological performance test.

C8.3.17.3. Only authorized Incinerators shall be used to treat medical waste. The incinerator must be designed and operated to maintain a minimum temperature and retention time sufficient to destroy all infectious agents and pathogens, and must meet applicable criteria in Chapter 2, "Air Emissions."

C8.3.17.4. Installations that intend to operate a medical waste incinerator shall provide sufficient information to the Spanish Base Commander for authorization (see Chapter 1 for the process).

C8.3.17.5. Ash or residue from the incineration of infectious medical waste must be assessed for classification as hazardous waste in accordance with the criteria in Chapter 6, "Hazardous Waste." Ash that is determined to be hazardous waste must be managed in accordance with Chapter 6. All other residue will be disposed of in a landfill that complies with the criteria of Chapter 7, "Solid Waste."

C8.3.17.6. Chemical disinfection must be conducted using procedures and compounds approved by appropriate DoD medical authority for use on any pathogen or infectious agent suspected to be present in the waste.

C8.3.18. Installations will develop contingency plans for treatment or disposal of infectious medical waste should the primary means become inoperable.

C8.3.19. Spills of infectious medical waste will be cleaned up as soon as possible in accordance with the following:

C8.3.19.1. Response personnel must comply with paragraph C8.3.13.

C8.3.19.2. Blood, body fluid, and other infectious fluid spills must be removed with an absorbent material that must then be managed as infectious medical waste.

C8.3.19.3. Surfaces contacted by infectious medical waste must be washed with soap and water and chemically decontaminated in accordance with subparagraph C8.3.17.6.

C8.3.20. Installations will keep records of the following information concerning infectious medical waste for at least five years after the date of disposal:

C8.3.20.1. Type and origin of waste;

C8.3.20.2. Amount of waste (volume or weight) and waste composition;

C8.3.20.3. A loading/unloading inventory (including accumulation beginning and ending dates);

C8.3.20.4. Treatment, if any, including date of treatment; and

C8.3.20.5. Disposition, including date of disposition, and if the waste was transferred to Spanish facilities, and copies of the waste manifest (*Hoja de Control y Seguimiento*) and request for acceptance (*Solicitud de Admisión*) for each transfer.

Table 8.1. Treatment and Disposal Method for Infectious Medical Waste

Type of Medical Waste	Method of Treatment	Method of Disposal
Microbiological	Steam sterilization[1]	Hazardous waste landfill (HWLF)[2]
	Chemical disinfection	HWLF[2]
	Incineration	HWLF[2]
Pathological	Incineration[3] (Unrecognizable body parts, e.g. human tissues)	HWLF[2]
	Cremation[3] (Recognizable, e.g. large body parts or entire corpses)	Burial
	None[3] (Recognizable, e.g. large body parts or entire corpses)	Burial
	Chemical sterilization[4]	HWLF[2]
	Steam sterilization[4]	HWLF[2]
Bulk blood & suction canister waste	Steam sterilization [4]	HWLF[2]
	Incineration [4]	HWLF[2]
Sharps in sharps containers	Steam sterilization	HWLF[2]
	Incineration	HWLF[2]

Notes:
1. Preferred method for cultures and stocks because they can be treated at point of generation.
2. As defined in Chapter 6. See C6.3.10.6 for the land disposal requirements.
3. All anatomical pathology waste (i.e., body parts) must be placed in containers lined with plastic bags that comply with paragraph C8.3.6, and may only be disposed of by burial in a designated area after being treated for disposal by incineration or cremation.
4. Bulk blood or suction canister waste known to be infectious must be treated by incineration or steam sterilization before disposal.

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C9. CHAPTER 9

PETROLEUM, OIL & LUBRICANTS

C9.1. SCOPE

This Chapter contains criteria to control and abate pollution resulting from the storage, transport, and distribution of petroleum products. Criteria for underground storage tanks (UST) containing POL or hazardous material products are addressed in Chapter 19, "Underground Storage Tanks." POL spill prevention and response planning criteria are contained in Chapter 18, "Spill Prevention and Response Planning."

C9.2. DEFINITIONS

C9.2.1. Aboveground Storage Container. POL storage containers, exempt from UST criteria, that are normally placed on or above the surface of the ground. POL storage containers located above the floor and contained in vaults or basements, bunkered containers, and also partially buried containers are considered aboveground storage containers. For the purposes of this Chapter, this includes any mobile or fixed structure, tank, equipment, pipe, or pipeline (other than a vessel or a public vessel) used in oil well drilling operations, oil production, oil refining, oil storage, oil gathering, oil processing, oil transfer, and oil distribution. This also includes equipment in which oil is used as an operating fluid, but excludes equipment in which oil is used solely for motive power.

C9.2.2. Below Ground Storage Container. Completely buried POL storage containers, including deferred USTs, that are exempt from all criteria in Chapter 19, "Underground Storage Tanks." For purposes of this paragraph, ONLY below ground storage containers that are exempt from requirements of Chapter 19 are counted toward the aggregate thresholds in subparagraph C9.2.7.2. below.

C9.2.3. Internal Inspector. Trained personnel designated by the manager of the POL facility to conduct the required inspections and testing of POL facilities.

C9.2.4. Loading/Unloading Racks. Location where tanker trucks/rail cars are loaded and unloaded by pipes, pumps, and loading arms.

C9.2.5. Loading/Unloading Areas. Any location where POL is authorized to be loaded or unloaded to or from a POL storage container.

C9.2.6. Pipeline Facility. Includes new and existing pipes, pipeline rights of way, auxiliary equipment (e.g., valves and manifolds), and buildings or other facilities used in the transportation of POL.

C9.2.7. POL. Refined petroleum, oils, and lubricants, including, but not limited to, petroleum, fuel, lubricant oils, synthetic oils, mineral oils, animal fats, vegetable oil, sludge, and POL mixed with wastes other than dredged spoil. Petroleum products are further classified as:

POL Class	Definition	Flashpoint	Common Examples
A	Liquefied hydrocarbons with an absolute vapor pressure > 98kPa (at 15ºC (59ºF))	NA	Butane, propane, etc.
A1	Stored at < 0ºC (32ºF)		
A2	Stored at ≥ 0ºC (32ºF)		
B	Based on flashpoint	< 55ºC (131ºF)	Gasoline, naptha, petroleum, etc.
B1		< 38ºC (100.4ºF)	
B2		≥ 38ºC (100.4ºF)	
C	Based on flashpoint	55ºC -100ºC (131º F-212º F)	Diesel, fuel oil, etc.
D	Based on flashpoint	>100ºC (212º F)	Asphalt, liquid paraffin, lubricants

C9.2.8. <u>POL Facility</u>. Area used to store or transfer POL products with a flashpoint ≤ 150ºC (302ºF).

C9.2.8.1. Types of POL facilities include the following:

C9.2.8.1.1. <u>POL Tank Farms</u>. Bulk storage facilities used to distribute POL products in bulk to storage facilities, filling stations, etc.

C9.2.8.1.2. <u>POL Storage Facilities for Internal Consumption</u>. POL containers that are intended for onsite use such as boilers, emergency generators, etc. (excludes tank farms and fillings stations).

C9.2.8.1.3. <u>Filling Stations</u>. A facility associated with the fueling of vehicles.

C9.2.8.1.4. A pipeline facility as identified in C9.2.6.

C9.2.8.2. The POL facility also comprises any ancillary piping associated with the POL storage containers.

C9.2.8.3. The storage and transfer areas of POL products with a flashpoint >150ºC are not subject to the requirements for POL facilities. However, they must comply with the requirements specified in Chapter 5, “Hazardous Material.”

C9.2.9. <u>POL Storage Container</u>. POL mobile/portable and fixed; and above and below ground storage containers with capacities GREATER than 55 gallons (mobile/portable and fixed; and above and below ground storage containers). USTs required to meet all requirements of Chapter 19 are EXCLUDED from the definition of POL storage containers.

C9.3. <u>CRITERIA</u>

C9.3.1. <u>Applicability</u>. The below criteria apply only at POL Facilities as defined in paragraph C9.2.8.

C9.3.2. Authorization of POL Facilities

C9.3.2.1. Installations shall provide sufficient information to the Spanish Base Commander for authorization of their POL tank farms, if requested (see Chapter 1 for the process).

C9.3.2.1.1. POL storage facilities for internal consumption and vehicle filling stations, which do not exceed the thresholds listed below, do not need authorization.

POL Class	POL Storage Facilities for Internal Consumption		Filling Stations	
	Indoor	Outdoor	Indoor	Outdoor
B	≤ 50 L (13.2 gal)	≤ 100 L (26.4 gal)	≤ 300 L (79.3 gal)	≤ 500 L (132.1 gal)
C and D	≤ 1,000 L(264.2 gal)	≤ 1,000 L (264.2 gal)	≤ 3,000 L (792 gal)	≤ 5,000 L (1,321 gal)

C9.3.3. General POL Storage Container Criteria

C9.3.3.1. Inspections and Testing. Inspections shall be conducted on all POL storage facilities and associated valves and piping in accordance with the criteria presented in Table 9.1. The scope and frequency of the inspections, as outlined in Table 9.1, shall be based on the type of POL facility as defined in C9.2.8. Testing shall be conducted on all POL storage facilities and associated valves and piping in accordance with recognized industry standards.

C9.3.3.2. Secondary Containment. POL storage containers must be provided with a secondary means of containment (e.g., dike) capable of holding the entire contents of the largest single tank plus sufficient freeboard to allow for precipitation and expansion of product. Alternatively, POL storage containers that are equipped with adequate technical spill and leak prevention options (such as overfill alarms and flow shutoff or restrictor devices) may provide secondary containment by use of a double wall container. New below ground storage containers must meet this criterion by use of a leak detection pipe and basin. A licensed technical authority may waive these secondary containment criteria for below ground storage containers.

C9.3.3.2.1. For POL storage containers >1,000 l (264 gallons):

C9.3.3.2.1.1. The total capacity of the secondary containment surrounding two or more bulk storage tanks shall not exceed 200,000 m^3 (52,800,000 gal). Secondary containment with multiple tanks shall be divided by dikes 0.7 m (2.29 ft) high, so that each compartment does not contain more than one tank with a capacity ≥ 20,000 m^3, or multiple tanks with a total capacity ≥ 20,000 m^3 (5,280,000 gal). The walls of the secondary containment shall be at least 1 m (3.28 ft) high.

C9.3.3.2.1.2. Secondary containment for POL storage containers or vehicle filling stations with aboveground storage tanks with an aggregate capacity >5,000 L (1,320 gal) shall have a slope with a 2% gradient towards a collection sump and discharge point.

C9.3.3.3. Permeability. Permeability for containment areas will be a maximum of 10^{-7} cm/sec.

C9.3.3.4. Containment Area Drainage. Drainage of stormwater from containment areas will be controlled by a valve that is locked closed when not in active use. Stormwater will be inspected for petroleum sheen before being drained from containment areas. If a petroleum sheen is present it must be collected with sorbent materials prior to drainage, or treated using an oil-water separator. Disposal of sorbent material exhibiting the hazardous characteristics in Appendix B.1 will be in accordance with Chapter 6, "Hazardous Waste."

C9.3.3.5. Valves and Piping. All aboveground valves, piping, and appurtenances associated with POL storage containers shall be periodically inspected in accordance with recognized industry standards.

C9.3.4. Additional POL Storage Container Criteria

C9.3.4.1. Testing. Buried piping associated with POL storage containers shall be tested for integrity and leaks at the time of installation, modification, construction, relocation, or replacement. New buried piping must be protected against corrosion in accordance with recognized industry standards.

C9.3.4.2. Storage Container Design. POL storage containers shall be designed or modernized in accordance with good engineering practice to prevent unintentional discharges by use of overflow prevention devices.

C9.3.4.3. Completely and Partially Buried Metallic POL Storage Containers. These must be protected from corrosion in accordance with recognized industry standards.

C9.3.5. Storage Container Wastes. POL container cleaning wastes frequently have hazardous characteristics (as defined in Appendix B.1) and must be handled and disposed of in accordance with requirements of Chapter 6, "Hazardous Waste." POL container waste and handling procedures include:

C9.3.5.1. POL container cleaning wastes (sludge and washwaters) must be disposed of in accordance with the criteria of Chapter 6.

C9.3.5.2. POL container bottom waters, which are periodically drained, must be collected and disposed of in accordance with Chapter 6.

C9.3.6. General Transport and Distribution Criteria

C9.3.6.1. Loading/Unloading Racks and Areas

C9.3.6.1.1. Secondary Containment. Loading/unloading racks shall be designed to handle discharges of at least the maximum capacity of any single compartment of a rail car or tank truck loaded or unloaded at the loading/unloading rack.

C9.3.6.1.2. Departing Vehicle Warning Systems. Provide an interlocked warning light or physical barrier system, warning signs, wheel chocks, or vehicle break interlock system at loading/unloading racks to prevent vehicles from departing before complete disconnection of flexible or fixed oil transfer lines.

C9.3.6.1.2.1. Engines shall be switched off during the loading or unloading of Class B POL products (e.g. gasoline or naphtha) into or from tank trucks.

C9.3.6.1.3. Vehicle Inspections. Prior to filling and prior to departure of any tank car or tank truck, closely inspect for discharges from the lowermost drain and all outlets of such vehicles, and if necessary, ensure that they are tightened, adjusted, or replaced to prevent liquid discharge while in transit.

C9.3.6.1.4. Loading/Unloading Areas. Provide appropriate containment and/or diversionary structures (dikes, berms, culverts, spill diversion ponds, etc.) or equipment (sorbent materials, wiers, booms, other barriers, etc.) at loading/unloading areas to prevent a discharge of POL, which reasonably could be expected to cause a sheen on Waters of Spain defined in Chapter 4, "Wastewater."

C9.3.6.2. POL Pipeline Facilities

C9.3.6.2.1. Provisions for Testing and Maintenance. All pipeline facilities carrying POL must be tested and maintained in accordance with recognized industry standards, including:

C9.3.6.2.1.1. Each pipeline operator handling POL will prepare and follow a procedural manual for operations, maintenance, and emergencies.

C9.3.6.2.1.2. Each new pipeline facility and each facility in which pipe has been repaired, replaced, or relocated must be tested in accordance with recognized industry standards, without leakage before being placed in-service.

C9.3.6.2.1.3. All new POL pipeline facilities must be designed and constructed to meet recognized industry construction standards.

C9.3.7. Personnel Training. At a minimum, all personnel handling POL shall be trained annually in the operation and maintenance of equipment to prevent discharges; discharge procedure protocols; general facility operations; and the applicable contents of the facility Spill Plan.

C9.3.8. Tank Closure. Closure activities for aboveground POL storage tanks, except those which contained POL class C or D and are <1,000 liters (264 gal), shall follow recognized industry standards when they are decommissioned. The tank closure shall be performed and certified by a DoD-approved or equivalent Spanish contractor.

Table 9.1. POL Storage Facility Periodic Inspection Requirements for ASTs and Associated Piping

		POL Tank Farm	POL Storage Facility for Internal Consumption	Filling Station
Inspections	Responsible Authority	Internal inspector	DoD approved or equivalent Spanish contractor	DoD approved or equivalent Spanish contractor
	Frequency	5 years, regardless of the storage capacity	5 years for POL facilities that exceed certain thresholds[1]. 10 years for all other POL facilities.	5 years for POL facilities that exceed certain thresholds[1]. 10 years for all other POL facilities.
	Scope of Inspection	1. Verify that periodic inspections and tests have been performed. 2. Verify the condition of security measures. 3. Inspect the condition of the tank (and associated piping), walls, foundations, fencing, lids, drainage valves, auxiliary equipment, etc. 4. For tanks that cannot be inspected visually, conduct tightness test. 5. Inspect the condition of the pumps, spouts, hoses and nozzles. 6. Inspect the cathodic protection. 7. Inspect the grounding system (if present), the electrical continuity of the pipelines, and other metallic elements if there is no documentation of periodic inspections by maintenance personnel 8. In tanks that can be visually inspected, measure the wall thickness.	1. Verify that periodic inspections and tests have been performed. 2. Verify the condition of security measures. 3. Inspect the condition of the tank (and associated piping), walls, foundations, fencing, lids, drainage valves, auxiliary equipment, etc. 4. For tanks that cannot be inspected visually, conduct tightness test. 5. Inspect the condition of the pumps, spouts, hoses and nozzles. 6. Inspect the grounding system (if present), the electrical continuity of the pipelines, and other metallic elements if there is no documentation of periodic inspections by maintenance personnel 7. In tanks that can be visually inspected, measure the wall thickness	1. Verify that periodic inspections and tests have been performed. 2. Verify the condition of security measures. 3. Inspect the condition of the tank (and associated piping), walls, foundations, fencing, lids, drainage valves, auxiliary equipment, etc. 4. For tanks that cannot be inspected visually, conduct tightness test. 5. Inspect the condition of the pumps, spouts, hoses and nozzles. 6. Inspect the grounding system (if present), the electrical continuity of the pipelines, and other metallic elements if there is no documentation of periodic inspections by maintenance personnel 7. In tanks that can be visually inspected, measure the wall thickness

Note:
[1] POL facilities that exceed the below thresholds must be visually inspected every 5 years:

POL Class	Indoor	Outdoor
B	>300 L (79.3 gal)	>500 L (132.1 gal)
C or D	>3,000 L (792 gal)	>5,000 L (1,321 gal)

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C10. CHAPTER 10

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C11. CHAPTER 11

PESTICIDES

C11.1. SCOPE

This Chapter contains criteria regulating the use, storage, and handling of pesticides, but does not address the use of these materials by individuals acting in an unofficial capacity in a residence or garden. The disposal of pesticides is covered in Chapter 6, "Hazardous Waste" and Chapter 7, "Solid Waste."

C11.2. DEFINITIONS

C11.2.1. Certified Pesticide Applicators. Personnel who apply pesticides and who have been formally certified in accordance with DoD 4150.07-M, Volume 1 "DoD Pest Management Training: The DoD Plan for the Certification of Pesticide Applicators" (which accepts Spanish certification in appropriate circumstances). Local nationals applying pesticides shall also hold a valid Spanish license for pesticide use.

C11.2.2. Integrated Pest Management (IPM). A planned program, incorporating continuous monitoring, education, record-keeping, and communication to prevent pests and disease vectors from causing unacceptable damage to operations, people, property, materiel, or the environment. IPM uses targeted, sustainable (effective, economical, environmentally sound) methods including education, habitat modification, biological control, genetic control, cultural control, mechanical control, physical control, regulatory control, and where necessary, the judicious use of least-hazardous pesticides.

C11.2.3. Pests. Arthropods, birds, rodents, nematodes, fungi, bacteria, viruses, algae, snails, marine borers, snakes, weeds, undesirable vegetation, and other organisms (except for microorganisms that cause human or animal disease) that adversely affect the well being of humans or animals, attack real property, supplies, equipment or vegetation, or are otherwise undesirable.

C11.2.4. Pest Management Consultant. Professional DoD pest management personnel located at component headquarters, field operating agencies, major commands, facilities engineering field divisions or activities, or area support activities who provide technical and management guidance for the conduct of installation pest management operations. Some pest management consultants may be designated by their component as certifying officials.

C11.2.5. Pesticide. Any substance or mixture of substances, including biological control agents, that may prevent, destroy, repel, or mitigate pests.

C11.2.6. Pesticide Waste. Materials subject to pesticide disposal restrictions including:

C11.2.6.1. Any pesticide that has been identified by the pest management consultant as cancelled under U.S. or Spanish authority;

C11.2.6.2. Any pesticide that does not meet specifications, is contaminated, has been improperly mixed, or otherwise unusable, whether concentrated or diluted;

C11.2.6.3. Any material used to clean up a pesticide spill; or

C11.2.6.4. Any containers, equipment, or material contaminated with pesticides.

C11.2.7. Registered Pesticide. A pesticide registered and approved for sale and use within the United States or Spain.

C11.3. CRITERIA

C11.3.1. All pesticide applications, excluding arthropod skin and clothing repellents, will be recorded using DD Form 1532-1, "Pest Management Maintenance Report", a computer-generated equivalent, or an Official Book of Hazardous Pesticides Movement (*Libro Oficial de Movimiento de Plaguicidas Peligrosos - LOM*). These records shall be maintained at the pesticide application facility. The Pest Management Maintenance Report has been assigned Report Control Symbol DD-A&T(A&AR)1080 in accordance with DoD 8910-M "DoD Procedures for Management of Information Requirements." These records shall be:

C11.3.1.1. Submitted at least quarterly to the Pest Management Consultant; and

C11.3.1.2. Archived for permanent retention in accordance with specific service procedures.

C11.3.2. The LOM shall include the following information:

C11.3.2.1. The date of the product purchase or transfer;

C11.3.2.2. The identification of the pesticide, including the commercial name, registration number in the corresponding official register, manufacturing lot reference number, and the quantity of the product;

C11.3.2.3. The identification of the supplier; and

C11.3.2.4. The signature of the buyer, taking responsibility for the proper use of the product.

C11.3.3. Installations will implement and maintain a current pest management plan that includes measures for all installation activities and satellite sites that perform pest control. This written plan will include IPM procedures for preventing pest problems in order to minimize the use of pesticides. The plan must be reviewed and approved in writing by the appropriate pest management consultant.

C11.3.4. All pesticide applications will be made by certified pesticide applicators, with the following exceptions:

C11.3.4.1. New DoD employees who are not certified may apply pesticides that are not classified as toxic, very toxic, or harmful during an apprenticeship period not to exceed 2 years and only under the supervision of a certified pesticide applicator;

C11.3.4.2. Arthropod skin and clothing repellents; and

C11.3.4.3. Pesticides applied as part of an installation's self help program that are not classified as toxic, very toxic, or harmful.

C11.3.5. All pesticide applicators will be included in a medical surveillance program to monitor the health and safety of persons occupationally exposed to pesticides.

C11.3.6. All pesticide applicators will be provided with personal protective equipment appropriate for the work they perform and the types of pesticides to which they may be exposed.

C11.3.7. Installations will only use registered pesticides that are on the list approved by the Armed Forces Pest Management Board (AFPMB) that have Spanish equivalents approved in writing by the appropriate pest management consultant. This may be documented as part of the approval of the pest management plan:

C11.3.8. Pesticides will be included in the installation spill contingency plan (See Chapter 18, "Spill Prevention and Response Planning.").

C11.3.9. Pest management facilities, including mixing and storage areas, will comply with the Armed Forces Pest Management Board Technical Guide No. 17, "Military Handbook - Design of Pest Management Facilities," and the following requirements:

C11.3.9.1. Constructed using non-flammable materials and protected against external extreme temperatures and humidity;

C11.3.9.2. Located at a safe distance from waterways and flood areas;

C11.3.9.3. Equipped with a ventilation system with external exhaust;

C11.3.9.4. Separated from housing or inhabited areas by means of walls;

C11.3.9.5. Include emergency equipment and personal protective equipment when storing or handling pesticide products classified as toxic or flammable.

C11.3.10. All pesticide applications will be in accordance with guidance given on the pesticide label. Labels will bear the appropriate use instructions and precautionary message based on the toxicity category of the pesticide ("danger", "warning", or "caution"). If foreign nationals will be using the pesticides or are present during mixing activities, the precautionary messages and use instructions will be in English and Spanish. In addition, the phytosanitary product labels shall also bear the risks and safety precaution phrases, as specified in Table 11.1.

C11.3.11. SDSs and labels for all pesticides will be available at the storage and holding facility, in accordance with Chapter 5, “Hazardous Material.”

C11.3.12. Pesticide storage areas will contain a readily-visible current inventory of all items in storage and shall be regularly inspected and secured to prevent unauthorized access.

C11.3.13. Pesticide derived waste awaiting disposal shall be stored and labeled in accordance with Chapter 6, “Hazardous Waste,” and either segregated from pesticide products or transferred to a hazardous waste accumulation point or storage area (see Chapter 6).

C11.3.14. Unless otherwise restricted or canceled, pesticides in excess of installation needs will be redistributed within the supply system or disposed of in accordance with procedures outlined below:

C11.3.14.1. The generator of pesticide wastes will determine whether or not the waste is hazardous, in accordance with Chapter 6, “Hazardous Waste.”

C11.3.14.2. Pesticide waste determined to be hazardous waste will be disposed of in accordance with the criteria for hazardous waste disposal in Chapter 6, “Hazardous Waste.”

C11.3.14.3. Pesticide waste that is determined not to be a hazardous waste will be disposed of in accordance with the label instructions, through DLA Disposition Services. Empty pesticide containers shall not be reused.

Table 11.1. Phytosanitary Product Labels

Standard phrases for safety precautions	
General (SP 1)	- Do not contaminate water with the product or its container. (Do not clean application equipment near surface water / Avoid contamination via drains from farmyards and roads) [*No contaminar el agua con el producto ni con su envase. (No limpiar el equipo de aplicación de producto cerca de aguas superficiales/Evítese la contaminación a través de los sistemas de evacuación de aguas de las explotaciones o de los caminos*)]
Safety precautions for operators (SPo)	- After contact with skin, first remove product with a dry cloth and then wash the skin with plenty of water. [*En caso de contacto con la piel, elimínese primero el producto con un paño seco y después lávese la piel con agua abundante*]. - Wash all protective clothing after use. [*Lávese toda la ropa de protección después de usarla*]. - After igniting the product, do not inhale smoke and leave the treated area immediately. [*Tras el inicio de la combustión del producto, abandónese inmediatamente la zona tratada sin inhalar el humo*]. - The container must be opened outdoors and in dry conditions. [*El recipiente debe abrirse al aire libre y en tiempo seco*]. - Ventilate treated areas/greenhouses thoroughly/time to be specified/until spray has dried before re-entry. [*Ventilar las zonas/los invernaderos tratados bien/durante un tiempo especificado/hasta que se haya secado la pulverización antes de volver a entrar*].
Safety precautions related to the environment (SPE)	- To protect groundwater/soil organisms do not apply this or any other product containing (identify active substance or class of substances, as appropriate) more than (time period or frequency to be specified). [*Para proteger las aguas subterráneas/los organismos del suelo, no aplicar este producto ni ningún otro que contenga (precísese la sustancia o la familia de sustancias, según corresponda) más de (indíquese el tiempo o la frecuencia)*]. - To protect groundwater/aquatic organisms do not apply to (soil type or situation to be specified) soils. [*Para proteger las aguas subterráneas/los organismos acuáticos, no aplicar en suelos (precísese la situación o el tipo de suelos)*]. - To protect aquatic organisms/non-target plants/non-target arthropods/insects respect an unsprayed buffer zone of (distance to be specified) to non-agricultural land/surface water bodies. [*Para proteger los organismos acuáticos/las plantas no objetivo/los artrópodos no objetivo/los insectos, respétese sin tratar una banda de seguridad de (indíquese la distancia) hasta la zona no cultivada/ las masas de agua superficial*].

Standard phrases for safety precautions	
	- To protect aquatic organisms/non-target plants do not apply on impermeable surfaces such as asphalt, concrete, cobblestones, railway tracks and other situations with a high risk of run-off. [*Para proteger los organismos acuáticos/las plantas no objetivo, no aplicar sobre superficies impermeables como el asfalto, el cemento, los adoquines, las vías del ferrocarril ni en otras situaciones con elevado riesgo de escorrentía*]. - To protect birds/wild mammals the product must be entirely incorporated in the soil; ensure that the product is also fully incorporated at the end of rows. [*Para proteger las aves/los mamíferos silvestres, el producto debe incorporarse completamente al suelo; asegurarse de que se incorpora al suelo totalmente al final de los surcos*]. - To protect birds/wild mammals remove spillages. [*Para proteger las aves/los mamíferos silvestres, recójase todo derrame accidental*]. - Do not apply during the bird breeding period. [*No aplicar durante el período de reproducción de las aves*]. - Dangerous to bees/To protect bees and other pollinating insects do not apply to crop plants when in flower/Do not use where bees are actively foraging/Remove or cover beehives during application and for (state time) after treatment/Do not apply when flowering weeds are present/Remove weeds before flowering/Do not apply before (state time). [*Peligroso para las abejas./Para proteger las abejas y otros insectos polinizadores, no aplicar durante la floración de los cultivos./No utilizar donde haya abejas en pecoreo activo./Retírense o cúbranse las colmenas durante el tratamiento y durante (indíquese el tiempo) después del mismo./No aplicar cuando las malas hierbas estén en floración./Elimínense las malas hierbas antes de su floración./No aplicar antes de (indíquese el tiempo*)].
Safety precautions related to good agricultural practice (SPa)	- To avoid the build-up of resistance do not apply this or any other product containing (identify active substance or class of substances, as appropriate) more than (number of applications or time period to be specified). [*Para evitar la aparición de resistencias, no aplicar este producto ni ningún otro que contenga (indíquese la sustancia activa o la clase de sustancias, según corresponda) más de (indíquese el número de aplicaciones o el plazo*)].

Standard phrases for special risks

- Contact with eyes is harmful (*Tóxico en contacto con los ojos*)
- May cause photosensitization (*Puede causar fotosensibilización*)
- Contact with vapor causes burns to skin and eyes and contact with liquid causes freezing (*El contacto con los vapores provoca quemaduras de la piel y de los ojos, el contacto con el producto líquido provoca congelación*)

C12. CHAPTER 12

HISTORIC & CULTURAL RESOURCES

C12.1. SCOPE

This Chapter contains criteria for required plans and programs needed to ensure proper protection and management of historic and cultural resources, such as properties on the World Heritage List or the Spanish Heritage lists (Spain's equivalent to the U.S. National Register of Historic Places).

C12.2. DEFINITIONS

C12.2.1. Adverse Effect. Changes that diminish the quality or significant value of historic or cultural resources.

C12.2.2. Archeological Resource. Any material remains of prehistoric or historic human life or activities. Such resources include, but are not limited to: pottery, basketry, bottles, weapons, weapon projectiles, tools, structures or portions of structures, pit houses, rock paintings, rock carvings, intaglios, graves, human skeletal remains, or any portion of any of the foregoing items.

C12.2.3. Cultural Mitigation. Specific steps designed to lessen the adverse effects of a DoD action on a historical or cultural resource, including:

C12.2.3.1. Limiting the magnitude of the action

C12.2.3.2. Relocating the action in whole or in part

C12.2.3.3. Repairing, rehabilitating, or restoring the affected resources, effected property

C12.2.3.4. Recovering and recording data from cultural properties that may be destroyed or substantially altered

C12.2.4. Historic and Cultural Resources Program. Identification, evaluation, documentation, curation, acquisition, protection, rehabilitation, restoration, management, stabilization, maintenance, recording, and reconstruction of historic and cultural resources and any combination of the foregoing.

C12.2.5. Historic or Cultural Resource. Physical remains of any prehistoric or historic district, site, building, structure, or object significant in world, national, or local history, architecture, archeology, engineering, or culture. The term includes artifacts, archeological resources, records, and material remains that are related to such a district, site, building, structure, or object, and also includes natural resources (plants, animals, landscape features, etc.) that may be considered important as a part of a country's traditional culture and history. The term also includes any property listed on the World Heritage List or the Spanish Heritage lists (Spain's equivalent of the National Register of Historic Places). Spanish lists of properties should be evaluated to determine if they are equivalent with the National Register of Historic Places prior to application.

C12.2.6. Inventory. To determine the location of historic and cultural resources that may have world, national, or local significance.

C12.2.7. Material Remains. Physical evidence of human habitation, occupation, use, or activity, including the site, loci, or context in which such evidence is situated including:

C12.2.7.1. Surface or subsurface structures;

C12.2.7.2. Surface or subsurface artifact concentrations or scatters;

C12.2.7.3. Whole or fragmentary tools, implements, containers, weapons, clothing, and ornaments;

C12.2.7.4. By-products, waste products, or debris resulting from manufacture or use;

C12.2.7.5. Organic waste;

C12.2.7.6. Human remains;

C12.2.7.7. Rock carvings, rock paintings, and intaglios;

C12.2.7.8. Rock shelters and caves;

C12.2.7.9. All portions of shipwrecks; or

C12.2.7.10. Any portion or piece of any of the foregoing.

C12.2.8. Preservation. The act or process of applying measures to sustain the existing form, integrity, and material of a building or structure, and the existing form and vegetative cover of a site. It may include initial stabilization work where necessary, as well as ongoing maintenance of the historic building materials.

C12.2.9. Protection. The act or process of applying measures designed to affect the physical condition of a property by safeguarding it from deterioration, loss, attack, or alteration, or to cover or shield the property from danger or injury. In the case of buildings and structures, such treatment is generally temporary and anticipates future historic preservation treatment; in the case of archaeological sites, the protective measure may be temporary or permanent.

C12.3. CRITERIA

C12.3.1. U.S. Installation commanders shall take into account the effect of any action on any property listed on the World Heritage List or on Spain's Heritage List for purposes of avoiding or mitigating any adverse effects.

C12.3.2. Installations shall have access to the World Heritage List and the Spanish equivalent of the National Register of Historic Places:

Cultural Heritage List (Real Estate):
http://www.mcu.es/bienes/cargarFiltroBienesInmuebles.do?layout=bienesInmuebles&cache=init&language=es

Cultural Heritage List (Movable Property):
http://www.mcu.es/bienes/cargarFiltroBienesMuebles.do?layout=bienesMuebles&cache=init&language=es

Catalogue of Historical Heritage for the Region of Andalucia:
http://www.juntadeandalucia.es/culturaydeporte/web/areas/bbcc/catalogo.

C12.3.3. U.S. Installation commanders shall ensure that personnel performing historic or cultural resource functions have the requisite expertise in world, national, and local history and culture. This may be in-house, contract, or through consultation with another agency. Government personnel directing such functions must have training in historic or cultural resource management.

C12.3.4. Installations shall, after coordination with the Spanish Base Commander, prepare, maintain, and implement a cultural resources management plan that contains information needed to make appropriate decisions about cultural and historic resources identified on the installation inventory, and for mitigation of any adverse effects.

C12.3.5. Installations shall, after coordination with the Spanish Base Commander, and if financially and otherwise practical:

C12.3.5.1. Inventory historic and cultural resources in areas under DoD control. An inventory shall be developed from a records search and visual survey.

C12.3.5.2. Establish measures sufficient to protect known historic or cultural resources until appropriate mitigation or preservation can be completed.

C12.3.5.3. Establish measures sufficient to protect known archeological resources until appropriate mitigation or preservation can be completed.

C12.3.5.4. Comply with the following restrictions regarding Historic and Cultural Resources in Andalusia:

C12.3.5.4.1. Advertisements and any type of aboveground cables and antennas are prohibited within historical gardens and on monuments.

C12.3.5.4.2. Construction of any type, that may alter the character of a historic and cultural resource or disturb its view, is prohibited.

C12.3.6. Installation Commanders shall establish measures to prevent DoD personnel from handling, disturbing or removing historic or cultural resources without permission of the Spanish Base Commander.

C12.3.7. Installation Commanders shall ensure that planning for major actions includes consideration of possible effects on historic or cultural resources.

C12.3.8. If potential historic or cultural resources not previously inventoried are discovered in the course of a DoD action, the U.S. Installation Commander and the Spanish Base Commander shall be notified immediately. The newly discovered items will be preserved and protected pending a decision on final disposition by the installation commander. The decision on final disposition will be made in consultation with the Spanish Base Commander or similar appropriate Spanish authorities.

C13. CHAPTER 13

NATURAL RESOURCES & ENDANGERED SPECIES

C13.1. SCOPE

This chapter establishes criteria for required plans and programs needed to ensure proper protection, enhancement, and management of natural resources and any species (flora or fauna) declared endangered or threatened by either the U.S. or Spanish governments.

C13.2. DEFINITIONS

C13.2.1. Adverse Effect. Changes that diminish the quality or significant value of natural resources. For biological resources, adverse effects include significant decreases in overall population diversity, abundance, and fitness.

C13.2.2. Bird Protection Areas. The following areas are included:

C13.2.2.1. Bird Special Protection Areas (*Zonas de Especial Protección para las Aves, ZEPA*).

C13.2.2.2. Areas considered in the recuperation and conservation plans for the species included in the Spanish lists.

C13.2.2.3. Priority areas for the reproduction, feeding, and local concentration of bird species included in the Spanish lists.

C13.2.3. Conservation. Planned management, use, and protection; continued benefit for present and future generations; and prevention of exploitation, destruction, and/or neglect of natural resources.

C13.2.4. Forested Land. Any piece of rural land covered by any kind of tree species (natural or planted) serving ecological, protection, production, recreation, or landscape functions.

C13.2.5. Harmful Organism. Any species, strain or plant biotype , animal or pathogenic agent harmful to plants or plant products.

C13.2.6. High Voltage Power Lines. Power lines with three phase alternative current of 50 Hz frequency, where effective nominal voltage between the phases is ≥1 KV.

C13.2.7. Management Plan. A document describing natural resources, their quantity, condition, and actions to ensure their conservation and good stewardship.

C13.2.8. National Parks (as designated by Spain). Natural spaces with a high ecological and cultural value and low interference with human activities. They possess ecological, aesthetic, cultural, educational, and scientific values.

C13.2.9. Natural Habitats of Community Interest. Those habitats that are in danger of disappearance in their natural range, have a small natural range due to their decline, are endangered because of their intrinsically restricted area, or display typical characteristics of one or more of the five following biogeographical regions: Alpine, Atlantic, Continental, Macaronesian and Mediterranean.

C13.2.10. Natural Resource. All living and inanimate materials supplied by nature that are of aesthetic, ecological, educational, historical, recreational, scientific, or other value.

C13.2.11. Natural Resources Management. Actions taken that combine science, economics, and policy, to study, manage, and restore natural resources to strike a balance with the needs of people and the ability of the ecosystem to support soil, water, forest, fish, wildlife, and coastal resources.

C13.2.12. Priority Areas for Bird Reproduction, Feeding and Shelter. Areas with a regular presence of protected species of birds during a period of 3 consecutive years.

C13.2.13. Protected Species. Any species of floura and fauna listed or designated by Spain, because continued existence of the species is, or is likely to be, threatened, and is therefore subject to special protection from destruction or adverse modification of associated habitat. Protected species include both those at risk of extinction and those vulnerable of becoming extinct if adverse impacts are not mitigated.

C13.2.14. Rare Species. Those species with small populations that could become vulnerable or endangered in the near future.

C13.2.15. Significant Land or Water Area. Land or water area that is normally ≥ 500 acres outside the cantonment area; areas of smaller size are included if they have natural resources that are especially vulnerable to disturbance.

C13.2.16. Site of Community Importance. A site identified by Spain that contributes significantly to the maintenance, restoration, or favorable conservation status of a natural habitat or species, and/or contributes significantly to the maintenance of biological diversity within the biogeographic region or regions concerned.

C13.2.17. Spanish Protected Species. Those species included in the Spanish List of Special Protection Wildlife Species and the Catalogue of Endangered Species.

C13.2.18. Susceptible Palm Species. Plants other than fruit and seeds, having a base diameter of > 5 cm, of *Areca catechu, Arenga pinnata, Borassus flabellifer, Calamus merillii, Caryota maxima, Caryota cumingii, Cocos nucifera, Corypha gebanga, Corypha elata, Elaeis guineensis, Livistona decipiens, Metroxylon sagu, Oreodoxa regia, Phoenix canariensis, Phoenix dactylifera, Phoenix theophrasti, Phoenix sylvestris, Sabal umbraculifera, Trachycarpus fortunei,* and *Washingtonia spp.*

C13.2.19. Threatened and Endangered Species. Any species of fauna or flora, listed in Table~~s~~ 13.1, "Threatened and Endangered Fauna or included in the European or Spanish lists of specially protected and endangered species. The websites below provide links to the European and Spanish list of protected species and protected habitats:

C13.2.19.1. EU Red List for Species:

C13.2.19.1.1. Mammals - http://ec.europa.eu/environment/nature/conservation/species/redlist/downloads/European_mammals.pdf

C13.2.19.1.2. Amphibians - http://ec.europa.eu/environment/nature/conservation/species/redlist/downloads/European_amphibians.pdf

C13.2.19.1.3. Reptiles - http://ec.europa.eu/environment/nature/conservation/species/redlist/downloads/European_reptiles.pdf

C13.2.19.1.4. Freshwater Fishes - http://ec.europa.eu/environment/nature/conservation/species/redlist/downloads/European_freshwater_fishes.pdf

C13.2.19.1.5. Butterflies - http://ec.europa.eu/environment/nature/conservation/species/redlist/downloads/European_butterflies.pdf

C13.2.19.1.6. Dragonflies - http://ec.europa.eu/environment/nature/conservation/species/redlist/downloads/European_dragonflies.pdf

C13.2.19.1.7. Saproxylic Beetles - http://ec.europa.eu/environment/nature/conservation/species/redlist/downloads/European_saproxylic_beetles.pdf

C13.2.19.1.8. Non-marine Molluscs - http://ec.europa.eu/environment/nature/conservation/species/redlist/downloads/European_molluscs.pdf

C13.2.19.1.9. Vascular Plants - http://ec.europa.eu/environment/nature/conservation/species/redlist/downloads/European_vascular_plants.pdf

C13.2.19.2. Spanish National List - http://www.magrama.gob.es/es/biodiversidad/temas/conservacion-de-especies-amenazadas/tabla_especies_LESPE_CEA_tcm7-203160.xls

C13.2.19.3. Spanish Regional List - http://www.juntadeandalucia.es/medioambiente/web/Bloques_Tematicos/Estadisticas_e_Indicadores/Estadisticas_Oficiales_de_la_Consejeria_de_Medio_Ambiente/FloraFauna/fyf_gradoamenaza_situacionespecies__2005.xls

C13.3. CRITERIA

C13.3.1. Protection of Species & Habitats

C13.3.1.1. Installations that have land and water areas shall take reasonable steps to protect and enhance known endangered or threatened species and Spanish protected species and their habitats.

C13.3.1.2. An assessment of potential impacts shall be completed for proposed projects located within or likely to have a significant effect on a Site of Community Importance. The assessment of potential impacts shall be submitted to the Spanish Base Commander prior to execution of the project (see Chapter 1 for the process).

C13.3.1.3. The following activities are prohibited for Spanish Animal Protected Species:

C13.3.1.3.1. Capturing or killing of the species in the wild;

C13.3.1.3.2. Disturbing species during their reproductive cycle, hibernation, or migration;

C13.3.1.3.3. Destroying or collecting eggs or nests in the wild; and

C13.3.1.3.4. Deteriorating or destroying breeding sites or resting places.

C13.3.1.4. The following activities are prohibited for Spanish Plant Protected Species:

C13.3.1.4.1. Picking, collecting, cutting, uprooting, or destroying plants in the wild; and

C13.3.1.4.2. Possessing, transporting, selling, or trading plant species collected in the wild.

C13.3.1.5. The following activities are prohibited in Spanish National Parks:

C13.3.1.5.1. Sport and commercial fishing and hunting;

C13.3.1.5.2. Siting hydroelectrical and mining operations, communication and power lines;

C13.3.1.5.3. Exploiting natural and agricultural resources; and

C13.3.1.5.4. Flyovers < 3,000 m (9,842.5 ft) high, unless authorized.

C13.3.2. Installations shall maintain, or have access to Table 13.1 as well as current list of Spanish protected species and protected habitats listed in C13.2.19.

C13.3.3. Installations with significant land or water areas shall, after coordination with the Spanish Base Commander or similar appropriate Spanish authorities, develop natural resources management plans.

C13.3.4. Installations with natural resources management plans shall, after coordination with the Spanish Base Commander or similar appropriate Spanish authorities, and if financially and otherwise practical, and in such a way that there is no net loss of mission capability:

C13.3.4.1. Conduct a survey to determine the presence of any threatened or endangered species or Spanish-protected species and habitats, or support Spanish surveys; and

C13.3.4.2. Implement natural resources management plans.

C13.3.5. The Spanish Base Commander, or if there is no Spanish Base Commander, the U.S. Ambassador will be notified (via the Component chain of command, who shall also notify the Environmental Executive Agent) of the discovery of any endangered or threatened species and Spanish protected species not previously known to be present on the installation.

C13.3.6. Installations shall maintain grounds to meet designated mission use and ensure harmony with the natural landscape and/or the adjacent Spanish facilities where practical.

C13.3.7. Installations shall ensure that personnel performing natural resource functions have the requisite expertise in the management of their discipline (i.e., endangered or threatened species, Spanish protected species, wetlands, soil stabilization). This may be in-house, contract, or through consultation with another agency. Government personnel directing such functions must have training in natural resources management.

C13.3.8. Installations shall place emphasis on the maintenance and protection of habitats favorable to the reproduction and survival of Spanish protected species and indigenous flora and fauna.

C13.3.8.1. Installations with Andaluisan-designated forests that intend to conduct the following activities shall provide sufficient information to the Spanish Base Commander for authorization (see Chapter 1 for the process):

C13.3.8.1.1. Harvesting of wood or other tree-derived materials (e.g., cork, tree fruits, etc);

C13.3.8.1.2. Utilization of areas impacted by wildfires;

C13.3.8.1.3. Substitution of tree or shrub species if not indicated in the Natural Resources Management Plan;

C13.3.8.1.4. Reforestation of deforested areas;

C13.3.8.1.5. Planting of rapid growth tree species;

C13.3.8.1.6. Cutting, burning, relocating, or partial harming of protected Andalusian forest species;

C13.3.8.1.7. Conducting potentially-eroding activities;

C13.3.8.1.8. Change in land uses; and

C13.3.8.1.9. Circulation of motor vehicle on forest trails outside of the established road network.

C13.3.8.2. Prevention activities shall be implemented in order to protect the forests from fires during periods of high risk.

C13.3.9. Land and vegetative management activities will be consistent with current conservation and land use principles (e.g., ecosystem protection, biodiversity conservation, and mission-integrated land use).

C13.3.10. Installations shall utilize protective vegetative cover or other standard soil erosion/sediment control practices to control dust, stabilize sites, and avoid silting of streams.

C13.3.11. Introduction of Harmful Organisms to Plants and Plant Products

C13.3.11.1. Organisms that are harmful to plants and plant products shall not be introduced or handled within the Spanish territory

C13.3.11.2. Imported plants and plant products shall be required to be accompanied by a phytosanitary certificate demonstrating they are free of harmful organisms.

C13.3.11.3. Special Provisions for Palm Species. Susceptible palm species shall only be introduced into the Spanish territory, under the following conditions, in order to prevent the spread of *Rhynchophorus ferrugineus (Olivier)*:

C13.3.11.3.1. Observe a one year quarantine, during which:

C13.3.11.3.1.1. The susceptible palm species shall be isolated at an authorized location to prevent the introduction of the *Rhynchophorus ferrugineus* or be subject to the application of an appropriate preventive treatmens; and

C13.3.11.3.1.2. Show no signs of the *Rhynchophorus ferrugineus* during official inspections which shall be carried out every three months.

C13.3.11.3.2. The importer shall be responsible for obtaining an authorization to operate the quarantine site and to maintain the documentation of the inspections.

C13.3.12. Measures Against Bird Electrocution and Collision with High Voltage Power Lines. Protective measures shall be taken to prevent bird electrocution and collision with high voltage power lines with exposed electrical wires during construction, modification, or expansion projects within Andalusian priority areas for reproduction, feeding, shelter, and local concentration of endangered bird species.

Table 13.1. Selected European-Listed Endangered & Threatened Species

Common Name	Scientific Name	OCONUS Country of Listing
	Mammals	
Bear, brown	Ursus arctos arctos	Italy
Chamois, Apennine	Rupicapra rupicapra ornata	Italy
Chimpanzee	Pan troglodytes	Wherever found in the wild
Ibex, Pyrenean	Capra pyrenaica pyrenaica	Spain
Lion, Asiatic	Panthera leo persica	Turkey
Lynx, Spanish	Felis pardina	Spain
Urial	Ovis musimon ophion	Cyprus
	Birds	
Bullfinch, Sao Miguel (finch)	Pyrrhula pyrrhula murina	Eastern Atlantic Ocean - Azores
Eagle, Spanish imperial	Aquila heliaca adalberti	Spain
Falcon, Eurasian peregrine	Falco peregrinus peregrinus	Europe, Eurasia south to Africa and Mideast
Gull, Audouin’s	Larus audouinii	Mediterranean Sea
Ibis, northern bald	Geronticus eremita	Southern Europe
Pigeon, Azores wood	Columba palumbus azorica	East Atlantic Ocean - Azores
Tern, roseate	Sterna dougallii dougallii	Tropical and temperate coasts of Atlantic Basin and East Africa
	Reptiles	
Lizard, Hierro giant	Gallotia simonyi simonyi	Spain (Canary Islands)
Lizard, Ibiza wall	Podarcis pityusensis	Spain (Balearic Islands)
Sea turtle, green	Chelonia mydas	circumglobal in tropical and temperate seas and oceans
Sea turtle, Kemp’s ridley	Lepidochelys kempii	Tropical and temperate seas in Atlantic Basin
Sea turtle, leatherback	Dermochelys coriacea	Tropical, temperate, and subpolar seas
Sea turtle, loggerhead	Caretta caretta	Circumglobal in tropical and temperate seas and oceans
Sea turtle, olive ridley	Lepidochelys olivacea	Circumglobal in tropical and temperate seas
	Fish	
Ala Balik (trout)	Salmo platycephalus	Turkey
Cicek (minnow)	Acanthorutilus handlirschi	Turkey

Note: This table does not include a complete list of Spanish-designated threatened and endangered species. The links to the European and Spanish and lists are provided in C13.2.19.

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C14. CHAPTER 14

POLYCHLORINATED BIPHENYLS

C14.1. SCOPE

This Chapter contains criteria to control and abate threats to human health and the environment from the handling, use, storage, and disposal of polychlorinated biphenyls (PCB). These criteria include specific requirements for most uses of PCBs, including, but not limited to, transformers, capacitors, heat transfer systems, hydraulic systems, electromagnets, switches and voltage regulators, circuit breakers, reclosers, and cables.

C14.2. DEFINITIONS

C14.2.1. Capacitor. A device for accumulating and holding a charge of electricity and consisting of conducting surfaces separated by a dielectric.

C14.2.2. In or Near Commercial Buildings. Within the interior of, on the roof of, attached to the exterior wall of, in the parking area serving, or within 30 meters of a non-industrial, non-substation building.

C14.2.3. Incinerator. An engineered device using controlled-flame combustion to thermally degrade PCBs and PCB items. Examples include rotary kilns, liquid injection incinerators, cement kilns, and high temperature boilers.

C14.2.4. Leak or Leaking. Any instance in which a PCB article, PCB container, or PCB equipment has any PCBs on any portion of its external surface.

C14.2.5. Mark. The descriptive name, instructions, cautions, or other information applied to PCBs and PCB items, or other objects subject to this Guide.

C14.2.6. Marked. PCB items and PCB storage areas and transport vehicles marked by applying a legible mark by painting, fixation of an adhesive label, or by any other method that meets these criteria.

C14.2.7. Non-PCB Transformers. Any transformer that contains <50 ppm PCB.

C14.2.8. PCB. Includes any combination of the following:

C14.2.8.1. Polychlorinated biphenyls;

C14.2.8.2. Polychlorinated triphenyls (PCTs);

C14.2.8.3. Monomethyltetrachlorodiphenyl methane;

C14.2.8.4. Monomethyldichlorodiphenyl methane; or

C14.2.8.5. Monomethyldibromodiphenyl methane.

C14.2.9. PCB Article. Any manufactured article, other than a PCB container, that contains PCBs and whose surface(s) has been in direct contact with PCB. This includes capacitors, transformers, electric motors, pumps, and pipes.

C14.2.10. PCB Article Container. Any package, can, bottle, bag, barrel, drum, tank, or other device used to contain PCB articles or PCB equipment, and whose surface(s) has not been in direct contact with PCBs.

C14.2.11. PCB Container. Any package, can, bottle, bag, barrel, drum, tank, or other device that contains PCBs or PCB articles, and whose surface(s) has been in direct contact with PCBs.

C14.2.12. PCB-Contaminated Electrical Equipment. Any electrical equipment including, but not limited to, transformers, capacitors, circuit breakers, reclosers, voltage regulators, switches, electromagnets, and cable that contain ≥ 50 ppm PCB, but < 500 ppm PCB.

C14.2.13. PCB Equipment. Any manufactured item, other than a PCB container or a PCB article container, which contains a PCB article or other PCB equipment, and includes microwave ovens, electronic equipment, and fluorescent light ballasts and fixtures.

C14.2.14. PCB Item. Any PCB article, PCB article container, PCB container, or PCB equipment that deliberately or unintentionally contains or has as a part of it any PCB, or PCBs at a concentration ≥ 50 ppm. Any equipment suspected to contain PCBs at ≥ 50 ppm is considered a PCB item, unless there are records (i.e., manufacturer's warranty or analytical data) to the contrary.

C14.2.15. Restricted Access Area. Areas where access by unauthorized personnel is controlled by fences, other man-made structures, or naturally occurring barriers such as mountains, cliffs, or rough terrain.

C14.2.16. Substantial Contact Area. An area that is subject to public access on a routine basis or which could result in substantial dermal contact by employees.

C14.2.17. PCB Large High Voltage Capacitor. A capacitor that contains ≥ 1.36 kg (3 lbs.) of dielectric fluid and which operates at 2,000 volts (alternating current (ac) or direct current (dc)) or above.

C14.2.18. PCB Large Low Voltage Capacitor. A capacitor that contains ≥ 1.36 kg (3 lbs.) of dielectric fluid and which operates below 2,000 volts (ac or dc).

C14.3. CRITERIA

C14.3.1. General

C14.3.1.1. The installation spill contingency plan will address PCB items, including temporary storage items. Chapter 18, "Spill Prevention and Response Planning," provides criteria on how to prepare these plans.

C14.3.1.2. Spills of PCB liquids will be responded to immediately upon discovery and cleaned up in accordance with the following:

C14.3.1.2.1. Surfaces that are located in substantial contact areas will be cleaned to 10 micrograms (µg) per 100 square centimeters (cm^2).

C14.3.1.2.2. Surfaces in all other contact areas will be cleaned to 100 µg per 100 cm^2.

C14.3.1.2.3. Contaminated soil located in access areas will be removed until the concentration of PCB remaining in the soil is ≤ 0.8 ppm for industrial areas, ≤ 0.08 ppm for residential areas, or ≤ 0.01 ppm for agricultural and livestock grazing areas and will be backfilled with clean soil with a PCB concentration < the method detection limit Restricted access areas in which PCB spills have been cleaned up shall have annotated on installation real property records the level of PCBs remaining in the soil, including-the extent, date and type of sampling, and a reference to any reports documenting the site conditions.

C14.3.1.2.4. Contaminated soil, PCB waste, and decontamination waste removed from the spill site will be disposed of in accordance with Chapter 6, “Hazardous Waste.”

C14.3.1.3. PCB items containing PCB concentrations ≥ 50 ppm or ≥ 1 liter of PCB-contaminated liquid by volume (i.e., electric motors using PCB coolants, hydraulic systems using PCB hydraulic fluid, and heat transfer systems using PCBs), as well as any PCB article containers used to store the preceding items, must be prominently marked in English and Spanish. The marking must identify the item as containing PCBs, warn against improper disposal and handling, and provide a phone number in case of spills or if questions arise about disposal. This marking criteria also applies to rooms (a warning label shall be applied to doors), vaults, and storage areas containing PCB transformers or storing PCBs or PCB items for disposal. In addition, the following PCB items must be marked at the time of items’ removal from use if not already marked: PCB large low voltage capacitors and equipment containing a PCB transformer or PCB large high voltage capacitor.

C14.3.1.4. Decontaminated PCB items shall be labeled as such. The label shall contain the following information:

C14.3.1.4.1. The type of fluid used as replacement;

C14.3.1.4.2. The date that the PCB fluid was replaced;

C14.3.1.4.3. The name of the authorized decontamination company;

C14.3.1.4.4. The PCB concentration (% by weight) of the replaced fluid; and

C14.3.1.4.5. The new fluid PCB concentration (% by weight).

C14.3.1.5. PCB equipment containing >1 liter of PCB-contaminated liquid shall be labeled as such. At a minimum, labels shall contain the following information:

C14.3.1.5.1. Date of labeling (day, month and year);

C14.3.1.5.2. Identification of the equipment (number and model);

C14.3.1.5.3. Type of equipment (transformer, capacitor, etc);

C14.3.1.5.4. Manufacturing date (day, month and year or “unknown”);

C14.3.1.5.5. Volume of fluid/PCB expressed in dm^3;

C14.3.1.5.6. PCBs concentration in ppm (actual concentration or > 500 ppm for equipment that may contain PCB);

C14.3.1.5.7. Group (manufactured with PCBs, contaminated with PCBs or equipment that may contain PCBs); and

C14.3.1.5.8. Total weight of the equipment in kilograms (solid and liquid).

C14.3.1.6. PCB liquid or PCB items shall not be stored or handled in proximity of explosive, flammable, oxidant, or corrosive substances.

C14.3.1.7. Each installation having PCB items will maintain a written inventory that includes a current list by type of all PCB items in use and all PCB items placed into storage for disposal or disposed of for that year. Inventory records shall be maintained for a period of time at least 3 years after the last item on the list is disposed of.

C14.3.1.8. Prior to 1 March of each year, installations having PCB items containing >1 liter of PCB by volume shall submit to the Spanish Base Commander, if requested, an annual declaration containing the inventory of PCB items and those suspected to containing PCB present as of 31 December of the previous year (see Chapter 1 for the process). The declaration shall also include records pertaining to the decontamination or disposal of PCB items. Elements of the declaration shall include:

C14.3.1.8.1. Year of applicability;

C14.3.1.8.2. Date and location;

C14.3.1.8.3. Name and address of the installation;

C14.3.1.8.4. Inventory of PCB equipment;

C14.3.1.8.5. Total weight of PCB equipment present;

C14.3.1.8.6. Total weight of PCB equipment scheduled to be decontaminated or disposed of;

C14.3.1.8.7. List of equipment containing ≥ 50 ppm of PCB;

C14.3.1.8.8. List of equipment suspected to contain <50 ppm of PCB;

C14.3.1.8.9. List of transformers where the PCB concentration was reduced to > 50 and <500 ppm;

C14.3.1.8.10. Disposal documentation of any PCB equipment included in previous declarations;

C14.3.1.8.11. Certificates for the elimination or destruction of PCB equipment;

C14.3.1.8.12. Certificates for the decontamination of PCB equipment; and

C14.3.1.8.13. Records of visual inspections of the PCB equipment.

C14.3.1.9. Disposal of PCB items will only be through DLA Disposition Services in accordance with DoD 4160.21-M "Defense Materiel Disposition Manual") or paragraph C14.3.5. and the hazardous waste disposal requirements as established in Chapter 6, "Hazardous Waste."

C14.3.1.10. All periodic inspections as required in this Chapter will be documented at the installation. Records of inspections and maintenance history will be maintained for three years after disposal of the transformer.

C14.3.2. PCB Transformers

C14.3.2.1. PCB transformers that are in use or in storage for reuse will not be used in any application that poses a risk of contamination to food or feed.

C14.3.2.2. All PCB transformers, including those in storage for reuse, will be registered with the servicing fire department/emergency service provider.

C14.3.2.3. PCB transformers in use in or near commercial buildings or located in sidewalk vaults will be equipped with electrical protection to minimize transformer failure that would result in the release of PCBs.

C14.3.2.4. PCB transformers removed and stored for reuse will only be returned to their original application and location and will not be used at another location unless there is no practical alternative; and any such alternative use will not exceed one year.

C14.3.2.5. PCB transformers will be managed as follows:

C14.3.2.5.1. Servicing of PCB transformers with PCB fluids is prohibited. PCB transformers, with a concentration of > 500 ppm and with a volume >1 dm^3 shall be be decontaminated or disposed of.

C14.3.2.6. All in-service PCB transformers (>500 ppm) will be inspected at least every 3 months except that PCB transformers with impervious, undrained secondary containment capacity of 100% of dielectric fluid or PCB transformers tested and found to contain <60,000 ppm PCBs will be inspected at least every 12 months.

C14.3.2.7. If any PCB transformer is involved in a fire and was subjected to heat and/or pressure sufficient to result in violent or nonviolent rupture, the installation will take measures to control water runoff, such as blocking floor drains. Runoff water will be tested and treated if required in accordance with Chapter 18, "Spill Prevention and Response Planning."

C14.3.2.8. Leaking PCB transformers shall be decontaminated and disposed of as soon as possible after discovery of the leak.

C14.3.2.9. All transformers will be considered and treated as PCB transformers unless information to the contrary exists.

C14.3.3. Other PCB Items

C14.3.3.1. All equipment containing >1 dm^3 (1 liter) PCB fluids (including, but not limited to, electromagnets, switches, and voltage regulators) shall be decontaminated or disposed of. Decontamination activities shall be conducted by DoD-approved contractors or the Spanish equivalent. The decontaminated equipment shall be serviced with PCB-free dielectric fluid and the equipment shall be appropriately labeled (following C14.3.1.4).

C14.3.3.2. Capacitors containing PCBs at any concentration must be managed as follows:

C14.3.3.2.1. Use and storage for reuse of PCB large high-voltage capacitors and PCB large low-voltage capacitors that pose an exposure risk to food or feed is prohibited.

C14.3.3.2.2. Use of PCB large high-voltage and PCB large low-voltage capacitors is prohibited unless the capacitor is used within a restricted-access electrical substation or in a contained and restricted-access indoor installation. The indoor installation will not have public access and will have an adequate roof, walls, and floor to contain any release of PCBs; and

C14.3.3.3. Any PCB item removed from service will be marked with the date it is removed from service.

C14.3.4. Storage

C14.3.4.1. PCBs and PCB items that are to be stored before disposal shall be labeled in accordance with C14.3.1.3 and Chapter 6, "Hazardous Waste." The storage facility shall assure the containment of PCBs, including:

C14.3.4.1.1. Roofs and walls of storage buildings that exclude rainfall;

C14.3.4.1.2. A containment berm, ≥ 6 inches high, sufficient to contain twice the internal volume of the largest PCB article or ≥ 50% of the total internal volume of all PCB articles or containers stored, whichever is greater;

C14.3.4.1.3. Drains, valves, floor drains, expansion joints, sewer lines, or other openings constructed to prevent any release from the bermed area;

C14.3.4.1.4. Continuous, smooth, and impervious flooring material; and

C14.3.4.1.5. To the maximum extent possible, a new PCB storage area will be located to minimize the risk of release due to seismic activity, floods, or other natural events. For facilities located where there is a high possibility of such risks, the installation spill prevention and control plan will address the risk.

C14.3.4.2. The temporary storage of PCB items in facilities that do not comply with the above requirements is prohibited.

C14.3.4.3. All PCB storage areas will be inspected at least monthly.

C14.3.4.4. Containers used for the storage of PCBs will be at least as secure as those required for their transport for disposal by DLA Disposition Services and, at a minimum, shall be impervious, double-walled, and labeled according to C14.3.1.3.

C14.3.5. Disposal

C14.3.5.1. Installations that generate PCB waste of ≥ 50 ppm PCB will maintain an audit trail for the wastes at least as stringent as that required under the criteria in Chapter 6 "Hazardous Waste." Installations shall dispose of PCB items either:

C14.3.5.1.1. Through DLA Disposition Services in accordance with DoD 4160.21-M; or

C14.3.5.1.2. For in-country disposal, PCB items shall be delivered to a Spanish-authorized company for disposal via incineration (which must meet a 99.9% combustion efficiency).

C14.3.5.2. Retrogrades of PCB Items DoD-generated PCB items manufactured in the United States will be returned to the United States via DLA Disposition Services for delivery to a permitted disposal facility if Spanish or third country disposal is not possible, is prohibited, or would not be managed in an environmentally sound manner. Ensure that all PCB items and equipment are marked in accordance with criteria in subparagraph C14.3.1.3.

C14.3.6. Elimination of PCB Products

C14.3.6.1. Installations shall minimize the use of PCBs and PCB items without degrading mission performance. Any device containing >1dm^3 of PCB fluid shall be decontaminated or disposed of, except for transformers with PCB concentrations <500 ppm, which may be used until the end of their life cycle.

C14.3.6.2. Installations shall not purchase or otherwise take control of PCBs or PCB items for use.

C14.3.6.3. All procurement of transformers or any other equipment containing dielectric or hydraulic fluid shall be accompanied by a manufacturer's certification that the equipment contains no detectable PCBs (<2 ppm) at the time of shipment.

C14.3.6.4. Such newly procured transformers and equipment shall have permanent labels affixed stating they are PCB-free (no detectable PCBs).

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C15. CHAPTER 15

ASBESTOS

C15.1. SCOPE

This Chapter contains criteria to control and abate threats to human health and the environment from asbestos, and describes management of asbestos during removal and disposal. Policy requirements for a comprehensive Occupational Health and Safety program are not covered in this Chapter. To protect personnel from asbestos exposure, refer to DoDI 6055.1, "DoD Safety and Occupational Health (SOH) Program"and DoDI 6055.5, "Industrial Hygiene and Occupational Health," and relevant service instructions.

C15.2. DEFINITIONS

C15.2.1. Adequately Wet. Sufficiently mix or penetrate with liquid to prevent the release of particulates. If visible emissions coming from ACM are observed, then that material has not been adequately wetted. However, the absence of visible emissions is not sufficient evidence of being adequately wet.

C15.2.2. Asbestos. Generic term used to describe six distinctive varieties of fibrous mineral silicates, including chrysotile, amosite, crocidolite, tremolite asbestos, anthophyllite asbestos, actinolite asbestos, and any other of these materials that have been chemically treated and/or altered.

C15.2.3. Asbestos-Containing Material (ACM). Any material containing asbestos (> 1 fiber).

C15.2.4. Asbestos Fibers. Particles of asbestos in any variety, with length > 5 micrometers, diameter < 3 micrometers, and the length to diameter ratio < 3:1.

C15.2.5. Category I Nonfriable ACM. Means asbestos containing packings, gaskets, resilient floor covering, and asphalt roofing products containing asbestos.

C15.2.6. Category II Nonfriable ACM. Means any material, excluding Category I nonfriable ACM, containing asbestos that, when dry, cannot be crumbled, pulverized, or reduced to powder by hand pressure.

C15.2.7. Friable Asbestos. Any material containing asbestos that, when dry, can be crumbled, pulverized or reduced to powder by hand pressure.

C15.2.8. Regulated ACM. Means (a) Friable asbestos material, (b) Category I nonfriable ACM that has become friable, (c) Category I nonfriable ACM that will be or has been subjected to sanding grinding, cutting, or abrading, or (d) Category II nonfriable ACM that has a high probability of becoming or has become crumbled, pulverized, or reduced to powder by the forces expected to act on the material in the course of demolition or renovation operations.

C15.3. CRITERIA

C15.3.1. Installations will appoint an asbestos program manager to serve as the single point of contact for all asbestos-related activities.

C15.3.2. Installations will prepare and implement an asbestos management plan. As a minimum, the plan will include the following:

C15.3.2.1. An ACM inventory, conducted by sample and analysis or visual determination;

C15.3.2.2. A notification and education program to tell workers, tenants, and building occupants where potentially friable ACM is located, and how and why to avoid disturbing the ACM; all persons affected shall be properly informed;

C15.3.2.3. Regular ACM surveillance to note, assess, and document any changes in the ACM's condition;

C15.3.2.4. Work control/permit systems to control activities that might disturb ACM;

C15.3.2.5. Operations and maintenance (O&M) work practices to avoid or minimize fiber release during activities affecting ACM;

C15.3.2.6. Record keeping to document O&M activities related to asbestos identification management and abatement. These records shall be maintained for forty years;

C15.3.2.7. Training for the asbestos program manager as well as custodial and maintenance staff. At a minimum, the training program shall address:

C15.3.2.7.1. The properties of asbestos and its effects on health;

C15.3.2.7.2. The types of products or materials likely to contain asbestos;

C15.3.2.7.3. The operations that could result in asbestos exposure and the importance of preventive controls to minimize exposure;

C15.3.2.7.4. Safe work practices, controls, and personal protective equipment;

C15.3.2.7.5. The appropriate role, choice, selection, limitations, and proper use of respiratory equipment;

C15.3.2.7.6. The means for determination of the correct functioning of the respiratory equipment;

C15.3.2.7.7. Emergency procedures;

C15.3.2.7.8. Decontamination procedures;

C15.3.2.7.9. Waste disposal; and

C15.3.2.7.10. Medical examination requirements.

C15.3.2.8. Certification for the asbestos program manager that meets the requirements of a recognized U.S. certification body and/or the Spanish equivalent;

C15.3.2.9. Procedures to assess and prioritize identified hazards for abatement; and

C15.3.2.10. Procedures to prevent the use of ACM in new construction. The use of crocidolite (blue) asbestos, amosite, anthophyllite asbestos, actinolite asbestos, tremolite asbestos, chrysotile, and fibers that contain any of these products is prohibited in new construction or maintenance work.

C15.3.3. The use of asbestos-containing building materials, which were installed prior to June 2002, may continue until they reach the end of their service life.

C15.3.4. Prior to demolition, renovation, or repair of a facility, the installation will make a determination whether or not the activity will remove or disturb ACM, and will record this determination on the project authorization document (e.g., work order).

C15.3.5. Prior to demolition or renovation of a facility that involves removing or disturbing ACM, a written assessment of the action will be prepared and furnished to the installation commander and the Spanish Base Commander (see Chapter 1 for the process). A copy of the assessment will also be kept on permanent file.

C15.3.6. Installations will remove friable ACM when the ACM poses a threat to release airborne asbestos fibers and cannot be reliably repaired or isolated.

C15.3.7. Before disturbing or demolishing a facility or part of a facility, installations will remove all potentially-impacted ACM.

C15.3.7.1. Air sampling shall be conducted during activities that involve removing or disturbing ACM to ensure that the concentration in the work area is < 0.1 fibers per cm^3 as an eight-hour time-weighted average (TWA). The sampling shall be repeated periodically during the demolition or renovation, and whenever a change of work procedures or conditions occurs.

C15.3.7.2. A final evaluation shall be conducted after removal or demolition work involving ACM to ensure the area is safe for re-occupancy.

C15.3.8. When disposing of asbestos waste, installations shall adequately wet all ACM waste and seal it in a leak-proof container. In-country disposal of friable ACM shall be conducted by an authorized Spanish company, in accordance with the procedures in Chapter 6 “Hazardous Waste.” Non-friable ACM may be disposed of in a non-hazardous waste landfill. Containers will be labeled in English and Spanish:

"DANGER - CONTAINS ASBESTOS FIBERS - AVOID CREATING DUST -
CANCER AND LUNG DISEASE HAZARD.
Follow safety instructions"

"ATENCIÓN CONTIENE AMIANTO
Respirar el polvo de amianto es peligroso para la salud
Seguir las normas de seguridad"

C15.3.8.1. The label shall also have a large white "a" on a black background above the text (which is printed against a red background).

C15.3.8.2. In accordance with Chapter 6, "Hazardous Waste" labeling requirements, all asbestos waste containers shall also have a second label (at least 10 x 10 cm) with the following information:

C15.3.8.2.1. The waste identification code;

C15.3.8.2.2. The packaging date;

C15.3.8.2.3. The nature of the risks associated with the waste; and

C15.3.8.2.4. The name, address, and telephone number of the owner of the waste.

C15.3.8.3. The recipient of the asbestos waste shall also be identified in a clear and legible manner in Spanish.

C15.3.8.4. Permanent records documenting the disposal action and site, including waste transport and disposal manifests, will be maintained for at least five years.

C15.3.9. DoD schools will comply with applicable requirements of 15 U.S.C. 2643(l) and implementing regulations in 40 CFR Part 763, Subpart E.

C16. CHAPTER 16

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C17. CHAPTER 17

LEAD-BASED PAINT

C17.1. SCOPE

C17.1.1. This Chapter contains criteria to establish and implement a lead hazard management program to identify, control, or eliminate lead-based paint hazards, through interim controls or abatement, in child-occupied facilities and military family housing, in a manner protective of human health and the environment. Policy requirements for a comprehensive Occupational Health and Safety program are not covered in this Chapter. To protect personnel from lead exposure, refer to DoDI 6055.1, DoDI 6055.5, and relevant service instructions.

C17.2. DEFINITIONS

C17.2.1. Abatement. Any set of measures designed to permanently eliminate lead-based paint or lead-based paint hazards. Abatement includes the removal of lead-based paint and lead-contaminated dust, the permanent enclosure or encapsulation of lead-based paint, the replacement of components or fixtures painted with lead-based paint, and the removal or covering of lead-contaminated soil. Abatement also includes all preparation, cleanup, disposal, and post-abatement clearance activities associated with such measures.

C17.2.2. Accessible Surface. An interior or exterior surface painted with lead-based paint that is accessible for a young child to mouth or chew.

C17.2.3. Bare Soil. Soil, including sand, not covered by grass, sod, or other live ground covers, or by wood chips, gravel, artificial turf, or similar covering.

C17.2.4. Child-Occupied Facility. A facility, or portion of a facility, visited regularly by the same child, 6 years of age or under, on at least two different days within any week, provided that each day’s visit lasts at least 3 hours and the combined weekly visits last at least 6 hours, and the combined annual visits last at least 60 hours. Child-occupied facilities may include, but are not limited to, day-care centers, preschools, playgrounds, and kindergarten classrooms.

C17.2.5. Clearance. Visual evaluation and testing (collection and analysis of environmental samples) conducted after lead-based paint hazard reduction activities, interim controls, and standard treatments to determine that the work is complete and no lead-contaminated bare soil or lead-contaminated settled dust exists in a facility frequented by children under the age of 6.

C17.2.6. Deteriorated Paint. Any interior or exterior paint or other coating that is peeling, chipping, chalking, cracking, or is otherwise damaged or separated from the substrate.

C17.2.7. Elevated Blood Lead Level. A confirmed concentration of lead in whole blood of 20 µg Pb/dl (micrograms of lead per deciliter) for a single test, or of 15-19 µg/dl in two tests taken at least 3 months apart.

C17.2.8. Encapsulation. The application of any covering or coating that acts as a barrier between the lead-based paint and the environment. Encapsulation may be used as a method of abatement if it is designed to be permanent.

C17.2.9. Enclosure. The use of rigid, durable construction materials that are mechanically fastened to the substrate to act as a barrier between lead-based paint and the environment. Enclosure may be used as a method of abatement if it is designed to be permanent.

C17.2.10. Evaluation. A visual evaluation, risk assessment, risk assessment screen, paint inspection, paint testing, or a combination of risk assessment and paint inspection to determine the presence of deteriorated paint, lead-based paint, or a lead-based paint hazard.

C17.2.11. Friction Surface. An interior or exterior surface that is subject to abrasion or friction, including but not limited to, window, floor, and stair surfaces.

C17.2.12. Hazard Reduction. Measures designed to reduce or eliminate human exposure to lead-based paint hazards through various methods including interim controls or abatement or a combination of the two.

C17.2.13. Impact Surface. An interior or exterior surface that is subject to damage by repeated sudden force, such as certain parts of doorframes.

C17.2.14. Interim Controls. A set of measures designed to temporarily reduce human exposure or likely exposure to lead-based paint hazards. Interim controls include, but are not limited to, repairs, occasional and ongoing maintenance, painting, temporary containment, specialized cleaning, clearance, ongoing activities, and the establishment and operation of management and resident education programs.

C17.2.15. Lead-Based Paint (LBP). Paint or other surface coatings that contain lead ≥ 1.0 milligrams per cm^2, or 0.5 percent by weight of 5,000 ppm by weight.

C17.2.16. Lead-Based Paint Hazard. Paint-lead hazard, dust-lead hazard or soil-lead hazard, also defined below:

C17.2.16.1. Paint-Lead Hazard. A paint-lead hazard is any of the following:

C17.2.16.1.1. Any lead-based paint on a friction surface that is subject to abrasion and where the lead dust levels on the nearest horizontal surface underneath the friction surface (e.g., the window sill, or floor) are equal to or greater than the dust-lead hazard levels identified in subparagraph C17.2.16.2.

C17.2.16.1.2. Any damaged or otherwise deteriorated lead-based paint on an impact surface that is caused by impact from a related building component (such as a doorknob that knocks into a wall or a door that knocks against its doorframe).

C17.2.16.1.3. Any chewable lead-based painted surface on which there is evidence of teeth marks.

C17.2.16.1.4. Any other deteriorated lead-based paint in any residential building or child-occupied facility or on the exterior of any residential building or child-occupied facility.

C17.2.16.2. Dust-Lead Hazard (previously defined as lead-contaminated dust). Surface dust in a residential dwelling or child-occupied facility that contains a mass-per-area concentration of lead equal to or exceeding 40 $\mu g/ft^2$ on floors or 250 $\mu g/ft^2$ on interior window sills based on wipe samples.

C17.2.16.3. Soil-Lead Hazard (previously defined as lead-contaminated soil). Bare soil on residential real property or on the property of a child-occupied facility that contains total lead $\geq$ 400 ppm ($\mu g/g$) in a play area, or an average of 1,200 ppm of bare soil in the rest of the yard based on soil samples.

C17.2.17. Lead-Based Paint Inspection. A surface-by-surface investigation to determine the presence of lead-based paint, and the provision of a report explaining the results of the investigation.

C17.2.18. Permanent. An expected design life of at least 20 years.

C17.2.19. Reevaluation. A visual evaluation of painted surfaces and limited dust and soil sampling conducted periodically following lead-based paint hazard reduction where lead-based paint is still present.

C17.2.20. Replacement. A strategy of abatement that entails removing building components that have surfaces coated with lead-based paint (such as windows, doors, and trim) and installing new components free of lead-based paint.

C17.2.21. Risk Assessment. An on-site investigation to determine the existence, nature, severity, and location of lead-based paint hazards and the provision of a report explaining the results of the investigation and options for reducing lead-based paint hazards.

C17.2.22. Risk Assessment Screen. A sampling protocol that is used in dwellings that are in relatively good condition and where the probability of finding lead-based hazards are low. The protocol involves inspecting such dwellings and collecting samples from representative locations on the floor, interior window sills, and window troughs to determine whether conducting a risk assessment is warranted.

C17.3. CRITERIA

C17.3.1. Installations shall:

C17.3.1.1. Develop and implement a multi-disciplinary lead-based paint hazard management program to identify, evaluate, and reduce lead-based paint hazards in child-occupied facilities and military family housing.

C17.3.1.2. Manage identified lead-based paint hazards through interim controls or abatement.

C17.3.1.3. Identify lead-based paint hazards in child-occupied facilities and military family housing using any or all of the following methods:

C17.3.1.3.1. Lead-based paint risk assessment screen. If the screen identifies dust-lead levels >25 μg Pb/ft^2 (2.32 μg Pb/m^2) for floors or >125 μg Pb/ft^2 (11.61 μg Pb/m^2) for interior window sills, a lead-based paint risk assessment shall be performed.

C17.3.1.3.2. Lead-based paint risk assessments.

C17.3.1.3.3. Routine facility inspection for fire and safety.

C17.3.1.3.4. Occupant, facility manager, and worker reports of deteriorated paint.

C17.3.1.3.5. Results of childhood blood lead screening or reports of children identified to have elevated blood lead levels.

C17.3.1.3.6. Lead-based paint reevaluations.

C17.3.1.3.7. Review of construction, painting, and maintenance histories.

C17.3.1.4. Ensure occupants and worker protection measures are taken during all maintenance, repair, and renovation activities that disturb areas known or assumed to have lead-based paint.

C17.3.1.5. Disclose the presence of any known lead-based paint or lead-based paint hazards to occupants of child-occupied facilities and military family housing and provide information on lead-based paint hazard reduction. In addition, inform occupants of military family housing, prior to conducting remodeling or renovation projects, of the hazards associated with these activities, and provide information on protecting family members from the hazards of lead-based paint.

C17.3.1.6. Ensure that all personnel involved in lead-based activities, including paint inspection, risk assessment, specification or design, supervision, and abatement, are properly trained.

C17.3.1.7. Dispose of lead-contaminated waste that meets the definition of a hazardous waste in accordance with Chapter 6, “Hazardous Waste.”

C18. CHAPTER 18

SPILL PREVENTION & RESPONSE PLANNING

C18.1. SCOPE

This Chapter contains criteria to plan for, prevent, control, and report spills of POL and hazardous substances. It is DoD policy to prevent spills of these substances due to DoD activities and to provide for prompt, coordinated response to contain and clean up spills that might occur. Remediation beyond that required for the initial response is conducted pursuant to DoDI 4715.8, ‘‘Environmental Remediation for DoD Activities Overseas.”

C18.2. DEFINITIONS

C18.2.1. Aboveground Storage Container. POL storage containers, exempt from UST criteria, that are normally placed on or above the surface of the ground. POL storage containers located above the floor and contained in vaults or basements, bunkered containers, and also partially buried containers are considered aboveground storage containers. For the purposes of this Chapter, this includes any mobile or fixed structure, tank, equipment, pipe, or pipeline (other than a vessel or a public vessel) used in oil well drilling operations, oil production, oil refining, oil storage, oil gathering, oil processing, oil transfer, and oil distribution. This also includes equipment in which oil is used as an operating fluid but excludes equipment in which oil is used solely for motive power.

C18.2.2. Decontamination Wastes. Waste materials generated during the decontamination of equipment and personnel used during spill response including but not limited to purging water, rinsing water, plastic containers, rags, gloves, and other personal protective equipment.

C18.2.3. Hazardous Substance. Any substance having the potential to do serious harm to human health or the environment if spilled or released in reportable quantity. A list of these substances and the corresponding reportable quantities is contained in Appendix A. Hazardous substances do not include:

C18.2.3.1. Petroleum, including crude POL or any fraction thereof, that is not otherwise specifically listed or designated as a hazardous substance above.

C18.2.3.2. Natural gas, natural gas liquids, liquefied natural gas, or synthetic gas usable for fuel (or mixtures of natural gas and such synthetic gas).

C18.2.4. Facility Incident Commander (FIC) (previously known as the Installation On-scene Coordinator). The official who coordinates and directs DoD control and cleanup efforts at the scene of a POL or hazardous substance spill due to DoD activities on or near the installation. This official is designated by the installation commander.

C18.2.5. Facility Response Team (FRT) (previously known as the Installation Response Team). A team performing emergency functions as defined and directed by the FIC.

C18.2.6. Oil. Oil of any kind or in any form, including, but not limited to, petroleum, fuel POL, lube oils, animal fats, vegetable oil, sludge, POL refuse, and POL mixed with wastes other than dredged spoil.

C18.2.7. POL. Refined petroleum, oils, and lubricants. (See also definition in Chapter 9, "Petroleum, Oil, and Lubricants.")

C18.2.8. Significant Spill. An uncontained release to the land or water in excess of any of the following quantities:

C18.2.8.1. For hazardous wastes or hazardous substances identified as a result of inclusion in Appendix A, any quantity in excess of the reportable quantity listed in Appendix A;

C18.2.8.2. For POL or liquid or semi-liquid hazardous material, hazardous waste or hazardous substances, in excess of 400 liters (110 gallons);

C18.2.8.3. For other solid hazardous material in excess of 225 Kg (500 pounds);

C18.2.8.4. For combinations of POL and liquid, semi-liquid, and solid hazardous materials, hazardous waste or hazardous substance, in excess of 340 Kg (750 pounds); or

C18.2.8.5. If a spill is contained inside an impervious berm, or on a nonporous surface, or inside a building and is not volatilized and is cleaned up, the spill is considered a contained release and is not considered a significant spill.

C18.2.9. Worst Case Discharge. The largest foreseeable discharge from the facility, under adverse weather conditions, as determined using as a guide the worst case discharge planning volume criteria in Appendix C, "Determination of Worst Case Discharge Planning Volume."

C18.3. CRITERIA

C18.3.1. Spill Prevention Control and Reporting Plan Requirement. All DoD installations will prepare, maintain, and implement a Spill Prevention and Response Plan, which provides for the prevention, control, and reporting of all spills of POL and hazardous substances. The plan will provide measures to prevent, and to the maximum extent practicable, to remove a worst case discharge from the facility. The plan shall be kept in a location easily accessible to the FIC and FRT.

C18.3.1.1. The plan will be updated at least every 5 years or:

C18.3.1.1.1. Within 6 months of any significant changes to operations.

C18.3.1.1.2. When there have been two significant spills to navigable waters in any 12-month period;

C18.3.1.1.3. When there has been a spill of ≥ 1,000 gallons.

C18.3.1.2. The plan shall be certified by an appropriately licensed or certified technical authority ensuring that the plan considers applicable industry standards for spill prevention and environmental protection, that the plan is prepared in accordance with good engineering practice, and is adequate for the facility. Technical changes (i.e., non-administrative) to the plan require recertification.

C18.3.2. Prevention Section. The prevention section of the plan will, at a minimum, contain the following:

C18.3.2.1. Name, title, responsibilities, duties, and 24-hour telephone number of the designated FIC and an alternate.

C18.3.2.2. General information on the installation including name, type or function, location and address, charts of drainage patterns, designated water protection areas, maps showing locations of facilities described in subparagraph C18.3.2.3, critical water resources, land uses, and possible migration pathways.

C18.3.2.3. An inventory of storage, handling, and transfer sites that could possibly produce a significant spill. For each listing, using maps as appropriate, a prediction of the direction and rate of flow shall be included, as well as the total quantity of POL or hazardous substances that might be spilled as a result of a major failure.

C18.3.2.4. An inventory of all POL and hazardous substances at storage, handling, and transfer facilities described in subparagraph C18.3.2.3.

C18.3.2.5. Procedures for the periodic integrity testing of all aboveground storage containers, including visual inspection and where deemed appropriate, another form of non-destructive testing. The frequency and type of inspection and testing must take into account container size and design (floating/fixed roof, skid-mounted, elevated, cut-and cover, partially buried, vaulted above-ground, etc.) and industry standards.

C18.3.2.6. Procedures for periodic inspection for all above ground valves, piping, and appurtenances associated with POL storage containers, in accordance with Chapter 9, "Petroleum, Oil, and Lubricants," subparagraph C9.3.3.1.

C18.3.2.7. Arrangements for Emergency Services. The plan will describe arrangements with installation and/or local police departments, fire departments, hospitals, contractors, and emergency response teams to coordinate emergency services for both on- and off-installation spills.

C18.3.2.8. Means to Contact Emergency Services. The plan will include a telephone number or other means to contact the appropriate emergency service provider (e.g., installation fire department) on a 24-hour basis.

C18.3.2.9. A detailed description of the facility's prevention, control, and countermeasures (including structures and equipment for diversion and containment of spills) for each site listed in the inventory. Measures shall permit, as far as practical, reclamation of spilled substances. Chapters governing hazardous materials, hazardous waste, POL, underground storage tanks, pesticides, and PCBs provide specific criteria for containment structure requirements.

C18.3.2.10. When secondary containment is not feasible for any container listed in the inventory, the plan shall include a detailed explanation of measures that will be taken to prevent spills (e.g., pre-booming, integrity testing, frequent inspection), as determined by the licensed or certified technical authority.

C18.3.2.11. A list of all emergency equipment (such as fire extinguishing systems, spill control equipment, communications and alarm systems [internal and external], personal protective equipment [PPE], and decontamination equipment) at each site listed in the inventory where this equipment is required. This list will be kept up-to-date. In addition, the plan will include the location and a physical description of each item on the list, and a brief outline of its capabilities.

C18.3.2.12. An evacuation plan for each site listed in the inventory, where there is a possibility that evacuation would be necessary. This plan will describe signal(s) to be used to begin evacuation, evacuation routes, alternate evacuation routes (in cases where the primary routes could be blocked by releases of hazardous waste or fires), and a designated meeting place.

C18.3.2.13. A description of deficiencies in spill prevention and control measures at each site listed in the inventory, to include corrective measures required, procedures to be followed to correct listed deficiencies and any interim control measures in place. Corrective actions must be implemented within 24 months of the date of plan preparation or revision.

C18.3.2.14. Written procedures for:

C18.3.2.14.1. Operations to preclude spills of POLs and hazardous substances;

C18.3.2.14.2. Inspections; and

C18.3.2.14.3. Record keeping requirements.

C18.3.2.15. Site-specific procedures shall be maintained at each site on the facility where significant spills could occur.

C18.3.3. <u>Spill Control Section</u>. The control section of the plan (which may be considered a contingency plan) will identify resources for cleaning up spills at installations and activities, and to provide assistance to other agencies when requested. At a minimum, this section of the plan will contain:

C18.3.3.1. Provisions specifying the responsibilities, duties, procedures, and resources to be used to contain and clean up spills.

C18.3.3.2. A description of immediate response actions that shall be taken when a spill is first discovered.

C18.3.3.3. The responsibilities, composition, and training requirements of the FRT.

C18.3.3.4. The command structure that will be established to manage a worst case discharge. Include an organization chart and the responsibilities and composition of the organization.

C18.3.3.5. Procedures for FRT alert and response to include provisions for:

C18.3.3.5.1. Access to a reliable communications system for timely notification of a POL spill or hazardous substance spill.

C18.3.3.5.2. Public affairs involvement.

C18.3.3.6. A current roster of the persons, and alternates, who must receive notice of a POL or hazardous substance spill, including a Defense Logistics Agency (DLA) Energy representative if applicable. The roster will include name, organization mailing address, and work and home telephone numbers. Without compromising security, the plan will include provisions for the notification of the emergency coordinator after normal working hours.

C18.3.3.7. The plan will provide for the notification of the FIC, installation commander, and local authorities in the event of hazard to human health or environment.

C18.3.3.8. Assignment of responsibilities for making the necessary notifications, including notification to the emergency services providers.

C18.3.3.9. Surveillance procedures for early detection of POL and hazardous substance spills.

C18.3.3.10. A prioritized list of various critical water and natural resources that will be protected in the event of a spill.

C18.3.3.11. Other resources addressed in prearranged agreements that are available to the installation to clean up or reclaim a large spill due to DoD activities, if such a spill exceeds the response capability of the installation.

C18.3.3.12. Cleanup methods, including procedures and techniques used to identify, contain, disperse, reclaim, and remove POL, and hazardous substances used in bulk quantity on the installation.

C18.3.3.13. Procedures for the proper reuse and disposal of recovered substances, decontamination wastes, contaminated POL and absorbent materials, and procedures to be accomplished prior to resumption of operations.

C18.3.3.14. A description of general health, safety, and fire prevention precautions for spill cleanup actions.

C18.3.3.15. A public affairs section that describes the procedures, responsibilities, and methods for releasing information in the event of a spill.

C18.3.4. Reporting Section. The reporting section of the spill plan will address the following:

C18.3.4.1. Recordkeeping when emergency procedures are invoked.

C18.3.4.2. Notification of spill responders when emergency procedures are invoked.

C18.3.4.3. Any significant spill will be reported to the FIC immediately. Immediate actions will be taken to eliminate the source and contain the spill.

C18.3.4.4. The FIC will immediately notify the appropriate In-Theater Component Commander and/or Defense Agency and the EEA and submit a follow-up written report when:

C18.3.4.4.1. The spill occurs inside a DoD installation and cannot be contained within any required berm or secondary containment;

C18.3.4.4.2. The spill exceeds 400 liters (110 gallons) of POLs;

C18.3.4.4.3. A water resource has been polluted; or

C18.3.4.4.4. The FIC has determined that the spill is significant.

C18.3.4.5. When a significant spill occurs inside a DoD installation and cannot be contained within the installation boundaries or threatens the local Spanish drinking water resource, the appropriate in-theater component commander and/or Defense Agency, EEA, and Spanish Base Commander will be notified immediately.

C18.3.4.6. If a significant spill occurs outside of a DoD installation, the person in charge at the scene will immediately notify via telephone the authorities listed in subparagraph C18.3.4.5, and additionally will notify the local fire departments and obtain necessary assistance.

C18.3.4.7. The following information shall be provided as necessary to the Spanish Base Commander for significant spills occurring outside of a DoD installation (see Chapter 1 for the process):

C18.3.4.7.1. Date, time, and location (road number, municipality, and province) of the event;

C18.3.4.7.2. Type of transport vehicle and name of transporter;

C18.3.4.7.3. Type of accident or spill;

C18.3.4.7.4. Information on the spilled material (material name, code, danger code, danger label);

C18.3.4.7.5. Atmospheric conditions (rain, snow, ice, wind, etc.);

C18.3.4.7.6. Consequences of the accident (spill/leak, fire, explosion, etc. and affected media [soil, air, surface water, etc.]);

C18.3.4.7.7. Casualties and injuries, if any (driver, other transporters, or bystanders);

C18.3.4.7.8. Emergency response actions, including any need to transfer the spilled material to another vehicle; and

C18.3.4.7.9. Additional data (duration of any road detours, etc.).

C18.3.5. Installations shall provide and document necessary training and spill response drills to ensure the effectiveness of personnel and equipment. The training and exercise program shall include the following elements:

C18.3.5.1. Use of personal protective equipment;

C18.3.5.2. Decontamination procedures, as necessary;

C18.3.5.3. Hazard communication training;

C18.3.5.4. FIC Notification Exercise;

C18.3.5.5. Emergency Procedures Exercise;

C18.3.5.6. Spill Management Team Tabletop Exercise; and

C18.3.5.7. Equipment Deployment Exercise.

C18.3.6. After completion of the initial response, any remaining free product and/or obviously contaminated soil will be appropriately removed and managed. Further action will be governed by DoDI 4715.8, "Environmental Remediation for DoD Activities Overseas" and EUCOM Directive 80-2, "Environmental Executive Agent Remediation Policy."

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C19. CHAPTER 19

UNDERGROUND STORAGE TANKS

C19.1. SCOPE

This Chapter contains criteria to control and abate pollution resulting from POL products and hazardous materials stored in USTs. Standards for USTs containing hazardous wastes are covered in Chapter 6, “Hazardous Waste.” Criteria for aboveground and below ground POL storage containers are addressed in Chapter 9, “Petroleum, Oil, and Lubricants.”

C19.2. DEFINITIONS

C19.2.1. Deferred UST. A deferred UST is an underground tank system that fits into one of the following categories:

C19.2.1.1. A hydrant fuel distribution system; or

C19.2.1.2. A field-constructed tank.

C19.2.2. Hazardous Material. Any material defined as a hazardous material in Chapter 5, “Hazardous Material.” The term does not include:

C19.2.2.1. Petroleum, including crude POL or any fraction thereof, that is not otherwise specifically listed or designated as a hazardous material above.

C19.2.2.2. Natural gas, natural gas liquids, liquefied natural gas, or synthetic gas usable for fuel (or mixtures of natural gas and such synthetic gas).

C19.2.3. Hazardous Material UST. A UST that contains a hazardous material (but not including hazardous waste as defined in Chapter 6, “Hazardous Waste,” or any mixture of such hazardous materials and petroleum, and which is not a petroleum UST.

C19.2.4. Internal Inspector. Trained personnel designated by the manager of the USTs to conduct the required inspections and testing.

C19.2.5. POL. Refined petroleum, oils, and lubricants, as defined in Chapter 9, “Petroleum, Oil and Lubricants.”

C19.2.6. POL Storage Facility. Area used to store or transfer POL products as defined in Chapter 9 “Petroleum, Oil and Lubricants.”

C19.2.7. Tank Tightness Testing. A test that must be capable of detecting a 0.1 liter (0.026 gallon) per hour leak from any portion of the tank that routinely contains product while accounting for the effects of thermal expansion or contraction of the product, vapor pockets, tank deformation, evaporation or condensation, and the location of the water table.

C19.2.8. Underground Storage Tank (UST). Any tank, including underground piping connected thereto, larger than 416 liters (110 gallons), that is used to contain POL products or hazardous material and which is completely or partially buried but does not include:

C19.2.8.1. Septic tanks;

C19.2.8.2. Stormwater or wastewater collection systems;

C19.2.8.3. Flow through process tanks;

C19.2.8.4. Surface impoundments, pits, ponds, or lagoons;

C19.2.8.5. Field constructed tanks;

C19.2.8.6. Hydrant fueling systems;

C19.2.8.7. Storage tanks located in an accessible underground area (such as a basement or vault) if the storage tank is situated upon or above the surface of the floor;

C19.2.8.8. USTs containing *de minimis* concentrations of regulated substances, except where C19.3.2.7. is applicable; and

C19.2.8.9. Emergency spill or overflow containment UST systems that are expeditiously emptied after use.

C19.3. CRITERIA

C19.3.1. All installations will maintain a UST inventory.

C19.3.1.1. Installations shall provide the Spanish Base Commander with a copy of their UST inventory (and subsequent updates or revisions) for authorization (see Chapter 1 for the process).

C19.3.2. POL USTs. All petroleum UST systems will be properly installed, protected from corrosion, provided with spill/overfill prevention, and will incorporate leak detection as described below in C19.3.2.3, “Leak Detection.” The UST and associated equipment shall be initially tested and certified by the installation company. Required periodic inspections and testing of POL USTs shall be performed in accordance with Table 19.1.

C19.3.2.1. Corrosion Protection. USTs and piping must be provided with corrosion protection unless constructed of fiberglass or other non-corrodible materials. The corrosion protection system must be certified by a DoD-approved or equivalent Spanish contractor.

C19.3.2.1.1. Filling station and internal fuel consumption USTs shall be equipped with active and passive corrosion protection systems. Single-walled USTs shall utilize one of the following corrosion protection systems:

C19.3.2.1.1.1. Appropriate painting or coating;

C19.3.2.1.1.2. Application of anti-corrosive materials;

C19.3.2.1.1.3. Use of cathodic protection in addition to anti-corrosion coating;or

C19.3.2.1.1.4. Other equivalent protection systems.

C19.3.2.1.2. If cathodic protection is utilized for filling station and internal fuel consumption USTs, the cathodic protection system shall be tested at the following frequency:

C19.3.2.1.2.1. Every 5 years for USTs with a capacity <10 m^3 (2,642 gal);

C19.3.2.1.2.2. Every 2 years for USTs with a capacity >10 and < 60 m^3 (2,642 to 15,852 gal); and

C19.3.2.1.2.3. Annually for USTs or tank farms with a capacity >60 m^3 (15,852 gal).

C19.3.2.1.3. If cathodic protection is provided by impressed current, the system shall also be inspected quarterly.

C19.3.2.2. <u>Spill/Overflow Protection</u>. USTs will be provided with spill and overfill prevention equipment, except where transfers are made in the amounts of ≤ 95 liters (25 gallons). USTs with a capacity > 3,000 liters (792.5 gal) shall be equipped with spill and overfill protection regardless of the volume transferred. Where spill and over-fill protection are required, a spill containment box must be installed around the fill pipe. Furthermore, all filling stations shall be equipped with an underground secondary containment for UST loading and vehicle refueling areas. Overfill prevention will be provided by one of the following methods:

C19.3.2.2.1. Automatic shut-off device (set at 95% of tank capacity).

C19.3.2.2.2. High level alarm (set at 90% of tank capacity).

C19.3.2.3. <u>Leak Detection</u>. Continuous leak detection systems must be capable of detecting a 0.38-liter (0.1-gallon) per hour leak rate or a release of 568 liters (150 gallons) (or 1% of tank volume, whichever is less) within 30 days with a probability of detection of 0.95 and a probability of false alarm of not more than 0.05. During periodic tightness tests, the detection system must be capable of detecting a 0.1-liter (0.026-gallon) per hour leak rate.

C19.3.2.3.1. USTs will use at least one of the following leak detection methods:

C19.3.2.3.1.1. Automatic tank gauging;

C19.3.2.3.1.2. Vapor monitoring;

C19.3.2.3.1.3. Groundwater monitoring;

C19.3.2.3.1.4. Interstitial monitoring (for double-walled USTs); or

C19.3.2.3.1.5. Secondary containment with automatic leak sensor (for single-walled USTs).

C19.3.2.3.2. All pressurized UST piping must be equipped with automatic line leak detectors and utilize either an annual tightness test or monthly monitoring.

C19.3.2.3.3. Suction piping will either have a line tightness test conducted every three years or use monthly monitoring.

C19.3.2.4. USTs and piping will be properly closed if not needed, or be upgraded or replaced. The following table shows the USTs closure requirements depending on the volume and class of POL stored:

POL Class (see Chapter 9)	Tank Capacity	Closure Requirements
A, B	Any volume	Performed by a DoD-approved or equivalent Spanish contractor
C, D	<416 L (110 gal)	No requirement
	416 to 1,000 L (110 to 264 gal)	Properly closed, upgraded or replaced
	>1,000 L (>264 gal)	Performed by a DoD-approved or equivalent Spanish contractor

C19.3.2.5. Any UST and piping not incorporating a functioning leak detection system will require immediate corrective action. Such systems will be tightness tested annually in accordance with recognized industry standards, and inventoried monthly to determine system tightness. The tightness test shall be performed by a DoD-approved or equivalent Spanish contractor.

C19.3.2.6. Any verified leaking UST or UST piping will be immediately removed from service. Any UST and piping suspected of leaking (e.g., leak detection equipment), will be verified for leakage to ensure there is not a false positive, or alternately, will immediately be removed from service. If the UST is still required, it will be repaired or replaced by an accredited company. If repairs are conducted, a tightness test shall be performed to confirm the integrity of the tank prior to its return to service. If the UST is no longer required it will be removed from the ground by a DoD-approved or equivalent Spanish contractor. When a leaking UST is removed, exposed free product and/or obviously contaminated soil in the immediate vicinity of the tank will be appropriately removed and managed in accordance with Chapter 6, “Hazardous Waste.” Additional action will be governed by DoDI 4715.8, “Environmental Remediation for DoD Activities Overseas.” Under extenuating circumstances (e.g., where the UST is located under a building), the UST will be cleaned and filled with an inert substance, and left in place.

C19.3.2.7. When a UST has not been used for one year, or is determined to no longer be required, all of the product and sludges must be removed. Subsequently, the UST must be either cleaned and filled with an inert substance, or removed by a DoD-approved or equivalent Spanish contractor. UST wastes must be sampled and tested in accordance with Chapter 9, "Petroleum, Oil, and Lubricants," paragraph C9.3.5., and managed accordingly.

C19.3.2.8. When the product stored in a UST is changed, the UST must be emptied and cleaned by removing all liquid and accumulated sludge. The removed liquid and sludge shall be managed as hazardous waste in accordance with Chapter 6, "Hazardous Waste."

C19.3.2.9. When a UST system is temporarily closed, corrosion protection and leak detection systems (if the UST is not empty) must be operated and maintained. If a UST system is temporarily closed for 3 months or greater, the following must be complied with :

C19.3.2.9.1. Vent lines must be left open and functioning; and

C19.3.2.9.2. All other lines, pumps, manways, and ancillary equipment must be secured and capped.

C19.3.3. UST Recordkeeping. Installations will maintain a tank system inventory to include tank system installation, repair, testing, inspections, removal, replacement, or upgrade, and operation of corrosion protection equipment for the life of the tank.

C19.3.4. Hazardous Material USTs

C19.3.4.1. All hazardous material USTs and piping must meet the same design and construction standards as required for petroleum USTs and piping, and in addition must be provided with secondary containment for both tank and piping. Secondary containment can be met by using double-walled tanks and piping, liners, or vaults.

C19.3.4.2. Leak Detection. The interstitial space (space between the primary and and secondary containment) for tanks and piping must be monitored monthly for liquids or vapors.

C19.3.4.3. Hazardous material USTs and piping that do not incorporate the criteria contained in subparagraph C19.3.4.1. shall be immediately removed from service and upgraded or replaced as necessary.

C19.3.4.4. All hazardous material USTs and associated piping shall be tightness tested prior to being commissioned and then every 5 years if storing corrosive materials and every 10 years if storing toxic, very toxic and noxious materials. Additionally, the USTs shall undergo an annual visual inspection of the tank, piping system, secondary containment and corrosion protection system.

C19.3.5. Deferred USTs. Deferred USTs constructed after 8 May 1985 must be designed and constructed with corrosion protection, non-corrodible materials, or be otherwise designed and constructed to prevent releases from corrosion or structural failure. UST materials must be compatible with the substance(s) to be stored.

Table 19.1. Periodic Testing and Inspection Requirements for POL USTs and Associated Piping

		POL Tank Farm	POL Storage Facility for Internal Consumption	Filling Station
Inspections	Responsible Authority	Internal inspector	Not required for underground storage tanks used for internal consumption	Not required for underground storage tanks at filling stations
	Frequency	5 years, regardless of the storage capacity	Not required for underground storage tanks used for internal consumption	Not required for underground storage tanks at filling stations
	Scope of Inspection	1. Verify that periodic inspections and tests have been performed. 2. Verify the condition of security measures. 3. Inspect the condition of the tank (and associated piping), walls, foundations, fencing, lids, drainage valves, auxiliary equipment, etc. 4. For tanks that cannot be inspected visually, conduct tightness test. 5. Inspect the condition of the pumps, spouts, hoses and nozzles. 6. Inspect the cathodic protection. 7. Inspect the grounding system (if present), the electrical continuity of the pipelines, and other metallic elements if there is no documentation of periodic inspections by maintenance personnel 8. In tanks that can be visually inspected, measure the wall thickness.	Not required for underground storage tanks used for internal consumption	Not required for underground storage tanks at filling stations
Tightness Testing	Responsible Authority	Internal inspector	DoD approved or equivalent Spanish contractor	DoD approved or equivalent Spanish contractor
	Frequency	5 years for tanks containing POL Class B.	5 years for operational tanks containing POL.	5 years for active tanks.

		POL Tank Farm	POL Storage Facility for Internal Consumption	Filling Station
		10 years for tanks containing POL class C or D. 3 years for tanks installed before 9 August 1988 that are not double-walled or have a spill retention volume and do not have a leak detection system.	10 years for decommissioned tanks.	10 years for decommissioned tanks.
	Scope of Tightness Test	Tanks: Test shall ensure a leak detection of 100 ml/h and shall be evaluated in accordance with recognized industry standards. Piping: Test shall be conducted in accordance with recognized industry standards or a pressure test shall be conducted at 2 bar (relative measure) during 1 hour. For piping associated to pumps, the test shall be carried out at a pressure of 1.5 times the maximum operating pressure of the pump during 1 hour.	Tanks: Only tanks that are not double-walled and do not have a leak detection system are required to undergo a tightness test. The test shall ensure a leak detection of 100 ml/h and shall be evaluated in accordance with recognized industry standards. Tightness tests are not required for any USTs containing fuel oil. Piping: Test shall be conducted in accordance with recognized industry standards.	Tanks: Only tanks that are not double-walled and do not have a leak detection system shall undergo a tightness test. The test shall ensure a leak detection of 100 ml/h and shall be evaluated in accordance with recognized industry standards. Piping: Test shall be conducted in accordance with recognized industry standards or a pressure test shall be conducted at 2 bar (relative measure) during 1 hour. For piping associated to pumps, the test shall be carried out at a pressure of 1.5 times the maximum operating pressure of the pump during 1 hour.

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All notes appear at the end of the table.

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Acenaphthene	83329		100	
Acenaphthylene	208968		5,000	
Acetaldehyde (I)	75070		1,000	
Acetaldehyde, chloro-	107200		1,000	P
Acetaldehyde, trichloro-	75876		5,000	P
Acetamide	60355		100	
Acetamide, N-(aminothioxomethyl)-	591082		1,000	
Acetamide, N-(4-ethoxyphenyl)-	62442		100	
Acetamide, 2-fluoro-	640197		100	P
Acetamide, N-9H-fluoren-2-yl-	53963		1	
Acetic acid	64197		5,000	
Acetic acid (2,4-dichlorophenoxy)-salts and esters	94757		100	
Acetic acid, lead(2+) salt	301042		10	
Acetic acid, thallium(1+) salt	563688		1000	
Acetic acid, (2,4,5-trichlorophenoxy)	93765		1,000	
Acetic acid, ethyl ester (I)	141786		5,000	
Acetic acid, fluoro-, sodium salt	62748		10	P
Acetic anhydride	108247		5,000	
Acetone (I)	67641		5,000	
Acetone cyanohydrin	75865	1,000	10	P
Acetone thiosemicarbazide	1752303	1,000/10,000	1	
Acetonitrile (I,T)	75058		5,000	
Acetophenone	98862		5,000	
2-Acetylaminofluorene	53963		1	
Acetyl bromide	506967		5,000	
Acetyl chloride (C,R,T)	75365		5,000	
1-Acetyl-2-thiourea	591082		1	P
Acrolein	107028	500	1	P
Acrylamide	79061	1,000/10,000	5,000	
Acrylic acid (I)	79107		5,000	
Acrylonitrile	107131	10,000	100	
Acrylyl chloride	814686	100	1	
Adipic acid	124049		5,000	
Adiponitrile	111693	1,000	1	
Aldicarb	116063	100/10,000	1	P
Aldrin	309002	500/10,000	1	P
Allyl alcohol	107186	1,000	100	P
Allylamine	107119	500	1	
Allyl chloride	107051		1,000	
Aluminum phosphide (R,T)	20859738	500	100	P
Aluminum sulfate	10043013		5,000	
4-Aminobiphenyl	92671		1	
5-(Aminomethyl)-3-isoxazolol	2763964		1,000	P
Aminopterin	54626	500/10,000	1	
4-Aminopyridine	504245		1,000	P

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Amiton	78535	500	1	
Amiton oxalate	3734972	100/10,000	1	
Amitrole	61825		10	
Ammonia	7664417	500	100	
Ammonium acetate	631618		5,000	
Ammonium benzoate	1863634		5,000	
Ammonium bicarbonate	1066337		5,000	
Ammonium bichromate	7789095		10	
Ammonium bifluoride	1341497		100	
Ammonium bisulfite	10192300		5,000	
Ammonium carbamate	1111780		5,000	
Ammonium carbonate	506876		5,000	
Ammonium chloride	12125029		5,000	
Ammonium chromate	7788989		10	
Ammonium citrate, dibasic	3012655		5,000	
Ammonium fluoborate	13826830		5,000	
Ammonium fluoride	12125018		100	
Ammonium hydroxide	1336216		1,000	
Ammonium oxalate	6009707 5972736 14258492		5,000	
Ammonium picrate (R)	131748		10	P
Ammonium silicofluoride	16919190		1,000	
Ammonium sulfamate	7773060		5,000	
Ammonium sulfide	12135761		100	
Ammonium sulfite	10196040		5,000	
Ammonium tartrate	14307438 3164292		5,000	
Ammonium thiocyanate	1762954		5,000	
Ammonium vanadate	7803556		1,000	P
Amphetamlne	300629	1,000	1	
Amyl acetate: Iso-Amyl acetate Sec-Amyl acetate Tert-Amyl acetate	628637 123922 626380 625161		5,000	
Aniline (I,T)	62533	1,000	5,000	
Aniline, 2,4,6- trimethyl	88051	500	1	
o-Anisidine	90040		100	
Anthracene	120127		5,000	
Antimony [4]	7440360		5,000	
Antimony pentachloride	7647189		1,000	
Antimony pentafluoride	7783702	500	1	
Antimony potassium tartrate	28300745		100	
Antimony tribromide	7789619		1,000	
Antimony trichloride	10025919		1,000	
Antimony trifluoride	7783564		1,000	
Antimony trioxide	1309644		1,000	
Antimycin A	1397940	1,000/10,000	1	
ANTU (Thiourea 1-Naphthalenyl)	86884	500/10,000	100	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Argentate(1-), bis(cyano-C)-, potassium	506616		1	P
Aroclor 1016	12674112		1	
Aroclor 1221	11104282		1	
Aroclor 1232	11141165		1	
Aroclor 1242	53469219		1	
Aroclor 1248	12672296		1	
Aroclor 1254	11097691		1	
Aroclor 1260	11096825		1	
Aroclors	1336363		1	
Arsenic [4]	7440382		1	
Arsenic acid H_3AsO_4	1327522 7778394		1	P
Arsenic disulfide	1303328		1	
Arsenic oxide As_2O_3	1327533		1	P
Arsenic oxide As_2O_5	1303282		1	P
Arsenic pentoxide	1303282	100/10,000	1	P
Arsenic trichloride	7784341		1	
Arsenic trioxide	1327533		1	P
Arsenic trisulfide	1303339		1	
Arsenous oxide	1327533	100/10,000	1	P
Arsenous trichloride	7784341	500	5,000	
Arsine	7784421	100	1	
Arsine, diethyl-	692422		1	P
Arsinic acid, dimethyl-	75605		1	
Arsorous dichloride, phenyl-	696286		1	P
Asbestos [5]	1332214		1	
Auramine	492808		100	
Azaserine	115026		1	
Aziridine	151564		1	P
Azindine, 2-methyl-	75558		1	P
Azirino[2',3',3,4]pyrrolo[1,2-a]indole-4, 7-dione,6-amino-8-[[aminocarbonylooxy) methyl]-1,1a,2,8,8a,8b-hexahydro-8a-methoxy-5-methyl-,[1aS-(1a-alpha, 8-beta, 8a-alpha, 8b-alpha)]-	50077		10	
Azinphos-ethyl	2642719	100/10,000	100	
Azinphos-methyl	86500	10/10,000	1	
Barium cyanide	542621		10	P
Benz[j]aceanthrylene, 1,2-dihydro-3-methyl-	56495		10	
Benz[c]acridine	225514		100	
Benzal chloride	98873	500	5,000	
Benzamide, 3,5-dichloro-N-(1,1-dimethyl-2-propynyl)-	23950585		5,000	
Benz[a]anthracene	56553		10	
1,2-Benzanthracene	56553		10	
Benz[a]anthracene, 7,12-dimethyl-	57976		1	
Benzenamine (I,T)	62533		5,000	
Benzenamine, 3-(Trifluoromethyl)	98168	500	1	
Benzenamine, 4,4'-carbonimidoylbis (N,N-	492808		100	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
dimethyl-				
Benzenamine, 4-chloro-	106478		1,000	P
Benzenamine, 4-chloro-2-methyl-, hydrochloride	3165933		100	
Benzenamine, N,N-dimethyl-4-(phenylazo-)	60117		10	
Benzenamine, 2-methyl-	95534		100	
Benzenamine, 4-methyl-	106490		100	
Benzenamine, 4,4'-methylenebis(2-chloro-	101144		10	
Benzenamine, 2-methyl-, hydrochloride	636215		100	
Benzenamine, 2-methyl-5-nitro-	99558		100	
Benzenamine, 4-nitro-	100016		5,000	P
Benzene (I,T)	71432		10	
Benzene, 1-(Chloromethyl)-4-Nitro-	100141	500/10,000	1	
Benzeneacetic acid, 4-chloro-alpha- (4-chlorophenyl)-alpha-hydroxy-, ethyl ester	510156		10	
Benzene, 1-bromo-4-phenoxy-	101553		100	
Benzenearsonic Acid	98055	10/10,000	1	
Benzenebutanoic acid, 4-[bis(2-chloroethyl)amino]-	305033		10	
Benzene, chloro-	108907		100	
Benzene, chloromethyl-	100447		100	P
Benzenediamin, ar-methyl-	25376458 95807 496720 823405		10	
1,2-Benzenedicarboxylic acid, dioctyl ester	117840		5,000	
1,2-Benzenedicarboxylic acid, [bis(2-ethylhexyl)]-ester	117817		100	
1,2-Benzenedicarboxylic acid, dibutyl ester	84742		10	
1,2-Benzenedicarboxylic acid, diethyl ester	84662		1,000	
1,2-Benzenedicarboxylic acid, dimethyl ester	131113		5,000	
Benzene, 1,2-dichloro-	95501		100	
Benzene, 1,3-dichloro-	541731		100	
Benzene, 1,4-dichloro-	106467		100	
Benzene, 1,1'-(2,2-dichloroethylidene)bis[4-chloro-	72548		1	
Benzene, dichloromethyl-	98873		5,000	
Benzene, 1,3-diisocyanotomethyl- (R,T)	584849 91087 264716254		100	
Benzene, dimethyl (I,T)	1330207		100	
m-Benzene, dimethyl	108383		1,000	
o-Benzene, dimethyl	95476		1,000	
p-Benzene, dimethyl	106423		100	
1,3-Benzenediol	108463		5,000	
1,2-Benzenediol, 4-[1-hydroxy-2-(methylamino)ethyl]- (R) -	51434		1,000	P
Benzeneethanamine, alpha, alpha-dimethyl-	122098		5,000	P
Benzene, hexachloro-	118741		10	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Benzene, hexahydro- (I)	110827		1,000	
Benzene, hydroxy-	108952		1,000	
Benzene, methyl-	108883		1,000	
Benzene, 2-methyl-1,3-dinitro-	606202		100	
Benzene, 1-methyl-2,4-dinitro-	121142		10	
Benzene, 1-methylethyl- (I)	98828		5,000	
Benzene, nitro-	98953		1,000	
Benzene, pentachloro-	608935		10	
Benzene, pentachloronitro-	82688		100	
Benzenesulfonic acid chloride (C,R)	98099		100	
Benzenesulfonyl chloride	98099		100	
Benzene, 1,2,4,5-tetrachloro-	95943		5,000	
Benzenethiol	108985		100	P
Benzene, 1,1'-(2,2,2-tri-chloroethylidene)bis[4-chloro-	50293		1	
Benzene, 1,1'-(2,2,2-trichloroethylidene) bis[4-methoxy-	72435		1	
Benzene, (trichloromethyl)-	98077		10	
Benzene, 1,3,5-trinitro-	99354		10	
Benzidine	92875		1	
Benzimidazole, 4,5-Dichloro-2-(Trifluoromethyl)-	3615212	500/10,000	1	
1,2-Benzisothiazol-3(2H)-one, 1,1-dioxide	81072		100	
Benzo[a]anthracene	56553		10	
Benzo[b]fluoranthene	205992		1	
Benzo[k]fluoranthene	207089		5,000	
Benzo[j,k]fluorene	206440		100	
1,3-Benzodioxole, 5-(1-propenyl)-	120581		100	
1,3-Benzodioxole, 5-(2-propenyl)-	94597		100	
1,3-Benzodioxole, 5-propyl-	94586		10	
Benzoic acid	65850		5,000	
Benzonitrile	100470		5,000	
Benzo[rst]pentaphene	189559		10	
Benzo[ghi]perylene	191242		5,000	
2H-1-Benzopyran-2-one, 4-hydroxy-3-(3-oxo-1-phenyl-butyl)-, & salts, when present at concentrations greater than 0.3%	81812		100	P
Benzo[a]pyrene	50328		1	
3,4-Benzopyrene	50328		1	
p-Benzoquinone	106514		10	
Benzotrichloride (C,R,T)	98077	500	10	
Benzoyl chloride	98884		1,000	
1,2-Benzphenanthrene	218019		100	
Benzyl chloride	100447	500	100	P
Benzyl cyanide	140294	500	1	
Beryllium [4]	7440417		10	P
Beryllium chloride	7787475		1	
Beryllium fluoride	7787497		1	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Beryllium nitrate	13597994 7787555		1	
alpha-BHC	319846		10	
beta-BHC	319857		1	
delta-BHC	319868		1	
gamma-BHC	58899		1	
Bicyclo [2,2,1]Heptane-2-carbonitrile, 5-chloro-6-(((Methylamino)Carbonyl) Oxy)Imino)-,(1s-(1-alpha, 2-beta, 4-alpha, 5-alpha, 6E))-	15271417	500/10,000	1	
2,2'-Bioxirane	1464535		10	
Biphenyl	92524		100	
(1,1'-Biphenyl)-4,4'diamine	92875		1	
(1,1'-Biphenyl)-4,4'diamine, 3,3'dichloro-	91941		1	
(1,1'-Biphenyl)-4,4'diamine, 3,3'dimethoxy-	119904		10	
(1,1'-Biphenyl)-4,4'diamine, 3,3'dimethyl-	119937		10	
Bis(chloromethyl) ketone	534076	10/10,000	1	
Bis(2-chloroethyl)ether	111444		10	
Bis(2-chloroethoxy)methane	111911		1,000	
Bis(2-ethylhexyl)phthalate	117817		100	
Bitoscanate	4044659	500/10,000	1	
Boron trichloride	10294345	500	1	
Boron trifluoride	7637072	500	1	
Boron trifluoride compound with methyl ether (1:1)	353424	1,000	1	
Bromoacetone	598312		1,000	P
Bromadiolone	28772567	100/10,000	1	
Bromine	7726956	500	1	
Bromoform	75252		100	
4-Bromophenyl phenyl ether	101553		100	
Brucine	357573		100	P
1,3-Butadiene	106990		10	
1,3-Butadiene, 1,1,2,3,4,4-hexachloro-	87683		1	
1-Butanamine, N-butyl-N-nitroso-	924163		10	
1-Butanol	71363		5,000	
2-Butanone	78933		5,000	
2-Butanone peroxide (R,T)	1338234		10	
2-Butanone, 3,3-dimethyl-1-(methylthio)-, O[(methylamno)carbonyl] oxime	39196184		100	P
2-Butenal	123739 4170303		100	
2-Butene, 1,4-dichloro- (I,T)	764410		1	
2-Butenoic acid, 2-methyl-, 7[[2,3-dihydroxy-2-(1-meth- oxyethyl)-3-methyl-1-oxobutoxy] methyl]-2,3,5,7a-tetrahydro-1H-pyrrolizin-1-yl ester, [1S-[1-alpha(Z),7(2S*,3R*), 7a-alpha]]-	303344		10	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Butyl acetate: iso-Butyl acetate sec-Butyl acetate tert-Butyl acetate	123864 110190 105464 540885		5,000	
n-Butyl alcohol (I)	71363		5,000	
Butylamine: iso-Butylamine sec-Butylamine tert-Butylamine	109739 78819 513495 13952846 75649		1,000	
Butyl benzyl phthalate	85687		100	
n-Butyl phthalate	84742		10	
Butyric acid	107926		5,000	
iso-Butyric acid	79312			
Cacodylic acid	75605		1	
Cadmium (2+) [4]	7440439		10	
Cadmium acetate	543908		10	
Cadmium bromide	7789426		10	
Cadmium chloride	10108642		10	
Cadmium oxide	1306190	100/10,000	1	
Cadmium stearate	2223930	1,000/10,000	1	
Calcium arsenate	7778441	500/10,000	1	
Calcium arsenite	52740166		1	
Calcium carbide	75207		10	
Calcium chromate	13765190		10	
Calcium cyanamide	156627		1,000	
Calcium cyanide Ca(CN)2	592018		10	P
Calcium dodecylbenzenesulfonate	26264062		1,000	
Calcium hypochlorite	7778543		10	
Camphechlor	8001352	500/10,000	1	
Camphene, octachloro-	8001352		1	P
Cantharidin	56257	100/10,000	1	
Carbachol chloride	51832	500/10,000	1	
Captan	133062		10	
Carbamic acid, ethyl ester	51796		100	
Carbamic acid, methylnitroso-, ethyl ester	615532		1	
Carbamic acid, Methyl-, 0-(((2,4-Dimethyl-1, 3-Dithiolan-2-yl)Methyllene)Amino)-	26419738	100/10,000	1	
Carbamic chloride, dimethyl-	79447		1	
Carbamodithioic acid, 1,2-ethaneiylbis, salts & esters	111546		5,000	
Carbamothioic acid, bis(1-methylethyl)-, S-(2,3-dichloro-2-propenyl) ester	2303164		100	
Carbaryl	63252		100	
Carbofuran	1563662	10/10,000	10	
Carbon disulfide	75150	10,000	100	P
Carbon oxyfluoride (R,T)	353504		1,000	
Carbon tetrachloride	56235		10	
Carbonic acid, dithallium(1+) salt	6533739		100	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Carbonic dichloride	75445		10	P
Carbonic difluoride	353504		1,000	
Carbonochloridic acid, methyl ester	79221		1,000	
Carbonyl Sulfide	463581		100	
Carbophenothion	786196	500	1	
Catechol	120809		100	
Chloral	75876		5,000	
Chlorambem	133904		100	
Chlorambucil	305033		10	
Chlordane	57749	1,000	1	
Chlordane, alpha & gamma isomers	57749		1	
Chlordane, technical	57749		1	
Chlorfenvinfos	470906	500	1	
Chlorinated champhene (Campheclor)	8001352		1	
Chlorine	7782505	100	10	
Chlormephos	24934916	500	1	
Chlormequat chloride	999815	100/10,000	1	
Chlornaphazine	494031		100	
Choroacetaldehyde	107200		1,000	P
Chloroacetophenone	532274		100	
Chloroacetic acid	79118	100/10,000	100	
p-Chloroaniline	106478		1,000	P
Chlorobenzene	108907		100	
Chlorobenzilate	510156		10	
p-Chloro-m-cresol (4)	59507		5,000	
1-Chloro-2,3-epoxypropane	106898		100	
Chlorodibromomethane	124481		100	
Chloroethane	75003		100	
Chloroethanol	107073	500	1	
Chloroethyl chlorofomate	627112	1,000	1	
2-Chloroethyl vinyl ether	110758		1,000	
Chloroform	67663	10,000	10	
Chloromethane	74873		100	
Chloromethyl ether	542881	100	1	P
Chloromethyl methyl ether	107302	100	1	
beta-Chloronaphthalene	91587		5,000	
2-Chloronaphthalene	91587		5,000	
Chlorophacinone	3691358	100/10,000	1	
o-Chlorophenol (2)	95578		100	
4-Chlorophenyl phenyl ether	7005723		5,000	
1-(o-Chlorophenyl)thiourea	5344821		100	P
Chloroprene	126998		100	
3-Chloropropionitrile	542767		1,000	P
Chlorosulfonic acid	7790945		1,000	
4-Chloro-o-toluidine, hydrochloride	3165933		100	
Chlorpyrifos	2921882		1	
Chloroxuron	1982474	500/10,000	1	
Chlorthiophos	21923239	500	1	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Chromic acetate	1066304		1,000	
Chromic acid	11115745 7738945		10	
Chromic acid H_2CrO_4, calcium salt	13765190		10	
Chromic chloride (Chromium chloride)	10025737	1/10,000	1	
Chromic sulfate	10101538		1,000	
Chromium [4]	7440473		5,000	
Chromous chloride	10049055		1,000	
Chrysene	218019		100	
Cobalt, ((2,2'-(1,2-ethanediylbis (Nitrilo-methylidyne))Bis(6-fluoro-phenolato))(2-)-N,N',O,O')-,	62207765	100/10,000	1	
Cobaltous bromide	7789437		1,000	
Cobalt carbonyl	10210681	10/10,000	1	
Cobaltous formate	544183		1,000	
Cobaltous sulfamate	14017415		1,000	
Coke Oven Emissions	NA		1	
Colchicine	64868	10/10,000	1	
Copper [4]	7440508		5,000	
Copper cyanide	544923		10	P
Coumaphos	56724	100/10,000	10	
Coumatetralyl	5836293	500/10,000	1	
Creosote	8001589		1	
Cresol(s) (Phenol, Methyl): m-Cresol o-Cresol p-Cresol	1319773 108394 95487 106445	 1,000/10,000	100 100 100 100	
Cresylic acid: m-Cresylic acid o-Cresylic acid p-Cresylic acid	1319773 108394 95487 106445		100 100 100 100	
Crimidine	535897	100/10,000	1	
Crotonaldehyde	123739 4170303	1,000 1,000	100 100	
Cumene (I)	98828		5,000	
Cupric acetate	142712		100	
Cupric acetoarsenite	12002038		1	
Cupric chloride	7447394		10	
Cupric nitrate	3251238		100	
Cupric oxalate	5893663		100	
Cupric sulfate	7758987		10	
Cupric sulfate, ammoniated	10380297		100	
Cupric tartrate	815827		100	
Cyanides (soluble salts and complexes) not otherwise specified	57125		10	P
Cyanogen	460195		100	P
Cyanogen bromide	506683	500/10,000	1,000	
Cyanogen chloride	506774		10	P
Cyanogen iodide (Iodine cyanide)	506785	1,000/10,000	1	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Cyanophos	2636262	1,000	1	
Cyanuric fluoride	675149	100	1	
2,5-Cyclohexadiene-1,4-dione	106514		10	
Cyclohexane (I)	110827		1,000	
Cyclohexane, 1,2,3,4,5,6-hexachloro, (1-alpha, 2-alpha, 3-beta, 4-alpha, 5-alpha, 6-beta)-	58899		1	
Cyclohexanone (I)	108941		5,000	
2-Cyclohexanone	131895		100	P
Cycloheximide	66819	100/10,000	1	
Cyclohexylamine	108918	10,000	1	
1,3-Cyclopentadiene, 1,2,3,4,5,5-hexachloro-	77474		10	
Cyclophosphamide	50180		10	
2,4-D Acid	94757		100	
2,4-D Ester	94111 94791 94804 1320189 1928387 1928616 1929733 2971382 25168267 53467111		100	
2,4-D, salts & esters (2,4-Dichlorophenoxyacetic Acid)	94757		100	
Daunomycin	20830813		10	
Decarborane(14)	17702419	500/10,000	1	
Demeton	8065483	500	1	
Demeton-S-Methyl	919868	500	1	
DDD, 4,4'DDD	72548		1	
DDE, 4,4'DDE	72559		1	
DDT, 4,4'DDT	50293		1	
DEHP (Diethylhexyl phthalate)	117817		100	
Diallate	2303164		100	
Dialifor	10311849	100/10,000	1	
Diazinon	333415		1	
Diazomethane	334883		100	
Dibenz[a,h]anthracene	53703		1	
1,2:5,6-Dibenzanthracene	53703		1	
Dibenzo[a,h]anthracene	53703		1	
Dibenzofuran	132649		100	
Dibenz[a,i]pyrene	189559		10	
1,2-Dibromo-3-chloropropane	96128		1	
Dibromoethane	106934		1	
Diborane	19287457	100	1	
Dibutyl phthalate	84742		10	
Di-n-butyl phthalate	84742		10	
Dicamba	1918009		1,000	
Dichlobenil	1194656		100	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Dichlone	117806		1	
Dichlorobenzene	25321226		100	
m-Dichlorobenzene (1,3)	541731		100	
o-Dichlorobenzene (1,2)	95501		100	
p-Dichlorobenzene (1,4)	106467		100	
3,3'-Dichlorobenzidine	91941		1	
Dichlorobromomethane	75274		5,000	
1,4-Dichloro-2-butene (I,T)	764410		1	
Dichlorodifluoromethane	75718		5,000	
1,1-Dichloroethane	75343		1,000	
1,2-Dichloroethane	107062		100	
1,1-Dichloroethylene	75354		100	
1,2-Dichloroethylene	156605		1,000	
Dichloroethyl ether	11444	10,000	10	
Dichloroisopropyl ether	108601		1,000	
Dichloromethoxy ethane	111911		1,000	
Dichloromethyl ether	542881		1	P
Dichloromethylphenylsilane	149746	1,000	1	
2,4-Dichlorophenol	120832		100	
2,6-Dichlorophenol	87650		100	
Dichlorophenylarsine	696286		1	P
Dichloropropane	26638197		1,000	
1,1-Dichloropropane	78999			
1,3-Dichloropropane	142289			
1,2-Dichloropropane	78875		1,000	
Dichloropropane--Dichloropropene (mixture)	8003198		100	
Dichloropropene	26952238		100	
2,3-Dichloropropene	78886			
1,3-Dichloropropene	542756		100	
2,2-Dichloropropionic acid	75990		5,000	
Dichlorvos	62737	1,000	10	
Dicofol	115322		10	
Dicrotophos	141662	100	1	
Dieldrin	60571		1	P
1,2:3,4-Diepoxybutane (I,T)	1464535	500	10	
Diethanolamine	111422		100	
Diethyl chlorophosphate	814493	500	1	
Diethylamine	109897		1,000	
Diethylarsine	692422		1	P
Diethylcarbmazine citrate	1642542	100/10,000	1	
1,4-Diethylenedioxide	123911		100	
Diethylhexyl phthalate	117817		100	
N,N-Diethylaniline	91667		1,000	
N,N'-Diethylhydrazine	1615801		10	
O,O-Diethyl S-methyl dithiophosphate	3288582		5,000	
Diethyl-p-nitrophenyl phosphate	311455		100	P
Diethyl phthalate	84662		1,000	
O,O-Diethyl O-pyrazinyl phosphorothioate	297972		100	P

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Diethylstilbestrol	56531		1	
Diethyl sulfate	64675		10	
Digitoxin	71636	100/10,000	1	
Diglycidyl ether	2238075	1,000	1	
Digoxin	20830755	10/10,000	1	
Dihydrosafrole	94586		10	
Diisopropyfluorophosphate	55914		100	P
Diisopropylfluorophosphate, 1,4,5,8-Dimethanonaphthalene, 1,2,3,4,10,10-10-hexachloro-1,4,4a,5,8,8a-hexahydro-, (1-alpha, 4-alpha, 4a-beta, 5-alpha, 8-alpha, 8a-beta)-	309002		1	P
1,4,5,8-Dimethanonaphthalene, 1,2,3,4,10,10-hexachloro-1,4,4a,5,8,8a-hexahydro, (1-alpha, 4-alpha, 4a-beta, 5a-beta, 8-beta, 8a-beta)-	465736		1	P
2,7:3,6-Dimethanonaphth[2,3 b]oxirene,3,4,5,6,9,9-hexachloro-1a,2,2a,3,6,6a,7,7a-octahydro-,(1a-alpha, 2-beta, 2a-alpha, 3-beta, 6-beta, 6a-alpha, 7beta, 7aalpha)-	60571		1	P
2,7:3,6 Dimethanonaphth[2,3-b]oxirene, 3,4,5,6,9,9-hexachloro-1a,2,2a,3,6,6a,7,7a-octa-hydro-, (1a-alpha, 2-beta, 2a-beta, 3-alpha, 6-alpha, 6a-beta, 7-beta, 7a-alpha)-	72208		1	P
Dimethoate	60515		10	P
3,3'-Dimethoxybenzidine	119904		10	
Dimefox	115264	500	1	
Dimethoate	60515	500/10,000	10	
Dimethyl Phosphorochloridothioate	2524030	500	1	
Dimethyl sulfate	77781	500	100	
Dimethylamine (I)	124403		1,000	
p-Dimethylaminoazobenzene	60117		10	
7,12-Dimethylbenz[a]anthracene	57976		1	
3,3'-Dimethylbenzidine	119937		10	
alpha,alpha-Dimethylbenzylhydroperoxide(R)	80159		10	
Dimethylcarbamoyl chloride	79447		1	
Dimethylformamide	68122		100	
Dimethyldichlorosilane	75785	500	1	
1,1-Dimethylhydrazine	57147	1,000	10	
1,2-Dimethylhydrazine	540738		1	
alpha, alpha-Dimethylphenethylamine	122098		5,000	P
Dimethyl-p-phenylenediamine	99989	10/10,000	1	
2,4-Dimethylphenol	105679		100	
Dimethyl phthalate	131113		5,000	
Dimethyl sulfate	77781		100	
Dimetilan	644644	500/10,000	1	
Dinitrobenzene (mixed): m-Dinitrobenzene o-Dinitrobenzene p-Dinitrobenzene	25154545 99650 528290 100254		100	
4,6-Dinitro-o-cresol and salts	534521	10/10,000	10	P

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Dinitrophenol: 2,5-Dinitrophenol 2,6-Dinitrophenol	25550587 329715 573568		10	
2,4-Dinitrophenol	51285		10	P
Dinitrotoluene	25321146		10	
3,4-Dinitrotoluene	610399			
2,4-Dinitrotoluene	121142		10	
2,6-Dinitrotoluene	606202		100	
Dinoseb	88857	100/10,000	1,000	P
Dinoterb	1420071	500/10,000	1	
Di-n-octyl phthalate	117840		5,000	
1,4-Dioxane	123911		100	
Dioxathion	78342	500	1	
Diphacinone	82666	10/10,000	1	
1,2-Diphenylhydrazine	122667		10	
Diphosphoramide, octamethyl-	152169	100	100	P
Diphosphoric acid, tetraethyl ester	107493		10	P
Dipropylamine	142847		5,000	
Di-n-propylnitrosamine	621647		10	
Diquat	85007 2764729		1,000	
Disulfoton	298044	500	1	P
Dithiazanine iodide	514738	500/10,000	1	
Dithiobiuret	541537	100/10,000	100	P
Diuron	330541		100	
Dodecylbenzenesulfonic acid	27176870		1,000	
Emetine, Dihydrochloride	316427	1/10,000	1	
Endosulfan	115297	10/10,000	1	P
alpha-Endosulfan	959988		1	
beta-Endosulfan	33213659		1	
Endosulfant sulfate	1031078		1	
Endothall	145733		1,000	P
Endothion	2778043	500/10,000	1	
Endrin	72208	500/10,000	1	P
Endrin aldehyde	7421934		1	
Endrin & metabolites	72208		1	P
Epichlorohydrin	106898	1,000	100	
Epinephrine	51434		1,000	P
EPN	2104645	100/10,000	1	
1,2-Epoxybutane	106887		100	
Ergocalciferol	50146	1,000/10,000	1	
Ergotamine tartrate	379793	500/10,000	1	
Ethanal	75070		1,000	
Ethanamine, N-ethyl-N-nitroso-	55185		1	
1,2-Ethanediamine, N,N-dimethyl-N'-2-pyridinyl-N'-(2-thienylmethyl)-	91805		5,000	
Ethane, 1,2-dibromo-	106934		1	
Ethane, 1,1-dichloro-	75343		1,000	
Ethane, 1,2-dichloro-	107062		100	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Ethanedinitrile	460195		100	P
Ethane, hexachloro-	67721		100	
Ethane, 1,1'-[methylenebis(oxy)]bis(2-chloro-	111911		1,000	
Ethane, 1,1'-oxybis-	60297		100	
Ethane, 1,1'-oxybis(2-chloro-	111444		10	
Ethane, pentachloro-	76017		10	
Ethanesulfonyl chloride, 2-chloro	1622328	500	1	
Ethane, 1,1,1,2-tetrachloro-	630206		100	
Ethane, 1,1,2,2-tetrachloro-	79345		100	
Ethanethioamide	62555		10	
Ethane, 1,1,1-trichloro-	71556		1,000	
Ethane, 1,1,2-trichloro-	79005		100	
Ethanimidothioic acid, N-[[(methylamino) carbonyl]oxy]-, methyl ester	16752775		100	P
Ethanol, 1,2-Dichloro-, acetate	10140871	1,000	1	
Ethanol, 2-ethoxy-	110805		1,000	
Ethanol, 2,2'-(nitrosoimino)bis-	1116547		1	
Ethanone, 1-phenyl-	98862		5,000	
Ethene, chloro-	75014		1	
Ethene, 2-chloroethoxy-	110758		1,000	
Ethene, 1,1-dichloro-	75354		100	
Ethene, 1,2-dichloro- (E)	156605		1,000	
Ethene, tetrachloro-	127184		100	
Ethene, trichloro-	79016		100	
Ethion	563122	1,000	10	
Ethoprophos	13194484	1,000	1	
Ethyl acetate (I)	141786		5,000	
Ethyl acrylate (I)	140885		1,000	
Ethylbenzene	100414		1,000	
Ethylbis(2-Chloroethyl)amine	538078	500	1	
Ethyl carbamate (urethane)	51796		100	
Ethyl chloride	75003		100	
Ethyl cyanide	107120		10	P
Ethylenebisdithiocarbamic acid, salts & esters	111546		5,000	
Ethylenediamine	107153		5,000	
Ethylenediamine-tetraacetic acid (EDTA)	60004		5,000	
Ethylene dibromide	106934		1	
Ethylene dichloride	107062		100	
Ethylene fluorohydrin	371620	10	1	
Ethylene glycol	107211		5,000	
Ethylene glycol monoethyl ether	110805		1,000	
Ethylene oxide (I,T)	75218	1,000	10	
Ethylenediamine	107153	10,000	5,000	
Ethylenethiourea	96457		10	
Ethyleneimine	151564	500	1	P
Ethyl ether (I)	60297		100	
Ethylthiocyanate	542905	10,000	1	
Ethylidene dichloride	75343		1,000	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Ethyl methacrylate	97632		1,000	
Ethyl methanesulfonate	62500		1	
Famphur	52857		1,000	P
Fenamlphos	22224926	10/10,000	1	
Fenltrothion	122145	500	1	
Fensulfothion	115902	500	1	
Ferric ammonium citrate	1185575		1,000	
Ferric ammonium oxalate	2944674 55488874		1,000	
Ferric chloride	7705080		1,000	
Ferric fluoride	7783508		100	
Ferric nitrate	10421484		1,000	
Ferric sulfate	10028225		1,000	
Ferrous ammonium sulfate	10045893		1,000	
Ferrous chloride	7758943		100	
Ferrous sulfate	7720787 7782630		1,000	
Fluenetil	4301502	100/10,000	1	
Fluoranthene	206440		100	
Fluorene	86737		5,000	
Fluorine	7782414	500	10	P
Fluoroacetamide	640197	100/10,000	100	P
Fluoracetic acid	144490	10/10,000	1	
Fluoroacetic acid, sodium salt	62786		10	P
Fluoroacetyl chloride	359068	10	1	
Fluorouracil	51218	500/10,000	1	
Fonofos	944229	500	1	
Formaldehyde	50000	500	100	
Formaldehyde cyanohydrin	107164	1,000	1	
Formetanate hydrochloride	23422539	500/10,000	1	
Formothion	2540821	100	1	
Formparanate	17702577	100/10,000	1	
Formic acid (C,T)	64186		5,000	
Fosthletan	21548323	500	1	
Fubendazole	3878191	100/10,000	1	
Fulminic acid, mercury(2^-) salt (R,T)	628864		10	P
Fumaric acid	110178		5,000	
Furan (I)	110009	500	100	
Furan, tetrahydro- (I)	109999		1,000	
2-Furancarboxaldehyde (I)	98011		5,000	
2,5-Furandione	108316		5,000	
Furfural (I)	98011		5,000	
Furfuran (I)	110009		100	
Gallium trichloride	13450903	500/10,000	1	
Glucopyranose, 2-deoxy-2-(3-methyl-3-nitrosoureido)-	18883664		1	
D-Glucose, 2-deoxy-2-[[(methylnitrosoamino)-carbonyl]amino]-	18883664		1	
Glycidylaldehyde	765344		10	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Glycol ethers[6]				**
Guanidine, N-methyl-N'-nitro-N-nitroso-	70257		10	
Guthion	86500		1	
Heptachlor	76448		1	P
Heptachlor epoxide	1024573		1	
Hexachlorobenzene	118741		10	
Hexachlorobutadiene	87683		1	
Hexachlorocyclohexane (gamma isomer)	58899		1	
Hexachlorocyclopentadiene	77474	100	10	
Hexachloroethane	67721		100	
Hexachlorophene	70304		100	
Hexachloropropene	1888717		1,000	
Hexaethyl tetraphosphate	757584		100	P
Hexamethylene-1, 6-diisocyanate	822060		100	
Hexamethylphosphoramide	680319		1	
Hexamethylenediamine, N,N'-Dibutyl	4835114	500	1	
Hexane	110543		5,000	
Hexone (Methyl isobutyl ketone)	108101		5,000	
Hydrazine (R,T)	302012	1,000	1	
Hydrazine, 1,2-diethyl-	1615801		10	
Hydrazine, 1,1-dimethyl-	57147		10	
Hydrazine, 1,2-dimethyl-	540738		1	
Hydrazine, 1,2-diphenyl-	122667		10	
Hydrazine, methyl-	60344		10	P
Hydrazinecarbothioamide	79196		100	P
Hydrochloric acid	7647010		5,000	
Hydrocyanic acid	74908	100	10	P
Hydrofluoric acid	7664393		100	
Hydrogen chloride (gas only)	7647010	500	5,000	
Hydrogen cyanide	74908		10	P
Hydrogen fluoride	7664393	100	100	
Hydrogen peroxide (Conc. >52%)	7722841	1,000	1	
Hydrogen phosphide	7803512		100	P
Hydrogen selenide	7783075	10	1	
Hydrogen sulfide	7783064	500	100	
Hydroperoxide, 1-methyl-1-phenylethyl-	80159		10	
Hydroquinone	123319	500/10,000	100	
2-Imidazolidinethione	96457		10	
Indeno(1,2,3-cd)pyrene	193395		100	
Iodomethane	74884		100	
Iron, Pentacarbonyl-	13463406	100	1	
Isobenzan	297789	100/10,000	1	
1,3-Isobenzofurandione	85449		5,000	
Isobutyronitrile	78820	1,000	1	
Isobutyl alcohol (I,T)	78831		5,000	
Isocyanic acid, 3,4-Dichlorophenyl ester	102363	500/10,000	1	
Isodrin	465736	100/10,000	1	P
Isofluorphate	55914	100	100	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Isophorone	78591		5,000	
Isophorone Diisocyanate	4098719	100	1	
Isoprene	78795		100	
Isopropanolamine dodecylbenzene sulfonate	42504461		1,000	
Isopropyl chloroformate	108236	1,000	1	
Isopropylmethylpryrazolyl dimethylcarbamate	119380	500	1	
Isosafrole	120581		100	
3(2H)-Isoxazolone, 5-(aminomethyl)-	2763964		1,000	P
Kepone	143500		1	
Lactonitrile	78977	1,000	1	
Lasiocarpine	303344		10	
Lead acetate	301042			
Lead arsenate	7784409 7645252 10102484		1	
Lead, bis(acetato-O)tetrahydroxytri	1335326		10	
Lead chloride	7758954		10	
Lead fluoborate	13814965		10	
Lead fluoride	7783462		10	
Lead iodide	10101630		10	
Lead nitrate	10099748		10	
Lead phosphate	7446277		10	
Lead stearate	7428480 1072351 52652592 56189094		10	
Lead subacetate	1335326		10	
Lead sulfate	15739807 7446142		10	
Lead sulfide	1314870		10	
Lead thiocyanate	592870		10	
Leptophos	21609905	500/10,000	1	
Lewisite	541253	10	1	
Lindane	58899	1,000/10,000	1	
Lithium chromate	14307358		10	
Lithium hydride	7580678	100	1	
Malathion	121755		100	
Maleic acid	110167		5,000	
Maleic anhydride	108316		5,000	
Maleic hydrazide	123331		5,000	
Malononitrile	109773	500/10,000	1,000	
Manganese, tricarbonyl methylcyclopentadienyl	12108133	100	1	
MDI (Methylene diphenyl diisocyanate)	101688		5,000	
Mechlorethamine	51752	10	1	
MEK (Methyl ethyl ketone)	78933		5,000	
Melphalan	148823		1	
Mephosfolan	950107	500	1	
Mercaptodimethur	2032657		10	
Mercuric acetate	1600277	500/10,000	1	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Mercuric chloride	7487947	500/10,000	1	
Mercuric cyanide	592041		1	
Mercuric nitrate	10045940		10	
Mercuric oxide	21908532	500/10,000	1	
Mercuric sulfate	7783359		10	
Mercuric thiocyanate	592858		10	
Mercurous nitrate	10415755 7782867		10	
Mercury	7439976		1	
Mercury (acetate-O)phenyl-	62384		100	P
Mercury fulminate	628864		10	P
Methacrolein diacetate	10476956	1,000	1	
Methacrylic anhydride	760930	500	1	
Methacrylonitrile (I,T)	126987	500	1,000	
Methacryloyl chloride	920467	100	1	
Methacryloyloxyethyl isocyanate	30674807	100	1	
Methamidophos	10265926	100/10,000	1	
Methanamine, N-methyl-	124403		1,000	
Methanamine, N-methyl-N-nitroso-	62759		10	P
Methane, bromo-	74839		1,000	
Methane, chloro- (I,T)	74873		100	
Methane, chloromethoxy-	107302		1	
Methane, dibromo-	74953		1,000	
Methane, dichloro-	75092		1,000	
Methane, dichlorodifluoro-	75718		5,000	
Methane, iodo-	74884		100	
Methane, isocyanato-	624839		10	P
Methane, oxybis(chloro-	542881		1	P
Methanesulfenyl chloride, trichloro-	594423		100	P
Methanesulfonyl fluoride	558258	1,000	1	
Methanesulfonic acid, ethyl ester	62500		1	
Methane, tetrachloro-	56235		10	
Methane, tetranitro- (R)	509148		10	P
Methane, tribromo-	75252		100	
Methane, trichloro-	67663		10	
Methane, trichlorofluoro-	75694		5,000	
Methanethiol (I,T)	74931		100	
6,9-Methano-2,4,3-benzodioxathiepin, 6,7,8,9,10, 10-hexa-chloro-1,5,5a,6,9,9a-hexahydro-, 3-oxide	115297		1	P
1,3,4-Metheno-2H-cyclobutal[cd]pentalen-2-one,1,1a,3,3a,4,5,5a,5b,6-decachloroctahydro-	143500		1	
4,7-Methano-1H-indene, 1,4,5,6,7,8,8 heptachloro-3a,4,7,7a-tetrahydro-	76448		1	P
4,7-Methano-1H-indene, 1,2,4,5,6,7,8,8 octachloro-2,3,3a,4,7,7a-hexahydro-	57749		1	
Methanol (I)	67561		5,000	
Methapyrilene	91805		5,000	
Methidathion	950378	500/10,000	1	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Methiocarb	2032657	500/10,000	10	P
Methomyl	16752775	500/10,000	100	P
Methoxychlor	72435		1	
Methoxyethylmercuric acetate	151382	500/10,000	1	
Methyl alcohol (I)	67561		5,000	
Methyl aziridine	75558		1	P
Methyl bromide	74839	1,000	1,000	
1-Methylbutadiene (I)	504609		100	
Methyl chloride (I,T)	74873		100	
Methyl 2-chloroacrylate	80637	500	1	
Methyl chlorocarbonate (I,T)	79221		1,000	
Methyl chloroform	71556		1,000	
Methyl chloroformate	79221	500	1,000	
3-Methylcholanthrene	56495		10	
4,4'-Methylenebis(2-chloroaniline)	101144		10	
Methylene bromide	74953		1,000	
Methylene chloride	75092		1,000	
4,4'-Methylenedianiline	101779		10	
Methylene diphenyl diisocyanate (MDI)	101688		5,000	
Methyl ethyl ketone (MEK) (I,T)	78933		5,000	
Methyl ethyl ketone peroxide (R,T)	1338234		10	
Methyl hydrazine	60344	500	10	P
Methyl iodide	74884		100	
Methyl isobutyl ketone	108101		5,000	
Methyl isocyanate	624839	500	10	P
Methyl isothiocyanate	556616	500	1	
2-Methyllactonitrile	75865		10	P
Methyl mercaptan	74931	500	100	
Methyl methacrylate (I,T)	80626		1,000	
Methyl parathion	298000		100	P
Methyl phenkapton	3735237	500	1	
Methyl phosphonic dichloride	676971	100	1	
4-Methyl-2-pentanone (I)	108101		5,000	
Methyl tert-butyl ether	1634044		1,000	
Methyl thiocyanate	556649	10,000	1	
Methylthiouracil	56042		10	
Methyl vinyl ketone	78944	10	1	
Methylmercuric dicyanamide	502396	500/10,000	1	
Methyltrichlorosilane	75796	500	1	
Metolcarb	1129415	100/10,000	1	
Mevinphos	7786347	500	10	
Mexacarbate	315184	500/10,000	1,000	
Mitomycin C	50077	500/10,000	10	
MNNG	70257		10	
Monocrotophos	6923224	10/10,000	1	
Monoethylamine	75047		100	
Monomethylamine	74895		100	
Muscimol	2763964	500/10,000	1,000	P

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Mustard gas	505602	500	1	
Naled	300765		10	
5,12-Naphthaacenedione, 8-acetyl-10-[3 amino-2,3,6-tri-deoxy-alpha-L-lyxo-hexopyranosyl)oxy]-7,8,9,10-tetrahydro-6,8,11-trihydroxy-1-methoxy-, (8S-cis)-	20830813		10	
1-Naphthalenamine	134327		100	
2-Naphthalenamine (beta-Naphthylamine)	91598		1	
Naphthalenamine, N,N'-bis(2-chloroethyl)-	494031		100	
Naphthalene	91203		100	
Naphthalene, 2-chloro-	91587		5,000	
1,4-Naphthalenedione	130154		5,000	
2,7-Naphthalenedisulfonic acid, 3,3' [(3,3'-dimethyl-(1,1'-biphenyl)-4,4'-dryl)-bis(azo)] bis(5-amino-4-hydroxy)-tetrasodium salt	72571		10	
Naphthenic acid	1338245		100	
1,4-Naphthoquinone	130154		5,000	
alpha-Naphthylamine	134327		100	
beta-Naphthylamine (2-Naphthalenamine)	91598		1	
alpha-Naphthylthiourea	86884		100	P
Nickel [4]	7440020		100	
Nickel ammonium sulfate	15699180		100	
Nickel carbonyl	13463393	1	10	P
Nickel carbonyl Ni(CO)4, (T-4)-	13463393		10	P
Nickel chloride	7718549 37211055		100	
Nickel cyanide	557197		10	P
Nickel hydroxide	12054487		10	
Nickel nitrate	14216752		100	
Nickel sulfate	7786814		100	
Nicotine & salts	54115	100	100	P
Nicotine sulfate	65305	100/10,000	1	
Nitric acid	7697372	1,000	1,000	
Nitric acid, thallium(1+) salt	10102451		100	
Nitric oxide	10102439	100	10	P
p-Nitroaniline	100016		5,000	P
Nitrobenzene (I,T)	98953	10,000	1,000	
4-Nitrobiphenyl	92933		10	
Nitrocyclohexane	1122607	500	1	
Nitrogen dioxide	10102440 10544726	100	10	P
Nitrogen oxide	10102439		10	P
Nitroglycerine	55630		10	P
Nitrophenol (mixed): m-Nitrophenol o-Nitrophenol (2) p-Nitrophenol (4)	25154556 554847 88755 100027		100 100 100 100	
2-Nitropropane (I,T)	79469		10	
N-Nitrosodi-n-butylamine	924163		10	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
N-Nitrosodiethanolamine	1116547		1	
N-Nitrosodiethylamine	55185		1	
N-Nitrosodimethylamine	62759	1,000	10	P
N-Nitrosodiphenylamine	86306		100	
N-Nitroso-N-ethylurea	759739		1	
N-Nitroso-N-methylurea	684935		1	
N-Nitroso-N-methylurethane	615532		1	
N-Nitrosomethylvinylamine	4549400		10	P
N-Nitrosomorpholine	59892		1	
N-Nitrosopiperidine	100754		10	
N-Nitrosopyrrolidine	930552		1	
Nitrotoluene: m-Nitrotoluene o-Nitrotoluene p-Nitrotoluene	1321126 99081 88722 99990		1,000	
5-Nitro-o-toluidine	99558		100	
Norbromide	991424	100/10,000	1	
Octamethylpyrophosphoramide	152169		100	P
Organorhodium complex (PMN-82-147)	0	10/10,000	1	
Osmium tetroxide	20816120		1,000	P
Ouabain	630604	100/10,000	1	
7-Oxabicyclo[2,2,1]heptane-2,3-dicarboxylic acid	145733		1,000	P
Oxamyl	23135220	100/10,000	1	P
1,2-Oxathiolane, 2,2-dioxide	1120714		10	
2H-1,3,2-Oxazaphosphorin-2-amine, N,N bis (2-chloroethyl)tetrahydro-, 2-oxide	50180		10	
Oxetane, 3,3-bis(chloromethyl)-	78717	500	1	
Oxirane (I,T)	75218		10	
Oxiranecarboxyaldehyde	765344		10	
Oxirane, (chloromethyl)-	106898		100	
Oxydisulfoton	2497076	500	1	
Ozone	10028156	100	1	
Paraformaldehyde	30525894		1,000	
Paraldehyde	123637		1,000	
Paraquat	1910425	10/10,000	1	
Paraquat methosulfate	2074502	10/10,000	1	
Parathion	56382	100	10	P
Parathion-methyl	298000	100/10,000	100	
Paris green	12002038	500/10,000	100	
PCBs: Aroclor 1016 Aroclor 1221 Aroclor 1232 Aroclor 1242 Aroclor 1248 Aroclor 1254 Aroclor 1260	1336363 12674112 11104282 11141165 53469219 12672296 11097691 11096825		 1 1 1 1 1 1 1	
PCNB (Pentachloronitrobenzene)	82688		100	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Pentaborane	19624227	500	1	
Pentachlorobenzene	608935		10	
Pentachloroethane	76017		10	
Pentachlorophenol	87865		10	
Pentachloronitrobenzene (PCNB)	82688		100	
Pentadecylamine	2570265	100/10,000	1	
Paracetic acid	79210	500	1	
1,3-Pentadiene (I)	504609		100	
Perachloroethylene	127184		100	
Perchloromethylmercaptan	594423	500	100	
Phenacetin	62442		100	
Phenanthrene	85018		5,000	
Phenol	108952	500/10,000	1,000	
Phenol, 2-chloro-	95578		100	
Phenol, 4-chloro-3-methyl-	59507		5,000	
Phenol, 2-cyclohexyl-4,6-dinitro-	131895		100	P
Phenol, 2,4-dichloro-	120832		100	
Phenol, 2,6-dichloro-	87650		100	
Phenol, 4,4'-(1,2-diethyl-1,2-ethenediyl)bis-, (E)	56531		1	
Phenol, 2,4-dimethyl-	105679		100	
Phenol, 2,4-dinitro-	51285		10	P
Phenol, methyl-: m-Cresol o-Cresol p-Cresol	1319773 108394 95487 106445		1,000	
Phenol, 2-methyl-4,6-dinitro-and salts	534521		10	P
Phenol, 2,2'-methylenebis[3,4,6-trichloro-	70304		100	
Phenol, 2,2'-thiobis(4-chloro-6-methyl)-	4418660	100/10,000	1	
Phenol, 2-(1-methylpropyl)-4,6-dinitro	88857		1,000	P
Phenol, 3-(1-methylethyl)-, methylcarbamate	64006	500/10,000	1	
Phenol, 4-nitro-	100027		100	
Phenol, pentachloro-	87865		10	
Phenol, 2,3,4,6-tetrachloro-	58902		10	
Phenol, 2,4,5-trichloro-	95954		10	
Phenol, 2,4,6-trichloro-	88062		10	
Phenol, 2,4,6-trinitro-, ammonium salt	131748		10	P
Phenoxarsine, 10,10'-oxydi-	58366	500/10,000	1	
L-Phenylalanine, 4-[bis(2-chloroethyl)aminol]	148823		1	
Phenyl dichloroarsine	696286	500	1	
1,10-(1,2-Phenylene)pyrene	193395		100	
p-Phenylenediamine	106503		5,000	
Phenylhydrazine hydrochloride	59881	1,000/10,000	1	
Phenylmercury acetate	62384	500/10,000	100	P
Phenylsilatrane	2097190	100/10,000	1	
Phenylthiourea	103855	100/10,000	100	P
Phorate	298022	10	10	P
Phosacetim	4104147	100/10,000	1	
Phosfolan	947024	100/10,000	1	
Phosgene	75445	10	10	P

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Phosmet	732116	10/10,000	1	
Phosphamidon	13171216	100	1	
Phosphine	7803512	500	100	
Phosphorothioic acid, o,o-Dimethyl-s (2-Methylthio) ethyl ester	2587908	500	1	
Phosphorothioic acid, methyl-, o-ethyl o-(4-(methylthio)phenyl) ester	2703131	500	1	
Phosphorothioic acid, methyl-, s-(2-(bis(1-methylethyl)amino)ethyl o-ethyl ester	50782699	100	1	
Phosphorothioic acid, methyl-, 0-(4-nitrophenyl) o-phenyl ester	2665307	500	1	
Phosphoric acid	7664382		5,000	
Phosphoric acid, diethyl 4-nitrophenyl ester	311455		100	P
Phosphoric acid, dimethyl 4-(methylthio) phenyl ester	3254635	500	1	
Phosphoric acid, lead(2+) salt (2:3)	7446277	500	10	
Phosphorodithioic acid, O,O-diethyl S-[2 (ethylthio)ethyl]ester	298044		1	P
Phosphorodithioic acid, O,O-diethyl S-(ethylthio), methyl ester	298022		10	P
Phosphorodithioic acid, O,O-diethyl S-methyl ester	3288582		5,000	
Phosphorodithoic acid, O,O-dimethyl S-[2(methyl-amino)-2-oxoethyl] ester	60515		10	P
Phosphorofluondic acid, bis(1-methylethyl) ester	55914		100	P
Phosphorothioic acid, O,O-diethyl O-(4-nitrophenyl) ester	56382		10	P
Phosphorothioic acid, O,[4-[(dime-thylamino)sulfonyl]phenyl]O,O-dimethyl ester	52857		1,000	P
Phosphorothioic acid, O,O-dimethyl O-(4-nitrophenyl) ester	298000		100	P
Phosphorothioic acid, 0,0-diethyl 0 pyrazinyl ester	297972		100	P
Phosphorus	7723140	100	1	
Phosphorus oxychloride	10025873	500	1,000	
Phosphorous pentachloride	10026138	500	1	
Phosphorus pentasulfide (R)	1314803		100	
Phosphorus pentoxide	1314563	10	1	
Phosphorus trichloride	7719122	1,000	1,000	
Phthalic anhydride	85449		5,000	
Physostigmine	57476	100/10,000	1	P
Phosostigmine, salicylate (1:1)	57647	100/10,000	1	
2-Picoline	109068		5,000	
Picotoxin	124878	500/10,000	1	
Piperidine	110894	1,000	1	
Piperidine, 1-nitroso-	100754		10	
Pirimifos-ethyl	23505411	1,000	1	
Plumbane, tetraethyl-	78002		10	P

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Polychlorinated biphenyls (See PCBs or Aroclor)	1336363		1	
Potassium arsenate	7784410		1	
Potassium arsenite	10124502	500/10,000	1	
Potassium bichromate	7778509		10	
Potassium chromate	7789006		10	
Potassium cyanide	151508	100	10	P
Potassium hydroxide	1310583		1,000	
Potassium permanganate	7722647		100	
Potassium silver cyanide	506616	500	1	P
Promecarb	2631370	500/10,000	1	
Pronamide	23950585		5,000	
Propanal, 2-methyl-2-(methylthio)-, O-[(methylamino)carbonyl]oxime	116063		1	P
1-Propanamine (I,T)	107108		5,000	
1-Propanamine, N-propyl-	142847		5,000	
1-Propanamine, N-nitroso-N-propyl-	621647		10	
Propane, 1,2-dibromo-3-chloro	96128		1	
Propane, 2-nitro- (I,T)	79469		10	
1,3-Propane sultone	1120714		10	
Propane 1,2-dichloro-	78875		1,000	
Propanedinitrile	109773		1,000	
Propanenitrile	107120		10	P
Propanenitrile, 3-chloro-	542767		1,000	P
Propanenitrile, 2-hydroxy-2-methyl-	75865		10	P
Propane, 2,2'-oxybis[2-chloro-	108601		1,000	
1,2,3-Propanetnol, trinitrate- (R)	55630		10	P
1-Propanol, 2,3-dibromo-, phosphate (3:1)	126727		10	
1-Propanol, 2-methyl- (I,T)	78831		5,000	
2-Propanone (I)	67641		5,000	
2-Propanone, 1-bromo-	598312		1,000	P
Propargite	2312358		10	
Propargyl alcohol	107197		1,000	P
Propargyl bromide	106967	10	1	
2-Propenal	107028		1	P
2-Propenamide	79061		5,000	
1-Propene, 1,1,2,3,3,3-hexachloro-	1888717		1,000	
1-Propene, 1,3-dichloro-	542756		100	
2-Propenenitrile	107131		100	
2-Propenenitrile, 2-methyl- (I,T)	126987		1,000	
2-Propenoic acid (I)	79107		5,000	
2-Prepenoic acid, ethyl ester (I)	140885		1,000	
2-Prepenoic acid, 2-methyl-, ethyl ester	97632		1,000	
2-Prepenoic acid, 2-methyl-, methyl ester (I,T)	80626		1,000	
2-Propen-1-o1	107186		100	P
Propiolactone, beta-	57578	500	1	
Propionaldehyde	123386		1,000	
Propionic acid	79094		5,000	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Propionic acid, 2-(2,4,5-trichlorophenoxyl)-	93721		100	
Propionic anhydride	123626		5,000	
Propoxor (Baygon)	114261		100	
Propionitrile	107120	500	10	
Propionitrile, 3-chloro-	542767	1,000	1,000	
Propiophenone, 1, 4-amino phenyl	70699	100/10,000	1	
n-Propylamine	107108		5,000	
Propyl chloroformate	109615	500	1	
Propylene dichloride	78875		1,000	
Propylene oxide	75569	10,000	100	
1,2-Propylenimine	75558	10,000	1	P
2-Propyn-1-o1	107197		1,000	P
Prothoate	2275185	100/10,000	1	
Pyrene	129000	1,000/10,000	5,000	
Pyrethrins	121299 121211 8003347		1	
3,6-Pyridazinedione, 1,2-dihydro-	123331		5,000	
4-Pyridinamine	504245		1,000	P
Pyridine	110861		1,000	
Pyridine, 2-methyl-	109068		5,000	
Pyridine, 2-methyl-5-vinyl-	140761	500	1	
Pyridine, 4-amino-	504245	500/10,000	1,000	
Pyridine, 4-nitro-, 1-oxide	1124330	500/10,000	1	
Pyridine, 3-(1-methyl-2-pyrrolidinyl)-, (S)	54115		100	P
2,4-(1H,3H)-Pyrimidinedione, 5-[bis(2-chloroethyl)amino]-	66751		10	
4(1H)-Pyrimidinone, 2,3-dihydro-6-methyl-2-thioxo-	56042		10	
Pyriminil	53558251	100/10,000	1	
Pyrrolidine, 1-nitroso-	930552		1	
Quinoline	91225		5,000	
Quinone (p-Benzoquinone)	106514		10	
Quintobenzene	82688		100	
Reserpine	50555		5,000	
Resorcinol	108463		5,000	
Saccharin and salts	81072		100	
Salcomine	14167181	500/10,000	1	
Sarin	107448	10	1	
Safrole	94597		100	
Selenious acid	7783008	1,000/10,000	10	
Selenious acid, dithallium (1+) salt	12039520		1,000	P
Selenium [4]	7782492		100	
Selenium dioxide	7446084		10	
Selenium oxychloride	7791233	500	1	
Selenium sulfide (R,T)	7488564		10	
Selenourea	630104		1,000	P
Semicarbazide hydrochloride	563417	1,000/10,000	1	
L-Serine, diazoacetate (ester)	115026		1	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Silane, (4-aminobutyl)diethoxymethyl-	3037727	1,000	1	
Silver [4]	7440224		1,000	
Silver cyanide	506649		1	P
Silver nitrate	7761888		1	
Silvex (2,4,5-TP)	93721		100	
Sodium	7440235		10	
Sodium arsenate	7631892	1,000/10,000	1	
Sodium arsenite	7784465	500/10,000	1	
Sodium azide	26628228	500	1,000	P
Sodium bichromate	10588019		10	
Sodium bifluoride	1333831		100	
Sodium bisulfite	7631905		5,000	
Sodium cacodylate	124652	100/10,000	1	
Sodium chromate	7775113		10	
Sodium cyanide	143339	100	10	P
Sodium dodecylbenzenesulfonate	25155300		1,000	
Sodium fluoride	7681494		1,000	
Sodium fluoroacetate	62748	10/10,000	10	
Sodium hydrosulfide	16721805		5,000	
Sodium hydroxide	1310732		1,000	
Sodium hypochlorite	7681529 10022705		100	
Sodium methylate	124414		1,000	
Sodium nitrite	7632000		100	
Sodium prentachlorophenate	131522	100/10,000	1	
Sodium phosphate, dibasic	7558794 10039324 10140655		5,000	
Sodium phosphate, tribasic	7601549 7758294 7785844 10101890 10124568 10361894		5,000	
Sodium selenate	13410010	100/10,000	1	
Sodium selenite	10102188 7782823	100/10,000	100	
Sodium tellurite	10102202	500/10,000	1	
Stannane, acetoxytriphenyl	900958	500/10,000	1	
Streptozotocin	18883664		1	
Strontium chromate	7789062		10	
Strychnidin-10-one	57249		10	P
Strychnidin-10-one, 2,3-dimethoxy-	357573		100	P
Strychnine, & salts	572494	100/10,000	10	P
Strychnine sulfate	60413	100/10,000	1	
Styrene	100425		1,000	
Styrene oxide	96093		100	
Sulfotep	3689245	500	100	
Sulfoxide, 3-chloropropyl octyl	3569571	500	1	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Sulfur monochloride	12771083		1,000	
Sulfur dioxide	7446095	500	1	
Sulfur phosphide (R)	1314803		100	
Sulfur tetrafluoride	7783600	100	1	
Sulfur trioxide	7446119	100	1	
Sulfuric acid	7664939 8014957	1,000	1,000	
Sulfuric acid, dithallium (1+) salt	7446186 10031591		100	P
Sulfuric acid, dimethyl ester	77781		100	
Tabun	77816	10	1	
2,4,5-T acid	93765		1,000	
2,4,5-T amines	2008460 1319728 3813147 6369966 6369977		5,000	
Tellurium	13494809	500/10,000	1	
Tellurium hexafluoride	7783804	100	1	
2,4,5-T esters	93798 1928478 2545597 25168154 61792072		1,000	
2,4,5-T salts	13560991		1,000	
2,4,5-T	93765		1,000	
TDE (Dichloro diphenyl dichloroethane)	72548		1	
TEPP (Tetraethyl ester diphosphoric acid)	107493	100	10	
Terbufos	13071799	100	1	
1,2,4,5-Tetrachlorobenzene	95943		5,000	
2,3,7,8-Tetrachlorodibenzo-p-dioxin (TCDD)	1746016		1	
1,1,1,2-Tetrachloroethane	630206		100	
1,1,2,2-Tetrachloroethane	79345		100	
Tetrachloroethene	127184		100	
Tetrachloroethylene	127184		100	
2,3,4,6-Tetrachlorophenol	58902		10	
Tetraethyl lead	78002	100	10	P
Tetraethyl pyrophosphate	107493		10	P
Tetraethyldithiopyrophosphate	3689245		100	P
Tetraethyltin	597648	100	1	
Tetramethyllead	75741	100	1	
Tetrahydrofuran (I)	109999		1,000	
Tetranitromethane (R)	509148	500	10	P
Tetraphosphoric acid, hexaethyl ester	757584		100	P
Thallic oxide	1314325		100	P
Thallium [4]	7440280		1,000	
Thallium acetate	563688		100	
Thallium carbonate	6533739		100	
Thallium chloride	7791120		100	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Thallium nitrate	10102451		100	
Thallium oxide	1314325		100	P
Thallium selenite	12039520		1,000	P
Thallium sulfate	7446186 10031591	100/10,000	100	P
Thallous carbonate (Thallium (I) carbonate)	6533739	100/10,000	100	
Thallous chloride (Thallium (I) chloride)	7791120	100/10,000	100	
Thallous malonate (Thallium (I) malonate)	2757188	100/10,000	1	
Thallous sulfate (Thallium (I) sulfate)	7446186	100/10,000	100	P
Thioacetamide	62555		10	
Thiocarbazide	2231574	1,000/10,000	1	
Thiodiphosphoric acid, tetraethyl ester	3689245		100	P
Thiofanox	39196184	100/10,000	100	P
Thioimidodicarbonic diamide [(H2N)C(S)] 2NH	541537		100	P
Thiomethanol (I,T)	74931		100	
Thionazin	297972	500	100	
Thioperoxydicarbonic diamide [(H2N)C(S)] 2S2, tetra-methyl-	137268		10	
Thiophenol	108985	500	100	P
Thiosemicarbazide	79196	100/10,000	100	P
Thiourea	62566		10	
Thiourea, (2-chlorophenyl)-	5344821	100/10,000	100	P
Thiourea, (2-methylphenyl)-	614788	500/10,000	1	
Thiourea, 1-naphthalenyl-	86884		100	P
Thiourea, phenyl-	103855		100	P
Thiram	137268		10	
Titanium tetrachloride	7550450	100	1,000	
Toluene	108883		1,000	
Toluenediamine	95807 496720 823405 25376458		10	
Toluene diisocyanate (R,T)	584849 91087 26471625	500	100	
o-Toluidine	95534		100	
p-Toluidine	106490		100	
o-Toluidine hydrochloride	636215		100	
Toxaphene	8001352		1	P
2,4,5-TP acid	93721		100	
2,4,5-TP acid esters	32534955		100	
1H-1,2,4-Triazol-3-amine	61825		10	
Trans-1,4-dichlorobutene	110576	500	1	
Triamiphos	1031476	500/10,000	1	
Triazofos	24017478	500	1	
Trichloroacetyl chloride	76028	500	1	
Trichlorfon	52686		100	
1,2,4-Trichlorobenzene	120821		100	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
1,1,1-Trichloroethane	71556		1,000	
1,1,2-Trichloroethane	79005		100	
Trichloroethene	79016		100	
Trichloroethylene	79016		100	
Trichloroethylsilane	115219	500	1	
Trichloronate	327980	500	1	
Trichloromethanesulfenyl chloride	594423		100	P
Trichloromonofluoromethane	75694		5,000	
Trichlorophenol: 2,3,4-Trichlorophenol 2,3,5-Trichlorophenol 2,3,6-Trichlorophenol 2,4,5-Trichlorophenol 2,4,6-Trichlorophenol 3,4,5-Trichlorophenol	21567822 15950660 933788 933755 95954 88062 609198		10 10 10 	
Trichlorophenylsilane	98135	500	1	
Trichloro(chloromethyl)silane	1558254	100	1	
Trichloro(dichlorophenyl)silane	27137855	500	1	
Triethanolamine dodecylbenzene-sulfonate	27323417		1,000	
Triethoxysilane	998301	500	1	
Trifluralin	1582098		10	
Triethylamine	121448		5,000	
Trimethylamine	75503		100	
Trimethylchlorsilane	75774	1,000	1	
2,2,4-Trimethylpentane	540841		1,000	
Trimethylolpropane phosphite	824113	100/10,000	1	
Trimethyiltin chloride	1066451	500/10,000	1	
1,3,5-Trinitrobenzene (R,T)	99354		10	
1,3,5-Trioxane, 2,4,6-trimethyl-	123637		1,000	
Triphenyltin chloride	639587	500/10,000	1	
Tris(2-chloroethyl)amine	555771	100	1	
Tris(2,3-dibromopropyl) phosphate	126727		10	
Trypan blue	72571		10	
Unlisted Hazardous Wastes Characteristic of **Ignitability**	NA		100	
Unlisted Hazardous Wastes Characteristic of **Corrosivity**	NA		100	
Unlisted Hazardous Wastes Characteristic of **Reactivity**	NA		100	
Unlisted Hazardous Wastes Characteristic of **Toxicity:**				
Arsenic			1	
Barium			1000	
Benzene			10	
Cadmium			10	
Carbon Tetrachloride			10	
Chlordane			1	
Chlorobenzene			100	
Chloroform			10	

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Chromium			10	
o-Cresol			100	
m-Cresol			100	
p-Cresol			100	
Cresol			100	
2,4-D (Dichlorophenoxyacetic acid)			100	
1,4-Dichlorobenzene			100	
1,2-Dichloroethane			100	
1,1-Dichloroethylene			100	
2,4-Dinitrotoluene			10	
Endrin			1	
Heptachlor (and epoxide)			1	
Hexachlorobenzene			10	
Hexachlorobutadiene			1	
Hexachloroethane			100	
Lead			10	
Lindane			1	
Mercury			1	
Methoxychlor			1	
Methyl ethyl ketone			5,000	
Nitrobenzene			1,000	
Pentachlorophenol			10	
Pyridine			1,000	
Selenium			10	
Silver			1	
Tetrachloroethylene			100	
Toxaphene			1	
Trichloroethylene			100	
2,4,5 Trichlorophenol			10	
2,4,5-TP			100	
Vinyl chloride			1	
Uracil mustard	66751		10	
Uranyl acetate	541093		100	
Uranyl nitrate	10102064 36478769		100	
Urea, N-ethyl-N-nitroso	759739		1	
Urea, N-methyl-N-nitroso	684935		1	
Urethane (Carbamic acid ethyl ester)	51796		100	
Valinomycin	2001958	1,000/10,000	1	
Vanadic acid, ammonium salt	7803556		1,000	P
Vanadic oxide V_20_5	1314621		1,000	P
Vanadic pentoxide	1314621		1,000	P
Vanadium pentoxide	1314621	100/10,000	1,000	
Vanadyl sulfate	27774136		1,000	
Vinyl chloride	75014		1	
Vinyl acetate	108054		5,000	
Vinyl acetate monomer	108054	1,000	5,000	
Vinylamine, N-methyl-N-nitroso-	4549400		10	P

Hazardous Substance/Material	CAS No.[1]	Threshold Planning Quantity (Pounds)[2]	RQ (Pounds)[3]	HW Designator
Vinyl bromide	593602		100	
Vinylidene chloride	75354		100	
Warfarin, & salts, when present at concentrations greater than 0.3%	81812	500/10,000	100	P
Warfarin sodium	129066	100/10,000	100	
Xylene (mixed): m-Benzene, dimethyl o-Benzene, dimethyl p-Benzene, dimethyl	1330207 108383 95476 106423		100 1,000 1,000 100	
Xylenol	1300716		1,000	
Xylylene dichloride	28347139	100/10,000	1	
Yohimban-16-carboxylic acid, 11,17 dimethoxy-18-[(3,4,5-trimethoxy-benzoyl)oxy]-, methyl ester (3-beta, 16-beta,17-alpha,18-beta,20-alpha)-	50555		5,000	
Zinc [4]	7440666		1,000	
Zinc acetate	557346		1,000	
Zinc ammonium chloride	52628258 14639975 14639986		1,000	
Zinc borate	1332076		1,000	
Zinc bromide	7699458		1,000	
Zinc carbonate	3486359		1,000	
Zinc chloride	7646857		1,000	
Zinc cyanide	557211		10	P
Zinc, dichloro(4,4-dimethyl-5((((methyl-amino)carbonyl)oxy)imino)pentaenitrile)-,(t-4)-	58270089	100/10,000	1	
Zinc fluoride	7783495		1,000	
Zinc formate	557415		1,000	
Zinc hydrosulfite	7779864		1,000	
Zinc nitrate	7779886		1,000	
Zinc phenosulfonate	127822		5,000	
Zinc phosphide	1314847	500	100	P
Zinc phosphide Zn_3P_2, when present at concentrations greater than 10%	1314847		100	P
Zinc silicofluoride	16871719		5,000	
Zinc sulfate	7733020		1,000	
Zirconium nitrate	13746899		5,000	
Zirconium potassium fluoride	16923958		1,000	
Zirconium sulfate	14644612		5,000	
Zirconium tetrachloride	10026116		5,000	

Notes:

1. Chemical Abstract Service (CAS) Registry Number.

2. Quantity in storage above which Environmental Executive Agent must be notified (See Chapter 5, "Hazardous Materials").

3. Reportable quantity release which requires notification (See Chapter 18, "Spill Prevention and Response Planning").

4. No reporting of releases of this hazardous substance is required if the diameter of the pieces of the solid metal released is ≥ 100 micrometers (0.004 inches).

5. The RQ for asbestos is limited to friable forms only.

6. Includes mono- and di-ethers of ethylene glycol, diethylene glycol, and triethylene glycol R-(OCH2CH2)n-OR'. Where n = 1, 2, or 3; R = alkyl C7 or less; or R = phenyl or akyl substituted phenyl; R' = H or alkyl C7 or less; or OR' consisting of carboxylic acid ester, sulfate, phosphate, nitrate, or sulfonate.

** Indicates that no RQ is being assigned to the generic or broad class.

APB1. APPENDIX B.1 – HAZARDOUS PROPERTIES

H1	**EXPLOSIVE**: substances and preparations which may explode under the effect of flame or which are more sensitive to shocks or friction than dinitrobenzene.
H2	**OXIDIZING**: substances and preparations which exhibit highly exothermic reactions when in contact with other substances, particularly flammable substances.
H3-A	**HIGHLY FLAMMABLE**: - liquid substances and preparations having a flash point < 21 ^{o}C (including extremely flammable liquids), or - substances and preparations which may become hot and finally catch fire in contact with air at ambient temperature without any application of energy, or - solid substances and preparations which may readily catch fire
H3-B	**FLAMMABLE**: liquid substances and preparations having a flash point ≥ 21 ^{o}C and ≤ 55 ^{o}C.
H4	**IRRITANT**: non-corrosive substances and preparations which, through immediate, prolonged or repeated contact with skin or mucus membrane, can cause inflammation.
H5	**HARMFUL**; substances and preparations which, if they are inhaled or ingested or if they penetrate the skin, may involve limited health risks.
H6	**TOXIC**; substances and preparations (including very toxic substances and preparations) which, if they are inhaled or ingested or if they penetrate the skin, may involve serious, acute or chronic health risks and even death.
H7	**CARCINOGENIC**: substances and preparations which, if they are inhaled or ingested or if they penetrate the skin, may induce cancer or increase its incidence.
H8	**CORROSIVE**: substances and preparations which may destroy living tissue on contact.
H9	**INFECTIOUS**: substances containing viable micro-organisms or their toxins which are known or reliably believed to cause disease in man or other living organisms.
H10	**TERATOGENIC**: substances and preparations which, if they are inhaled or ingested or if they penetrate the skin, may induce non-hereditary congenital malformations or increase their incidence.
H11	**MUTAGENIC**: substances and preparations which, if they are inhaled or ingested or if they penetrate the skin, may induce hereditary genetic defects or increase their incidence.
H12	Substances and preparations which release toxic or very toxic gases in contact with water, air or an acid.
H13	**SENSITIZING**: substances and preparations which, if they are inhaled or if they penetrate the skin, are capable of eliciting a reaction of hypersensitization such that on further exposure to the substance or preparation, characteristic adverse effects are produced.
H14	**ECOTOXIC**: substances and preparations which present or may present immediate or delayed risks for one or more sectors of the environment.
H15	Substances and preparations capable by any means, after disposal, of yielding another substance (e.g. a leachate), which possesses any of the characteristics listed

	above.

APB1.2 THRESHOLDS FOR CERTAIN HAZARDOUS PROPERTIES

Wastes are classified as hazardous if they have one or more of the following characteristics:

(a)	Flash point ≤ 55 ^{o}C
(b)	One or more substances classified as very toxic at a total concentration ≥ 0.1 %
(c)	One or more substances classified as toxic at a total concentration ≥ 3 %
(d)	One or more substances classified as harmful at a total concentration ≥ 25 %
(e)	One or more corrosive substances classified as R35 at a total concentration ≥ 1 %
(f)	One or more corrosive substances classified as R34 at a total concentration ≥ 5 %
(g)	One or more irritant substances classified as R41 at a total concentration ≥ 10 %
(h)	One or more irritant substances classified as R36, R37, R38 at a total concentration ≥ 20 %
(i)	One substance known to be carcinogenic of category 1 or 2 at a concentration ≥ 0.1 %
(j)	One substance known to be carcinogenic of category 3 at a concentration ≥ 1 %
(k)	One substance toxic for reproduction of category 1 or 2 classified as R60, R61 at a concentration ≥ 0.5 %
(l)	One substance toxic for reproduction of category 3 classified as R62, R63 at a concentration ≥ 5 %
(m)	One mutagenic substance of category 1 or 2 classified as R46 at a concentration ≥ 0.1 %
(n)	One mutagenic substance of category 3 classified as R40 at a concentration ≥ 1 %

Category	Description
D1	Open ground storage (e.g., landfills)
D2	Treatment on land (e.g., liquid wastes or sludge biodegradation on soils)
D3	Deep injection (e.g., waste injection into wells, salt domes, or natural geological faults)
D4	Surface impoundments (e.g., discharge of liquid waste or sludge into wells, ponds, lagoons, etc.)
D5	Specially equipped landfills (e.g., placing waste in separate airtight cells, covered and isolated from each other and from the environment)
D6	Discharge of solid waste into water, except for immersion
D7	Immersion, including landfills under the sea subsoil
D8	Biological treatment not specified in this Appendix, resulting in compounds and mixtures eliminated according to one of the processes included in Categories D1 to D12
D9	Physical and chemical treatment not specified in this Appendix, resulting in compounds and mixtures eliminated according to one of the processes included in Categories D1 to D12 (e.g., evaporation, drying processes, cementation, etc.)
D10	Incineration on land
D11	Incineration at sea – This activity is not permitted.
D12	Permanent storage (e.g., emplacement of containers in a mine, etc.)
D13	Preliminary collection prior to one of the processes listed in Categories D1 to D 12; to be used in the cases when a different D-code suitable for the activity is not available (D13 can be used to cover preliminary processing of waste such as sorting, fragmentation, compaction, pelletizing, drying, shredding, conditioning
D14	Preliminary reconditioning prior to one of the processes listed in Categories D1 to D12
D15	Preliminary storage prior to one of the processes listed in D1 to D14 (except for temporary storage before waste collection at the site where it is produced)

Category	Description
R1	Main use as fuel or as other means of producing energy. In case of incineration, the activity is classified as recovery "R" provided that the incinerator meets energy efficiency targets of 0.6 for plants installed before year 2009, or of 0.65 for plants installed after year 2009
R2	Solvent regeneration and recovery
R3	Organic substance recycling and recovery not used as solvents (including composting processes and other biological transformations). Includes gasification and pyrolysis where the substances are used as chemicals
R4	Metal or metallic component recycling or recovery
R5	Other inorganic substance recycling and recovery. It includes the cleaning activities resulting in the re-use of soil, and the recycling of inorganic construction material
R6	Acid or base regeneration
R7	Recovery of products useful to capture pollutants
R8	Recovery of products from catalysts
R9	Oil regeneration and other re-utilization of oils
R10	Spreading on soil for the benefit of agriculture or ecology
R11	Use of wastes from one of the processes in R1 to R10
R12	Exchanging wastes in order to subject them to one of the processes in R1 to R11; to be used in the cases when a different R-code suitable for the activity is not available (R12 can be used to cover preliminary processing of waste such as sorting, fragmentation, compaction, pelletizing, drying, shredding, conditioning
R13	Waste stockpiling in order to subject them to one of the processes in R1 to R12 (except for temporary storage before waste collection at the site where it is produced)

APC1. APPENDIX C

DETERMINATION OF WORST CASE DISCHARGE PLANNING VOLUME

APC1.1. This appendix provides criteria to determine, on an installation-specific basis, the extent of a worst-case discharge (WCD).

APC1.2. This Appendix provides criteria to determine the volume of oil or hazardous substance to be used in planning for a WCD. Installations should calculate both WCD volumes that apply to the installation's design and operation and use the larger volume as the WCD planning volume.

APC1.3. For installations transferring oil to and from vessels with tank capacities of 10,500 gallons (250 barrels) or more, the WCD planning volume is calculated as follows:

APC1.3.1. Where applicable, the loss of the entire capacity of all in-line and break out tank(s) needed for the continuous operation of the pipelines used for the purposes of handling or transporting oil, in bulk, to or from a vessel regardless of the presence of secondary containment; plus

APC1.3.2. The discharge from all piping carrying oil between the marine transfer manifold and the valve or manifold adjacent to the POL storage container. The discharge from each pipe is calculated as follows: The maximum time to discover the release from the pipe in hours, plus the maximum time to shut down flow from the pipe in hours (based on historic discharge data or the best estimate in the absence of historic discharge data for the installation) multiplied by the maximum flow rate expressed in gallons per hour (based on the maximum relief valve setting or maximum system pressure when relief valves are not provided) plus the total line drainage volume expressed in gallons for the pipe between the marine transfer manifold and the valve or manifold adjacent to the POL storage container.

APC1.4. For installations with POL Storage Containers:

APC1.4.1. Single POL Storage Container Facilities. For facilities containing only one aboveground oil or hazardous substance storage container, the WCD planning volume equals the capacity of the oil or hazardous substance storage container. If adequate secondary containment (sufficiently large to contain the capacity of the above ground oil or hazardous substance storage container plus sufficient freeboard to allow for precipitation) exists for the oil storage container, multiply the capacity of the container by 0.8.

APC1.4.2. Multiple POL Storage Container Facilities

APC1.4.2.1. Facilities having no secondary containment. If none of the above ground storage containers at the facility have adequate secondary containment, the worst case planning volume equals the total above ground oil and hazardous substance storage capacity at the facility.

APC1.4.2.2. Facilities having complete secondary containment. If every above ground storage container at the facility has adequate secondary containment, the WCD planning volume equals the capacity of the largest single above ground oil or hazardous substance storage container.

APC1.4.2.3. Facilities having partial secondary containment. If some, but not all above ground storage containers at the facility have adequate secondary containment, the WCD planning volume equals the sum of:

APC1.4.2.3.1. The total capacity of the above ground oil and hazardous substance storage container that lacks adequate secondary containment; plus

APC1.4.2.3.2. The capacity of the largest single above ground oil or hazardous substance storage container that has adequate secondary containment.

APC1.4.3. For purposes of this Appendix, the term “adequate secondary containment” means an impervious containment system such as a dike, berm, containment curb, drainage system or other device that will prevent the escape of spilled material into the surrounding soil.